

Krishna's
B.Sc. Objective



GEOMETRY and VECTOR CALCULUS

(For I year B.Sc. Students of all Colleges affiliated to Universities in U.P.)

As Per U.P. Unified Syllabus

By

Solved Question Papers

2018 & 2019

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KRISHNA HOUSE, 11, Shivaji Road, Meerut-250 001 (U.P.), India

Syllabus

GEOMETRY and VECTOR CALCULUS

B.Sc. I Year; III Paper

U.P. UNIFIED Syllabus (w.e.f. 2011-12)

GEOMETRY

Unit 1

General equation of second degree, Tracing of conics, System of conics, Confocal conics, Polar equation of a conic and its properties.

Unit 2

Three dimensional system of co-ordinates, Projection and direction cosines, Plane, Straight line.

Unit 3

Sphere, cone and cylinder.

Unit 4

Central conicoids, Reduction of general equation of second degree, Tangent plane and normal to a conicoid, Pole and polar, Conjugate diameters, Generating lines, Plane sections.

VECTOR CALCULUS

Unit 5

Vector differentiation and integration, Gradient, divergence and curl and their properties, Line integrals, Theorems of Gauss, Green and Stokes and problems based on these.

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B.Sc. Objective Mathematics

Book-3

GEOMETRY AND VECTOR CALCULUS

Unit-I

Chapter -1 : The General Equation of the Second Degree (Tracing of the Conics)

Chapter -2 : General Conics, Contacts and Confocals

Chapter -3 : Polar Equation of a Conic

Chapter -4 : Systems of Co-ordinates

Unit-II

Chapter -5 : Directions Cosines and Projections

Chapter -6 : The Plane

Chapter -7 : The Straight Line

Unit-III

Chapter -8 : The Sphere

Chapter -9 : The Cone

Chapter-10: The Cylinder

Unit-IV

Chapter -11: The Central Conicoids

Chapter -12: The Parabooids

Chapter -13: Generating Lines

Chapter -14: The Plane Sections of Conicoids

Chapter -15: Reduction of General Equations of Second Degree

Chapter -16: Vector Calculus

1

The General Equation of the Second Degree (Tracing of the Conics)

CONIC SECTIONS

1. A conic section or conic is the locus of a point which moves so that its distance from a fixed point (focus) is in a constant ratio to its perpendicular distance from a fixed straight line (directrix). The constant ratio is called the eccentricity and is denoted by e .
2. If the focus does not lie upon the directrix, the conic is a parabola, an ellipse or a hyperbola according as $e = , < \text{ or } > 1$. The circle is a special case of an ellipse when $e = 0$ and the focus is the centre of the circle and directrix is at an infinite distance from the focus. If the focus lies on the directrix, the conic section is a pair of straight line.
3. The general equation of the second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$$
 in every case represents a conic section.

CENTRE

1. The centre of a conic section is a point which bisects all those chords of the conic that pass through it.
2. If the equation of the conic is $ax^2 + 2hxy + by^2 + c = 0$ then the origin is the centre of the conic.
3. The general equation of the second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent a conic with centre at the origin only if $f = g = 0$ i.e. only if the first degree terms are absent from the equation of the conic.

CENTRE OF A CONIC

1. The general equation of the conic is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and its discriminant is

$$\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2$$

Then equation of conic with respect to centre of origin is

Case I: If $ab - h^2 = 0$ i.e. $h^2 = ab$ i.e. if the second degree terms in the general equation of the conic are not in a perfect square then conic is either a pair of intersecting straight lines, or a circle, or an ellipse or a hyperbola.

Case II: If $ab - h^2 \neq 0$ and $\Delta = 0$ then the conic always represents a pair of straight lines.

Case III: If $ab - h^2 = 0$ and $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$ then the conic always represents a pair of parallel straight lines.

Case IV: If $ab - h^2 = 0$ and $hf - bg$ or $gh - af$ is not zero, the conic represents a parabola which has no centre.

TO FIND THE COORDINATES OF THE CENTRE

The general equation of the conic is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0.$$

Its centre is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$. Now consider

$F(x, y) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then $\frac{\partial F}{\partial x} = 2(ax + hy + g)$ and $\frac{\partial F}{\partial y} = 2(hx + by + f)$. The centre of the conic $F(x, y) = 0$ is obtained by solving the equation $\frac{\partial F}{\partial x} = 0$ and $\frac{\partial F}{\partial y} = 0$ i.e. $ax + hy + g = 0$ and $hx + by + f = 0$.

ASYMPTOTES

The general equation of the conic is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

and its discriminant is

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \text{ then}$$

Case I: The equation of the asymptotes of the general conic is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c - \frac{\Delta}{ab - h^2} = 0$$

Case II: The equation of the hyperbolic conjugate of the general conic is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c - \frac{2\Delta}{ab - h^2} = 0$$

Case III: If (x, y) be the coordinates of the centre of the hyperbola

$$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ then}$$

(A) The equation of the asymptotes is $f(x, y) = f(x_1, y_1)$.

(B) The equation of the conjugate hyperbola is $f(x, y) = 2f(x_1, y_1)$.

NATURE OF A CONIC

The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents

1. A pair of straight lines if $\Delta = 0$.
2. A pair of parallel straight lines if $\Delta = 0$ and $h^2 = ab$.
3. A circle if $a = b$ and $h = 0$.
4. A parabola if $h^2 = ab$ and $\Delta \neq 0$.
5. An ellipse if $h^2 < ab$ and $\Delta \neq 0$.
6. A hyperbola if $h^2 > ab$ and $\Delta \neq 0$.
7. A rectangular hyperbola if $\Delta \neq 0$, $h^2 > ab$ and $a + b = 0$.

LENGTH OF SEMI-AXES

1. The length of semi-axes of the conic $Ax^2 + 2Hxy + By^2 = 1$

is $\left(A - \frac{1}{r^2}\right)\left(B - \frac{1}{r^2}\right) = H^2$, it gives two roots of r^2 say r_1^2 and r_2^2 .

Case I: If both r_1^2 and r_2^2 are positive i.e. $AB > H^2$ then the conic is an ellipse. If $r_1 > r_2$ then

Semi-major axes $= r_1$

Semi-minor axes $= r_2$

Case II: If r_1^2 is positive and r_2^2 is negative i.e. $AB < H^2$ then the conic is hyperbola.

Semi-transverse axes $= r_1$

Semi-conjugate axes $= \sqrt{|r_2^2|}$

2. The eccentricity of the conic is e then

$$e = \sqrt{1 - \frac{r_2^2}{r_1^2}}.$$

3. The semi-latus rectum is $\frac{|r_2^2|}{r_1}$.

4. If (x_1, y_1) be the centre of the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then the co-ordinate of foci are

$(x_1 \pm er_1 \cos \theta, y_1 \pm er_1 \sin \theta)$ where

$er_1 = \sqrt{r_1^2 - r_2^2}$ and θ be the angle which the major axes (ellipse) or transverse axes (hyperbola) makes with x-axis.

5. The equation of the directrix of the conic is $(x - x_1) \cos \theta + (y - y_1) \sin \theta = \pm \frac{r_1}{e}$.
6. The distance of the directrix from the centre C is $\frac{r_1}{e}$.

EXERCISE

MULTIPLE CHOICE QUESTIONS

- The co-ordinate of the centre of the conic $ax^2 + 2hxy + by^2 + c = 0$ are
 - (a, b, c)
 - $(-a, -b, -c)$
 - (a, h, c)
 - $(0, 0, 0)$
- The conic $2x^2 + xy - y^2 - x + 5y + 6 = 0$ represents
 - a parabola
 - a circle
 - a hyperbola
 - an ellipse
- The general equation of the second degree $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents a conic with centre at origin only when
 - $f = g = h = 0$
 - $f = g = 0$
 - $f = 0$
 - $a = b = c = 0$
- The centre of the conic $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$ is
 - $(2, 3)$
 - $(3, 2)$
 - $(14, 11)$
 - $(14, -4)$
- The asymptotes of the hyperbola $xy = hx + ky$ are
 - $x = k, y = h$
 - $x = h, y = k$
 - $x = h + k = y$
 - $x = 0, y = 0$
- The eccentricity e of the ellipse is
 - $\sqrt{1 + \frac{b^2}{a^2}}$
 - $\sqrt{1 + \frac{b}{a}}$
 - $\sqrt{1 - \frac{b^2}{a^2}}$
 - $\sqrt{1 - \frac{b}{a}}$
- If the focus does not lie upon the directrix, the conic is circle when
 - $e > 1$
 - $e < 1$
 - $e = 1$
 - $e = 0$
- If the equation of hyperbola and asymptotes are $H = 0$ and $A = 0$ respectively then the equation of conjugate hyperbola is
 - $2A - H = 0$
 - $A + H = 0$
 - $A - 2H = 0$
 - $A - H = 0$
- $2x^2 - 3xy - 2y^2 + 2x + y + 1 = 0$ always represents
 - an ellipse
 - a hyperbola
 - a parabola
 - a rectangular hyperbola
- If the equation of hyperbola is $2x^2 + xy - y^2 - x + 5y + 6 = 0$ and its asymptote is $y^2 - xy - 2x^2 - 5y + x + 6 = 0$ then equation of conjugate hyperbola is
 - $y^2 - xy - 2x^2 - 5y + x + 12 = 0$
 - $y^2 - xy - 2x^2 + 5y + x + 18 = 0$
 - $y^2 - xy - 2x^2 - 5y + x = 0$
 - $y^2 - xy - 2x^2 - 5y + x + 6 = 0$
- If the roots r_1^2 and r_2^2 of the quadratic $\left(A - \frac{1}{r^2}\right)\left(B - \frac{1}{r^2}\right) = H^2$ in r^2 are of the signs then the conic $Ax^2 + 2Hxy + By^2 = 1$ is
 - parabola
 - hyperbola
 - ellipse
 - circle
- In two dimensional geometry, the general equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ always represents a
 - coincide
 - two planes
 - conic
 - central coincide
- The co-ordinate of the centre of the conic $x^2 - 4xy - 2y^2 + 10x + 4y = 0$ are
 - $(1, 2)$
 - $(-1, 2)$
 - $(1, -2)$
 - $(-1, -2)$
- The conic $ax^2 + 2hxy + by^2 = 1$ is an ellipse of
 - $h^2 < ab$
 - $h^2 > ab$
 - $h^2 = ab$
 - $h = 0$

15. The conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight line only when
 (a) $\Delta = 0, h^2 = ab$ (b) $\Delta \neq 0, h^2 = ab$
 (c) $\Delta = 0, h^2 \neq ab$ (d) $\Delta \neq 0, h^2 \neq ab$
16. If the conic $x^2 + 2xy + (1 + \lambda)y^2 + 2\lambda y - 1 = 0$ represents an ellipse then the value of λ is
 (a) $\lambda = 0$ (b) $\lambda < 0$
 (c) $\lambda > 0$ (d) any real number
17. The length of the transverse axis of the hyperbola $x^2 - 3xy + y^2 + 10x - 10y + 21 = 0$ is
 (a) $2\sqrt{2}$ (b) 2
 (c) $\sqrt{2}$ (d) $2\sqrt{3}$
18. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a circle if
 (a) $a \neq b, h = 0$ (b) $a = b, h \neq 0$
 (c) $a = b, h = 0$ (d) $a \neq b, h \neq 0$
19. The centre of the conic $9x^2 - 24xy + 16y^2 - 18x - 101y + 19 = 0$ is
 (a) $(3, -5)$ (b) $(5, -3)$
 (c) $(-3, -5)$ (d) no centre
20. If in a conic, the focus lies on the directrix, the conic rectum formed
 (a) an ellipse
 (b) a parabola
 (c) a pair of straight lines
 (d) a hyperbola
21. The centre of the conic $ax^2 + 2hxy + by^2 = 1$ is
 (a) $(0, 0)$ (b) (a, b)
 (c) $(h, 0)$ (d) $(0, h)$
22. The eccentricity of the conic $x^2 - 3xy + y^2 - 10x - 10y + 21 = 0$ is
 (a) $\sqrt{\frac{5}{6}}$ (b) $\sqrt{\frac{6}{5}}$
 (c) $\sqrt{\frac{10}{3}}$ (d) $\sqrt{\frac{21}{10}}$
23. The minor axis of the ellipse $x^2 + xy + y^2 + x + y - 1 = 0$ is
 (a) $3x + 3y + 2 = 0$ (b) $x - y = 2$
 (c) $3x + y - 2 = 0$ (d) $x - y = 0$
24. If $h^2 = ab$ for the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then it represents
 (a) only parabola
 (b) only straight lines
 (c) a parabola or a pair of straight lines
 (d) none of these
25. The conic $\sqrt{ax} + \sqrt{by} = 1$ represents a
 (a) hyperbola (b) ellipse
 (c) circle (d) parabola
26. The length of the major axis of the ellipse $x^2 + xy + y^2 + x + y - 1 = 0$ is
 (a) $\frac{2\sqrt{6}}{3}$ (b) $\frac{3\sqrt{6}}{2}$
 (c) $\frac{4\sqrt{6}}{3}$ (d) $\frac{3\sqrt{6}}{4}$
27. The conic $5x^2 - 2xy + 5y^2 - 10y - 7 = 0$ represents **[Meerut 2009, 10 B, Avadh 2014]**
 (a) parabola (b) ellipse
 (c) hyperbola (d) straight lines
28. The equation of the transverse axis of the hyperbola $x^2 + 4xy + y^2 = 4$ is
 (a) $x - y = 0$ (b) $x - y = 1$
 (c) $x + y = 2$ (d) not exist
29. If the roots r_1^2 and r_2^2 of the quadratic $\left(A - \frac{1}{r^2}\right)\left(B - \frac{1}{r^2}\right) = H^2$ in r^2 are of opposite signs, then the conic $Ax^2 + 2Hxy + By^2 = 1$ is a
 (a) parabola (b) ellipse
 (c) hyperbola (d) circle
30. The conic $x^2 + xy + y^2 + x + y = 4$ is
 (a) a parabola (b) an ellipse
 (c) a hyperbola (d) none of these

31. The length of the conjugate axis of the conic $7x^2 + 52xy - 32y^2 + 125 = 0$ is
 (a) 5 (b) 3
 (c) 1 (d) 7
32. The intersection point of the conic $x^2 + y^2 + 2x + 3y + 2 = 0$ with y-axis are
 (a) (0, 2) (0, -1) (b) (0, -1) (0, -2)
 (c) (0, 0) (-1, -2) (d) none of these
33. If the equation $ax^2 - y^2 = 0$ represents a pair of perpendicular straight lines then a is
 (a) -1 (b) $\sqrt{2}$
 (c) $\sqrt{3}$ (d) 1
34. If the length of the semi-axes of an ellipse are 3 and 5 then its eccentricity is
 (a) $\frac{1}{5}$ (b) $\frac{2}{5}$
 (c) $\frac{3}{5}$ (d) $\frac{4}{5}$
35. The eccentricity of the conic $5x^2 - 6xy + 5y^2 - 8 = 0$ is
 (a) $\frac{\sqrt{3}}{2}$ (b) $\sqrt{\frac{1}{2}}$
 (c) $\frac{2}{\sqrt{3}}$ (d) $\sqrt{\frac{2}{3}}$
36. If the equation $ax^2 - 17xy + 6y^2 + 23x - 2y = 20$ represents a pair of straight lines then a is
 (a) 17 (b) 6
 (c) 7 (d) 2
37. The equation $x^2 + y^2 - 6x - 8y = 25$ represents
 (a) circle (b) parabola
 (c) hyperbola (d) ellipse
38. If the curve $px^2 - 6xy + 5y^2 + 22x - 26y + 29 = 0$ represents an ellipse then p is
 (a) $\geq \frac{3}{5}$ (b) $> \frac{3}{5}$
 (c) $< \frac{3}{5}$ (d) $= \frac{3}{5}$
39. The intersect point of the conic $xy - x^2 = 3y$ with co-ordinate axes are
 (a) (1, 3) only (b) (0, 0) (1, 3)
 (c) (0, 3) only (d) (0, 0) only
40. The latus rectum of the conic whose length of semi-axes are given by $r_1^2 = 25$ and $r_2^2 = 4$ is
 (a) $\frac{8}{5}$ (b) $\frac{5}{8}$
 (c) $\frac{2}{5}$ (d) $\frac{4}{25}$
41. The eccentricity of the conic $x^2 - 3xy + y^2 + 1 = 0$ is
 (a) $\frac{1}{\sqrt{5}}$ (b) $\sqrt{\frac{6}{5}}$
 (c) $\sqrt{6}$ (d) $\sqrt{\frac{5}{6}}$
42. If for a conic $r_1^2 = -16$ and $r_2^2 = 25$ then its semi latus rectum is
 (a) $\frac{32}{5}$ (b) $\frac{16}{5}$
 (c) $\frac{48}{5}$ (d) $\frac{64}{5}$
43. If $4x^2 + 12xy + 9y^2 + 6x + 9y + 2 = 0$ represent two straight lines then angle between them is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{3}$ (d) 0°
44. Conjugate hyperbola + hyperbola is equal to
 (a) asymptote (b) 2 asymptotes
 (c) 4 asymptotes (d) 0
45. Latus rectum of the conic is
 (a) $\frac{2|r_2^2|}{r_1}$ (b) $\frac{2|r_1|}{r_2}$
 (c) $2r_1|r_2|$ (d) $\frac{2|r_2|}{r_1}$
46. The eccentricity of the conic is
 (a) $\frac{r_2}{r_1}$ (b) $\sqrt{1 + \frac{r_2^2}{r_1^2}}$
 (c) $\sqrt{1 - \frac{r_2^2}{r_1^2}}$ (d) $\sqrt{1 + \frac{r_1^2}{r_2^2}}$

47. The equation $x^2 + 2xy + y^2 - 2x = 1$ always represents a
 (a) parabola (b) hyperbola
 (c) ellipse (d) circle
48. The centre of the conic $8x^2 + 4xy + 5y^2 - 24x - 24y = 0$ is
 (a) (2, 1) (b) (-1, 2)
 (c) (1, 2) (d) (-1, -2)
49. If r_1 and r_2 be the lengths of semi-axes of the conic such that $r_1^2 = 9$ and $r_2^2 = 4$ then conic is
 (a) parabola (b) ellipse
 (c) hyperbola (d) none of these
50. If for a conic $r_1^2 = \frac{4}{3}$ and $r_2^2 = -4$ then its eccentricity is
 (a) 4 (b) 1
 (c) 2 (d) $\frac{2}{3}$
51. If for a conic r_1^2 is 4 and eccentricity is $\frac{\sqrt{3}}{2}$ then the distance of directrix from its centre is
 (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}}$
 (c) $\frac{4}{\sqrt{3}}$ (d) $\frac{8}{\sqrt{3}}$
52. The coordinates of the centre of the conic $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$ are
[Kanpur 2018]
 (a) (-2, 3) (b) (3, -3)
 (c) (2, 3) (d) (-2, -3)
53. The coordinates of focus of conic $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ are
[Kanpur 2018]
 (a) (2, 1) (b) (2, 2)
 (c) (1, 1) (d) (1, 2)
54. Conic $3x^2 - 4xy + 9y = 0$ represents
[Agra 2017]
 (a) Parabola (b) Ellipse
 (c) Hyperbola (d) None of these
55. The centre of the conic $x^2 - xy + y^2 + 5x - 5y + 11 = 0$ is
[Avadh 2018]
 (a) (1, 0) (b) (2, -2)
 (c) (-1, 1) (d) (-5/3, 5/3)
56. The conic $x^2 + xy + y^2 + x + y = 1$ is
[Avadh 2018]
 (a) Ellipse (b) Parabola
 (c) Hyperbola (d) None of these
57. The eccentricity of the conic $3/r = 1 + \sqrt{3} \cos \theta + \sin \theta$ is
[Avadh 2018]
 (a) 1 (b) 2
 (c) 1/2 (d) 3/4
58. The centre of the circle $ax^2 + (2a - 3)y^2 - 4x - 1 = 0$ is
[Avadh 2018]
 (a) $\left(\frac{2}{3}, 0\right)$ (b) $\left(-\frac{2}{3}, 0\right)$
 (c) $\left(0, \frac{2}{3}\right)$ (d) $\left(0, -\frac{2}{3}\right)$
59. Conic $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ represents
 (a) Ellipse
 (b) A straight line
 (c) Pair of straight line
 (d) Hyperbola
60. $x^2 - 3xy + y^2 + 10x - 10y + c = 0$ represents a hyperbola, when c is equal to
 (a) $c = 20$ (b) $c \neq 20$
 (c) $c = 16$ (d) $c \neq 16$

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(d)	2.	(c)	3.	(b)	4.	(a)	5.	(a)	6.	(c)	7.	(d)	8.	(c)	9.	(d)	10.	(b)
11.	(c)	12.	(c)	13.	(b)	14.	(a)	15.	(c)	16.	(c)	17.	(a)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(b)	23.	(d)	24.	(c)	25.	(d)	26.	(c)	27.	(b)	28.	(a)	29.	(c)	30.	(b)
31.	(a)	32.	(b)	33.	(d)	34.	(d)	35.	(a)	36.	(c)	37.	(d)	38.	(b)	39.	(d)	40.	(a)
41.	(b)	42.	(d)	43.	(d)	44.	(b)	45.	(a)	46.	(c)	47.	(a)	48.	(c)	49.	(b)	50.	(c)
51.	(c)	52.	(c)	53.	(d)	54.	(c)	55.	(d)	56.	(a)	57.	(b)	58.	(a)	59.	(c)	60.	(b)

HINTS AND SOLUTIONS

2. Here $a = 2, b = -1, h = \frac{1}{2}$ so $h^2 > ab$ i.e. hyperbola.

4. Let $F(x, y) = 14x^2 - 4xy + 11y^2 - 44x - 58y + 71$

$$\frac{\partial F}{\partial x} = 28x - 4y - 44 = 0 \text{ i.e. } 7x - y = 11$$

$$\frac{\partial F}{\partial y} = -4x + 22y - 58 = 0 \text{ i.e. } 2x - 11y = -29$$

After solving these equations we get $x = 2, y = 3$ which is centre.

10. Let $F(x, y) = 2x^2 + xy - y^2 - x + 5y + 6$

$$\frac{\partial F}{\partial x} = 4x + y - 1 = 0, \frac{\partial F}{\partial y} = x - 2y + 5 = 0$$

After solving we get

$$x = -\frac{1}{3}, y = \frac{7}{3} \text{ so } F(x_1, y_1) = 12$$

So, asymptote is $F(x, y) = F(x_1, y_1)$ i.e.

$$2x^2 + xy - y^2 - x + 5y - 6 = 0.$$

Thus, equation of conjugate hyperbola is

$$F(x, y) = 2F(x_1, y_1) \text{ i.e.}$$

$$2x^2 + xy - y^2 - x - 5y = 18$$

16. Here $a = 1, b = 1 + \lambda, h = 1$ so for ellipse $h^2 < ab$ i.e. $1 < 1 + \lambda$ i.e. $\lambda > 0$.

17. Let $F(x, y) = x^2 - 3xy + y^2 + 10x - 10y + 21 = 0$

then its centre is $(-2, 2)$ also $c_1 = yx_1 + fy_1 + c = 1$
Thus equation of the conic referred to the centre c as origin is $-x^2 + 3xy - y^2 = 1$ reducing to the standard form $Ax^2 + 2Hxy + By^2 = 1$. The square of the semi-axes are given by

$$\left(A - \frac{1}{r^2}\right)\left(B - \frac{1}{r^2}\right) = H^2$$

$$\text{i.e. } \left(-1 - \frac{1}{r^2}\right)\left(-1 - \frac{1}{r^2}\right) = \left(\frac{3}{2}\right)^2$$

After solving we get $r_1^2 = 2$ and $r_2^2 = \frac{-2}{3}$, so the curve

is hyperbola and length of transverse axes $= 2r_1 = 2\sqrt{2}$

22. See the solution of (17) and find r_1^2, r_2^2 then

eccentricity is defined by $e = \sqrt{1 - \frac{r_2^2}{r_1^2}}$

28. $F = x^2 + 4xy + y^2 - 4$, centre is $(0, 0)$ and $c_1 = gx_1 + fy_1 + c = -y$. So equation of the conic referred to the centre C as origin is $x^2 + 4xy + y^2 = 4$ i.e. $\frac{1}{4}x^2 + xy + \frac{1}{4}y^2 = 1$

Compare $Ax^2 + 2Hxy + By^2 = 1$. The square of the semi-axes are $\left(A - \frac{1}{r^2}\right)\left(B - \frac{1}{r^2}\right) = H^2$. So we get

$$r_2^2 = -4 \text{ and } r_1^2 = \frac{4}{3}.$$

So equation of transverse axes is

$$\left(A - \frac{1}{r_1^2}\right)x + Hy = 0 \text{ i.e. } x - y = 0.$$

33. Lines are $y = \sqrt{a}x$ and $y = -\sqrt{a}x$ for perpendicularity $m_1m_2 = -1$ i.e. $-a = -1$ i.e. $a = 1$.

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EQUATION OF A CONIC SECTION

1. The equation of a conic section is given by the general equation of the second degree in x and y i.e.

$$F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

2. The general equation of a conic is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, which may be

$$\frac{a}{c}x^2 + \frac{2h}{c}xy + \frac{b}{c}y^2 + \frac{2g}{c}x + \frac{2f}{c}y + 1 = 0$$

Thus five conditions are required to determine five contacts namely $\frac{a}{c}, \frac{h}{c}, \frac{b}{c}, \frac{g}{c}$ and $\frac{f}{c}$.

3. Every straight line through a given point cuts the conic in two points.

TANGENT TO THE GENERAL CONIC

1. The equation of the tangent at any point (x_1, y_1) on the conic

$$F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{is } axx_1 + h(x_1y + xy_1) + hyy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

2. The condition that the line $lx + my + n = 0$ may touch the conic

$$F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$+ 2fy + c = 0$$

$$\text{is } Al^2 + Bm^2 + Cn^2 + 2Fmn + 2Gnl$$

$$+ 2Hlm = 0$$

where A, B, C, F, G, H are the cofactors of $a, b,$

$$c, f, g, h \text{ in the determinant } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}.$$

3. The co-ordinates (x_1, y_1) of the point of contact are obtained from the relations

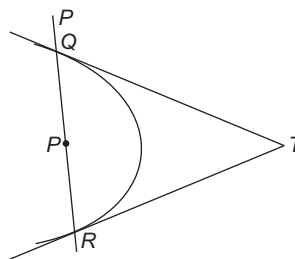
$$\frac{ax_1 + hy_1 + g}{l} = \frac{hx_1 + by_1 + f}{m}$$

$$= \frac{gx_1 + fy_1 + c}{n}$$

POLAR OF A POINT OR CHORD OF CONTACT

1. The equation of the chord of contact of the point $P(x_1, y_1)$ with respect to the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $axx_1 + h(xy_1 + x_1y) + byy_1 + g(x + x_1) + f(y + y_1) + c = 0$

2. Let P be a given point in the plane of a conic. Draw a straight line through the point P to meet the conic in the point Q and R . Draw the tangents at the points Q and R on the conic and let these tangents intersect at the point T . The locus of point T is called the polar of a point P with respect to the given conic and point P is called the pole of the polar.



3. The equation of the polar of the point $P(x_1, y_1)$ with respect to the conic

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$\text{is } axx_1 + h(xy_1 + x_1y) + byy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

4. If the polar of a point P with respect to a given conic passes through a point Q , then the polar

of Q will also pass through P . Then points P and Q are called conjugate points.

- Two lines are called conjugate lines if the pole of one lies on the other.
- The equation of the chord with (x_1, y_1) as the middle point of the conic

$F(x, y) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is

$$x \left(\frac{1}{2} \frac{\partial F}{\partial x} \right)_{(x_1, y_1)} + y \left(\frac{1}{2} \frac{\partial F}{\partial y} \right)_{(x_1, y_1)} + c_1 = F(x_1, y_1)$$

where $c_1 = gx_1 + fy_1 + c$.

DIAMETER AND CONJUGATE DIAMETERS

- The locus of the middle points of a system of parallel chords is called a diameter.
- The equation of a diameter of the conic $F(x, y) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $(a + hm)x + (h + bm)y + g + mf = 0$
- If two diameters are such that each diameter bisects the chords parallel to the other, they are called conjugate diameters.
- The condition that $y = mx$ and $y = m^1x$ be parallel to conjugate diameters of the conic $F(x, y) = 0$ is $a + h(m + m^1) + bmm^1 = 0$
- The condition that the two straight lines $Ax^2 + 2Hxy + By^2 = 0$ may be conjugate diameters of the conic $ax^2 + 2hxy + by^2 = 1$ is $aB - 2hH + bA = 0$

DIRECTOR CIRCLE OF A CONIC AND FOCI

- The locus of the points of intersection of perpendicular tangents, is called a director circle.

- The equation of the director circle of the conic

$$F(x, y) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

is $(ab - h^2)(x^2 + y^2) + 2x(bg - hf) + 2y(af - gh) + c(a + b) - g^2 - f^2 = 0$.

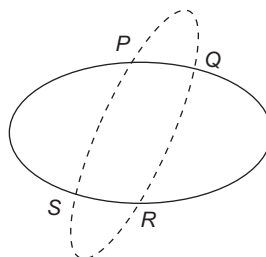
If $ab = a^2$, the conic $F(x, y) = 0$ represents a parabola then equation of director circle is $2Gx + 2Fy - (A + B) = 0$.

- Every central conic has four and only four foci, two of which are real and two imaginary.
- The tangents from a focus to a conic satisfy the conditions for a circle, conversely if the tangents from a point to a conic satisfy the conditions of a circle, the point must be a focus of the conic.

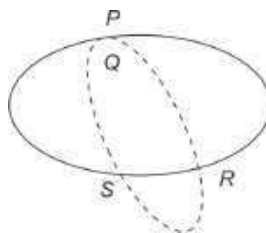
CONTACTS OF CONIC

Two conics intersect in four points which are all real, or two real and two imaginary, or all the four imaginary. We will discuss the contacts of different orders :

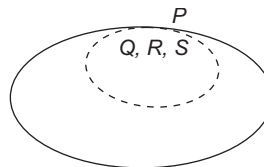
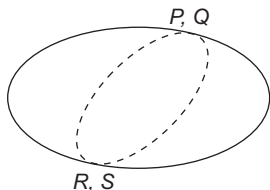
- The contact is said to be of zeroth order if all the four points of intersection namely P, Q, R and S are distinct.



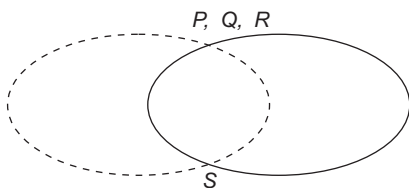
- If two of the four points say P and Q , coincide at P and the other two points (R and S) are distinct, the contact at P is said to be of the first order.



- If the points P and Q coincide, as also R and S , but P and R do not coincide. The conics then touch at two points P and R and have a double contacts.



4. If three points P, Q, R coincide at P and the fourth point S is distinct from them, the contact is said to be of the second order.



5. If all the four points P, Q, R, S coincide at P , the contact is said to be of the third order.

CONFOCAL CONICS

1. All those conics which have the same two points or their foci are called confocal conics.
2. The equation of confocal to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$.
3. Through any given point in the plane of a conic, two confocals can be drawn, one of which is a hyperbola and the other an ellipse.
4. Confocals cut at right angles.
5. One and only one members of confocals touches a given line.

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The independent contacts in the general conic equation of the second degree $F(x, y) = 0$ are
(a) 3 (b) 4
(c) 5 (d) 6
2. The equation of the tangent at $(1, 0)$ on the conic $x^2 + 2xy + y^2 - 1 = 0$ is
(a) $x + 2y = 2$ (b) $x + y = 1$
(c) $x - 2y = 2$ (d) $2x + y = 1$
3. The locus of the middle points of a system of parallel chords of a conic is called
(a) chord (b) radius
(c) diameter (d) conic
4. Confocal conics have
(a) same axes only
(b) same centre only
(c) same axes and centre both
(d) none of these
5. In a conic if all the four points of intersection are distinct then the contact is of
(a) zeroth order (b) first order
(c) second order (d) third order
6. If one confocal of the conic is ellipse then other one is
(a) parabola (b) circle
(c) hyperbola (d) ellipse
7. The angle between two confocals is
[Kanpur 2018, Avadh 2018]
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) 0
8. If all the four points of two conics coincide at one point then contact is said to be of
(a) first order (b) second order
(c) third order (d) fourth order

9. The equation of the tangent at (1, 1) of the conic $2x + 2y + 1 = 0$ is
 (a) $x + y + 1 = 0$ (b) $x + y + 2 = 0$
 (c) $x + y + 3 = 0$ (d) $x + y = 0$
10. If the equation of the system of parallel chords of the conic $x^2 + 2xy + y^2 + 1 = 0$ is $y = x + 1$ then equation of diameter is
 (a) $x + y + 1 = 0$ (b) $x + y = 0$
 (c) $x + y + 3 = 0$ (d) $x + y + 2 = 0$
11. If a central conic has four and only four foci and these are **[Kanpur 2018]**
 (a) all reals
 (b) all imaginary
 (c) two real and two imaginary
 (d) none of these
12. If two diameters are such that each diameter bisects the chords parallel to the other then these diameters are called
 (a) parallel diameters
 (b) bisected diameters
 (c) conjugate diameters
 (d) perpendicular diameters
13. Through any given point in the plane of conic, two confocals can be drawn which are
 (a) circle and ellipse
 (b) circle and hyperbola
 (c) ellipse and hyperbola
 (d) plane and ellipse
14. In the plane of the conics, confocal cuts at
 (a) acute angle (b) obtuse angle
 (c) right angle (d) none of these
15. The locus of a point from which two perpendicular tangents can be drawn one to each of two given confocals is
 (a) an ellipse (b) a circle
 (c) a hyperbola (d) a plane
16. If two conic have their axes parallel then the conic that pass through their point of intersection is
 (a) parabola (b) hyperbola
 (c) ellipse (d) circle
17. The condition that two straight lines $x^2 + y^2 + 2xy = 0$ be the conjugate diameters of the conic $ax^2 + 2hxy + by^2 = 1$ is
 (a) $a + h + b = 0$ (b) $a - h + b = 0$
 (c) $a + b = 0$ (d) $a - 2h + b = 0$
18. The condition that $y = mx$ and $y = m^1x$ be parallel to conjugate diameters of the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is
 (a) $a + hmm^1 + b = 0$
 (b) $a + h(m + m^1) + b = 0$
 (c) $a + h(m + m^1) + bmm^1 = 0$
 (d) $a + hmm^1 + b(m + m^1) = 0$
19. If $x + \lambda = 0$ be the tangent of the conic $x^2 + 2xy - y^2 + 1 = 0$ then the value of λ is
 (a) 2 (b) -2
 (c) $\sqrt{2}$ (d) not exist
20. The system of conic confocal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
 (a) a parabola (b) an ellipse
 (c) a hyperbola (d) a plane
21. The locus of the pole of a given straight line w.r.t. a series of confocal conics is a
 (a) circle (b) parabola
 (c) ellipse (d) straight line
22. The locus of the points of contact of the tangents drawn from a given point to a system of confocal is a :
 (a) linear curve (b) quadratic curve
 (c) cubic curve (d) none of these
23. The two parabolas which have common focus and their axes in opposite directions always intersect at
 (a) 0 (b) $\frac{\pi}{6}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

24. The equation of the circle which has contact of the second order with the conic $ax^2 + 2hxy + by^2 + 2cx = 0$ at the origin is

(a) $bx^2 + by^2 + 2cx = 0$ (b) $ax^2 + ay^2 + 2bx = 0$
 (c) $ax^2 + ay^2 + 2x = 0$ (d) $by^2 + ax^2 + 2cx = 0$

25. The confocal conic of the parabola $y^2 = 4ax$ is

(a) $y^2 = \lambda(x + \lambda)$ (b) $y^2 = 2\lambda(x + \lambda)$
 (c) $y^2 = 4\lambda(x + \lambda)$ (d) $y^2 = 4\lambda x$

26. The number of confocal conics through a given point of an ellipse are

(a) 3 (b) 4 (c) 2 (d) infinite

27. The confocal conic of an ellipse is

(a) ellipse only (b) hyperbola only
 (c) ellipse and hyperbola both
 (d) not exist

28. The confocal conic of $x^2 + 2y^2 = 4$ is

(a) $\frac{x^2}{2} + y^2 = 4$ (b) $\frac{x^2}{4} + \frac{y^2}{48} = 1$
 (c) $\frac{x^2}{45} + \frac{y^2}{48} = 1$ (d) $\frac{x^2}{48} + \frac{y^2}{45} = 1$

29. Two conics are confocal if they have common

(a) foci (b) centre
 (c) axes (d) latus rectum

30. The confocal conic of the conic $y^2 = 8(x + 2)$ is

(a) $y^2 = 8(x + 2)$ (b) $y^2 = 12(x + 3)$
 (c) $y^2 = 8(x - 2)$ (d) $y^2 = 12(x - 3)$

31. The equation of a diameter of the conic $F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is

[Kanpur 2018]

(a) $(a + hm)x + (h + bm)y + g + mf = 0$
 (b) $(a + hm)x - (h + bm)y + g - mf = 0$
 (c) $(a + hm)x + (h + bm)y - g - mf = 0$
 (d) None of these

32. The polar equation of the chord of contact of the point (r_1, θ_1) with respect to the conic $\frac{l}{r} = 1 + e \cos \theta$ is

[Kanpur 2018]

(a) $\left(\frac{l}{r} - e \cos \theta\right) \left(\frac{l}{r_1} - e \cos \theta_1\right) = \cos(\theta - \theta_1)$
 (b) $\left(\frac{l}{r} + e \cos \theta\right) \left(\frac{l}{r_1} + e \cos \theta_1\right) = \cos(\theta - \theta_1)$
 (c) $\left(\frac{l}{r} + e \cos \theta\right) \left(\frac{l}{r_1} - e \cos \theta_1\right) = \cos(\theta + \theta_1)$
 (d) $\left(\frac{l}{r} - e \cos \theta\right) \left(\frac{l}{r_1} + e \cos \theta_1\right) = \cos(\theta + \theta_1)$

33. Let $P(a \cos \alpha, b \sin \alpha)$ be a part on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. If P and Q are extremities of conjugate diameter, then the coordinate of Q is

[Avadh 2018]

(a) $(a \sin \alpha, b \cos \alpha)$ (b) $(a \cos \alpha, b \sin \alpha)$
 (c) $(-a \sin \alpha, b \cos \alpha)$ (d) $(a \sin \alpha, -b \cos \alpha)$

ANSWERS

MULTIPLE CHOICE QUESTIONS

[illegible]

HINTS AND SOLUTIONS

2. Tangent is $axx_1 + h(x_1y + xy_1) + hyy_1 + g(x + x_1)$
 $+ f(y + y_1) + c = 0$
i.e. $x + 1(y + 0) + 0 + 0 + 0 - 1 = 0$ *i.e.* $x + y = 1$.

10. The equation of diameter is
 $(a + hm)x + (h + bm)y + g + mf = 0$ where $a = 1$,
 $h = 1, b = 1, m = 1, g = 0, f = 0$ so we get $x + y = 0$

19. The condition of tangents is

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & o \end{vmatrix} = 0$$

$$\text{i.e. } \begin{vmatrix} 1 & 1 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & \lambda \\ 1 & 0 & \lambda & 0 \end{vmatrix} = 0$$

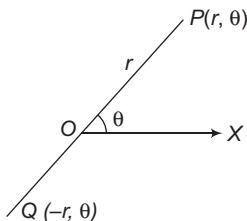
It gives an imaginary value of λ .

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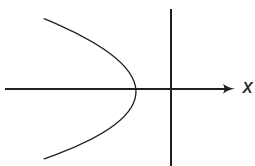
Polar Equation of a Conic

CONIC SECTION

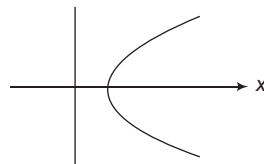
1. A conic section is the locus of a point which moves so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.
2. In polar system, the position of a point in a plane is determined by its distance r from a fixed point O , called the pole or origin and the angle θ that the line OP makes with OX is called vectorial angle. Fixed line OX is called initial line. The distance r is called the radius vector and is taken to be positive if measured along the line bonding the vectorial angle, and negative if measured in opposite direction. In polar system the co-ordinates of a point are not unique.



3. The polar equation of a conic with its latus rectum of length $2l$, eccentricity e and the focus being the pole is $\frac{l}{r} = 1 + e \cos \theta$



4. If we take the positive direction of the initial line opposite to the direction directed from the focus towards the directrix, the equation of the conic is $\frac{l}{r} = 1 - e \cos \theta$



5. If the conic is a parabola i.e. $e = 1$ then equation of parabola is $r = \frac{l}{2} \sec^2 \frac{\theta}{2}$
6. The equation of the directrix of the conic $\frac{l}{r} = 1 + e \cos \theta$ is $\frac{l}{r} = e \cos \theta$
7. The polar equation of a conic with its focus as the pole and its axis inclined at angle α to the initial line is $\frac{l}{r} = 1 + e \cos(\theta - \alpha)$
8. The equation of directrix in case 7 is $\frac{l}{r} = e \cos(\theta - \alpha)$

CHORD AND TANGENT OF THE CONIC

1. The equation of the chord of the conic $\frac{l}{r} = 1 + e \cos \theta$, whose extremities are $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$ is $\frac{l}{r} = e \cos \theta + \sec \left(\frac{\theta - \theta_1}{2} \right) \cos \left\{ \theta - \left(\frac{\theta_1 + \theta_2}{2} \right) \right\}$
2. If the points P and Q are such that their vectorial angles are $\alpha - \beta$ and $\alpha + \beta$, so that the sum of the angles is 2α and their difference is 2β then the chord PQ in case (1) is

$$\frac{l}{r} = e \cos \theta + \sec \beta \cos(\theta - \alpha)$$

3. The equation of the tangent at the point (r_1, θ_1) of the conic $\frac{l}{r} = 1 + e \cos \theta$ is

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \theta_1)$$

4. If the conic is $\frac{l}{r} = 1 + e \cos(\theta - \alpha)$ then the tangent at θ_1 is

$$\frac{l}{r} = e \cos(\theta - \alpha) + \cos(\theta - \theta_1)$$

5. If the conic is $\frac{l}{r} = 1 - e \cos \theta$ then the tangent at θ_1 is

$$\frac{l}{r} = -e \cos \theta + \cos(\theta - \theta_1)$$

ASYMPTOTES

1. The equation of the asymptotes of the conic $\frac{l}{r} = 1 + e \cos \theta$ are

$$\frac{le}{r} = (e^2 - 1) \cos \theta \pm \sqrt{(e^2 - 1)} \cdot \sin \theta$$

2. These two asymptotes are real only when $e > 1$ and then conic is a hyperbola.

AUXILIARY CIRCLE AND DIRECTOR CIRCLE

1. The locus of the foot of the perpendicular from the focus on any tangent to a conic is a circle called the auxiliary circle of the conic.

2. The polar equation of the auxiliary circle of the conic $\frac{l}{r} = 1 + e \cos \theta$ is

$$(e^2 - 1)r^2 - 2ler \cos \theta + l^2 = 0$$

3. If the conic is parabola i.e. $e = 1$ then by point 2 auxiliary circle is $\frac{l}{r} = 2 \cos \theta$

4. The locus of the point of intersection of two perpendicular tangents to a conic is called the director circle of the conic.

5. The equation of the director circle of the conic $\frac{l}{r} = 1 + e \cos \theta$ is

$$(1 - e^2)r^2 + 2ler \cos \theta - 2l^2 = 0$$

6. If the conic is parabola i.e. $e = 1$ then by 5 the director circle is $\frac{l}{r} = \cos \theta$

7. The polar equation of the chord of contact of the point (r', θ') w.r.t the conic $\frac{l}{r} = 1 + e \cos \theta$ is

$$\left(\frac{l}{r} - e \cos \theta\right) \left(\frac{l}{r'} - e \cos \theta'\right) = \cos(\theta - \theta')$$

POLAR AND NORMAL

1. The equation of the polar of a point (r^1, θ^1) w.r.t. the conic $\frac{l}{r} = 1 + e \cos \theta$ is

$$\left(\frac{l}{r} - e \cos \theta\right) \left(\frac{l}{r'} - e \cos \theta'\right) = \cos(\theta - \theta').$$

2. The pole of a line is the point of intersection of the tangents at its extremities.

3. The equation of any line perpendicular to $\frac{l}{r} = A \cos \theta + B \sin \theta$ is obtained by writing $\theta + \frac{\pi}{2}$ for θ and changing l to a new constant L .

4. The equation of the normal at point α on the conic $\frac{l}{r} = 1 + e \cos \theta$ is

$$\frac{le \sin \alpha}{(1 + e \cos \alpha)} \cdot \frac{1}{r} = \sin(\theta - \alpha) + e \sin \theta$$

5. In case of parabola i.e. $e = 1$ by point 4 equation of the normal at a point α is $\frac{l \sin \alpha}{1 + \cos \alpha} \cdot \frac{1}{r} = \sin(\theta - \alpha) + \sin \theta$

6. The equation of the normal at the point (r_1, α) to the conic $\frac{l}{r} = 1 - e \cos \theta$ is

$$-\frac{le \sin \alpha}{1 - e \cos \alpha} \cdot \frac{1}{r} = \sin(\theta - \alpha) - e \sin \theta$$

TRANSFORMATION OF CO-ORDINATES

1. If the cartesian co-ordinate of a point P is (x, y) and the corresponding polar co-ordinate is (r, θ) then

$$x = r \cos \theta, \quad y = r \sin \theta \quad \text{so} \quad r = \sqrt{x^2 + y^2}$$

$$\text{and} \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

2. The distance between two points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ is

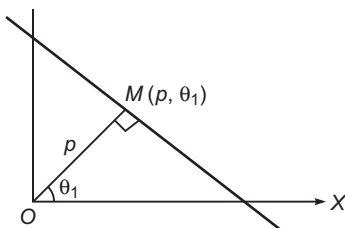
$$AB = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1)}$$

GENERAL EQUATION OF LINE

1. The general equation of the straight line in polar co-ordinate system is

$$r \cos(\theta - \theta_1) = p$$

2. The equation of straight line passing through (r_1, θ_1) and (r_2, θ_2) is



$$\frac{\sin(\theta_2 - \theta_1)}{r} = \frac{\sin(\theta - \theta_1)}{r_2} + \frac{\sin(\theta_2 - \theta)}{r_1}$$

3. Area of a triangle whose vertices are (r_1, θ_1) , (r_2, θ_2) and (r_3, θ_3) is

$$D = \frac{1}{2} \{r_1r_2 \sin(\theta_2 - \theta_1) + r_2r_3 \sin(\theta_3 - \theta_2) + r_1r_3 \sin(\theta_1 - \theta_3)\}$$

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The conic $\frac{l}{r} = 1 + e \cos \theta$ represents a parabola if

(a) $e = 1$ (b) $e > 1$ (c) $e < 1$ (d) $e = 0$

2. The equation of the directrix of the conic $\frac{l}{r} = 1 + e \cos \theta$ is

(a) $\frac{l}{r} = -e \cos \theta$ (b) $\frac{r}{l} = e \cos \theta$

(c) $\frac{r}{l} = -e \cos \theta$ (d) $\frac{l}{r} = e \cos \theta$

3. The semi latus rectum between the segments of a focal chord is always a

(a) arithmetic mean (b) geometric mean
(c) harmonic mean (d) none of these

4. The equation of the conic $\frac{l}{r} = 1 + e \cos \theta$ is equivalent to the following equation [Avadh 2018]

(a) $\frac{l}{r} = 1 - e \cos \theta$ (b) $\frac{l}{r} = 1 + e \cos \theta$

(c) $\frac{-l}{r} = 1 + e \sin \theta$ (d) $\frac{l}{r} = -1 + e \cos \theta$

5. The equation of the tangent at the point $(r_1, \frac{\pi}{2})$ of the conic $\frac{l}{r} = 1 + e \cos \theta$ is

(a) $\frac{l}{r} = \cos \theta + \sin \theta$ (b) $\frac{l}{r} = e \cos \theta - \sin \theta$

(c) $\frac{l}{r} = e \cos \theta + \sin \theta$ (d) $\frac{l}{r} = \cos \theta - \sin \theta$

6. The slope of the tangent to the conic $\frac{l}{r} = 1 + e \cos \theta$ at the point α on it is given by

(a) $\frac{e + \cos \alpha}{\sin \alpha}$ (b) $\frac{e - \cos \alpha}{\sin \alpha}$

(c) $-\frac{e + \cos \alpha}{\sin \alpha}$ (d) $\frac{e^2 - \cos \alpha}{\sin \alpha}$

7. The number of normals that can be drawn from a point to a parabola are

(a) 1 (b) 2
(c) 3 (d) 6

8. In a polar system the representation of the co-ordinates of a point are

(a) unique (b) finite
(c) infinite (d) none of these

9. If l be the semi latus rectum of a conic then the sum of the reciprocals of the segments of any focal chord is
- (a) $\frac{e}{l}$ (b) $\frac{2}{l}$
 (c) $\frac{l}{e}$ (d) $\frac{l}{2}$
10. The condition that the line $\frac{l}{r} = A \cos \theta + B \sin \theta$ to be a tangent to the conic $\frac{l}{r} = 1 + e \cos \theta$ is **[Avadh 2018]**
- (a) $(A + e)^2 + B^2 = 1$ (b) $(B - e)^2 + A^2 = 1$
 (c) $(B + e)^2 + A^2 = 1$ (d) $(A - e)^2 + B^2 = 1$
11. The locus of the point of intersection of two tangents to the parabola, which cut one another at a constant angle is a
- (a) circle (b) parabola
 (c) hyperbola (d) ellipse
12. The nature of the conic $\frac{15}{r} = 3 - 4 \cos \theta$ is
- (a) parabola
 (b) ellipse
 (c) hyperbola
 (d) rectangular hyperbola
13. Two real asymptotes of the conic $\frac{l}{r} = 1 + e \cos \theta$ exist only when
- (a) $e < 1$ (b) $e > 1$
 (c) $0 < e < 1$ (d) e is real
14. The locus of the point of intersection of two perpendicular tangents to a conic is called
- (a) circle (b) director circle
 (c) auxiliary circle (d) parabola
15. The equation of the director circle of the conic $\frac{l}{r} = 1 + \cos \theta$ is
- (a) $2r^2 + 2rl \cos \theta - 2l^2 = 0$
 (b) $r^2 + lr \cos \theta + l^2 = 0$
 (c) $\frac{l}{r} = 1 + \cos \theta$
 (d) $\frac{l}{r} = \cos \theta$
16. The equation of the normal at the point $\left(r, \frac{\pi}{2}\right)$ of the parabola $\frac{l}{r} = 1 + \cos \theta$ is
- (a) $\frac{1}{r} = \sin \theta - \cos \theta$ (b) $\frac{1}{r} = 1 - \cos \theta$
 (c) $\frac{1}{r} = \sin \theta + \cos \theta$ (d) $\frac{1}{r} = 1 - \sin \theta$
17. The equation of the auxiliary circle of the conic $\frac{l}{r} = 1 + \cos \theta$ is given by
- (a) $(e^2 - 1)r^2 + 2lr \cos \theta + l^2 = 0$
 (b) $\frac{l}{r} = \cos \theta$
 (c) $\frac{l}{r} = 2 \cos \theta$
 (d) $\frac{l}{r} = 1 + \cos \theta$
18. The conic $r = \frac{l}{2} \sec^2 \frac{\theta}{2}$ always represents a
- (a) parabola (b) hyperbola
 (c) ellipse (d) circle
19. The asymptotes of the conic $\frac{1}{r} = 1 + 2 \cos \theta$ are
- (a) $r(\sqrt{3} \cos \theta \pm \sin \theta) = 2$
 (b) $\sqrt{3}(\sqrt{3} \cos \theta \pm \sin \theta) = 2r$
 (c) $r(\sqrt{3} \cos \theta \pm \sin \theta) = 3$
 (d) $r\sqrt{3}(\sqrt{3} \cos \theta \pm \sin \theta) = 2$
20. The coincident point of (r, θ) is
- (a) $(r, \theta + 2\pi)$ (b) $(-r, \theta + \pi)$
 (c) $(-r, \theta)$ (d) $(-r, -\theta)$
21. The equation of the directrix of the conic $\frac{2}{r} = 1 + \frac{1}{2} \cos \theta$ is
- (a) $\frac{4}{r} = \cos \theta$ (b) $\frac{2}{r} = \cos \theta$
 (c) $\frac{1}{r} = 2 \cos \theta$ (d) none of these
22. The conic $\frac{l}{r} = 2 + 3 \cos \theta$ represents a
- (a) parabola (b) circle
 (c) hyperbola (d) ellipse

23. The point on the curve $\frac{14}{r} = 3 - 8\cos\theta$ whose radius vector is 2, is represented by
 (a) $\left(3, \frac{2\pi}{3}\right)$ (b) $\left(2, \frac{\pi}{3}\right)$
 (c) $\left(3, \frac{\pi}{3}\right)$ (d) $\left(2, \frac{2\pi}{3}\right)$
24. The directrix of the conic $\frac{4}{r} = \frac{2}{3} + \cos\theta$ is
 (a) $\frac{4}{r} = \cos\theta$ (b) $\frac{6}{r} = \cos\theta$
 (c) $\frac{1}{r} = \frac{2}{3} + \cos\theta$ (d) none of these
25. The semi latus rectum of the conic $\frac{1}{2r} = \frac{5}{3} + 2\cos\theta$ is
 (a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $\frac{3}{10}$ (d) $\frac{10}{3}$
26. $r = \frac{l}{2} \operatorname{cosec}^2 \frac{\theta}{2}$ represents a **[Agra 2017]**
 (a) parabola (b) hyperbola
 (c) ellipse (d) circle
27. The polar equation of the auxiliary circle of the conic $\frac{1}{r} = 1 + e\cos\theta$ is
 (a) $(e^2 - 1)r^2 - 2er\cos\theta + 1 = 0$
 (b) $(e^2 + 1)r^2 - 2er\cos\theta + 1 = 0$
 (c) $(e^2 + 1)r^2 + 2er\cos\theta + 1 = 0$
 (d) $(e^2 - 1)r^2 + 2er\cos\theta - 1 = 0$
28. If two real asymptotes exist for a conic then it is
 (a) parabola (b) hyperbola
 (c) ellipse (d) circle
29. $r = a$ represents a
 (a) circle (b) parabola
 (c) ellipse (d) hyperbola
30. The condition that the line $\frac{l}{r} = A\cos\theta + B\sin\theta$ be a tangent to the conic $\frac{l}{r} = 1 + \cos\theta$ is
 (a) $A^2 + B^2 + 2A = 0$ (b) $A + B = 0$
 (c) $A^2 + B^2 - 2A = 0$ (d) $A^2 + B^2 = 0$
31. The equation of the director circle of the conic $\frac{1}{r} = 1 + e\cos\theta$ is
 (a) $(1 + e^2)r^2 + 2er\cos\theta - 2l^2 = 0$
 (b) $(1 - e^2)r^2 + 2er\cos\theta - 2l^2 = 0$
 (c) $(1 + e^2)r^2 + 2er\cos\theta + 2l^2 = 0$
 (d) none of these
32. The slope of the tangent at the point (r_1, θ_1) of the conic $\frac{l}{r} = 1 + \cos\theta$ is
 (a) $\frac{\cos\theta_1}{\sin\theta_1}$ (b) $\frac{1 + \cos\theta_1}{\sin\theta_1}$
 (c) $-\frac{1 + \cos\theta_1}{\sin\theta_1}$ (d) $-\frac{1 + \sin\theta_1}{\cos\theta_1}$
33. The locus of the foot of the perpendicular from the focus of any tangent to a conic is called
 (a) auxiliary circle (b) director circle
 (c) ellipse (d) parabola
34. The condition that the line $\frac{l}{r} = 2\cos\theta + \sin\theta$ be a tangent to the conic $\frac{l}{r} = 1 + e\cos\theta$ is
 (a) $e = 1$ (b) $e = 2$
 (c) $e = \frac{3}{2}$ (d) $e = 3$
35. The equation of the director circle of the conic $\frac{3}{r} = 1 + 4\cos\theta$
 (a) $5r^2 - 8r\cos\theta + 6 = 0$
 (b) $5r^2 + 8r\cos\theta + 6 = 0$
 (c) $5r^2 + 8r\cos\theta - 6 = 0$
 (d) $5r^2 - 8r\cos\theta - 6 = 0$
36. If $e_1^2 + e_2^2 = 1$ for two different conics of same nature then conic is
 (a) ellipse (b) parabola
 (c) circle (d) hyperbola
37. The eccentricity of the conic $\frac{1}{r} = \frac{1}{6} + \frac{1}{18}\cos\theta$ is
 (a) $\frac{1}{18}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{6}$ (d) $\frac{1}{2}$

38. For the rectangular hyperbola, eccentricity is
 (a) 3 (b) 2
 (c) $\sqrt{3}$ (d) $\sqrt{2}$
39. The perpendicular focal chords of a rectangular hyperbola are
 (a) unequal
 (b) equal
 (c) may be equal or unequal
 (d) none of these
40. If $e = \frac{\pi}{2}$, then the conic is
 (a) circle (b) parabola
 (c) ellipse (d) hyperbola
41. If $e = \cos\theta$ for $0 < \theta < \frac{\pi}{2}$ then the conic is
 (a) parabola (b) ellipse
 (c) hyperbola (d) circle
42. The semi latus rectum of the conic $r(\cos\theta + \cos\alpha) = 2$ is
 (a) $2\sin\alpha$ (b) $\sin\alpha$
 (c) $4\sin\alpha$ (d) $\csc\alpha$
43. If e and e' be the eccentricities of ellipse and hyperbola respectively then $e^2 + \frac{1}{e'^2}$ is
 (a) > 2 (b) < 2
 (c) $= 2$ (d) none of these
44. If for a conic e is given by $2e^2 - 3e - 1 = 0$ then the conic is
 (a) circle (b) parabola
 (c) ellipse (d) hyperbola
45. The conic $\cos\frac{\theta}{2} + \sin\frac{\theta}{3} = \frac{1}{\sqrt{2}}$ represents a
 (a) ellipse (b) parabola
 (c) hyperbola (d) circle
46. Which of the following curve has an asymptotes?
 (a) circle (b) ellipse
 (c) rectangle (d) hyperbola
47. The auxiliary circle of the conic $\frac{2}{r} = 1 + \cos\theta$ is
 (a) $r = \sec\theta$ (b) $r = \csc\theta$
 (c) $r = \sin\theta$ (d) $r = \cos\theta$
48. The equation of director circle of the conic $\frac{3}{r} = 1 + \cos\theta$ is
 (a) $r \sec\theta = 3$ (b) $r \cos\theta = 3$
 (c) $r \csc\theta = 3$ (d) none of these
49. Equation of normal of $\frac{2}{r} = 1 + \cos\theta$ at $\left(r, \frac{\pi}{2}\right)$ is
 (a) $\frac{2}{r} = \sin\theta + \cos\theta$ (b) $\frac{2}{r} = \sin\theta$
 (c) $\frac{2}{r} = \sin\theta \cos\theta$ (d) $\frac{2}{r} = \cos\theta$
50. The maximum asymptotes of the hyperbola are
 (a) 1 (b) 2
 (c) 3 (d) 4
51. The asymptotes of the conic $\frac{2}{r} = 1 + 3\cos\theta$ are
 (a) $\frac{1}{r} = 4\cos\theta \pm \sin\theta$
 (b) $\frac{1}{r} = 4\cos\theta \pm \sqrt{2}\sin\theta$
 (c) $\frac{3}{r} = \cos\theta \pm \sin\theta$
 (d) $\frac{3}{r} = 4\cos\theta \pm \sqrt{2}\sin\theta$
52. The equation of the director circle of the conic $\frac{1}{r} = \frac{1}{2} + 3\cos\theta$ is
 (a) $35r^2 - 24r\cos\theta + 8 = 0$
 (b) $35r^2 - 16r\cos\theta + 8 = 0$
 (c) $5r^2 - 24r\cos\theta - 8 = 0$
 (d) $3r^2 - 16r\cos\theta - 8 = 0$
53. Tangents from the focus to a conic satisfy the conditions for an **[Kanpur 2018]**
 (a) ellipse (b) hyperbola
 (c) circle (d) None of these
54. The equation of the director circle of the conic $\frac{l}{r} = 1 + e \cos\theta$ is
 (a) $(1 - e^2) r^2 + 2ler \cos\theta + 2l^2 = 0$
 (b) $(1 + e^2) r^2 + 2ler \cos\theta - 2l^2 = 0$
 (c) $(1 - e^2) r^2 + 2ler \cos\theta - 2l^2 = 0$
 (d) None of these

55. The equation of the auxiliary circle of the conic $\frac{l}{r} = 1 + e \cos \theta$ is
[Kanpur 2018]
(a) $(e^2 - 1)r^2 + 2ler \cos \theta + l^2 = 0$
(b) $(e^2 + 1)r^2 + 2ler \cos \theta - l^2 = 0$
(c) $(e^2 + 1)r^2 - 2ler \cos \theta + l^2 = 0$
(d) $(e^2 + 1)r^2 + 2ler \cos \theta + l^2 = 0$
56. The equation of tangent at the point (r_1, θ_1) of the conic $l/r = 1 + e \cos \theta$ is
[Kanpur 2018, Avadh 2018]
(a) $l/r = e \cos(\theta - \theta_1) + \cos \theta$
(b) $l/r = e \cos \theta + \cos(\theta - \theta_1)$
(c) $l/r = e \cos(\theta + \theta_1) + \cos \theta$
(d) $l/r = e \cos \theta + \cos(\theta + \theta_1)$
57. The equation of normal at a point α on the conic $l/r = 1 + \cos \theta$ is
[Kanpur 2018]
(a) $\frac{l \sin \alpha}{r(1 + \cos \alpha)} = \sin(\theta + \alpha) + \sin \alpha$
(b) $\frac{l \sin \alpha}{r(1 + \cos \alpha)} = \sin(\theta - \alpha) + \sin \theta$
(c) $\frac{l \sin \alpha}{r(1 + \cos \alpha)} = \sin(\theta - \alpha) - \sin \theta$
(d) $\frac{l \sin \alpha}{r(1 - \cos \alpha)} = \sin(\theta + \alpha) - \sin \theta$
58. The conic $\frac{l}{r} = 1 + 3 \cos \theta$ is
[Avadh 2018]
(a) ellipse (b) parabola
(c) hyperbola (d) None of these
59. The conic $\frac{l}{r} = 1 + e \cos \theta$ is a hyperbola if
[Avadh 2018]
(a) $e < 1$ (b) $e = 1$
(c) $e > 1$ (d) $e = 0$
60. The distance between $\left(1, \frac{\pi}{6}\right)$ and $\left(2, \frac{\pi}{3}\right)$ is
[Avadh 2018]
(a) $\sqrt{5 - 2\sqrt{3}}$ (b) $\sqrt{5 + 2\sqrt{3}}$
(c) $2 + \sqrt{2}$ (d) $2 - \sqrt{2}$
61. The polar equation of a circle which is passing through pole and whose centre lies on the initial line is
[Avadh 2018]
(a) $r = 2a \sin \theta$
(b) $r = a \sin 2\theta$
(c) $r = a \cos \frac{\theta}{2}$
(d) $r = a \cos \theta$
62. Let PSQ be a chord of the conic $\frac{l}{r} = 1 + \cos \theta$ passing through the focus and inclined at an angle α with initial line then $SP \cdot SQ$ is
[Avadh 2018]
(a) $\frac{1}{1 - \cos^2 \alpha}$ (b) $\frac{l^2}{\sin^2 \alpha}$
(c) $\frac{l^2}{\cos^2 \alpha}$ (d) $\frac{1}{1 + \cos^2 \alpha}$
63. If e is the eccentricity of a parabola and e' is the eccentricity of conic such that $e + e'^2 = 5$ then conic represents a
(a) parabola (b) ellipse
(c) circle (d) hyperbola
64. The equation of the normal of the conic $\frac{5}{r} = 4 + 2 \cos \theta$ at $\left(r, \frac{\pi}{2}\right)$ is
(a) $\frac{5}{r} = 4 \sin \theta + 6 \cos \theta$
(b) $\frac{5}{r} = 4 \sin \theta - 8 \cos \theta$
(c) $\frac{5}{r} = 4 \cos \theta - 8 \sin \theta$
(d) $\frac{5}{r} = 4 \sin \theta + 8 \cos \theta$

ANSWERS

MULTIPLE CHOICE QUESTIONS

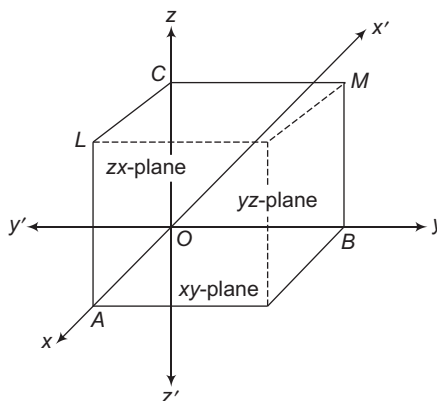
1.	(a)	2.	(d)	3.	(c)	4.	(d)	5.	(c)	6.	(c)	7.	(c)	8.	(c)	9.	(b)	10.	(d)
11.	(c)	12.	(c)	13.	(b)	14.	(b)	15.	(d)	16.	(a)	17.	(c)	18.	(a)	19.	(d)	20.	(b)
21.	(a)	22.	(c)	23.	(d)	24.	(a)	25.	(c)	26.	(a)	27.	(a)	28.	(b)	29.	(a)	30.	(c)
31.	(b)	32.	(c)	33.	(a)	34.	(b)	35.	(a)	36.	(a)	37.	(b)	38.	(d)	39.	(b)	40.	(d)
41.	(b)	42.	(a)	43.	(b)	44.	(d)	45.	(b)	46.	(d)	47.	(a)	48.	(b)	49.	(c)	50.	(b)
51.	(d)	52.	(a)	53.	(c)	54.	(c)	55.	(d)	56.	(b)	57.	(b)	58.	(c)	59.	(c)	60.	(a)

HINTS AND SOLUTIONS

5. The tangent of $\frac{l}{r} = 1 + e \cos \theta$ at (r_1, θ_1) is 25. $\frac{1}{2r} = \frac{5}{3} + 2 \cos \theta \Rightarrow \frac{3}{10r} = 1 + \frac{6}{5} \cos \theta$ here $l = \frac{3}{10}$.
- $\frac{l}{r} = e \cos \theta + \cos(\theta - \theta_1)$ so required tangent at 35. The director circle of $\frac{l}{r} = 1 + e \cos \theta$ is $(1 - e^2)r^2 + 2ler \cos \theta - 2l^2 = 0$ so director circle of $\frac{3}{r} = 1 + 4 \cos \theta$ is $l = 3, e = 4$
- $\left(r, \frac{\pi}{2}\right)$ is $\frac{l}{r} = e \cos \theta + \cos\left(\theta - \frac{\pi}{2}\right)$ i.e. $-15r^2 + 24r \cos \theta - 18 = 0 \Rightarrow 5r^2 - 8r \cos \theta + 6 = 0$.
- $\frac{l}{r} = e \cos \theta + \sin \theta$.
12. $\frac{15}{r} = 3 - 4 \cos \theta$ may be written as $\frac{5}{r} = 1 - \frac{4}{3} \cos \theta$ so 36. If $e_1^2 + e_2^2 = 1$ then e_1 and e_2 both are less than or equal to 1. So conics are ellipse.
- $e = \frac{-4}{3} < 1$ so it is hyperbola.
19. The asymptotes of $\frac{l}{r} = 1 + e \cos \theta$ are 37. Here $\frac{1}{r} = \frac{1}{6} + \frac{1}{18} \sin \theta$ i.e. $\frac{6}{r} = 1 + \frac{1}{3} \sin \theta$ so $e = \frac{1}{3}$.
- $\frac{le}{r} = (e^2 - 1) \cos \theta \pm \sqrt{(e^2 + 1)} \cdot \sin \theta$ so asymptotes 41. Since $\cos \theta$ lies between 0 to 1 when $r < \theta < \frac{\pi}{2}$ so it represents an ellipse.
- of $\frac{1}{r} = 1 + 2 \cos \theta$ are $e = 1, e = 2$
- $\frac{2}{r} = 3 \cos \theta \pm \sqrt{3} \sin \theta \Rightarrow 2 = r\sqrt{3}(\sqrt{3} \cos \theta \pm \sin \theta)$ 44. $2e^2 - 3e - 1 = 0$ i.e. $e = \frac{3 \pm \sqrt{9 + 8}}{4} = \frac{3 \pm \sqrt{17}}{4}$ i.e. $e > 1$ so it is a hyperbola.

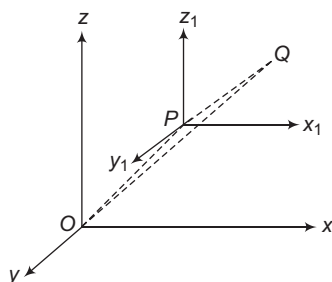
CARTESIAN CO-ORDINATES SYSTEM

1. In three dimension system we have three non-coplanar lines $X'OX$, $Y'OY$ and $Z'OZ$ which are called the x -axis, y -axis and z -axis respectively. The planes XOY , YOZ and ZOX are called coordinate planes. These planes divide the space into eight parts which are called the octants. The point O is called the origin.
2. The following table determine the signs in eight octants :



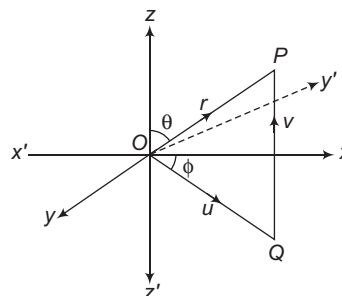
octant	$OXYZ$	$OXY'Z$	$OXY'Z'$	$OXYZ'$	$OX'YZ$	$OX'Y'Z$	$OX'YZ'$	$OX'Y'Z'$
X	+	+	+	+	-	-	-	-
Y	+	-	-	+	+	-	+	-
Z	+	+	-	-	+	+	-	-

3. Let OX , OY and OZ be a rectangular set of axes. Referred to these axes let the co-ordinates of two points P and Q be (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively then the co-ordinates of Q referred to P as origin are $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$.



SPHERICAL POLAR CO-ORDINATES

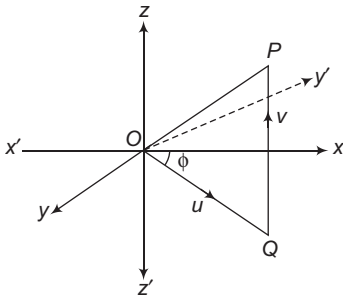
1. Let $X'OX$, $Y'OY$ and $Z'OZ$ be the set of rectangular axes. Let P be a point in space. Draw PN perpendicular from P to the xy plane. Suppose $OP = r$, $\angle ZOP = \theta$ and $\angle XON = \phi$ measured positively in the direction shown by arrows in the figure. The quantities r , θ , ϕ are called the spherical polar co-ordinates of P and denoted by (r, θ, ϕ) .



2. The relations between spherical polar co-ordinates and cartesian co-ordinates are $x = r \cos \phi \sin \theta$, $y = r \sin \phi \sin \theta$, $z = r \cos \theta$,
 $x^2 + y^2 + z^2 = r^2$, $\tan \theta = \frac{\sqrt{x^2 + y^2}}{z}$,
 $\tan \phi = \frac{y}{x}$

CYLINDRICAL CO-ORDINATES

1. Let P be a point in space. Suppose $OQ = u$, $\angle XOQ = \phi$. $QP = z$ then the quantities u , ϕ , z are called cylindrical co-ordinates of P and is denoted by (u, ϕ, z) .



2. The relation between the cylindrical and cartesian co-ordinates are $x = u \cos \phi$,
 $y = u \sin \phi$, $z = z$, $u^2 = x^2 + y^2$, $\tan \phi = \frac{y}{x}$.

DISTANCE BETWEEN TWO POINTS

1. The distance between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by
 $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
2. The distance between the points $P(x, y, z)$ and the origin $O(0, 0, 0)$ is $OP = \sqrt{x^2 + y^2 + z^2}$.

SECTION-FORMULA

1. The co-ordinates of a point say $R(\bar{x}, \bar{y}, \bar{z})$ dividing internally the join of two given points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio $m : n$ is

$$\bar{x} = \frac{mx_2 + nx_1}{m+n}, \quad \bar{y} = \frac{my_2 + ny_1}{m+n},$$

$$\bar{z} = \frac{mz_2 + nz_1}{m+n}.$$

2. The co-ordinates of the mid point of the join of $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are
 $\bar{x} = \frac{x_1 + x_2}{2}$, $\bar{y} = \frac{y_1 + y_2}{2}$, $\bar{z} = \frac{z_1 + z_2}{2}$.

3. If $m : n = \lambda : 1$ in (point 1) then the co-ordinates of the points are $\bar{x} = \frac{x_1 + \lambda x_2}{\lambda + 1}$, $\bar{y} = \frac{y_1 + \lambda y_2}{\lambda + 1}$,
 $\bar{z} = \frac{z_1 + \lambda z_2}{\lambda + 1}$.

4. If the ratio $\frac{m}{n}$ is positive, then the point R divides PQ internally and if it is negative then divides PQ externally.

5. The co-ordinates of point $R(\bar{x}, \bar{y}, \bar{z})$ which divides the join of the line joining the points P and Q externally in the ratio $m : n$ are

$$\bar{x} = \frac{mx_2 - nx_1}{m-n}, \quad \bar{y} = \frac{my_2 - ny_1}{m-n},$$

$$\bar{z} = \frac{mz_2 - nz_1}{m-n}.$$

CENTROID OF A TRIANGLE AND TETRAHEDRON

1. The centroid of the triangle with vertices (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) is
 $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$.

2. The centroid of the tetrahedron with vertices (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) and (x_4, y_4, z_4) is

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right).$$

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The octant position of the point $(1, -2, 3)$ is
 (a) $OXYZ$ (b) $OX'YZ$
 (c) $OXY'Z$ (d) $OXYZ'$
2. The distance between the points $(1, 2, 3)$ and $(2, 3, -1)$ is
 (a) $3\sqrt{2}$ (b) $2\sqrt{3}$
 (c) $\sqrt{6}$ (d) 0
3. The points $A(0, 1, 2)$, $B(2, -1, 3)$ and $C(1, -3, 1)$ forms an
 (a) isosceles triangle
 (b) equilateral triangle
 (c) isosceles right angle triangle
 (d) none of these
4. If A, B and C are the points $A(-1, -1, -1)$, $B(1, 3, 2)$ and $C(5, 11, 8)$ then the ratio in which B divides AC is
 (a) 2 : 1 (b) 3 : 1
 (c) 1 : 3 (d) 1 : 2
5. The equation $f(x, y, z) = 0$ represents a
 (a) curve (b) surface
 (c) line (d) circle
6. In three dimension geometry $x = a$ represents a
 (a) point (b) line
 (c) plane (d) cone
7. The locus of the point whose x -co-ordinate is a will be
 (a) $yz = a$ (b) $x = a$
 (c) $y = a$ (d) $z = a$
8. The three mutually perpendicular plane divide the space into
 (a) three parts (b) five parts
 (c) six parts (d) eight parts
9. The distance of the point (a, b, c) from x -axis is
 (a) $\sqrt{b^2 + c^2}$ (b) a
 (c) $\sqrt{a^2 + b^2}$ (d) $b^2 + c^2$
10. The distance of the point $(+1, -2, +3)$ from the yz -plane is
 (a) 1 (b) 2
 (c) 3 (d) -2
11. If (r, θ, ϕ) be the spherical polar co-ordinates of a point $P(x, y, z)$ then θ is equal to
 (a) $\tan^{-1}\left(\frac{y}{x}\right)$ (b) $\tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$
 (c) $\tan^{-1}\left(\frac{z}{x}\right)$ (d) $\tan^{-1}\left(\frac{\sqrt{y^2 + z^2}}{x}\right)$
12. If point $P(a, b, 7)$ lies on the line whose end points are $A(1, 2, 3)$ and $B(2, 10, 1)$ then b is
 (a) -1 (b) -14
 (c) -7 (d) 14
13. The cylindrical co-ordinates of the point $(3, 4, 5)$ is
 (a) $\left(5, \tan^{-1}\frac{4}{3}, 5\right)$ (b) $\left(3, \tan^{-1}\frac{4}{3}, 5\right)$
 (c) $\left(5, \tan^{-1}\frac{3}{4}, 5\right)$ (d) $\left(5, \tan^{-1}\frac{4}{3}, 3\right)$
14. The ratio in which the yz -plane divides the join of the points $(-2, 4, 7)$ and $(3, -5, 8)$ is
 (a) 4 : 3 (b) 3 : 2
 (c) 2 : 3 (d) 3 : 4
15. The octant position of the point $(-a, -b, +c)$ is
 (a) $OX'Y'Z'$ (b) $OXYZ$
 (c) $OXY'Z$ (d) $OXYZ'$
16. The distance between the points $(2, 3, 1)$ and $(-1, 3, -3)$ is
 (a) 5 (b) 4
 (c) $\sqrt{5}$ (d) 2
17. The locus of the point whose z -co-ordinate is a will be
 (a) $z = a$ (b) $xy = a$
 (c) $x = a$ (d) $y = a$
18. The distance of the point $(4, 5, 6)$ from y -axis is
 (a) $\sqrt{39}$ (b) $\sqrt{61}$
 (c) $\sqrt{52}$ (d) 5

19. The distance of the point $(2, 4, 6)$ from the xz -plane is :
 (a) 2 (b) 4
 (c) 6 (d) $\sqrt{40}$
20. The perpendicular distance of the point (a, b, c) from the z -axis is
 (a) $\sqrt{a^2 + b^2 + c^2}$ (b) $\sqrt{a^2 + b^2}$
 (c) c (d) $\sqrt{b^2 + c^2}$
21. If $A(2, 3, 4)$, $B(3, -2, 2)$ and $C(6, -17, -4)$ are three points then the ratio in which C divides AB is
 (a) 4 : 3 (b) 2 : 3
 (c) 3 : 4 (d) -4 : 3
22. The y -co-ordinate of the point which divides the join of $(2, 3, 4)$ and $(3, -4, 7)$ in the ratio 2 : -4 is
 (a) 21 (b) 14
 (c) 10 (d) 12
23. The centroid of the triangle whose vertices co-ordinates are $(1, 0, 2)$, $(2, 3, -1)$, $(3, 4, 2)$ is
 (a) $\left(-2, \frac{-7}{3}, 1\right)$ (b) $\left(2, \frac{-7}{3}, -1\right)$
 (c) $\left(2, \frac{7}{3}, 1\right)$ (d) $\left(2, \frac{7}{3}, -1\right)$
24. The plane $x + y + z = 1$ divide the points $A(1, 2, 4)$ and $B(2, 4, 5)$ in the ratio
 (a) -3 : 4 (b) 3 : 4
 (c) 4 : 3 (d) -4 : 3
25. The distance of the point (a, b, c) from the xy -plane is :
 (a) a (b) b
 (c) c (d) $\sqrt{a^2 + b^2}$
26. If $(4, \phi, z)$ be the cylindrical co-ordinates of the point $(2, 4, 6)$ then $\tan \phi$ is
 (a) 3 (b) 2
 (c) $\frac{3}{2}$ (d) $\frac{1}{2}$
27. The x -co-ordinate of the point which divides the joint of $(1, 2, 3)$ and $(3, 5, 7)$ in the ratio 1 : 2 is
 (a) $\frac{5}{3}$ (b) $\frac{1}{3}$
 (c) $\frac{13}{3}$ (d) $\frac{10}{3}$
28. If (r, θ, ϕ) be the spherical co-ordinate of the point $(1, 3, 6)$ then $\tan \theta$ is
 (a) 2 (b) $3\sqrt{5}$
 (c) $\sqrt{\frac{37}{3}}$ (d) $\frac{\sqrt{10}}{6}$
29. The y -co-ordinate of the point which divides the ratio of the points $A(1, 3, 2)$ and $B(2, 3, -2)$ externally in the ratio 2 : 3 is
 (a) -1 (b) 8
 (c) -3 (d) 3
30. The distance of the point (a, b, c) from z -axis is
 (a) $\sqrt{a^2 + c^2}$ (b) $\sqrt{a^2 + b^2}$
 (c) $\sqrt{b^2 + c^2}$ (d) $\sqrt{a^2 + b^2 + c^2}$
31. If (u, ϕ, z) be the cylindrical co-ordinate of a point $P(x, y, z)$ in 3D-geometry then u is
 (a) $x^2 + y^2 + z^2$ (b) $x^2 + z^2$
 (c) $x^2 + y^2$ (d) $y^2 + z^2$
32. If (x_1, y_1, z_1) and (x_2, y_2, z_2) be the co-ordinates of P and Q with respect to origin O then the co-ordinates of Q with respect to P are
 (a) (x_2, y_2, z_2)
 (b) $(x_1 + x_2, y_1 + y_2, z_1 + z_2)$
 (c) $(x_1 - x_2, y_1 - y_2, z_1 - z_2)$
 (d) $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$
33. If (r, θ, ϕ) be the spherical polar co-ordinates of $(1, 2, 3)$ then θ is
 (a) $\tan^{-1}(\sqrt{13})$ (b) $\tan^{-1}\left(\frac{\sqrt{10}}{2}\right)$
 (c) $\tan^{-1}\left(\frac{\sqrt{5}}{3}\right)$ (d) $\frac{\pi}{2}$
34. The co-ordinates of the point which divides the join of the points $A(3, 5, 7)$ and $(1, -1, -3)$ in equal ratio are
 (a) $(1, 3, 5)$ (b) $(2, 2, 2)$
 (c) $(2, 3, 5)$ (d) $(1, 3, 2)$
35. The centroid of the triangle with vertices $(1, -1, 2)$, $(2, 3, 5)$ and $(3, 2, 1)$ is
 (a) $\left(2, \frac{8}{3}, \frac{8}{3}\right)$ (b) $\left(2, \frac{-4}{3}, \frac{8}{3}\right)$
 (c) $\left(2, \frac{4}{3}, \frac{8}{3}\right)$ (d) $\left(2, \frac{8}{3}, \frac{4}{3}\right)$

36. If (u, ϕ, z) be the cylindrical co-ordinate of a point $P(3, 4, 5)$ in 3D-geometry then u is equal to
 (a) $\sqrt{34}$ (b) $\sqrt{41}$
 (c) 3 (d) 5
37. If $(1, 2, 3)$ and $(3, 4, 6)$ be the co-ordinates of P and Q with respect to origin $(0, 0, 0)$ then the co-ordinates of Q with respect to P are
 (a) $(-2, -2, -3)$ (b) $(2, 2, 3)$
 (c) $(1, 2, 3)$ (d) $(3, 4, 6)$
38. The locus of the point whose y -coordinate is Q will be
 (a) $x = Q$ (b) $y = b$
 (c) $z = b$ (d) $xz = a$
39. If $(0, 1, 2)$ and $(2, 3, 5)$ be the co-ordinates of P and Q with respect to origin O then y -coordinate of the point Q with respect to P is
 (a) 2 (b) -2
 (c) 1 (d) 3
40. If (r, θ, ϕ) be the spherical polar co-ordinates of $(3, 4, 2)$ then r is equal to
 (a) 5 (b) $2\sqrt{5}$
 (c) $\sqrt{29}$ (d) $\sqrt{13}$
41. If (u, ϕ, z) be the cylindrical co-ordinates of a point $(3, 4, 6)$ then z is
 (a) 5 (b) 3
 (c) 4 (d) 6
42. The distance between the points $(0, 0, 1)$ and $(1, 2, -1)$ is
 (a) 1 (b) 2
 (c) 3 (d) $\sqrt{2}$
43. The points $(1, 2, 3)$, $(2, 3, 1)$ and $(3, 1, 2)$ forms a triangle which is
 (a) isosceles right triangle
 (b) equilateral
 (c) isosceles
 (d) none of these
44. The cylindrical equation of the sphere whose centre is the origin and radius a is
 (a) $u^2 + a^2 = z^2$ (b) $u^2 + z^2 = a^2$
 (c) $z^2 + a^2 = u^2$ (d) $x^2 + y^2 + z^2 = a^2$
45. The locus of the points which are equidistance from the points $A(1, 2, 3)$ and $B(3, 2, 1)$ is
 (a) $x + y + z = 0$ (b) $x - z = 0$
 (c) $y - z = 0$ (d) $x + y - z = 0$
46. The perpendicular distance of the point $(2, 3, 4)$ from OX -axis is
 (a) 5 (b) $\sqrt{13}$
 (c) $\sqrt{20}$ (d) none of these
47. The plane $2x + 2y - 2z = 1$ divide the points $A(2, 1, 5)$ and $B(3, 4, 3)$ in the ratio
 (a) 7 : 5 (b) 7 : 3
 (c) 5 : 7 (d) 3 : 7
48. If the vertices of a triangle are the points $(1, -1, 2)$, $(2, 3, 4)$ and $(3, 2, 1)$ then the co-ordinates of its centroid are
 (a) $(2, 0, 1)$ (b) $(2, 3, 4)$
 (c) $(2, 1, 4)$ (d) $\left(2, \frac{4}{3}, \frac{7}{3}\right)$
49. The x -co-ordinate of the point which divides the join of the points $A(1, 2, 3)$ and $B(2, 4, -1)$ internally in the ratio 1 : 2 is
 (a) $\frac{4}{3}$ (b) $\frac{3}{4}$
 (c) 2 (d) $\frac{8}{3}$
50. If (r, θ, ϕ) be the spherical co-ordinate of the point $(2, 3, 4)$ then ϕ is
 (a) $\tan^{-1}(2)$ (b) $\tan^{-1}\left(\frac{3}{2}\right)$
 (c) $\tan^{-1}\left(\frac{4}{3}\right)$ (d) $\frac{\pi}{2}$
51. If (u, ϕ, z) be the cylindrical co-ordinates of the point $(1, 4, 6)$ then u is
 (a) $\sqrt{17}$ (b) $\sqrt{37}$
 (c) $\sqrt{52}$ (d) none of these
52. If the vertices of a triangle ABC are the points $A(2, -1, 0)$, $B(3, 3, -3)$, $C(0, 1, 4)$, then coordinates of its centroid are **[Kanpur 2018]**
 (a) $\left(\frac{5}{3}, 1, \frac{1}{3}\right)$ (b) $\left(\frac{-5}{3}, 2, \frac{-1}{3}\right)$
 (c) $(5, 2, 1)$ (d) $\left(4, \frac{1}{3}, \frac{2}{3}\right)$

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(c)	2.	(a)	3.	(c)	4.	(d)	5.	(b)	6.	(c)	7.	(b)	8.	(d)	9.	(a)	10.	(a)
11.	(b)	12.	(b)	13.	(a)	14.	(c)	15.	(c)	16.	(a)	17.	(a)	18.	(c)	19.	(b)	20.	(b)
21.	(d)	22.	(c)	23.	(c)	24.	(a)	25.	(c)	26.	(b)	27.	(a)	28.	(d)	29.	(d)	30.	(b)
31.	(c)	32.	(d)	33.	(c)	34.	(b)	35.	(c)	36.	(d)	37.	(b)	38.	(b)	39.	(a)	40.	(c)
41.	(d)	42.	(c)	43.	(b)	44.	(b)	45.	(b)	46.	(a)	47.	(c)	48.	(d)	49.	(a)	50.	(b)

HINTS AND SOLUTIONS

4.
$$\begin{array}{c} A \quad \lambda \quad B \quad 1 \quad C \\ (-1, -1, -1) \quad (1, 3, 2) \quad (5, 11, 8) \end{array}$$

$$\therefore \bar{x} = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

So $1 = \frac{5\lambda - 1}{\lambda + 1} \Rightarrow \lambda = \frac{1}{2}$

12.
$$\begin{array}{c} A \quad \lambda \quad P \quad 1 \quad B \\ (1, 2, 3) \quad (9, b, 7) \quad (2, 10, 1) \end{array}$$

$$\therefore \bar{z} = \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2} \Rightarrow 7 = \frac{3 + \lambda}{\lambda + 1} \Rightarrow 6\lambda = -4$$

$$\lambda = \frac{-2}{3} \text{ now } \bar{y} = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

$$\Rightarrow b = \frac{\lambda \cdot 10 + 2}{\lambda + 1} = \frac{\frac{-2}{3} \cdot 10 + 2}{\frac{-2}{3} + 1} = \frac{-14}{1} = -14$$

13. Here $x = 3, y = 4, z = 5$ so $u = \sqrt{x^2 + y^2} = 5,$

$$\tan \phi = \frac{y}{x} = \frac{4}{3},$$

$$\phi = \tan^{-1}\left(\frac{4}{3}\right) \text{ and } x = z = 5$$

16. $d = \sqrt{(2+1)^2 + (3-3)^2 + (1+3)^2} = 5$

23. The centroid of triangle is $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$

i.e. $\left(\frac{1+2+3}{3}, \frac{0+3+4}{3}, \frac{2-1+2}{3}\right)$ i.e. $\left(2, \frac{7}{3}, 1\right)$

28. $\tan \theta = \frac{\sqrt{x^2 + y^2}}{z} = \frac{\sqrt{1+9}}{6} = \frac{\sqrt{10}}{6}$

40. $r = \sqrt{x^2 + y^2 + z^2} = \sqrt{9+16+4} = \sqrt{29}$

45.
$$\begin{array}{c} A \quad P \quad B \\ (1, 2, 3) \quad (x, y, z) \quad (3, 2, 1) \end{array}$$

$$\text{Here } AP = PB \Rightarrow \sqrt{(x-1)^2 + (y-2)^2 + (z-3)^2} = \sqrt{(x-3)^2 + (y-2)^2 + (z-1)^2}$$

$$\Rightarrow x - z = 0$$

47. Let the point be $P(x, y, z)$ divide in ratio $\lambda : 1$ then

$$x = \frac{3\lambda + 2}{\lambda + 1}, y = \frac{4\lambda + 1}{\lambda + 1}, z = \frac{3\lambda + 5}{\lambda + 1}$$

$$\begin{array}{c} A \quad \lambda \quad P \quad 1 \quad B \\ (2, 1, 5) \quad (x, y, z) \quad (3, 4, 3) \end{array}$$

These lies on the plane $2x + 2y - 2z = 1.$

So $\frac{4\lambda - 2}{\lambda + 1} = \frac{1}{2}$

$$\Rightarrow 8\lambda - 4 = \lambda + 1 \Rightarrow 7\lambda = 5 \Rightarrow \lambda = \frac{5}{7}$$

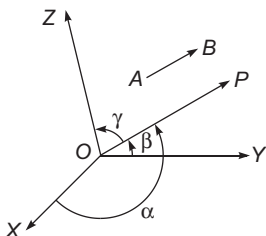
50. $\tan \phi = \frac{y}{x} = \frac{3}{2} \Rightarrow \phi = \tan^{-1}\left(\frac{3}{2}\right)$

51. $u = \sqrt{x^2 + y^2} = \sqrt{1+16} = \sqrt{17}$

Direction Cosines and Projections

DIRECTION COSINES OF A LINE OR VECTOR

1. If a directed line or a vector makes angles α , β and γ with positive directions of x -axis, y -axis and z -axis respectively then $\cos\alpha$, $\cos\beta$, $\cos\gamma$ are called the direction cosines (d.c.) of that directed line. These are usually denoted by l, m, n .
2. If the d.c.'s of a directed line AB are l, m, n then the d.c.'s of the directed line BA are $-l, -m, -n$.



3. The direction cosines of the x -axis are 1, 0, 0. Similarly the d.c.'s of the y -axis are 0, 1, 0 and d.c.'s of the z -axis are 0, 0, 1.
4. Suppose l, m, n are the d.c.'s of a line then any three numbers a, b, c which are proportional to l, m, n i.e. $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$ are called direction ratios of that line.
5. Direction cosines of a line are unique but direction ratio are not unique.
6. If l, m, n are the direction cosines of a line OP ($OP = r$), then the co-ordinates of P are (lr, mr, nr) .
7. If l, m, n are the direction cosines of a line then $l^2 + m^2 + n^2 = 1$
8. If α, β, γ are the angles which a line makes with the axes, then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$
9. The direction cosines of a line whose direction ratios are a, b, c are

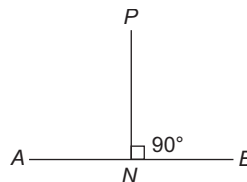
$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}}.$$

10. The direction ratios of a line PQ joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ and its direction cosines are

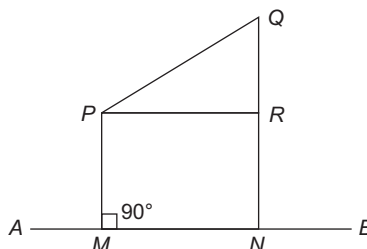
$$\frac{x_2 - x_1}{PQ}, \frac{y_2 - y_1}{PQ}, \frac{z_2 - z_1}{PQ}.$$

PROJECTIONS

1. Let P be a given point and AB the given straight line. Draw PN perpendicular from P to AB , meeting AB in N . Then the foot N of the perpendicular PN is called the projection of the given point P on the given line AB .



2. The projection of the line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line whose direction cosines are l, m, n is $l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$.



3. If the projection of PQ on AB is zero, then PQ is perpendicular to AB .
4. If O and P are two points $(0, 0, 0)$ and (x_1, y_1, z_1) , then the projection of OP on a line whose direction cosines are l, m, n is $lx_1 + my_1 + nz_1$.
5. The projection of the line joining the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on a line whose direction ratios are a, b, c is
$$\frac{(x_2 - x_1)a + (y_2 - y_1)b + (z_2 - z_1)c}{\sqrt{a^2 + b^2 + c^2}}$$
3. Two lines in point 1 are parallel if and only if
$$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$$
4. If θ be the angle between two lines whose direction ratios are a_1, a_2, a_3 and b_1, b_2, b_3 then
$$\cos \theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$
5. Two lines in point 4 are perpendicular iff $a_1 b_1 + a_2 b_2 + a_3 b_3 = 0$
6. Two lines in point 4 are parallel iff
$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$$

ANGLE BETWEEN TWO LINES

1. If θ is the angle between two lines or vectors whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 then $\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$
2. Two lines in point 1 are perpendicular if and only if $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$
7. Three points P, Q, R are collinear if direction ratios of PQ and RS are proportional and PQ and PR pass through P .

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The direction cosines of the y -axis are
 - (a) 1, 0, 0
 - (b) 0, 0, 1
 - (c) 0, 1, 0
 - (d) 0, 1, 1
2. If l, m, n are the direction cosines of a straight line then $l^2 + m^2 + n^2$ is
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
3. The direction ratios of the line joining the points $(1, 2, 3)$ and $(2, 3, 4)$ are
 - (a) 1, 1, 1
 - (b) 1, 2, 3
 - (c) $\frac{2}{5}, \frac{3}{5}, \frac{4}{5}$
 - (d) $\frac{1}{5}, \frac{2}{5}, \frac{3}{5}$
4. The length of a line whose projection on axes are 5, 10, 10 is
 - (a) 25
 - (b) 5
 - (c) 10
 - (d) 15
5. The angle between two diagonals of a cube is

[Avadh 2018]

 - (a) $\sin^{-1}\left(\frac{1}{3}\right)$
 - (b) $\cos^{-1}\left(\frac{1}{3}\right)$
 - (c) $\tan^{-1}\left(\frac{1}{3}\right)$
 - (d) $\sec^{-1}\left(\frac{1}{3}\right)$
6. The projection of the line joining the points $(1, 2, 4)$ and $(3, 5, 7)$ on a line whose direction cosines are 1, 2, 3 is
 - (a) 17
 - (b) -17
 - (c) 15
 - (d) 16
7. If ΔDEF is the projection of ΔABC on a given plane and θ be the angle between the triangle ABC plane and then ΔDEF is equal to
 - (a) ΔABC
 - (b) $\Delta ABC \sin \theta$
 - (c) $\Delta ABC \tan \theta$
 - (d) $\Delta ABC \cos \theta$
8. If α, β, γ be the angles which a line makes with positive direction of the co-ordinate axes then $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$ is
 - (a) 0
 - (b) 1
 - (c) -1
 - (d) 2

9. The angle between two lines whose direction cosines are given by $l + m + n = 0$, $2mn + 3ln - 5lm = 0$ is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) π
10. If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of two lines then these lines will be perpendicular if $l_1 l_2 + m_1 m_2 + n_1 n_2$ is
 (a) 0 (b) 1
 (c) -1 (d) 2
11. The co-ordinates of the point in which perpendicular from (0, 0, 0) to join of (-9, 4, 5) and (11, 0, -1) meets is
 (a) (1, 2, 1) (b) (1, 2, 2)
 (c) (2, 1, 1) (d) (2, 2, 1)
12. If $\sqrt{3}, \sqrt{3}, \sqrt{3}$ are the direction ratios of a line then its direction cosines are
 (a) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (b) 1, 1, 1
 (c) $\sqrt{3}, \sqrt{3}, \sqrt{3}$ (d) $\frac{1}{3\sqrt{3}}, \frac{1}{3\sqrt{3}}, \frac{1}{3\sqrt{3}}$
13. If the projection of a line on the co-ordinate axes are 5, 10, 10 then direction ratios of the line are
 (a) 1, 2, 3 (b) 1, 3, 2
 (c) 1, 2, 2 (d) 1, 3, 3
14. If $\frac{l}{2}, \frac{m}{2}, \frac{n}{2}$ are the direction cosines of a line then
 (a) $l^2 + m^2 + n^2 = 1$ (b) $l^2 + m^2 + n^2 = 0$
 (c) $l^2 + m^2 + n^2 = 4$ (d) $l^2 + m^2 + n^2 = 2$
15. The projection of line AB on the line CD where A, B, C, D are (1, 2, 4), (2, 0, 1), (-1, 0, 2), (1, 2, -1) respectively is
 (a) $\frac{7}{17}$ (b) $\frac{\sqrt{7}}{17}$
 (c) $\frac{\sqrt{7}}{\sqrt{17}}$ (d) $\frac{7}{\sqrt{17}}$
16. The co-ordinates of the foot of perpendicular drawn from the point (0, 0, 0) to the line $x - 1 = y = z$ are
 (a) $\left(\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{3}, -\frac{1}{3}\right)$
 (c) $\left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}\right)$ (d) $\left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$
17. The direction ratios of the x-axis are
 (a) 1, 0, 0 (b) 0, 1, 0
 (c) 0, 0, 1 (d) 1, 1, 0
18. If α, β, γ are the angles which a line makes with the axes then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ is
 (a) 1 (b) 2
 (c) 3 (d) 0
19. If the projection of a line on co-ordinate axes are 2, 3, 6 then its length is
 (a) 2 (b) 3
 (c) 6 (d) 7
20. If $a, \frac{1}{2}, \frac{-1}{2}$ are the direction cosines of a given line then the value of a is
 (a) $\pm \frac{1}{2}$ (b) $\pm \frac{1}{3}$
 (c) $\pm \frac{1}{\sqrt{2}}$ (d) $\pm \frac{1}{\sqrt{3}}$
21. The projection of the line AB, where the co-ordinates of A and B are (0, 0, 0) and (a, b, c) on a given line $x = y = z$ is
 (a) $\frac{a + b + c}{\sqrt{3}}$ (b) $\frac{a + b + c}{3}$
 (c) $\frac{a + b - c}{\sqrt{3}}$ (d) $\frac{a + b - c}{3}$
22. If the points are A(1, 1, 0) and B(0, 1, 1) then the angle between OA and OB, where O is origin is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
23. If θ be the angle between two diagonals of a cube then $\sin \theta$ is equal to
 (a) $\frac{\sqrt{2}}{\sqrt{3}}$ (b) $\frac{2}{3}$
 (c) $\frac{\sqrt{2}}{3}$ (d) $\frac{2\sqrt{2}}{3}$
24. The projection of a line on axes are 5, 10, 10 then its direction cosines are
 (a) $\frac{1}{15}, \frac{2}{15}, \frac{3}{5}$ (b) $\frac{2}{15}, \frac{1}{13}, \frac{3}{5}$
 (c) $\frac{1}{15}, \frac{2}{15}, \frac{3}{5}$ (d) $\frac{1}{15}, \frac{2}{15}, \frac{2}{15}$

[Avadh 2018]

[Avadh 2018]

25. If 1, -1, 0 and -1, 1, 2 are direction cosines of the two lines then the direction ratios of the line perpendicular to both are
 (a) 2, 2, 1 (b) -2, -2, -1
 (c) -2, -2, 0 (d) -2, +2, 1
26. The projection of a line on the plane is
 (a) a point (b) a line
 (c) a plane (d) none of these
27. The straight lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fmn + gln + hlm = 0$ are perpendicular if
 (a) $\frac{a}{f} + \frac{b}{g} + \frac{c}{h} = 0$ (b) $af + bg + ch = 0$
 (c) $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$ (d) $\frac{a}{f} = \frac{b}{g} = \frac{c}{h}$
28. The direction ratios of the line which is equally inclined to the axes are
 (a) 1, 1, 1 only (b) -1, -1, -1 only
 (c) $\pm 1, \pm 1, \pm 1$ (d) 1, 2, 3
29. The condition that the two lines whose direction ratios are $a, 0, 1$ and $2, b, c$ are perpendicular is
 (a) $2a + b + c = 0$ (b) $2a + c = 0$
 (c) $a + b + c = 0$ (d) $2a - c = 6$
30. If the vertices A, B, C of a triangle have co-ordinates (2, 3, 5), (-1, 3, 2) and (3, 5, -2) respectively then angle B is
 (a) $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$ (b) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 (c) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (d) $\frac{\pi}{2}$
31. The projection of a line on axes are 2, 4, 6 then the length of the line is
 (a) 2 (b) 4
 (c) $\sqrt{14}$ (d) $2\sqrt{14}$
32. The perpendicular distance of a point (1, 0, 2) from a line through origin and whose d.c.'s are 1, 1, 0 is
 (a) $\sqrt{\frac{3}{2}}$ (b) $\frac{1}{\sqrt{2}}$
 (c) $\sqrt{3}$ (d) $\frac{3}{\sqrt{2}}$
33. The sum of the direction cosines of a line whose direction ratios are 2, 3, -6 is
 (a) $\frac{1}{7}$ (b) 1
 (c) $-\frac{1}{7}$ (d) 0
34. If the length of a line OP is r and l, m, n are the direction cosines of the line OP then the co-ordinates of P are
 (a) (lr^2, mr^2, nr^2) (b) $\left(\frac{l}{r}, \frac{m}{r}, \frac{n}{r}\right)$
 (c) $\left(\frac{r}{l}, \frac{r}{m}, \frac{r}{n}\right)$ (d) (lr, mr, nr)
35. If O and P are the points (0, 0, 0) and (x, y, z) then the projection of OP on a line whose direction cosines are l, m, n is
 (a) $x\sqrt{l} + y\sqrt{m} + z\sqrt{n}$ (b) $l^2x + m^2y + n^2z$
 (c) $lx + my + nz$ (d) $\frac{x}{l} + \frac{y}{m} + \frac{z}{n}$
36. The condition that the two lines whose direction ratios are $a, -1, 1$ and $1, 3, c$ are perpendicular is
 (a) $2a + c = 1$ (b) $a + 3c = 2$
 (c) $a - c = 3$ (d) $a + c = 3$
37. If the projection of a line PQ on another line is zero then PQ is always
 (a) parallel to AB
 (b) perpendicular to AB
 (c) having some angle to AB
 (d) none of these
38. The projection of a line joining the points (1, 0, 2) and (2, 3, 4) on a line whose direction ratios are 1, 1, 1 is
 (a) 2 (b) $\sqrt{3}$
 (c) $2\sqrt{3}$ (d) $3\sqrt{2}$
39. If the d.c.'s of a line are 2, 3, 1 then which one is not its direction ratios
 (a) 4, 6, 2 (b) 12, 18, 6
 (c) 6, 9, 2 (d) 14, 21, 7

40. The sum of the direction cosines of a line whose direction ratios are 1, 2, 3 is
 (a) $\frac{1}{\sqrt{14}}$ (b) $\frac{2}{\sqrt{14}}$
 (c) $\frac{3}{\sqrt{14}}$ (d) $\frac{6}{\sqrt{14}}$
41. If the length of a line OP through the origin O is 4 and $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ are the direction cosines of the line OP then the co-ordinates of P are
 (a) $(4\sqrt{3}, 4\sqrt{3}, 4\sqrt{3})$ (b) $\left(\frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}\right)$
 (c) $\left(\frac{1}{4\sqrt{3}}, \frac{1}{4\sqrt{3}}, \frac{1}{4\sqrt{3}}\right)$ (d) $(12, 12, 12)$
42. The angle between the lines whose direction cosines are given by the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ is
[Avadh 2018]
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$
43. If the points are $A(1, 2, 3)$ and $B(2, 1, 2)$ then the angle between OA and OB , where O is origin is
 (a) $\cos^{-1}\left(\frac{10}{\sqrt{14}}\right)$ (b) $\cos^{-1}\left(\frac{10}{3}\right)$
 (c) $\cos^{-1}\left(\frac{10}{6\sqrt{14}}\right)$ (d) $\cos^{-1}\left(\frac{10}{3\sqrt{14}}\right)$
44. The co-ordinates of the foot of perpendicular drawn from the point $(1, 2, 1)$ to the line joining $(1, 4, 6)$ and $(5, 4, 4)$ are
 (a) $(3, 5, 4)$ (b) $(3, 4, 5)$
 (c) $(4, 3, 5)$ (d) $(5, 4, 3)$
45. If $A(6, 3, 2)$, $B(5, 1, 4)$, $C(3, -4, 7)$, $D(0, 2, 5)$ are four points then the projection of AB on CD is
[Kanpur 2018]
 (a) $\frac{13}{7}$ (b) $\frac{12}{7}$ (c) $\frac{15}{7}$ (d) $\frac{11}{7}$
46. The direction cosines of a segment of a line whose projections on the axes are 2, 3, 6 are given by
 (a) $\frac{2}{7}, \frac{3}{7}, \frac{4}{7}$ (b) $\frac{3}{7}, \frac{5}{7}, \frac{6}{7}$
 (c) $\frac{2}{7}, \frac{4}{7}, \frac{5}{7}$ (d) $\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$
47. If O, A, B are the points $(0, 0, 0)$, $(3, 3, 6)$, $(3, 5, -1)$ respectively then the angle between OA and OB is
 (a) 0 (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$
48. If the vertices A, B and C of a triangle ABC have co-ordinates $(2, 3, 5)$, $(-1, 3, 2)$ and $(3, 5, -2)$ respectively then angle B is
 (a) 0 (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{3}$
49. The projections of a line on co-ordinate axes are 1, 4, 5 then its length is
 (a) 1 (b) 4
 (c) 5 (d) $\sqrt{42}$
50. If two lines whose direction ratios are $a, 2, 4$ and $-1, 2, 2$ are perpendicular then a is
 (a) 4 (b) 8
 (c) 9 (d) 12
51. If a line makes equal angle with axes then its direction ratios are
 (a) 1, 1, 1 (b) 1, 0, 0
 (c) 1, 1, 0 (d) 0, 1, 1
52. If the direction cosines of a line are $\left(\frac{1}{a}, \frac{1}{a}, \frac{1}{a}\right)$ then a is
 (a) $\pm\sqrt{3}$ (b) $\sqrt{3}$ only
 (c) $-\sqrt{3}$ only (d) 1
53. If $a, 0, 0$ are the d.c.'s of a line then a is
 (a) +1 (b) -1
 (c) 0 (d) 1 or -1
54. The condition that the two lines whose d.c.'s are $0, a, \frac{1}{2}$ and $\frac{1}{3}, b, \frac{1}{2}$ are perpendicular is
 (a) $4ab + 1 = 0$ (b) $ab - 1 = 0$
 (c) $2ab + 1 = 0$ (d) $4ab - 1 = 0$
55. If the mid points of a triangle are $(3, 0, 1)$, $(1, 2, -3)$ and $(-1, 1, -4)$ then its centre is
 (a) $(1, 1, 2)$ (b) $(1, -1, 2)$
 (c) $(1, 1, -2)$ (d) $(-1, 1, 2)$

56. If a variable line in two adjacent positions has d.c.'s l, m, n and $l + \delta l, m + \delta m, n + \delta n$ the small angle between them is $\delta\theta$ then $(\delta\theta)^2$ is
- (a) $-(\delta l)^2 - (\delta m)^2 - (\delta n)^2$
 (b) $(\delta l)^2 + (\delta m)^2 + (\delta n)^2$
 (c) $(\delta l)^2 + (\delta m)^2 - (\delta n)^2$
 (d) $-(\delta l)^2 - (\delta m)^2 + \delta n^2$
57. The direction cosines of the line segment joining the points $P(-2, 1, -8)$ and $Q(4, 3, -5)$ are
[Kanpur 2018]
- (a) $\frac{5}{7}, \frac{2}{7}, \frac{3}{7}$ (b) $\frac{4}{7}, \frac{3}{7}, \frac{2}{7}$
 (c) $\frac{6}{7}, \frac{2}{7}, \frac{3}{7}$ (d) $\frac{5}{7}, \frac{6}{7}, \frac{3}{7}$
58. The projection of join of points $(0, 0, 0)$ and $(1, 1, 1)$ on a straight line with direction cosines $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ is
[Avadh 2018]
- (a) 1 (b) $\frac{1}{\sqrt{3}}$
 (c) $\sqrt{3}$ (d) $3\sqrt{3}$
59. The direction cosines of the line joining two points $(0, 1, 0)$ and $(2, 1, 1)$ are
[Avadh 2018]
- (a) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (b) 1, 0, 1
 (c) $\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}$ (d) $\frac{1}{\sqrt{5}}, 0, \frac{2}{\sqrt{5}}$
60. If α, β, γ be the angles which a line makes with the positive directions of the axes, then the value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is
[Avadh 2018]
- (a) -1 (b) +1
 (c) -2 (d) +2
61. The direction cosines of a line equally inclined with coordinate axes are
[Avadh 2018]
- (a) 1, 1, 1 (b) 1, 0, 0
 (c) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (d) 0, 1, 0
62. If the direction ratio of two lines intersect at $\frac{\pi}{4}$ are $2, -1, 2$ and $p, 3, 5$ then the value of p is
- (a) 1 (b) 2 (c) 4 (d) 8
63. If a line makes angle $\alpha, \beta, \gamma, \delta$ with four diagonal of a cube, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma + \sin^2\delta$ is equal to
- (a) $\frac{4}{3}$ (b) $\frac{7}{3}$ (c) $\frac{10}{3}$ (d) $\frac{8}{3}$
64. The triangle formed by the vertices $(a, b, c), (b, c, a)$ and (c, a, b) is
- (a) right angle triangle only
 (b) isoscale triangle only
 (c) equilateral triangle only
 (d) right angle isoscale triangle
65. The direction cosines of the line whose equations are $x + y = 3$ and $x + y + z = 0$ are
[Kanpur 2018]
- (a) $0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ (b) $0, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$
 (c) $1, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ (d) $1, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$
66. The direction cosines of the line joining the points $(1, 0, 1)$ and $(0, 1, 0)$ are
[Agra 2017]
- (a) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$
 (c) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$ (d) None of these

ANSWERS

MULTIPLE CHOICE QUESTIONS

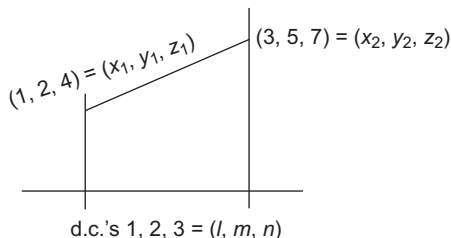
1.	(c)	2.	(b)	3.	(a)	4.	(d)	5.	(b)	6.	(a)	7.	(d)	8.	(c)	9.	(a)	10.	(a)
11.	(b)	12.	(a)	13.	(c)	14.	(c)	15.	(d)	16.	(c)	17.	(a)	18.	(b)	19.	(d)	20.	(c)
21.	(a)	22.	(b)	23.	(d)	24.	(d)	25.	(c)	26.	(b)	27.	(c)	28.	(c)	29.	(b)	30.	(d)
31.	(d)	32.	(d)	33.	(c)	34.	(d)	35.	(c)	36.	(d)	37.	(b)	38.	(c)	39.	(c)	40.	(d)
41.	(b)	42.	(b)	43.	(d)	44.	(b)	45.	(a)	46.	(d)	47.	(d)	48.	(c)	49.	(d)	50.	(d)

HINTS AND SOLUTIONS

4. Given that $lr = 5, mr = 10, nr = 10$
 So $l^2r^2 + m^2r^2 + n^2r^2 = 25 + 100 + 100$
 $\Rightarrow r^2 = 225 \Rightarrow r = 15$

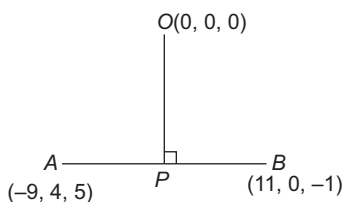
5. The direction ratios of the diagonals are 1, -1, -1 and 1, 1, -1 so angle between them is
 $\cos\theta = \frac{1-1+1}{\sqrt{3}\sqrt{3}} = \frac{1}{3}$ i.e. $\theta = \cos^{-1}\left(\frac{1}{3}\right)$

6. Required projection is
 $l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$
 $1(3 - 1) + 2(5 - 2) + 3(7 - 4)$
 $= 2 + 6 + 9 = 17$



8. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 - 2\sin^2\alpha + 1 - 2\sin^2\beta + 1 - 2\sin^2\gamma$
 $= 3 - 2(\sin^2\alpha + \sin^2\beta + \sin^2\gamma)$
 $= 3 - 2(2) = -1$
 $\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
 $\Rightarrow 1 - \sin^2\alpha + 1 - \sin^2\beta + 1 - \sin^2\gamma = 1$

- $\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$
 11. d.r's of AB is 20, -4, -6
 Equation of line AB is $\frac{x+9}{20} = \frac{y-4}{-4} = \frac{z-5}{-6} = r$
 Then point P is $(20r - 9, -4r + 4, -6r + 5)$

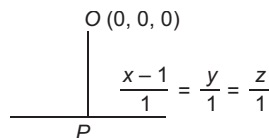


So direction ratio of OP is $20r - 9, -4r + 4, -6r + 5$
 Now OP and AB are perpendicular so
 $20(20r - 9) - 4(-4r + 4) - 6(-6r + 5) = 0$

After solving we get $r = \frac{113}{46}$

Put in P we get $P(1, 2, 2)$

16. Let the point P is given by $x - 1 = y = z = r$



So $P(r+1, r, r)$. Thus direction of P are $r+1, r, r$ and direction ratio of given line are $1, 1, 1$ but both are perpendicular so $r+1+r+r=0 \Rightarrow 3r=-1 \Rightarrow r = \frac{-1}{3}$

So required co-ordinate of P is $\left(\frac{2}{3}, \frac{-1}{3}, \frac{-1}{3}\right)$.

25. If a, b, c be the required direction ratios of perpendicular line then $a-b+c=0$ and $-a+b+2c=0$ then $\frac{a}{-2} = \frac{b}{-2} = \frac{c}{0}$

$$\Rightarrow a:b:c = -2:-2:0$$

27. The given relations are $al+bm+cn=0$ and $fmn+gln+hlm=0$

$$n = \frac{-(al+bm)}{c} \text{ put in other equation}$$

$$-hlm + (fm+gl) \frac{(al+bm)}{c} = 0$$

$$agl^2 + bfm^2 + (af+bg-ch)lm = 0$$

$$\text{i.e. } ag\left(\frac{l}{m}\right)^2 + (af+bg-ch)\left(\frac{l}{m}\right) + bf = 0$$

If l_1, m_1, n_1 and l_2, m_2, n_2 are the d.c.'s of two lines then $\frac{l_1 l_2}{m_1 m_2} = \frac{cbf}{cag}$ i.e. $\frac{l_1 l_2}{m_1 m_2} = \frac{n_1 n_2}{abh}$

The lines will be perpendicular if

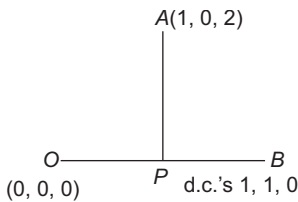
$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\text{i.e. } cbf + cag + abh = 0 \text{ i.e., } \frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$$

29. $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

$$\text{i.e. } 2a + 0b + c = 0 \Rightarrow 2a + c = 0$$

32. Equation of line OB is $\frac{x}{1} = \frac{y}{1} = \frac{z}{0} = r$



So co-ordinate of P is $(r, r, 0)$ d.r's of AP is $r-1, r, -2$ which is perpendicular to d.r's of OB i.e. $1(r-1) + 1(r) + 0(-2) = 0$

$$\Rightarrow r = \frac{1}{2}$$

So P is $\left(\frac{1}{2}, \frac{1}{2}, 0\right)$. Thus

$$AP = \sqrt{\left(\frac{1}{2}-1\right)^2 + \left(\frac{1}{2}-0\right)^2 + (-2)^2}$$

$$= \sqrt{\frac{1}{4} + \frac{1}{4} + 4}$$

$$\text{i.e. } AP = \frac{3}{\sqrt{2}}$$

$$36. (a) \cdot (1) + (-1)(3) + 1 \cdot c = 0 \Rightarrow a + c = 3$$

40. Since d.r.'s are $1, 2, 3$ so its d.c.'s are $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$ and their sums is $\frac{6}{\sqrt{14}}$.

46. Let l, m, n are the required d.c.'s then given that $lr = 2, mr = 3, nr = 6$ so

$$(lr)^2 + (mr)^2 + (nr)^2 = 2^2 + 3^2 + 6^2 = 49 \text{ i.e. } r = 7$$

So required d.c.'s are $\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$.

48. d.r.'s of BA are $3, 0, 3$ and d.r.'s of BC are $4, 2, -4$. If θ be the angle between BC and BA then

$$\cos \theta = \frac{12 + 0 - 12}{\sqrt{4+0+9}\sqrt{16+4+16}} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

52. Since d.c.'s are $\frac{1}{a}, \frac{1}{a}, \frac{1}{a}$ so $\frac{1}{a^2} + \frac{1}{a^2} + \frac{1}{a^2} = 1$

$$\text{i.e. } a^2 = 3 \text{ i.e. } a = \pm \sqrt{3}$$

$$56. \cos \delta \theta = l(l + \delta l) + m(m + \delta m) + n(n + \delta n)$$

$$1 - 2\sin^2 \frac{\theta}{2} = 1 + (l\delta l + m\delta m + n\delta n)$$

$$\text{i.e. } -2\sin^2 \frac{\theta}{2} = l\delta l + m\delta m + n\delta n \quad \dots(1)$$

$$\text{Also } l^2 + m^2 + n^2 = 1$$

$$\text{and } (l + \delta l)^2 + (m + \delta m)^2 + (n + \delta n)^2 = 1$$

Subtracting these two equations we get

$$\delta l^2 + \delta m^2 + \delta n^2 = -(2l\delta l + 2m\delta m + 2n\delta n) \quad \dots(2)$$

By (1) and (2)

$$-2\sin^2 \frac{\theta}{2} = -\frac{1}{2}(\delta l^2 + \delta m^2 + \delta n^2)$$

$$\text{Since } \delta \theta \text{ is small so } \sin \frac{\theta}{2} = \frac{\theta}{2}$$

$$\text{We get } -2\left(\frac{\delta \theta}{2}\right)^2 = -\frac{1}{2}(\delta l^2 + \delta m^2 + \delta n^2)$$

$$\text{i.e. } \delta \theta^2 = \delta l^2 + \delta m^2 + \delta n^2$$

PLANE

1. A plane is a surface such that every straight line joining any two points on it lies wholly on it.
2. A straight line which is perpendicular to every line lying in a plane is called a normal to that plane. All the normals to a plane are parallel lines.
3. The equation of a plane whose perpendicular distance from the origin is p and $\cos\alpha, \cos\beta, \cos\gamma$ are direction-cosines of this perpendicular, is $x \cos\alpha + y \cos\beta + z \cos\gamma = p$ i.e. $lx + my + nz = p$. This is known as the equation of a plane in normal form.
4. The general equation of a plane is $ax + by + cz + d = 0$. Here the coefficients a, b, c of x, y and z are direction ratios of normal to this plane.
5. The number of arbitrary constants in the general equation of the plane are three.
6. The equation of any plane passing through the origin is $ax + by + cz = 0$.
7. If the plane makes intercepts a, b and c on the axes of x, y and z respectively, then the equation of the plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
8. The equation of xy -plane is $z = 0$, the equation of yz -plane is $x = 0$ and the equation of zx -plane is $y = 0$.
9. The equation of a plane through a given point $A(x_1, y_1, z_1)$ and perpendicular to a line whose direction ratios are a, b, c is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
10. The equation of a plane which passes through three points whose co-ordinates are $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and (x_3, y_3, z_3) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

11. The condition that four points $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$ and (x_4, y_4, z_4) are coplanar is

$$\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

12. The equation of the plane parallel to the yz -plane (respectively xz -plane, xy -plane) and at a distance a from it is $x = a$ (respectively $y = a, z = a$).
13. The equation of the plane perpendicular to the x -axis (respectively y -axis and z -axis) is $x = a$ (respectively $y = a$ and $z = a$).
14. The equation of plane parallel to x -axis (respectively y -axis and z -axis) is $by + cz + d = 0$ (respectively $ax + cz + d = 0$ and $ax + by + d = 0$).
15. The angle between two planes is defined as the angle between their normal drawn from any point to the planes.
16. If the equation of the plane is $ax + by + cz + d = 0$ and $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be any two points.
 - (i) If the points A and B lie on the same side of the plane then $ax_1 + by_1 + cz_1 + d$ and $ax_2 + by_2 + cz_2 + d$ have the same signs.
 - (ii) If the points A and B lie on opposite sides of the plane then $ax_1 + by_1 + cz_1 + d$ and $ax_2 + by_2 + cz_2 + d$ have opposite signs.

17. The length of the perpendicular from the point (x_1, y_1, z_1) to a given plane $ax + by + cz + d = 0$ is

$$\pm \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$$

18. The distance between two planes first take a point on one plane and then find the length of perpendicular from this point to the other plane.

19. The equation of any plane passing through the line of intersection of two planes $P = 0$ and $Q = 0$ is $P + \lambda Q = 0$ where λ is a parameter.

20. The condition that the line whose d.c.'s are l, m, n is parallel to the plane $ax + by + cz + d = 0$ is $al + bm + cn = 0$.

21. The condition that the line whose d.c.'s are l, m, n is perpendicular to the plane $ax + by + cz + d = 0$ is $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$.

THE ANGLE BETWEEN A LINE AND A PLANE

- The angle between a line and a plane is defined to be the complement of the angle between the line and the normal to the plane.
- The equation of the planes bisecting the angles between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are

$$\begin{aligned} \frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} \\ = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}} \end{aligned}$$

- The equation of the plane (see point 2) bisecting the angle in which the origin lies is

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

- The equation of the plane (see point 2) bisecting the angle in which the origin does not lies is

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = - \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

5. If in (point 2), the constant terms d_1 and d_2 are of the same sign then :

- If $a_1a_2 + b_1b_2 + c_1c_2 < 0$, then the angle between the planes in which the origin lies is acute.
- If $a_1a_2 + b_1b_2 + c_1c_2 > 0$, then the angle between the planes in which the origin lies is obtuse.

6. The equation

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

represents a pair of lines if

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

7. The angle between two planes defined in (point 6) is

$$\tan^{-1} \left[\frac{2\sqrt{f^2 + g^2 + h^2 - ab - bc - ac}}{a + b + c} \right].$$

8. Two planes defined in (point 6) are perpendicular if $a + b + c = 0$.

PROJECTION ON A PLANE AND AREA OF A TRIANGLE

- If the projection of an area A on the co-ordinate planes yz , zx and xy by A_x , A_y and A_z respectively, then

$$A^2 = A_x^2 + A_y^2 + A_z^2$$

- The area of a triangle ABC , the co-ordinates of whose vertices are $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ is Δ where $\Delta = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$ and

$$\Delta x = \frac{1}{2} \begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix}, \Delta y = \frac{1}{2} \begin{vmatrix} x_1 & z_1 & 1 \\ x_2 & z_2 & 1 \\ x_3 & z_3 & 1 \end{vmatrix},$$

$$\Delta z = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

VOLUME OF A TETRAHEDRON

1. The volume of the tetrahedron $ABCD$, whose vertices are the points $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$ and $D(x_4, y_4, z_4)$ is

$$V = \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}.$$

2. If V is the volume of the tetrahedron $OABC$ whose vertices are the points $O(0, 0, 0)$, $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$

$$\text{then } V = \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}.$$

EXERCISE

MULTIPLE CHOICE QUESTIONS

- The equation of the co-ordinate plane xy is
 - $x = 0$
 - $y = 0$
 - $xy = 0$
 - $z = 0$
- The number of arbitrary constant in the general equation of the plane $ax + by + cz + d = 0$ are
 - 1
 - 2
 - 3
 - 4
- The equation of the plane through $(2, 0, 1)$ and perpendicular to a plane whose d.r. are $1, 1, 1$ is
 - $x + y + z = 3$
 - $x + y + z = 2$
 - $x + y + z = 0$
 - $2x + z = 1$
- The equation of the plane perpendicular to z -axis is :
 - $x = a$
 - $y = a$
 - $z = a$
 - $xy = a$
- $ax + cz + d = 0$ always represents a plane parallel to
 - x -axis
 - y -axis
 - z -axis
 - any axis
- The length of the perpendicular from the point $(1, 2, 3)$ to a plane $x - 2y + 2z = 0$ is
 - 0
 - 2
 - 3
 - 1
- The angle between the planes $2x - y + z = 7$ and $x + y + 2z = 9$ is
 - $\frac{\pi}{2}$
 - $\frac{\pi}{3}$
 - $\frac{\pi}{4}$
 - 0
- The equation of the plane which makes intercepts $1, 2, 3$ on the co-ordinate axes is
 - $x + 2y + 3z = 6$
 - $2x + y + 3z = 6$
 - $6x + 3y + 2z = 6$
 - none of these
- The equation of the plane through $A(2, 3, -1)$ at right angle to OA , where O is origin is
 - $2x + 3y - z = 14$
 - $2x + 3y - z = 12$
 - $2x + 3y + z = 14$
 - $2x + 3y + z = 12$
- The equation of the plane which makes equal intercepts on the axes and pass through the point $(2, 3, 4)$ is
 - $x + y + z = 1$
 - $x + y + z = 5$
 - $x + y + z = 7$
 - $x + y + z = 9$
- The distance between the planes $x + 2y + z = 1$ and $2x + 4y + 2z = 2$ is
 - 1
 - 2
 - 0
 - 3
- The foot of the perpendicular from the origin to a plane is $(2, 1, 3)$, then the equation of the plane is
 - $2x + y + 3z = 14$
 - $x + 2y + 3z = 14$
 - $3x + 2y + z = 14$
 - $x + 3y + 2z = 14$

13. The position of the points $(1, -1, 3)$ and $(3, 3, 3)$ from the plane $5x + 2y - 7z + 9 = 0$ is
 (a) equidistance and same side
 (b) only equidistance
 (c) only opposite side
 (d) equidistance and opposite side
14. The equation of the plane passing through the line of intersection of the planes $x + y + z = 1$, $2x - y + 2z = 2$ and the point $(0, 0, 2)$ is
 (a) $x + y + z = 3$ (b) $x = 0$
 (c) $z = 0$ (d) $y = 0$
15. If d_1 and d_2 are both positive and the origin lies in the obtuse angle between the two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ then the value of $a_1a_2 + b_1b_2 + c_1c_2$ is
 (a) positive
 (b) negative
 (c) 0
 (d) may be positive or negative
16. The equation of the plane through $P(a, b, c)$ and perpendicular to OP is
 (a) $ax + by + cz = a^2 + b^2 + c^2$
 (b) $ax + by + cz = 1$
 (c) $ax + by + cz = \sqrt{a^2 + b^2 + c^2}$
 (d) $ax + by + cz = 0$
17. The intercepts form of the plane is
 (a) $lx + my + nz = p$ (b) $ax + by + cz + d = 0$
 (c) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (d) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
18. The length of the perpendicular drawn from the origin to the plane $x + 4y - 8z + 18 = 0$ is
 (a) 2 (b) 3 (c) 4 (d) 6
19. The equation of the plane parallel to the xy -plane and at a distance a from it is
 (a) $x = a$ (b) $y = a$
 (c) $xy = a$ (d) $z = a$
20. The intercept made by the plane $ax + by + cz = d$ on x -axis is
 (a) $\frac{d}{b}$ (b) $\frac{d}{c}$ (c) $\frac{a}{d}$ (d) $\frac{d}{a}$
21. $a(x - \alpha) + b(y - \beta) + c(z - \gamma) = 0$ represents a plane passing through the point
 (a) (a, b, c) (b) (α, β, r)
 (c) $(a\alpha, b\beta, c\gamma)$ (d) $\left(\frac{a}{\alpha}, \frac{b}{\beta}, \frac{c}{\gamma}\right)$
22. The equation of the plane passing through the origin and parallel to the plane $x - 3y + 4z + 2 = 0$ is
 (a) $x - 3y + 4z = 0$ (b) $x - 3y + 4z = 2$
 (c) $x - 3y + 4z = 4$ (d) $x - 3y + 4z = -2$
23. The area of a triangle included between the plane $3x - 4y + z = 12$ and the coordinate planes is
 (a) $6\sqrt{26}$ (b) $2\sqrt{26}$ (c) $\sqrt{26}$ (d) $3\sqrt{26}$
24. The equation of the plane perpendicular to x -axis is :
 (a) $x = a$ (b) $y = a$
 (c) $z = a$ (d) $yz = 1$
25. The equation of the plane through $(0, 1, 0)$ and perpendicular to a line whose d.r. are $0, 1, 0$ is
 (a) $z = 1$ (b) $y = 1$
 (c) $x = 1$ (d) $x + z = 1$
26. The volume of the tetrahedron formed by the planes $x + y + z = 1$, $x + y = 0$, $y + z = 0$ and $z + x = 0$ is
 (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) 1 (d) 0
27. The equation of the plane through the line of intersection of the planes $x + y + z + 1 = 0$ and $-x + y + z - 1 = 0$ and parallel to x -axis is
 (a) $2y + 2z = 1$ (b) $y + z = 1$
 (c) $2y + z = 1$ (d) $y + z = 0$
28. The equation of the plane passes through the z -axis is
 (a) $x + \lambda y = 0$ (b) $x - \lambda y = 1$
 (c) $z = 0$ (d) $z = 1$
29. The condition that a line whose d.r.s are l, m, n is perpendicular to the plane $ax + by + cz + d = 0$ is
 (a) $al + bm + cn = 0$ (b) $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$
 (c) $al + bm + cn = 1$ (d) none of these
30. The angle between the planes $x = a$ and $y = b$ is
 (a) 0 (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$

31. The equation of a plane whose perpendicular distance from the origin is 2 and 1, -2, 2 are the d.r's of this perpendicular, is
 (a) $x - 2y + 2z = 0$ (b) $x - 2y + 2z = 2$
 (c) $x - 2y + 2z = 3$ (d) $x - 2y + 2z = 6$
32. The equation of the yz -plane at a distance a from the origin is
 (a) $y = a$ (b) $z = a$
 (c) $x = a$ (d) $y + z = a$
33. The equation of the plane which contains the line of intersection of the planes $3x - y + 2z + 1 = 0$ and $-x + y + 2z + 2 = 0$ and which is perpendicular to the plane $x + y + z = 0$ is
 (a) $7x - 4y - 3z + 4 = 0$
 (b) $11x - 3y - 4z - 4 = 0$
 (c) $6x + y - 7z + 1 = 0$
 (d) $2x + 4z + 3 = 0$
34. The perpendicular distance between the planes $x + y + z = 0$, $3x + 3y - 3z + 2 = 0$ is
 (a) $\frac{3}{2}$ (b) $\frac{3\sqrt{3}}{2}$ (c) $\frac{2}{3}$ (d) $\frac{2}{3\sqrt{3}}$
35. The equation of the plane through the line of intersection of the planes $x + 1 = 0$, $y + 3 = 0$ and passing through the origin is
 (a) $x + y = 2$ (b) $x - y + z = 1$
 (c) $x + y - z = 2$ (d) $x - y = 2$
36. The equation of the plane parallel to y -axis does not contain the variable
 (a) x (b) y
 (c) z (d) both x and z
37. All normals to a plane are always
 (a) perpendicular (b) intersect
 (c) parallel (d) none of these
38. In 3-dimension geometry $z = a$ represents a
 (a) plane (b) line
 (c) point (d) sphere
39. The plane $ax + cz + d = 0$ is parallel to
 (a) x -axis (b) y -axis
 (c) z -axis (d) yz -plane
40. The angle between the normals to the planes $2x - y + z = 13$ and $x + y + 2z = 9$ is
 (a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$
41. The bisector of the acute angle between the planes $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ is
 (a) $x - 13y + 32z = 0$
 (b) $23x - 13y + 32z + 45 = 0$
 (c) $23x - y + 32z = 0$
 (d) none of these
42. The area of the triangle whose vertices are $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ is
 (a) $3\sqrt{3}$ (b) $\frac{\sqrt{3}}{4}$ (c) $\sqrt{3}$ (d) $\frac{\sqrt{3}}{2}$
43. The area of the triangle included between the plane $2x - 3y + 4z = 12$ and the co-ordinate planes is
 (a) $2\sqrt{29}$ (b) $5\sqrt{29}$
 (c) $2\sqrt{29}$ (d) $4\sqrt{29}$
44. The angle between the planes formed by $x^2 + 4y^2 - z^2 + 4xy = 0$ is
 (a) $\tan^{-1}\left(\frac{5}{2}\right)$ (b) $\tan^{-1}\left(\frac{5}{\sqrt{2}}\right)$
 (c) $\tan^{-1}\left(\frac{\sqrt{5}}{2}\right)$ (d) $\tan^{-1}\left(\sqrt{\frac{5}{2}}\right)$
45. The volume of the tetrahedron whose vertices are the points $(0, 0, 0)$, $(1, 1, 1)$, $(1, 2, 3)$, $(2, 5, 6)$ is
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) 0
46. The planes represented by $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ will be at right angles if
 (a) $a + b + c = 1$ (b) $abc = 0$
 (c) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$ (d) $a + b + c = 0$
47. The equation of the plane perpendicular to z -axis is
 (a) $x = a$ (b) $y = a$
 (c) $z = a$ (d) $x + y = a$
48. The area of a triangle whose vertices are given by the co-ordinates $(0, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ is
 (a) 1 (b) $\frac{1}{2\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\sqrt{2}$

49. In 3-dimension geometry $y = a$ represents a plane parallel to
 (a) xy -plane (b) yz -plane
 (c) xz -plane (d) none of these
50. The number of arbitrary constants in the intercept form of the plane is
 (a) 4 (b) 3
 (c) 2 (d) 1
51. The planes $x - y + z = 7$ and $3x + 2y - z + 9 = 0$ are
 (a) parallel (b) intercept each other
 (c) perpendicular (d) none of these
52. If the area of projections of $\triangle ABC$ on the co-ordinate planes yz, zx and xy are respectively A_x, A_y and A_z then the area of $\triangle ABC$ is
 (a) $\sqrt{A_x + A_y + A_z}$ (b) $A_x^2 + A_y^2 + A_z^2$
 (c) $\sqrt{A_x^2 + A_y^2 + A_z^2}$ (d) $A_x + A_y + A_z$
53. If the foot of the perpendicular drawn from the origin to a plane is $(12, -4, 3)$ then the plane is
 (a) $4x - 3y + z = 103$
 (b) $12x - 4y + 3z = 169$
 (c) $12x - 4y + 3z = 103$
 (d) $4x - 3y + z = 169$
54. The distance between the planes $4x - 2y + 2z + 3 = 0$ and $2x - 2y + z + 1 = 0$ is
 (a) $\frac{2}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$
55. If the planes represented by $x^2 - 3y^2 + az^2 + 2yz + 3zx + xy = 0$ will be right angle then a is
 (a) 6 (b) 3
 (c) 2 (d) 0
56. The distance of the point $(1, 2, 0)$ from the plane $4x + 3y + 12z + 16 = 0$ is
 (a) 2 (b) 4
 (c) 12 (d) 16
57. The image of the point $(3, 5, 7)$ in the plane $2x + y + z = 6$ is
 (a) $(1, 3, 5)$ (b) $(1, -3, 5)$
 (c) $(1, -3, -5)$ (d) $(-5, 1, 3)$
58. The angle between the planes $x^2 + 4xy + 4y^2 - z^2 = 0$ is formed by
 (a) $\cos^{-1}\left(\frac{3}{2}\right)$ (b) $\cos^{-1}\left(\frac{2}{3}\right)$
 (c) $\cos^{-1}\left(\frac{4}{3}\right)$ (d) $\cos^{-1}\left(\frac{3}{4}\right)$
59. The angle between the planes $2x - y + z = 7$ and $x + y + 2z = 9$ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
60. The distance between two planes $2x - 2y + z + 1 = 0$ and $4x - 2y + 2z + 3 = 0$ is
 (a) $\frac{1}{3}$ (b) $\frac{5}{6}$
 (c) $\frac{7}{6}$ (d) $\frac{1}{6}$
61. If the plane $x + y + z = 5$ intersect the co-ordinate axes then the area of triangle formed by them is
 (a) $\frac{25}{2}$ (b) $\frac{25\sqrt{3}}{2}$
 (c) $\frac{15\sqrt{3}}{2}$ (d) $\frac{5\sqrt{3}}{2}$
62. The equation of the plane passing through $(2, 4, 6)$ and makes equal intercepts on co-ordinate axes is
 (a) $2x + 3y + 2z = 12$ (b) $x + 3y + 2z = 12$
 (c) $x + y + z = 12$ (d) $2x + 3y + z = 12$
63. The equation of the plane passing through the points $(1, 2, 1), (1, 1, 0)$ and $(-2, 2, -1)$ is
 (a) $2x + 3y - 3z = 5$ (b) $2x + 3y + 3z = 5$
 (c) $2x - 3y + 3z = 5$ (d) $-2x + 3y + 3z = 5$
64. The angle between the planes $3x - 4y + 5z = 0$ and $2x - y - 2z = 5$ is **[Kanpur 2018]**
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
 (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$
65. The angle between the planes $x + y + z = 1$ and $2x - y + z = 2$ is **[Avadh 2018]**
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
 (c) $\cos^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (d) $\cos^{-1}\left(\frac{2}{3}\right)$

- # ANSWERS

[illegible]

HINTS AND SOLUTIONS

3. Equation of plane passing through $(2, 0, 1)$ is
 $a(x - 2) + b(y - 0) + c(z - 1) = 0$ i.e.
 $ax + by + cz - 2a - c = 0$. Since this is perpendicular to plane whose d.r.'s are $1, 1, 1$ so
 $a = b = c = 1$ i.e. required plane is $x + y + z = 3$.

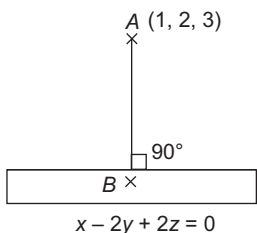
6. The direction ratio of line AB is $1, -2, 2$ so equation of line AB is $\frac{x-1}{1} = \frac{y-2}{-2} = \frac{z-3}{2} = r$

so co-ordinate of B is $(r + 1, -2r + 2, 2r + 3)$ which lies on plane so

$$1(r + 1) - 2(-2r + 2) + 2(2r + 3) = 0 \text{ i.e. } r = -\frac{1}{3} \text{ so } B$$

is $\left(\frac{2}{3}, \frac{8}{3}, \frac{7}{3}\right)$ thus AB is given by

$$AB = \sqrt{\left(1 - \frac{2}{3}\right)^2 + \left(2 - \frac{8}{3}\right)^2 + \left(3 - \frac{7}{3}\right)^2} = 1.$$



11. The point $(1, 0, 0)$ lies on first plane so its perpendicular distance from the plane $2x + 4y + 2z = 2$ is

$$p = \frac{2 + 0 + 0 - 2}{\sqrt{4 + 16 + 4}} = 0.$$

14. The equation of the plane passing through the line of intersection of given planes is $(x + y + z - 1) + \lambda(2x - y + 2z - 2) = 0$ i.e.
 $(1 + 2\lambda)x + (1 - \lambda)y + (1 + 2\lambda)z - 1 - 2\lambda = 0$ it pass through $(0, 0, 2)$ so $2 + 4\lambda - 1 - 2\lambda = 0 \Rightarrow 2\lambda = -1, \lambda = -\frac{1}{2}$ so required plane $y = 0$

23. The area of a triangle ABC whose intercepts a, b and c on the co-ordinate and is
 $A = \frac{1}{2} \sqrt{b^2c^2 + c^2a^2 + a^2b^2}$ so put $a = 4, b = -3, c = 12$ we get $A = 3\sqrt{26}$

30. $x + 0y + 0z = a, 0x + 1y + 0z = b$ so angle is
 $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2 = 1.0 + 1.0 + a.0 = 0 \Rightarrow \theta = \frac{\pi}{2}$

31. Required plane is $lx + my + nz = p$

Given d.r. are $1, -2, 2$ so d.c.'s are $\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}$ and

$$p = 2 \text{ so we get } \frac{1}{3}x - \frac{2}{3}y + \frac{2}{3}z = 2$$

$$\Rightarrow x - 2y + 2z = 6$$

$$\cos\theta = \frac{21 + (-1)(1) + 1(2)}{\sqrt{4 + 1 + 1}\sqrt{1 + 1 + 4}} = \frac{3}{6} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

44. Given that $x^2 + 4y^2 - z^2 + 4xy = 0$

Compare it with

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

$$a = 1, b = 4, c = -1, h = 2$$

The angle between two planes is

$$\theta = \tan^{-1} \left\{ \frac{2\sqrt{f^2 + g^2 + h^2 - ab - bc - ac}}{a + b + c} \right\}$$

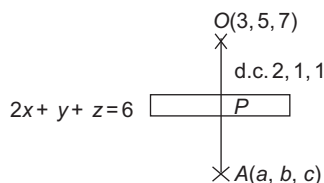
$$\theta = \tan^{-1} \left\{ \frac{2\sqrt{4 - 4 + 4 + 1}}{4} \right\} = \tan^{-1} \left\{ \frac{\sqrt{5}}{2} \right\}$$

54. The point $\left(0, \frac{3}{2}, 0\right)$ lies on first plane so its perpendicular distance from the plane $2x - 2y + z + 1 = 0$ is $p = \frac{0 - 3 + 0 + 1}{\sqrt{4 + 4 + 1}} = \frac{2}{3}$

55. $\therefore a + b + c = 0$ so $1 - 3 + a = 0 \Rightarrow a = 2$.

57. d.r.'s of OP are $2, 1, 1$ so its equation is
 $\frac{x-3}{2} = \frac{y-5}{1} = \frac{z-7}{1} = r$

Co-ordinates of P is $(2r + 3, r + 5, r + 7)$



Which lies on plane so $4r + 6 + r + 5 + r + 7 = 6 \Rightarrow 6r = -12, r = -2$

So, co-ordinate of P is $(-1, 3, 5)$. If (a, b, c) are the co-ordinate of image point A then $\frac{a+3}{2} = -1,$

$\frac{b+5}{2} = 3, \frac{c+7}{2} = 5$ i.e. co-ordinate of image point is $A(-5, 1, 3)$.

The Straight Line

THE EQUATIONS OF STRAIGHT LINE

1. The two non-parallel planes intersect in a line. Therefore, two equations of the first degree in x, y, z say, $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ together represent a straight line. These are called the general equations of a straight line of the unsymmetric form of the line.

2. The equation of straight line passing through a given point $A(\alpha, \beta, \gamma)$ and having d.c.'s l, m, n is

$$\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$$

3. The equation of straight line passing through a given point $A(\alpha, \beta, \gamma)$ and having d.r.'s a, b, c is

$$\frac{x - \alpha}{a} = \frac{y - \beta}{b} = \frac{z - \gamma}{c}$$

4. The equation of straight line through two given points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

5. The direction ratios of the line given in (point 1) are $b_1c_1 - b_2c_1, a_2c_1 - a_1c_2, a_1b_2 - a_2b_1$

THE STRAIGHT LINE AND THE PLANE

1. The conditions that the line $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ may be parallel to the plane $ax + by + cz + d = 0$ is $al + bm + cn = 0$ and $a\alpha + b\beta + c\gamma + d \neq 0$

2. The conditions that the line $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ is perpendicular to the plane $ax + by + cz + d = 0$ is

$$\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$$

3. The condition for the line $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ to lie in a plane $ax + by + cz + d = 0$ is

$$al + bm + cn = 0 \text{ and } a\alpha + b\beta + c\gamma + d = 0$$

4. The equation of any plane through the line

$$\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$$

$$\text{is } a(x - \alpha) + b(y - \beta) + c(z - \gamma) = 0$$

$$\text{where } al + bm + cn = 0$$

5. The equation of the plane through the line $\frac{x - \alpha}{l_1} = \frac{y - \beta}{m_1} = \frac{z - \gamma}{n_1}$ and parallel to the line

$$\frac{x}{l_2} = \frac{y}{m_2} = \frac{z}{n_2} \text{ is}$$

$$\begin{vmatrix} x - \alpha & y - \beta & z - \gamma \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

PROJECTION OF A LINE ON PLANE

1. The projection of a line on a given plane is the line of intersection of the two planes normally (i) the given plane and (ii) the plane through the given line and perpendicular to the given plane.
2. If P be the point of intersection of the given line with the given plane and Q be the foot of the perpendicular from any point on the line to the plane, then the line PQ is called the projection of the given line on the given plane.

COPLANAR LINES

1. The condition that two given lines

$$\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n} \text{ and}$$

$\frac{x - \alpha'}{l'} = \frac{y - \beta'}{m'} = \frac{z - \gamma'}{n'}$ be coplanar is

$$\begin{vmatrix} \alpha' - \alpha & \beta' - \beta & \gamma' - \gamma \\ l & m & n \\ l' & m' & n' \end{vmatrix} = 0$$

2. The equation of plane containing the lines given in (point 1) is

$$\begin{vmatrix} x - \alpha & y - \beta & z - \gamma \\ l & m & n \\ l' & m' & n' \end{vmatrix} = 0$$

3. The condition that two given lines

$$a_1x + b_1y + c_1z + d_1 = 0$$

$$= a_2x + b_2y + c_2z + d_2$$

$$\text{and } a_3x + b_3y + c_3z + d_3 = 0$$

$$= a_4x + b_4y + c_4z + d_4$$

is

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = 0$$

4. The condition that the lines

$$\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$$

$$\text{and } a_1x + b_1y + c_1z + d_1 = 0$$

$$= a_2x + b_2y + c_2z + d_2 \text{ may intersect is}$$

$$\frac{a_1\alpha + b_1\beta + c_1\gamma + d_1}{a_1l + b_1m + c_1n} = \frac{a_2\alpha + b_2\beta + c_2\gamma + d_2}{a_2l + b_2m + c_2n}$$

and the equation of plane containing them is

$$\frac{a_1x + b_1y + c_1z + d_1}{a_1l + b_1m + c_1n} = \frac{a_2x + b_2y + c_2z + d_2}{a_2l + b_2m + c_2n}.$$

5. If the equation of the two lines are $u_1 = 0 = v_1$ and $u_2 = 0 = v_2$ then the equation of the line intersecting both the given lines are

$$u_1 + \lambda_1 v_1 = 0 \quad \text{and} \quad u_2 + \lambda_2 v_2 = 0.$$

INTERSECTION OF THREE PLANES

1. Let the equation of three planes be given by

$$a_1x + b_1y + c_1z + d_1 = 0$$

$$a_2x + b_2y + c_2z + d_2 = 0$$

$$\text{And } a_3x + b_3y + c_3z + d_3 = 0$$

$$\text{And } \Delta_1 = \begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} a_1 & c_1 & d_1 \\ a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$\Delta_4 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Case I: If $\Delta_4 \neq 0$ then the three planes intersect in a point.

Case II: If $\Delta_4 = 0$ and $\Delta_3 \neq 0$ (or $\Delta_1 \neq 0$, or $\Delta_2 \neq 0$) then the three planes form a triangular prism.

Case III: If $\Delta_4 = 0 = \Delta_1 = \Delta_2 = \Delta_3$ then the three planes intersect in a common line.

SHORTEST DISTANCE

1. The lines which do not lie in a plane are called skew lines. The length of the line intercepted between two lines which is perpendicular to both is the shortest distance between them. The straight line which is perpendicular to each of the two skew lines is called the line of shortest distance (S.D.).

2. If two lines are coplanar then the shortest distance between them is zero.
3. By a proper choice of axes the equation of two skew lines can be given by the equations $y = x \tan \alpha$, $z = c$ and $y = -x \tan \alpha$, $z = -c$.

EXERCISE

MULTIPLE CHOICE QUESTIONS

- The equation of straight line through the points (1, 2, 3) and (3, 5, 7) is
 - $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-7}{4}$
 - $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-2}{4}$
 - $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-7}{2}$
 - $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$
- The direction ratio of the line $3x + 2y - z - 4 = 0 = 4x + y - 2z + 3$ is
 - 3, -2, 5
 - 3, 2, 5
 - 3, 2, -5
 - 3, 2, 5
- The lines $x = ay + b$, $z = cy + d$ and $x = a'y + b'$, $z = c'y + d'$ are perpendicular if $aa' + cc'$ is equal to :
 - 0
 - 1
 - 1
 - 4
- The equations $ax + by + cz + d = 0$ and $a'x + b'y + c'z + d' = 0$ together represents
 - a plane
 - two planes
 - a straight line
 - two straight lines
- The co-ordinates of the point of intersection of the line $6x = 3y = 2z$ with the plane $x + y + z = 6$ is
 - (2, 4, 6)
 - (2, 1, 3)
 - (3, 1, 2)
 - (1, 2, 3)
- The distance of the point (4, 6, 7) from the point where the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ meets the plane $x + y + z = 22$ is
 - 1
 - 2
 - 3
 - $2\sqrt{3}$
- The distance of the point (1, 1, 0) from the plane $x + y + z = 0$ measured parallel to the line $-x = y = z$ is
 - $3\sqrt{2}$
 - $2\sqrt{3}$
 - 2
 - 3
- The co-ordinates of the point where the line joining the points (2, -3, 1) and (3, -4, -5) meets the plane $2x + y + z = 7$ is
 - (2, 1, 1)
 - (1, -2, 7)
 - (1, -1, 2)
 - (1, 7, 2)
- The image of the point (1, 3, 4) in the plane $z - 3 = 0$ is
 - (2, -4, 7)
 - (3, -6, 4)
 - (1, 3, 2)
 - (1, -3, 7)
- Two non-parallel planes always intersect
 - at a point
 - in a plane
 - in a line
 - none of these
- The direction ratios of the line $2x - 1 = y - \frac{1}{2} = 3z - 2$ are
 - 3, 1, 2
 - 2, 3, 6
 - 3, 2, 1
 - 3, 6, 2
- The equation of the plane through the point (1, 2, -1) and perpendicular to the line $3x - y + z + 1 = 0 = 5x + y + 3z$ is
 - $x - y + 2z = 5$
 - $x + y + z = 5$
 - $x + y - 2z = 5$
 - $x + y - 2z = 0$
- If the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is perpendicular to the plane $ax + by + cz + d = 0$ then
 - $al + bm + cn = 0$
 - $al + bm + cn = 1$
 - $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$
 - $al^2 + bm^2 + cn^2 = 1$

14. The image of the point $(0, 0, 0)$ with respect to the plane $2x + y = 5$ is
 (a) $(2, 2, 2)$ (b) $(4, 2, 0)$
 (c) $(1, 2, 5)$ (d) $(2, 1, 5)$
15. The point in which the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ meets the plane $x - 2y + z = 20$ is
 (a) $(8, 7, 25)$ (b) $(8, 7, 24)$
 (c) $(8, 7, 26)$ (d) $(8, 7, 27)$
16. The distance of the point $(1, 3, 4)$ from the plane $2x - y + z = 3$ measured parallel to the line $\frac{x}{-2} = \frac{y}{1} = \frac{z}{1}$ is
[Kanpur 2018]
 (a) 0 (b) 1
 (c) $\frac{1}{2}$ (d) 2
17. The angle between the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and the plane $3x + 2y - 3z = 4$ is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{4}$ (d) 0
18. The equation to the plane through the line $\frac{x-2}{2} = \frac{y-3}{4} = \frac{z-4}{5}$ parallel to the x-axis is
 (a) $5y - 4z = 1$ (b) $5z - 4y = 1$
 (c) $4z - 5y = 1$ (d) $4y - 5z = 1$
19. The equation of the plane containing the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is
 (a) $x - 2y + z = 1$ (b) $x - 2y + z = 0$
 (c) $x + 2y - z = 0$ (d) $x + 2y - z = 1$
20. The plane through the line $3x - 4y + 5z = 10$, $2x + 2y - 3z = 4$ and parallel to the line $x = 2y = 3z$ is
 (a) $x + 20y + 27z = 14$ (b) $x - 20y + 27z = 14$
 (c) $x - 20y - 27z = 14$ (d) none of these
21. The sum of the d.c.'s of the line $3x - 3 = 3y - 1 = 6z - 9$ is
 (a) $\frac{1}{2}$ (b) 1
 (c) 0 (d) 3
22. The line $2x - 4 = 3y + 6 = 6z - 30$ meets the plane $y + z = 0$ at the point
 (a) $(1, 4, -4)$ (b) $(-1, 4, -4)$
 (c) $(-1, -4, 4)$ (d) $(-1, -4, -4)$
23. The direction ratios of the line $4x + 4y - 5z = 12$, $8x + 12y + 3z = 32$ are
 (a) 3, 4, 2 (b) 2, 3, 4
 (c) 4, 3, 2 (d) 2, 4, 3
24. The angle between the line $\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-5}{4}$ and the plane $4x + 4y - 5z = 0$ is
 (a) 0 (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$
25. The equation to the plane through the line $y + z = 1$, $x = 0$ and parallel to the line $x - z = 1, y = 0$ is
 (a) $x + y + z = 1$ (b) $x - y + z = 1$
 (c) $x + y - z = 1$ (d) $x - y - z = -1$
26. The image of the point $(0, 0, 0)$ in the plane $x + y = 2$ is
 (a) $(2, 2, 2)$ (b) $(1, 1, 0)$
 (c) $(1, 2, 1)$ (d) $(2, 2, 0)$
27. A straight line passing through a given point $A(\alpha, \beta, \gamma)$ and having d.c.'s l, m, n . If $P(x, y, z)$ be any point on it such that $AP = r$ then
 (a) $x + \alpha = lr$ (b) $x - \alpha = lx$
 (c) $x - r = \alpha l$ (d) $x + r = \alpha l$
28. The equation of straight line passing through $(1, -1, 3)$ and $(0, 2, 1)$ is
 (a) $\frac{x-1}{1} = \frac{y+1}{-3} = \frac{z-3}{2}$
 (b) $\frac{x}{-1} = \frac{y-2}{2} = \frac{z-1}{3}$
 (c) $\frac{x+1}{1} = \frac{y-1}{3} = \frac{z+3}{2}$
 (d) $\frac{x}{1} = \frac{y-2}{3} = \frac{z-1}{2}$
29. The sum of the d.c.'s of the line $3x + 2y - z - 4 = 0$, $4x + y - 2z + 3 = 0$ is
 (a) -3 (b) 2
 (c) -5 (d) -6

30. The lines $x = ay + 2$, $z = y + 3$ and $x = y + 1$, $z = y - 1$ are perpendicular if
 (a) $a = 0$ (b) $a = -1$
 (c) $a = -2$ (d) $a = -3$
31. One of the condition that the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ lie in the plane $ax + by + cz + d = 0$ is $a\alpha + b\beta + c\gamma + d = 0$ then the other is
 (a) $l + m + n = 0$
 (b) $al + bm + cn = 0$
 (c) $a + b + c = 0$
 (d) $al + bm + cn \neq 0$
32. The equation of the plane containing two parallel lines $\frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ and $\frac{x-1}{1} = \frac{y}{0} = \frac{z-1}{-1}$ is
 (a) $x = y$ (b) $x = z$
 (c) $y = z$ (d) $y = 0$
33. The equation of the plane through the line $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z-4}{5}$ and parallel to z -axis is
 (a) $3x = 2y$ (b) $2x = 3y$
 (c) $x = 2y$ (d) $2x = y$
34. The foot of the perpendicular from the point $(1, 6, 3)$ to the line $6x = 3y - 3 = 2z - 4$ is
 (a) $(6, 3, 3)$ (b) $(2, 4, 6)$
 (c) $(1, 3, 5)$ (d) $(0, 1, 2)$
35. The shortest distance between the lines $\frac{x-1}{1} = \frac{y-2}{0} = \frac{z-3}{0}$ and $x - 2 = y - 3 = z - 6$ is
 (a) $2\sqrt{2}$ (b) $\sqrt{6}$
 (c) $\sqrt{2}$ (d) $\sqrt{3}$
36. The surface generated by the lines which intersect the lines $y = x$, $z = 1$, $y = -x$, $z = -1$ and x -axis is
 (a) $y = xz$ (b) $x = yz$
 (c) $z = xy$ (d) $xyz = 0$
37. The direction cosines of any straight line perpendicular to x -axis are
 (a) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ (b) $\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$
 (c) $0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$
38. The angle between the straight line $\frac{x-3}{1} = \frac{y+4}{-1} = \frac{z}{0}$ and the plane $y - z + 2 = 0$ is
 (a) 0 (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
39. The point of intersection of the lines $4x - 3y + 1 = 0$ $= 5x - 3y + 2$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is
 (a) $(1, 1, 1)$ (b) $(1, -1, -1)$
 (c) $(-1, -1, -1)$ (d) $(-1, -1, 1)$
40. The condition that the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is parallel to the plane $ax + by + cz + d = 0$ is
 (a) $al + bm + cn = 0$
 (b) $a\alpha + b\beta + c\gamma + d \neq 0$
 (c) either (a) or (b)
 (d) (a) and (b) both
41. The equation of straight line passing through $(1, 0, 2)$ and whose d.c.'s are proportional to $2, 1, 3$ is
 (a) $\frac{x-1}{2} = \frac{y}{1} = \frac{z-2}{3}$ (b) $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{6}$
 (c) $\frac{x-1}{2} = \frac{y}{3} = \frac{z-2}{1}$ (d) none of these
42. The co-ordinate of the foot of the perpendicular from the point $(3, -1, 11)$ to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is
 (a) $(3, 2, 0)$ (b) $(0, 2, 3)$
 (c) $(2, 3, 4)$ (d) $(2, 5, 7)$
43. The lines $\frac{x}{a\alpha} = \frac{y}{b\beta} = \frac{z}{c\gamma}$, $\frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}$ and $\frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}$ are coplanar if
 (a) $a = b$ and $b = c$
 (b) $a = b$ and $a = c$
 (c) $b = c$ and $c = a$
 (d) $a = b$ or $b = c$ or $c = a$
44. The point of intersection of the lines $\frac{x-1}{2} = \frac{y-3}{3} = \frac{z-3}{4}$ and $4x - 3y + 1 = 0$ $= 5x - 3z + 2$ is
 (a) $(-1, -2, 1)$ (b) $(-1, -2, -2)$
 (c) $(-1, -1, -1)$ (d) $(1, 1, 1)$

45. The perpendicular distance between the point $(1, 2, 3)$ and the straight line $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is
- (a) $\frac{\sqrt{3}}{5}$ (b) $\frac{\sqrt{3}}{5}$
 (c) $\frac{\sqrt{5}}{3}$ (d) $\frac{\sqrt{5}}{3}$
46. The shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is :
- (a) $\frac{2}{\sqrt{6}}$ (b) $\frac{1}{\sqrt{6}}$
 (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$
47. The trace planes $x - y + z = 4$, $2x - y - z = -4$ and $x + y - 5z = -14$ intersect at
- (a) a point
 (b) common line of intersection
 (c) triangular prism
 (d) plane
48. The distance of the point $(4, 1, 1)$ from the line of intersection of $x + y + z - 4 = 0 = x - 2y - z - 4$ is :
- (a) $\frac{5\sqrt{42}}{14}$ (b) $\frac{3\sqrt{42}}{14}$
 (c) $\frac{\sqrt{42}}{14}$ (d) $\frac{\sqrt{42}}{7}$
49. The distance of the point $A(1, 0, 0)$ from the line $6x = 3y = 2z$ measured parallel to the plane $x - y + z = 0$ is
- (a) $\sqrt{\frac{11}{2}}$ (b) $\sqrt{\frac{9}{2}}$
 (c) $\sqrt{\frac{13}{2}}$ (d) 3
50. The shortest distance between the lines $2x = y = 2z - 2$ and $x + y + z = 1 = x - y - z$ is
- (a) $\frac{4}{\sqrt{11}}$ (b) $\frac{3}{\sqrt{11}}$
 (c) $\frac{2}{\sqrt{11}}$ (d) $\frac{1}{\sqrt{11}}$
51. The line $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$ is perpendicular to the plane $lx + my + nz + d = 0$ only when
- (a) $al + bm + cn = 0$ (b) $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$
 (c) $al + bm + cn = 1$ (d) $al + bm + cn \neq 0$
52. The image of the point $(-1, 2, 3)$ in the yz -plane is
- (a) $(1, 2, 3)$ (b) $(-1, 2, -3)$
 (c) $(1, 2, -3)$ (d) $(1, -2, 3)$
53. The point of intersection of the lines $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-1}{2}$ and $x - 2 = y - 2 = z - 2$ is
- (a) $(1, 2, 3)$ (b) $(2, 2, 2)$
 (c) $(1, 1, 1)$ (d) $(3, 3, 3)$
54. The distance of the point $(0, 0, 0)$ from the line through $(1, 1, 1)$ whose d.c.'s are proportional to $1, 0, 1$ is
- (a) $2\sqrt{2}$ (b) $\sqrt{3}$
 (c) $\sqrt{6}$ (d) 1
55. The shortest distance between the lines $x = y = z$ and $x = 1, y = 0, z$ is
- (a) 1 (b) $2\sqrt{2}$
 (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$
56. The d.c.'s of the straight line $y = 0 = z$ are
- (a) $0, 0, 1$ (b) $1, 0, 0$
 (c) $0, 1, 0$ (d) $1, 0, 1$
57. The equation of any plane through the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is only when $al + bm + cn$ is
- (a) equal to zero
 (b) equal to any number
 (c) not equal to zero
 (d) may or may not equal to zero
58. The length of the shortest distance between the lines $x + y + 2z - 3 = 0 = 2x + 3y + 4z - 4$ and z -axis is :
- (a) 3 (b) 4
 (c) 2 (d) 1

59. The shortest distance between the z -axis and the line $x - 3 = 0, y + z - 1 = 0$ is
 (a) 2 (b) 3
 (c) $\sqrt{3}$ (d) $\sqrt{2}$
60. One of the condition that the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is parallel to the plane $ax + by + cz + d = 0$ is $al + bm + cn = 0$, then the other is
 (a) $\alpha + \beta + \gamma + d \neq 0$
 (b) $a + b + c + d \neq 0$
 (c) $d \neq 0$
 (d) $a\alpha + b\beta + c\gamma + d \neq 0$
61. The equation of the plane which pass through the line of intersection of the planes $u_1 \equiv a_1x + b_1y + c_1z + d_1 = 0$ and $u_2 \equiv a_2x + b_2y + c_2z + d_2 = 0$ and is parallel to x -axis is
 (a) $a_2u_1 = a_1u_2$ (b) $b_2u_1 = b_1u_2$
 (c) $c_2u_1 = c_1u_2$ (d) $d_2u_1 = d_1u_2$
62. The equation of the plane through the points $(0, 0, 0), (1, 1, 1)$ and parallel to the line $3x = 2y = z$ is
 (a) $4x - 3y + z = 3$ (b) $3x - 4y + z = 0$
 (c) $x + 3y - 4z = 0$ (d) $x - 3y + 4z = 0$
63. The shortest distance between the lines $\frac{x-1}{0} = \frac{y}{1} = \frac{z}{0}$ and $\frac{x-2}{0} = \frac{y}{0} = \frac{z}{1}$ is
 (a) 0 (b) 1
 (c) 2 (d) 3
64. If the straight line $x = 2y = az$ is parallel to the plane $x + 2y + 3z + 4 = 0$ then a is
 (a) $-\frac{3}{2}$ (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) $\frac{3}{2}$
65. The image of the point $(-1, 2, 3)$ in the xz -plane is
 (a) $(-1, -2, 3)$ (b) $(1, 2, 3)$
 (c) $(1, -2, 3)$ (d) $(-1, -2, -3)$
66. The shortest distance between the lines $\frac{x-2}{0} = \frac{y}{0} = \frac{z}{1}$ and $\frac{x-1}{0} = \frac{y}{1} = \frac{z}{0}$ is
 (a) 0 (b) 1
 (c) 2 (d) $\sqrt{2}$
67. Which one of the following represents a straight line in three dimension geometry?
 (a) $x = 0$
 (b) $x + 2y + 3z = 1$
 (c) $x + 3y + 4z = 1, z = 3$
 (d) $x^2 + y^2 + z^2 = 2$
68. If the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{3}$ lies on the plane $x + 2y - pz - 3 = 0$ then p is
 (a) 1 (b) -1
 (c) 2 (d) 3
69. The shortest distance between any edges of a cube and diagonal which does not intersect it is
 (a) a (b) $\frac{a}{\sqrt{2}}$
 (c) $a\sqrt{2}$ (d) $2a$
70. The condition that the planes $x = cy + bz, y = az + cx$ and $z = bx + ay$ pass through a line is
 (a) $a^2 + b^2 + c^2 + 2abc = 1$
 (b) $a^2 + b^2 + c^2 = 1$
 (c) $2abc = 1$
 (d) $a^2 + b^2 + c^2 = 2abc$
71. The equation of the plane through the point $(2, -1, 1)$ and the line $4x - 3y + 5 = 0, y - 2z - 5 = 0$ is
[Kanpur 2018]
 (a) $2x - y - 3z = 5$ (b) $2x + 3y - z = 5$
 (c) $3x - y + 2z = 4$ (d) $4x - y - 4z = 5$
72. The equation of the line through the point $(1, 2, 3)$ and parallel to the line $x - y + 2z - 5 = 0, 3x + y + z - 6 = 0$
[Kanpur 2018]
 (a) $\frac{x+1}{3} = \frac{y-2}{-5} = \frac{z+3}{4}$
 (b) $\frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$
 (c) $\frac{x-1}{3} = \frac{y+2}{-5} = \frac{z+3}{4}$
 (d) $\frac{x-1}{3} = \frac{y+2}{5} = \frac{z-3}{4}$

73. The co-ordinates of the point of intersection of the line $\frac{x+1}{1} = \frac{y+3}{3} = \frac{z-2}{2}$ with the plane $3x + 4y + 5z = 20$ is **[Kanpur 2018]**
 (a) (0, 0, 3) (b) (0, 0, 4)
 (c) (0, 1, 4) (d) (1, 0, 4)
74. The shortest distance between the lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$; $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is **[Kanpur 2018]**
 (a) $3\sqrt{30}$ (b) $4\sqrt{30}$
 (c) $5\sqrt{30}$ (d) none of these
75. The equation of shortest distance between $3x - 9y + 5z = 0$ and $x + y - z = 0$ and $6x + 8y + 3z - 13 = 0 = x + 2y + z - 3$ is **[Kanpur 2018]**
 (a) $10x + 29y + 16z = 0 = 0$, $13x + 82y + 55z + 109 = 0$
 (b) $10x - 29y + 16z = 0 = 0$, $13x + 82y + 55z - 109 = 0$
 (c) $10x + 29y - 16z = 0$, $13x - 82y - 54z - 107 = 0$
 (d) none of these
76. The equation of the perpendicular from the point (1, 6, 3) to the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is **[Kanpur 2018]**
 (a) $x - 1 = 0, 2y + 3z = 21$
 (b) $x - 2 = 0, 3y + 2z = 21$
 (c) $x - 9 = 0, 2y + 3z = 23$
 (d) $x + 1 = 0, 2y + 3z = 21$
77. The shortest distance between the lines $y = mx, z = c$; and $y = -mx, z = -c$ is **[Avadh 2018]**
 (a) c (b) $c/2$ (c) $2c$ (d) 0
78. The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-y}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if **[Avadh 2018]**
 (a) $k = 1$ (b) $k = 2$
 (c) $k = 3$ (d) $k = -5$
79. The equation of perpendicular from the point (1, 6, 3) to the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is **[Avadh 2018]**
- (a) $\frac{x-1}{0} = \frac{y-6}{2} = \frac{z-3}{-3}$
 (b) $\frac{x-1}{2} = \frac{y-6}{-3} = \frac{z-3}{6}$
 (c) $\frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$
 (d) $\frac{x-1}{3} = \frac{y-6}{0} = \frac{z-3}{2}$
80. The angle between the plane $2x + 3y - z = 7$ and line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is **[Avadh 2018]**
 (a) $\cos^{-1}\left(\frac{5}{14}\right)$ (b) $\sin^{-1}\left(\frac{5}{14}\right)$
 (c) $\cos^{-1}\left(\frac{1}{14}\right)$ (d) $\sin^{-1}\left(\frac{1}{14}\right)$
81. The equation of the projection of the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}$ on the plane $x + 2y + z = 6$ is **[Kanpur 2018]**
 (a) $9x - 2y - 5z = 4$
 (b) $9x - 2y - 5z + 4 = 0$
 (c) $9x - 2y + 5z - 5 = 0$
 (d) $9x - 2y - 5z + 5 = 0$
82. The shortest distance between the line $x + a = 2y = -12z$ and $x = y + 2a = 6z - 6a$ is **[Kanpur 2018]**
 (a) 0 (b) a (c) $2a$ (d) $3a$
83. The equation of plane through (2, 1, 4) and perpendicular to the line of intersection of the plane $3x + 4y + 7z + 4 = 0$ and $x - y + 2z + 3 = 0$ is **[Kanpur 2018]**
 (a) $15x + y - 7z - 3 = 0$
 (b) $15x - y - 7z - 3 = 0$
 (c) $15x + y + 7z - 3 = 0$
 (d) $15x + y + 7z + 3 = 0$
84. The image of the point (1, 3, 4) in the plane $2x - y + z + 3 = 0$ is **[Agra 2017]**
 (a) (3, 5, 2) (b) (-3, 5, 2)
 (c) (3, -5, 2) (d) none of these
85. Two lines, which do not lie in the same plane are called **[Avadh 2018]**
 (a) parallel (b) intersecting
 (c) co-incident (d) skew

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(d)	2.	(a)	3.	(c)	4.	(c)	5.	(d)	6.	(c)	7.	(b)	8.	(b)	9.	(c)	10.	(c)
11.	(d)	12.	(c)	13.	(c)	14.	(b)	15.	(c)	16.	(a)	17.	(d)	18.	(c)	19.	(b)	20.	(b)
21.	(a)	22.	(c)	23.	(b)	24.	(a)	25.	(d)	26.	(d)	27.	(b)	28.	(a)	29.	(d)	30.	(c)
31.	(b)	32.	(d)	33.	(a)	34.	(c)	35.	(c)	36.	(a)	37.	(c)	38.	(d)	39.	(c)	40.	(d)
41.	(a)	42.	(d)	43.	(d)	44.	(c)	45.	(a)	46.	(b)	47.	(c)	48.	(b)	49.	(a)	50.	(a)
51.	(b)	52.	(a)	53.	(d)	54.	(d)	55.	(d)	56.	(b)	57.	(a)	58.	(c)	59.	(b)	60.	(d)
61.	(a)	62.	(b)	63.	(b)	64.	(a)	65.	(d)	66.	(b)	67.	(c)	68.	(a)	69.	(c)	70.	(a)
71.	(d)	72.	(b)	73.	(b)	74.	(a)	75.	(b)	76.	(a)	77.	(d)	78.	(c)	79.	(c)	80.	(b)
81.	(b)	82.	(c)	83.	(a)	84.	(b)	85.	(d)										

HINTS AND SOLUTIONS

3. Given lines are $\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$ and $\frac{x-b'}{a'} = \frac{y}{1} = \frac{z-d'}{c'}$ so these are perpendicular when $aa' + 1 + cc' = 0$ i.e. $aa' + cc' = -1$.

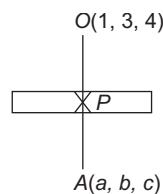
5. Let $6x = 3y = 2z = r$ so intersection point is $\left(\frac{r}{6}, \frac{r}{3}, \frac{r}{2}\right)$ which lies on $x + y + z = 6$ i.e. $\frac{r}{6} + \frac{r}{3} + \frac{r}{2} = 6$ i.e. $r = 6$ so required co-ordinate is $(1, 2, 3)$.

6. Let the point of intersection is given by $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2} = r$ so $P(r+3, 2r+4, 2r+5)$ it lies on given plane $x + y + z = 22$ so $r+3 + 2r+4 + 2r+5 = 22$ i.e. $r = 2$ so $P(5, 8, 9)$. Thus its distance from $A(4, 6, 7)$ is

$$AP = \sqrt{1 + 4 + 4} = 3$$

9. The d.r.'s of OP are 2, -1, 1 so its equation

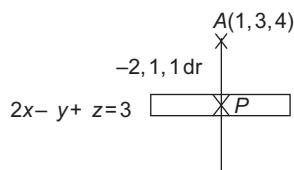
$$\frac{x-1}{0} = \frac{y-3}{0} = \frac{z-4}{1} = r$$



The co-ordinate of P is given by $(1, 3, r+4)$ it lies on given plane so $r+4-3=0 \Rightarrow r=-1$

So, P is $(1, 3, 3)$ thus A is $(1, 3, 2)$.

16. Since given line is $\frac{x}{-2} = \frac{y}{1} = \frac{z}{1}$



So equation of line AP is

$$\frac{x-1}{-2} = \frac{y-3}{1} = \frac{z-4}{1} = r$$

Co-ordinate of P is $(-2r + 1, r + 3, r + 4)$ which lies on given plane so

$$-4r + 2 - r - 3 + r + 4 = 3 \Rightarrow r = 0$$

So co-ordinate of P is $(1, 3, 4)$ thus $AP = 0$

18. Equation of plane containing the line is

$$a(x - 2) + b(y - 3) + c(z - 4) = 0$$

Since it is parallel to x -axis so

$$a \cdot 1 + b \cdot 0 + c \cdot 0 = 0 \quad \dots(1)$$

$$\text{Also } 2a + 4b + 5c = 0 \quad \dots(2)$$

After solving (1) and (2) we get $b = 4, c = -4$

Put it in plane the required plane is $5y - 4z + 1 = 0$ 54.

32. Equation of plane containing first line is $ax + by + cz = 0$ since it containing line (2) so plane pair through $(1, 0, 1)$ i.e.

$$\left. \begin{aligned} a + 0b + c &= 0 \\ \text{Also } a + 0b - c &= 0 \end{aligned} \right\}$$

After solving we get $a = 0, b = -2, c = 0$

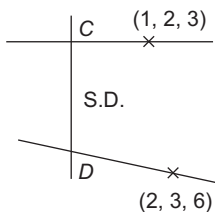
Thus, the plane is $y = 0$

35. Let a, b, c be the d.r. of shortest distance then

$$a + 0b + 0c = 0$$

with

$$a + b + c = 0$$



We get d.r.'s are $0, -1, 1$

So d.c. of S.D. are $\frac{0}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

$$\text{Thus } CD = \frac{0}{\sqrt{2}}(2 - 1) - \frac{1}{\sqrt{2}}(3 - 2) + \frac{1}{\sqrt{2}}(6 - 3)$$

$$= \frac{-1}{\sqrt{2}} + \frac{3}{\sqrt{2}} = \sqrt{2}$$

$$47. \Delta_4 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 1 \\ 2 & -1 & -1 \\ 1 & 1 & -5 \end{vmatrix} = 0$$

$$\text{Now } \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 4 \\ 2 & -1 & -4 \\ 1 & 1 & -14 \end{vmatrix} \neq 0$$

So it is a triangular prism.

P is $(r + 1, 1, r + 1)$

d.r. of OP are $r + 1, 1, r + 1$ which is perpendicular to $1, 0, 1$ (d.r. of line) so $r + 1 + r + 1 = 0 \Rightarrow r = -1$ so $P(0, 1, 0)$

$$\frac{x-1}{1} = \frac{y-1}{0} = \frac{z-1}{1} = r$$

Thus $OP = 1$

$$\frac{x = 2y = az}{x + 2y + 3z + 4 = 0}$$

$$64. \text{ Line is } \frac{x}{1} = \frac{y}{1/2} + \frac{z}{1/a} \quad \frac{x = 2y = az}{x + 2y + 3z + 4 = 0}$$

Its d.r. are $1, \frac{1}{2}, \frac{1}{a}$ which are perpendicular to the d.r.

of normal to the plane i.e. $1, 2, 3$ so

$$1 \cdot 1 + 2 \cdot \frac{1}{2} + \frac{3}{a} = 0$$

$$\Rightarrow a = \frac{-3}{2}$$

EQUATIONS OF A SPHERE

1. A sphere is a locus of a point in space which moves so that its distance from a fixed point (called centre) is constant. The constant distance is called the radius of sphere.

2. The equation of the sphere when centre is (x_1, y_1, z_1) and radius is r is

$$(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$$

3. If the centre is origin i.e. $(0, 0, 0)$ in (point 2) then the equation of the sphere is

$$x^2 + y^2 + z^2 = r^2$$

4. The general equation of the sphere is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$

where centre of the sphere is $C(-u, -v, -w)$

and radius is $r = \sqrt{u^2 + v^2 + w^2 - d}$

5. The diameter of a sphere always subtends a right angle at any point on the surface of the sphere.

6. The equation of the sphere when the end points of its diameter are (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$$

7. The equation of the sphere passing through four given points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) and (x_4, y_4, z_4) is

$$\begin{vmatrix} x^2 + y^2 + z^2 & x & y & z & 1 \\ x_1^2 + y_1^2 + z_1^2 & x_1 & y_1 & z_1 & 1 \\ x_2^2 + y_2^2 + z_2^2 & x_2 & y_2 & z_2 & 1 \\ x_3^2 + y_3^2 + z_3^2 & x_3 & y_3 & z_3 & 1 \\ x_4^2 + y_4^2 + z_4^2 & x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

8. If $C(-u, -v, -w)$ be the centre and $r = \sqrt{u^2 + v^2 + w^2 - d}$ be the radius of the sphere then :

Case I: If $u^2 + v^2 + w^2 > d$, the radius r is real and so the sphere is real.

Case II: If $u^2 + v^2 + w^2 = d$, then $r = 0$ and so the sphere reduce to a point, called a point sphere.

Case III: If $u^2 + v^2 + w^2 < d$, then r is imaginary and then the sphere is called a virtual sphere.

PLANE SECTION OF A SPHERE

1. The plane section of a sphere is a circle.
2. Let R and r be the radius of the sphere S and if plane section i.e. circle C respectively then

Case I: If $R < r$, the circle C is called a small circle.

Case II: If $R = r$, the circle C is called a great circle and centre of circle coincide with the centre of the sphere.

Case III: If $R > r$, then circle is a imaginary circle i.e. the sphere and the circle do not intersect.

3. The general equation of a circle is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ with $lx + my + nz = p$
4. The curve of intersection of two spheres is a circle. If sphere are $S_1 = 0$ and $S_2 = 0$ then the circle is $S_1 = 0, S_1 - S_2 = 0$ or $S_2 = 0, S_1 - S_2 = 0$
5. The equation of the sphere through the circle $S = 0, P = 0$ is $S + \lambda P = 0$, where λ is arbitrary constant.

TANGENT PLANE AND TANGENT LINE

1. A line which meets a sphere in two coincident points is called the tangent line to the sphere.
2. The locus of the tangent line to a sphere through a point lying on sphere is called its tangent plane at that point.
3. The equation of the tangent plane at the point (x_1, y_1, z_1) of the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is

$$xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0$$
4. The equation of the tangent plane at (x_1, y_1, z_1) to the sphere $x^2 + y^2 + z^2 = r^2$ is

$$xx_1 + yy_1 + zz_1 = r^2$$
5. The tangent line and the tangent plane both are perpendicular to the radius through the point of contact.
6. The condition that the plane $lx + my + nz = p$ is a tangent plane to the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is

$$(lu + mv + nw - p)^2 = (l^2 + m^2 + n^2)(u^2 + v^2 + w^2 - d).$$
7. The condition that the plane $lx + my + nz = p$, touches the sphere $x^2 + y^2 + z^2 = r^2$ is

$$p^2 = r^2(l^2 + m^2 + n^2)$$
8. If straight line in any direction and through a given point A meets the sphere in two points P and Q , then $AP \cdot AQ$ is constant and is called the power of the point A .
9. The plane of contact is the locus of the point of contact of the tangent planes which pass through a given point (not on the sphere). Its equation is same as tangent plane.

POLE AND POLAR PLANE

1. Consider a line through a fixed point A to intersect a given sphere in the points P and Q . Take a point R on this line such that

$$\frac{1}{AP} + \frac{1}{AQ} = \frac{2}{AR}, \text{ then the locus of the point } R$$

is a plane called the **polar plane** of the point A with respect to the given sphere and the fixed point A is called the pole of the polar plane.

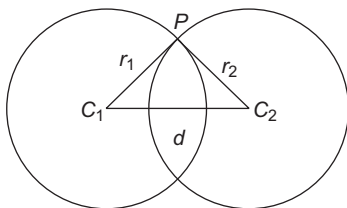
2. The equation of the polar plane of a point $A(x_1, y_1, z_1)$ with respect to a sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is

$$xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0$$
3. The pole of the plane $lx + my + nz = p$ with respect to the sphere $x^2 + y^2 + z^2 = r^2$ is

$$\left(\frac{l}{p} r^2, \frac{m}{p} r^2, \frac{n}{p} r^2 \right).$$
4. The distance of two points from the centre of a sphere and proportional to the distance of each from the polar plane of the other.
5. The polar plane of a point P with respect to a sphere is perpendicular to the line joining the point P to the centre of the sphere.
6. If the line joining the centre O of the sphere and a point P meets the polar plane of P in the point Q , then $OP \cdot OQ = (\text{radius})^2$
7. The two planes such that the pole of each lies on the other, are called conjugate planes.

THE ANGLE OF INTERSECTION OF TWO SPHERES

1. The angle of intersection of any two spheres at a common point in the angle between the tangent planes at that point. Two spheres are said to cut orthogonally if their angle of intersection is a right angle.
2. Let P be the point of intersection of the two spheres whose centres are C_1 and C_2 with radii r_1 and r_2 . If θ be the angle of intersection then



$$\theta = \cos^{-1} \left(\frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2} \right)$$

- Two spheres $x^2 + y^2 + z^2 + 2u_1x + 2v_1y + 2w_1z + d_1 = 0$ and $x^2 + y^2 + z^2 + 2u_2x + 2v_2y + 2w_2z + d_2 = 0$ cut orthogonally when $2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$
- If the distance between the centres of the two spheres is equal to the difference of their radii, then the two spheres touch internally. If the distance between the centres of the two spheres is equal to the sum of their radii, then the two spheres touch **externally**.
- The length of the tangent from the point (x_1, y_1, z_1) to the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is $x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d$.

THE RADICAL PLANES

- The radical plane of two spheres is the locus of points whose power with respect to two spheres are equal.

- The equation of the radical plane of two spheres $S_1 = 0$ and $S_2 = 0$ is $S_1 - S_2 = 0$.
- The radical plane of two spheres is perpendicular to the line joining their centre.
- The radical plane of two spheres passes through their points of intersection.
- The radical planes of three spheres taken two at a time pass through a common line which is said to be the radical line of three spheres and the radical lines of the four spheres taken three at a time meet in one point called radical centre of the four spheres.

COAXIAL SYSTEM OF SPHERES

- A system of sphere is called a co-axial system of sphere if for all spheres any two of them have same radical plane.
- The general equation of a co-axial system of sphere $S_1 = 0$ and $S_2 = 0$ is $S_1 + \lambda S_2 = 0$, and $S_1 + \mu(S_1 - S_2) = 0$ represent a co-axial system of sphere. Similarly if $S = 0$ be a sphere and $P = 0$ be a plane then $S + \lambda P = 0$ represents a co-axial system of sphere.
- The centres of the sphere of co-axial system which have zero radius are called the limiting points of co-axial system. The limiting point of $x^2 + y^2 + z^2 + 2ux + d = 0$ are $(\sqrt{d}, 0, 0)$ and $(-\sqrt{d}, 0, 0)$.

EXERCISE

MULTIPLE CHOICE QUESTIONS

- The co-ordinates of the centre of the sphere $(x - 1)^2 + (y + 2)^2 + (z - 3)^2 = 4$ are
 - (1, 2, 3)
 - (1, -2, -3)
 - (-1, -2, -3)
 - (1, -2, 3)
- The minimum independent condition that determined the sphere completely and uniquely are :
 - 5
 - 4
 - 3
 - 2
- The diameter of the sphere $(x - 1)(x - 2) + (y - 2)(y - 3) + (z - 3)(z - 4) = 0$ is
 - 2
 - 3
 - $\sqrt{2}$
 - $\sqrt{3}$
- If a sphere of constant radius $2k$ passes through the origin and meets the axes in A, B, C then the centroid of the tetrahedron OABC lies on
 - $x^2 + y^2 + z^2 = 4k^2$
 - $x^2 + y^2 + z^2 = k^2$
 - $9(x^2 + y^2 + z^2) = k^2$
 - $9(x^2 + y^2 + z^2) = 4k^2$

5. The radius of the sphere
 $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is
[Agra 2017]
 (a) $\sqrt{u^2 + v^2 + w^2 - d}$ (b) $u^2 + v^2 + w^2 - d$
 (c) $u + v + w - d$ (d) $\sqrt{u^2 + v^2 + w^2 - d}$
6. The angle at any point on the surface of sphere that the diameter subtends is
 (a) $\frac{\pi}{2}$ (b) π
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$
7. The equation of the sphere when the end points of its diameter are $(0, 0, 0)$ and $(-1, 2, 1)$ is
 (a) $x^2 + y^2 + z^2 + 2x - y + z = 0$
 (b) $x^2 + y^2 + z^2 + x - 2y - z = 0$
 (c) $x^2 + y^2 + z^2 - x - 2y - 2z = 0$
 (d) $x^2 + y^2 + z^2 + x + 2y + z = 0$
8. Which is same in the sphere and its great circle?
 (a) only centre
 (b) only radius
 (c) centre and radius both
 (d) neither centre nor radius
9. If the sphere $x^2 + y^2 + z^2 = 9$ touches the plane $x + y + z = a\sqrt{3}$, then the value of a is
 (a) $\sqrt{3}$ (b) $3\sqrt{3}$
 (c) 9 (d) 3
10. The radius of the sphere for which the circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0$, $2x + 3y + 4z = 8$ is a great circle is
 (a) 4 (b) 2
 (c) 1 (d) $2\sqrt{2}$
11. If a point moves so that the sum of the squares of its distance from the six faces of a cube is constant then its locus is
 (a) cylinder (b) circle
 (c) sphere (d) cone
12. The angle between the spheres
 $x^2 + y^2 + z^2 + 6y - u = 0$ and
 $x^2 + y^2 + z^2 - 2x - 5z + 4 = 0$ is
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
13. If the plane $2x + y + 3z = a$ touches the sphere $x^2 + y^2 + z^2 = 4$ then a is
 (a) $\sqrt{14}$ (b) $2\sqrt{14}$
 (c) $3\sqrt{14}$ (d) $5\sqrt{14}$
14. If any tangent plane to the sphere $x^2 + y^2 + z^2 = R^2$ makes intercepts a , b and c on the co-ordinate axes then
[Avadh 2018]
 (a) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{R}$ (b) $a + b + c = R$
 (c) $a^2 + b^2 + c^2 = R^2$ (d) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{R^2}$
15. The number of spheres touching the three co-ordinate axes of fixed radius r is
 (a) 8 (b) 6
 (c) 4 (d) 2
16. The pole of the plane $lx + my + nz = p$ with respect to the sphere $x^2 + y^2 + z^2 = a^2$ is **[Avadh 2018]**
 (a) $\left(\frac{al}{p}, \frac{am}{p}, \frac{an}{p}\right)$
 (b) $\left(\frac{pl}{a^2}, \frac{pm}{b^2}, \frac{pn}{c^2}\right)$
 (c) $\left(\frac{a^2l}{p}, \frac{a^2m}{p}, \frac{a^2n}{p}\right)$
 (d) $\left(\frac{al^2}{p}, \frac{am^2}{p}, \frac{an^2}{p}\right)$
17. If the line joining the centre of the sphere of radius r and a point P meets the polar plane of P in the point Q , then $OP \cdot OQ$ is
 (a) r (b) $2r$
 (c) $2r^2$ (d) r^2

18. If two spheres of radii 3 and 4 cut orthogonally then the radius of the common circle is
 (a) $\frac{3}{5}$ (b) $\frac{7}{5}$
 (c) $\frac{11}{5}$ (d) $\frac{12}{5}$
19. The angle between the radius through any point and the tangent plane at that point in a sphere is
 (a) 0 (b) π
 (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$
20. The locus of a point which moves so that its distance from a fixed point is constant in space is called
 (a) circle (b) plane
 (c) sphere (d) cone
21. The centre of the sphere $2x^2 + 2y^2 + 2z^2 - 4x - 3y + 4z + 2 = 0$ is
 (a) $\left(1, \frac{3}{4}, -1\right)$ (b) $\left(1, \frac{3}{4}, 1\right)$
 (c) $\left(2, \frac{3}{2}, -2\right)$ (d) $\left(2, \frac{3}{2}, 2\right)$
22. The radius of the sphere $x^2 + y^2 + z^2 + 6x - 8y - 10z + 1 = 0$ is
 (a) 7 (b) $\sqrt{7}$
 (c) 17 (d) $\sqrt{17}$
23. The curve of intersection of two spheres is
 (a) line (b) plane
 (c) circle (d) sphere
24. If a plane passes through a fixed point (a, b, c) and cuts the axes in A, B, C then the locus of the centre of the sphere $OABC$ is
 (a) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (b) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$
 (c) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$ (d) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = -1$
25. The radius of the circle $x^2 + y^2 + z^2 = 25$, $x + \frac{3}{2}y - 2z = 0$ is
 (a) 3 (b) 2
 (c) 5 (d) 25
26. If R and r be the radius of the sphere S and its plane section C respectively and $R = r$, then the circle C is
 (a) small circle (b) imaginary circle
 (c) large circle (d) great circle
27. The polar plane of $(1, -1, 1)$ with respect to the sphere $x^2 + y^2 + z^2 = 4$ is
 (a) $x - y + z = 0$ (b) $-x + y - z = 2$
 (c) $-x + y - z = -4$ (d) $-x + y - z = 4$
28. The pole of the plane $x + 2y + 3z = 3$ with respect to the sphere $x^2 + y^2 + z^2 = 9$ is
 (a) $(3, 6, 9)$ (b) $(1, 2, 3)$
 (c) $(1, 1, 1)$ (d) $(3, 3, 3)$
29. If the tangent plane to the sphere $x^2 + y^2 + z^2 = r^2$ makes intercept a, b, c on the axes then
 (a) $a + b + c = r$
 (b) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{r^2}$
 (c) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = r$
 (d) $a^2 + b^2 + c^2 = r^2$
30. If two spheres $x^2 + y^2 + z^2 + 2x + 1 = 0$ and $x^2 + y^2 + z^2 + ax + 2 = 0$ cut orthogonally then the value of a is
 (a) 1 (b) 2
 (c) 3 (d) 4
31. The number of spheres that can be made to pass through the three points $(a, 0, 0), (0, b, 0)$ and $(0, 0, c)$ are
 (a) 3 (b) 2
 (c) 1 (d) infinite
32. The equation $ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + d = 0$ represents a sphere of
 (a) $u = v = w$ (b) $a = b = d$
 (c) $a = b = c$ (d) $a = c = d$

33. The tangent plane of the sphere $x^2 + y^2 + z^2 = 25$ at the point $(2, -3, 4)$ is
 (a) $2x - 3y + 4z = 25$ (b) $2x - 3y + 4z = 1$
 (c) $2x - 3y + 4z = 5$ (d) none of these
34. If the plane $2x - 2y + z + 12 = 0$ touches the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z - 3 = 0$ then the point of contact is
 (a) $(1, 4, 2)$ (b) $(1, 4, -2)$
 (c) $(-1, 4 - 2)$ (d) $(-1, -4, -2)$
35. If the distance between the centres of two spheres is equal to the sum of their radii then the two spheres touches
 (a) internally
 (b) externally
 (c) may be internally or externally
 (d) none of these
36. The angle between two spheres $x^2 + y^2 + z^2 + 6y + 2z + 8 = 0$ and $x^2 + y^2 + z^2 + 6x + 8y + 4z + 20 = 0$ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$
37. The angle between the radical plane and the line joining their centres is
 (a) 0 (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
38. The limiting point of the co-axial system of sphere $x^2 + y^2 + z^2 + 24x + d = 0$ is
 (a) $(d, 0, 0)$ (b) $(0, d, 0)$
 (c) $(0, \sqrt{d}, 0)$ (d) $(\sqrt{d}, 0, 0)$
39. The locus of a point whose power with respect to the two given spheres are equal is called
 (a) conjugate planes (b) polar planes
 (c) radical planes (d) parallel planes
40. If two spheres of radius r_1 and r_2 cut orthogonally then radius of the common circle is [Avadh 2018]
 (a) $r_1 r_2$ (b) $r_1 r_2 \sqrt{r_1^2 + r_2^2}$
 (c) $\frac{r_1 r_2}{\sqrt{r_1^2 + r_2^2}}$ (d) $\sqrt{r_1^2 + r_2^2}$
41. The radius of the circle in which the sphere $x^2 + y^2 + z^2 - 2x - 4y - 6z = 0$ is cut by the plane $x + y + z = 0$ is
 (a) $\sqrt{5}$ (b) $2\sqrt{2}$
 (c) 2 (d) $\sqrt{2}$
42. The equation of the sphere which passes through the points $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ and radius as small or possible is
 (a) $x^2 + y^2 + z^2 - 2x - 2y - 2z = 1$
 (b) $3(x^2 + y^2 + z^2) - 3x - 3y - 3z = 1$
 (c) $3(x^2 + y^2 + z^2) - x - y - z = 1$
 (d) $3(x^2 + y^2 + z^2) - 2x - 2y - 2z = 1$
43. The members of the co-axial system $x^2 + y^2 + z^2 + 2ux + d = 0$ touch one another only when
 (a) $d > 0$ (b) $d < 0$
 (c) $d = 0$ (d) disimaginary
44. If a sphere of radius 3 passes through the origin and meets the axes in A, B, C then the centroid of the triangle ABC is
 (a) $a(x^2 + y^2 + z^2) = 4$
 (b) $x^2 + y^2 + z^2 = 9$
 (c) $x^2 + y^2 + z^2 = 4$
 (d) $x^2 + y^2 + z^2 = 3$
45. The equation of the sphere with centre at $(0, 0, 0)$ and touching the plane $x + y + z = 1$ is
 (a) $x^2 + y^2 + z^2 = 1$
 (b) $x^2 + y^2 + z^2 = 3$
 (c) $3(x^2 + y^2 + z^2) = 2$
 (d) $3(x^2 + y^2 + z^2) = 1$

46. If a point in 3-dimensional geometry moves so that the ratio of its distance from two fixed points is constant then its locus is
 (a) circle (b) sphere
 (c) cylinder (d) cone
47. The equation of the sphere circumscribing the tetrahedron whose faces are $x = 0$, $y = 0$, $z = 0$, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ is
 (a) $x^2 + y^2 + z^2 + ax + by + cz = 1$
 (b) $x^2 + y^2 + z^2 - x - y - z = \sqrt{a^2 + b^2 + c^2}$
 (c) $x^2 + y^2 + z^2 - ax - by - cz = 0$
 (d) $x^2 + y^2 + z^2 = \sqrt{a^2 + b^2 + c^2}$
48. The section of the sphere by a plane is
 (a) line (b) cone
 (c) cylinder (d) circle
49. If the distance between the centres of the two spheres is equal to the difference of their radius, then the two spheres touches
 (a) internally (b) externally
 (c) intersect (d) none of these
50. The radical planes of four spheres taken in pairs always meet in a
 (a) sphere (b) circle
 (c) plane (d) point
51. The power of the point $(2, -1, 3)$ with respect to the sphere $x^2 + y^2 + z^2 - 3x + y - z + 10 = 0$ is
 (a) 10 (b) 14
 (c) 8 (d) 6
52. The radical axes for the spheres $(x - 3)^2 + y^2 + z^2 = 11$, $x^2 + (y - 2)^2 + z^2 = 6$ and $(x + 1)^2 + y^2 + (z - 2)^2 = 7$ is
 (a) $\frac{x}{3} = \frac{y}{2} = \frac{z}{2}$ (b) $\frac{x}{3} = \frac{y}{3} = \frac{z}{2}$
 (c) $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ (d) $\frac{x}{3} = \frac{y}{3} = \frac{z}{4}$
53. The pole of the plane $x - y + 5z = 3$ with respect to the sphere $x^2 + y^2 + z^2 = 9$ is
 (a) $(3, -3, 15)$ (b) $(-3, 3, 15)$
 (c) $(3, 3, 15)$ (d) $(3, -3, -15)$
54. The radius of the circle $3x^2 + 3y^2 + 3z^2 + x - 5y - z = 0$, $x + y = 2$ is
 (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $2\sqrt{2}$ (d) 2
55. If $2x^2 + 2y^2 + cz^2 + 2x - 3y + 4z - 1 = 0$ represent a sphere then c is
 (a) 1 (b) 2 (c) 3 (d) not exist
56. If the plane $2x - 2y + z - 9 = 0$ touches the sphere $x^2 + y^2 + z^2 = a^2$ then a is **[Kanpur 2018]**
 (a) 1 (b) 2 (c) 9 (d) 3
57. The equation of sphere with centre at $(2, 3, -4)$ and touching the plane $2x + 6y - 3z + 15 = 0$ is **[Kanpur 2018]**
 (a) $x^2 + y^2 + z^2 + 4x - 6y + 8z + 20 = 0$
 (b) $x^2 + y^2 + z^2 + 4x + 6y + 8z + 20 = 0$
 (c) $x^2 + y^2 + z^2 + 4x - 6y - 8z + 20 = 0$
 (d) $x^2 + y^2 + z^2 - 4x - 6y + 8z - 20 = 0$
58. The equation of sphere having its centre on the line $2x - 3y = 0 = 5y + 2z$ and passing through the points $(0, -2, -4)$ and $(2, -1, -1)$ is **[Kanpur 2018]**
 (a) $x^2 + y^2 + z^2 - 6x - 4y - 10z + 12 = 0$
 (b) $x^2 + y^2 + z^2 - 6x - 4y + 10z + 12 = 0$
 (c) $x^2 + y^2 + z^2 + 6x + 4y - 10z + 12 = 0$
 (d) $x^2 + y^2 + z^2 + 6x - 4y - 10z - 12 = 0$
59. The angle of intersection of the spheres $x^2 + y^2 + z^2 - 2x - 4y - 6z + 10 = 0$ and $x^2 + y^2 + z^2 - 6x - 2y + 2z + 2 = 0$ is **[Kanpur 2018]**
 (a) $\sinh^{-1}\left(\frac{2}{3}\right)$ (b) $\cosh^{-1}\left(\frac{2}{3}\right)$
 (c) $\sin^{-1}\left(\frac{2}{3}\right)$ (d) $\cos^{-1}\left(\frac{2}{3}\right)$
60. Equation of the sphere through the circle $x^2 + y^2 + z^2 = 9$, $x + y - 2z + 4 = 0$ and the origin is **[Avadh 2018, Kanpur 2018]**
 (a) $4(x^2 + y^2 + z^2) - 9x + 9y - 18z = 0$
 (b) $4(x^2 + y^2 + z^2) + 9x - 9y - 18z = 0$
 (c) $4(x^2 + y^2 + z^2) + 9x + 9y - 18z = 0$
 (d) $4(x^2 + y^2 + z^2) - 9x - 9y + 18z = 0$

61. The equation of sphere which touches the plane $3x + 2y - z + 2 = 0$ at the point $(1, -2, 1)$ and also cuts orthogonally the sphere $x^2 + y^2 + z^2 - 4x + 6y + 4 = 0$ is [Kanpur 2018]
- (a) $x^2 + y^2 + z^2 - 7x + 10y - 5z + 24 = 0$
 (b) $x^2 + y^2 + z^2 + 7x + 10y - 5z + 12 = 0$
 (c) $x^2 + y^2 + z^2 + 7x - 10y - 5z + 12 = 0$
 (d) $x^2 + y^2 + z^2 + 7x - 10y + 5z - 12 = 0$
62. The equation of sphere joining $(2, -3, 4)$ and $(-5, 6, -7)$ as diameter is [Avadh 2018]
- (a) $x^2 + y^2 + z^2 + x + y + z = 0$
 (b) $x^2 + y^2 + z^2 + x - y + z = 1$
 (c) $x^2 + y^2 + z^2 + 3x - 3y + 3z - 56 = 0$
 (d) none of these
63. The equation of the sphere passing through the point $(0, 0, 0)$, $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ is [Avadh 2018]
- (a) $x^2 + y^2 + z^2 = a^2$
 (b) $x^2 + y^2 + z^2 = a^2 + b^2 + c^2$
 (c) $x^2 + y^2 + z^2 + ax + by + cz = 0$
 (d) $x^2 + y^2 + z^2 - ax - by - cz = 0$
64. The radius of the sphere $2x^2 + 2y^2 + 2z^2 + x + y + z = 0$ is [Avadh 2018]
- (a) $\frac{3}{4}$
 (b) $\frac{1}{4}$
 (c) $\frac{\sqrt{3}}{4}$
 (d) none of these
65. The equation of the sphere whose centre $(-3, 4, 5)$ and radius 7 is [Avadh 2018]
- (a) $x^2 + y^2 + z^2 - 6x - 8y - 10z + 1 = 0$
 (b) $x^2 + y^2 + z^2 - 6x + 8y + 10z + 1 = 0$
 (c) $x^2 + y^2 + z^2 + 6x - 8y - 10z + 1 = 0$
 (d) none of these
66. The centre of the sphere $(x + 1)(x + 3) + (y - 2)(y - 4) + (z + 1)(z + 3) = 0$ is [Avadh 2018]
- (a) $(-1, 1, -1)$
 (b) $(1, -1, 1)$
 (c) $(2, -3, 2)$
 (d) $(-2, 3, -2)$

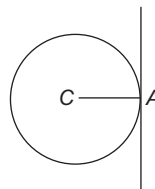
ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(d)	2.	(b)	3.	(d)	4.	(b)	5.	(d)	6.	(a)	7.	(b)	8.	(c)	9.	(d)	10.	(b)
11.	(c)	12.	(a)	13.	(b)	14.	(d)	15.	(a)	16.	(c)	17.	(d)	18.	(d)	19.	(c)	20.	(c)
21.	(a)	22.	(a)	23.	(c)	24.	(c)	25.	(c)	26.	(d)	27.	(c)	28.	(a)	29.	(b)	30.	(c)
31.	(d)	32.	(c)	33.	(c)	34.	(c)	35.	(b)	36.	(d)	37.	(b)	38.	(d)	39.	(c)	40.	(c)
41.	(d)	42.	(d)	43.	(c)	44.	(c)	45.	(d)	46.	(b)	47.	(c)	48.	(d)	49.	(a)	50.	(d)
51.	(b)	52.	(c)	53.	(a)	54.	(b)	55.	(b)	56.	(d)	57.	(d)	58.	(b)	59.	(d)	60.	(c)
61.	(b)	62.	(c)	63.	(d)	64.	(c)	65.	(c)	66.	(d)								

HINTS AND SOLUTIONS

4. The equation of sphere $OABC$ is $x^2 + y^2 + z^2 - ax - by - cz = 0$ its radius is $\sqrt{a^2 + b^2 + c^2} = 2k \Rightarrow a^2 + b^2 + c^2 = 16k^2$
- If x, y, z be the centroid then $x = \frac{a}{4}, y = \frac{b}{4}, z = \frac{c}{4}$
- So locus is $16(x^2 + y^2 + z^2) = 16k^2$
 $\Rightarrow x^2 + y^2 + z^2 = k^2$
10. The equation of sphere is $x^2 + y^2 + z^2 + 2\lambda x + (7 + 3\lambda)y - 2(1 - 2\lambda)z + 2 - 8\lambda = 0$
- and its centre is $\left(-\lambda, -\frac{3\lambda + 7}{2}, 1 - 2\lambda\right)$ which lies on given plane so $\lambda = -1$ thus sphere is $x^2 + y^2 + z^2 - 2x + 4y - 6z + 10 = 0$ and its radius = 2
13. The centre of the sphere is $(0, 0, 0)$ and radius is 2 so length of perpendicular from centre on tangent is equal to sphere
- i.e. $2 = \frac{a}{\sqrt{4 + 1 + 9}} = \frac{a}{\sqrt{14}} \Rightarrow a = 2\sqrt{14}$
14. Let the tangent plane in intercept form is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
- Since, it is tangent on given sphere s.
- $$\frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = \text{radius} = r$$
- $$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{r^2}$$
18. $r = \frac{r_1 r_2}{\sqrt{r_1^2 + r_2^2}} = \frac{3 \times 4}{5} = \frac{12}{5}$
24. Let the plane be $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 1$ it pass through (a, b, c) so $\frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 1$ equation of sphere is
- $x^2 + y^2 + z^2 - px - qy - rz = 0$ its centre is $\left(\frac{p}{2}, \frac{q}{2}, \frac{r}{2}\right)$. Let the centre is (x, y, z) then $p = 2x$, $q = 2y$ and $r = 2z$ their locus of centre is $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$
30. Two sphere cut orthogonally then $2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$
 $\Rightarrow 2(1)\left(\frac{a}{2}\right) + 2 \cdot 0 + 2 \cdot 0 \cdot 0 = 1 + 2 \Rightarrow a = 3$
34. The centre of the sphere is $(1, 2, -1)$ and d.r. of the line CA are $2, -2, 1$ so equation of line CA is $\frac{x-1}{2} = \frac{y-2}{-2} = \frac{z+1}{1} = r$



i.e. $A(2r + 1, -2r + 2, r - 1)$

A lies on tangent plane so

$$4r + 2 + 4r - 4 + r - 1 + 12 = 0 \Rightarrow r = -1$$

Thus co-ordinate of point of contact A is $(-1, 4, -2)$.

43. The equation of sphere is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad \dots(1)$$

It pass through $(1, 0, 0)$, so $1 + 2u + d = 0$

$$\Rightarrow u = -\left(\frac{1+d}{2}\right)$$

Similarly it pass through $(0, 1, 0)$ so $v = -\left(\frac{1+d}{2}\right)$

and pass through $(0, 0, 1)$ so $w = -\left(\frac{1+d}{2}\right)$

Thus radius of the sphere is $R = \sqrt{u^2 + v^2 + w^2 - d}$

$$\text{or } R^2 = 3\left(\frac{1+d}{2}\right)^2 - d = \frac{3d^2 + 2d + 3}{4}$$

For minima $D(R^2) = 0 \Rightarrow 6d + 2 = 0$ i.e.

$$d = -\frac{1}{3}$$

$$\therefore u = v = w = -\frac{1}{3}$$

So by equation (1) required sphere is

$$x^2 + y^2 + z^2 - \frac{2x}{3} - \frac{2y}{3} - \frac{2z}{3} - \frac{1}{3} = 0$$

$$\text{or } 3(x^2 + y^2 + z^2) - 2x - 2y - 2z = 1$$

48. In this case sphere pass through $(0, 0, 0)$ $(a, 0, 0)$ $(0, b, 0)$ and $(0, 0, c)$ so its equation is

$$x^2 + y^2 - ax - by - cz = 0$$

○○○

CONE

1. A cone is a surface generated by a moving straight line which passes through a fixed point and intersects a given curve or touches a given surface. The fixed point is called the vertex and the given curve is called the guiding curve of the cone. The moving straight line is called a generator of the cone.

2. A cone which is cut by a straight line other than the generators only at two points is called a quadratic cone.

3. The equation of a quadratic cone with its vertex at the origin is a homogeneous equation of second degree in x, y, z i.e.
 $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$.

Thus every homogeneous equation of second degree in x, y and z represent a cone whose vertex is at the origin.

4. The lines $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is a generator of the cone whose vertex at the origin i.e.
 $f(x, y, z) \equiv ax^2 + by^2 + cz^2 + 2fyz$

$+ 2gzx + 2hxy = 0$ iff its direction ratios l, m, n satisfy the equation of cone i.e. $f(l, m, n) = 0$

5. The general equation of a cone of second degree which passes through the co-ordinate axes, the axes being rectangular is
 $fyz + gzx + hxy = 0$.

6. The equation of a cone whose vertex is the point (α, β, γ) and base the conic

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, z = 0 \text{ is}$$

$$a(\alpha z - \gamma x)^2 + 2h(\alpha z - \gamma x)(\beta z - \gamma y)$$

$$+ b(\beta z - \gamma y)^2 + 2g(\alpha z - \gamma x)(z - \gamma)$$

$$+ 2f(\beta z - \gamma y)(z - \gamma) + c(z - \gamma)^2 = 0$$

7. The condition that the general equation of second degree $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0$ represents a cone is

$$\begin{vmatrix} a & h & g & u \\ h & b & f & v \\ g & f & c & w \\ u & v & w & d \end{vmatrix} = 0$$

TANGENT PLANE

1. The tangent plane at a point to the cone is the locus of the tangent lines through that point drawn in all directions.

2. The equation of the tangent plane at the point (x_1, y_1, z_1) of the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ is $x(ax_1 + hy_1 + gz_1) + y(hx_1 + by_1 + fz_1) + z(gx_1 + fy_1 + cz_1) = 0$
 The tangent plane at any point of a cone passes through its vertex.

3. The condition that the plane $lx + my + nz = 0$ may touch the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ is $Al^2 + Bm^2 + Cn^2 + 2Fmn + 2Gnl + 2Hlm = 0$ where

$$A = bc - f^2, \quad B = ca - g^2, \quad C = ab - h^2,$$

$$F = gh - af, \quad G = hf - bg, \quad H = fg - ch.$$

RECIPROCAL CONE

1. The reciprocal cone of a given cone is the locus of the normal through the vertex of the cone to the tangent planes of the given cone.
2. The equation of the cone reciprocal to the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ is $Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gzx + 2Hxy = 0$ where A, B, C, F, G, H are defined in point 3.
3. Two cones which are such that each is the locus of the normals through the vertex to the tangent planes to the other are called reciprocal cones.
4. The condition of perpendicularity between the lines in which the plane $ux + vy + wz = 0$ cuts the cone $f(x, y, z) \equiv ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ is $(a + b + c)(u^2 + v^2 + w^2) = f(u, v, w)$.
5. The condition of the tangency of a plane and a cone given in point 4 is $Au^2 + Bv^2 + Cw^2 + 2Fuv + 2Gwu + 2Huv = 0$
2. The equation of the right circular cone with vertex at (α, β, γ) and l, m, n as direction ratios of the axis is $[l(x - \alpha) + m(y - \beta) + n(z - \gamma)] = \cos^2 \theta (l^2 + m^2 + n^2)x$
 $[(x - \alpha)^2 + (y - \beta)^2 + (z - \gamma)^2]$
3. The equation of the right circular cone with vertex at origin, axis the z -axis and semi-vertical angle θ is $x^2 + y^2 = z^2 \tan^2 \theta$.
4. If the vertex at the origin, axis the y -axis (respectively x -axis) and semi-vertical angle is θ is $x^2 + z^2 = y^2 \tan^2 \theta$
(respectively $y^2 + z^2 = x^2 \tan^2 \theta$)

THREE MUTUALLY PERPENDICULAR GENERATORS

1. The condition that the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ may have three mutually perpendicular generators is $a + b + c = 0$.
2. The condition that the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ may have three mutually perpendicular tangent planes is $f^2 + g^2 + h^2 = ab + bc + ac$.

RIGHT CIRCULAR CONE

1. A right circular cone is a surface generated by a straight line which moves in such a way that it passes through a fixed point (called vertex) and

it makes a constant angle with a fixed line (called axis) through the fixed point. The constant angle is called the semi-vertical angle.

2. The equation of the right circular cone with vertex at (α, β, γ) and l, m, n as direction ratios of the axis is $[l(x - \alpha) + m(y - \beta) + n(z - \gamma)] = \cos^2 \theta (l^2 + m^2 + n^2)x$

$$[(x - \alpha)^2 + (y - \beta)^2 + (z - \gamma)^2]$$

3. The equation of the right circular cone with vertex at origin, axis the z -axis and semi-vertical angle θ is $x^2 + y^2 = z^2 \tan^2 \theta$.

4. If the vertex at the origin, axis the y -axis (respectively x -axis) and semi-vertical angle is θ is

$$x^2 + z^2 = y^2 \tan^2 \theta$$

$$\text{(respectively } y^2 + z^2 = x^2 \tan^2 \theta \text{)}$$

THE ENVELOPING CONE

1. The locus of the tangents from a given point to a given surface is a cone called the enveloping cone.
2. The equation of the (tangent cone or) enveloping cone from the point (x_1, y_1, z_1) to the sphere $x^2 + y^2 + z^2 = a^2$ is $SS_1 = T^2$ where $S = x^2 + y^2 + z^2 - a^2$,
 $S_1 = x_1^2 + y_1^2 + z_1^2 - a^2$
and $T = xx_1 + yy_1 + zz_1 - a^2$
3. The equation of the enveloping cone for the conicoid $ax^2 + by^2 + cz^2 = 1$ from the point (x_1, y_1, z_1) is $(ax^2 + by^2 + cz^2 - 1)(ax_1^2 + by_1^2 + cz_1^2 - 1) = (axx_1 + byy_1 + czz_1 - 1)^2$

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. Every homogeneous equation of second degree in x , y and z always represent a
 - (a) cone
 - (b) cylinder
 - (c) circle
 - (d) sphere
2. The equation of the cone whose vertex is the origin and base the curve $z = k$, $f(x, y) = 0$ is
[Avadh 2018]
 - (a) $f\left(\frac{xk}{z}, \frac{yk}{z}\right) = 0$
 - (b) $f\left(\frac{zk}{x}, \frac{zk}{y}\right) = 0$
 - (c) $f\left(\frac{xz}{k}, \frac{yz}{k}\right) = 0$
 - (d) $f\left(\frac{k}{xz}, \frac{k}{yz}\right) = 0$
3. The vertex of the cone $4x^2 - y^2 + 2z^2 + 2xy - 3yz + 12x - 11y + 6z + 4 = 0$ is
[Kanpur 2018]
 - (a) (1, 2, 3)
 - (b) (1, -2, 3)
 - (c) (1, -2, -3)
 - (d) (-1, -2, -3)
4. One of the equation of the line in which plane $2x + y - z = 0$ cuts the cone $4x^2 - y^2 + 3z^2 = 0$ is
 - (a) $\frac{x}{-1} = \frac{y}{2} = \frac{z}{0}$
 - (b) $\frac{x}{-1} = \frac{y}{2} = \frac{z}{1}$
 - (c) $\frac{x}{2} = \frac{y}{-1} = \frac{z}{0}$
 - (d) $\frac{x}{-1} = \frac{y}{0} = \frac{z}{2}$
5. The equation of reciprocal cone of the given cone $\sqrt{fx} \pm \sqrt{gy} \pm \sqrt{hz} = 0$ is
 - (a) $\sqrt{fyz} + \sqrt{gzx} + \sqrt{hxy} = 0$
 - (b) $y\sqrt{fx} + z\sqrt{gy} + x\sqrt{hz} = 0$
 - (c) $\sqrt{fx} + \sqrt{gy} + \sqrt{hz} = 0$
 - (d) $fyz + gzx + hxy = 0$
6. The reciprocal cone of the given cone $ax^2 + by^2 + cz^2 = 0$ is
 - (a) $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 0$
 - (b) $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1$
 - (c) $ax^2 + by^2 + cz^2 = 0$
 - (d) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$
7. If the cone $x^2 + ay^2 - 3z^2 - hz + zx - 2xy = 0$ have three mutually perpendicular generators then the value of a is
 - (a) -2
 - (b) -1
 - (c) 1
 - (d) 2
8. The equation $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + d = 0$ represents a cone only when
 - (a) $a = b = c = d$
 - (b) $f = g = h = d$
 - (c) $d = 0$
 - (d) $d = 1$
9. The general equation of second degree of a cone which pass through the co-ordinate axes is
[Agra 2018]
 - (a) $x^2 + y^2 + z^2 + fyz + gzx + hxy = 0$
 - (b) $fyz + gzx + hxy = 0$
 - (c) $y^2 + fyz + gzx = 0$
 - (d) $x^2 + fyz + gzx + hxy = 0$
10. The condition that the equation $ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + d = 0$ represents a cone is
 - (a) $\frac{u}{a} + \frac{v}{b} + \frac{w}{c} = d$
 - (b) $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = d$
 - (c) $\frac{a}{u} + \frac{b}{v} + \frac{c}{w} = d$
 - (d) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = d$

11. The plane $ax + by + cz = 0$ cuts the cone $yz + zx + xy = 0$ in perpendicular lines only when
[Kanpur 2018, Avadh 2018]
 (a) $a + b + c = 0$ (b) $(a + b + c)^{-1} = abc$
 (c) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$ (d) $abc = 0$
12. The angle between the lines of section of the plane $3x + y + 5z = 0$ and the cone $6yz - 2zx + 5xy = 0$ is
 (a) $\cos^{-1}\left(\frac{1}{3\sqrt{2}}\right)$ (b) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$
 (c) $\cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (d) $\cos^{-1}\left(\frac{1}{6}\right)$
13. The equation of cone whose vertex is $(1, 1, 0)$ and the guiding curve is $x^2 + z^2 = 4, y = 0$ is
 (a) $x^2 + 3y^2 + z^2 + 2yz + 6x = 4$
 (b) $x^2 - 3y^2 + z^2 - 2xy + 8y = 4$
 (c) $x^2 + 3y^2 - z^2 - 2yz + 6x = 4$
 (d) none of these
14. The equation of the cone with vertex at the origin, semi-vertical angle is θ and axis as x-axis is
[Avadh 2018]
 (a) $y^2 + z^2 = x^2$
 (b) $y^2 - z^2 = x^2$
 (c) $y^2 + z^2 = x^2 \tan^2 \theta$
 (d) $y^2 + z^2 = \tan^2 \theta$
15. If the cone $px^2 + y^2 + 2z^2 - 2yz + 4zx + 2xy$ have three mutually perpendicular tangent planes then the value of p is
 (a) $\frac{-4}{3}$ (b) $\frac{4}{3}$
 (c) $\frac{3}{4}$ (d) $\frac{-3}{4}$
16. The section of a right circular cone by any plane perpendicular to its axis is
 (a) circle (b) ellipse
 (c) triangle (d) square
17. If a right circular cone has three mutually perpendicular tangent planes then its semi-vertical angle is
 (a) $\tan^{-1}(\sqrt{2})$ (b) $\cot^{-1}(\sqrt{2})$
 (c) $\sin^{-1}(\sqrt{2})$ (d) $\cos^{-1}(\sqrt{2})$
18. The surface generated by a straight line which passes through a fixed point and intersects a given curve is called
 (a) cylinder (b) conicoid
 (c) sphere (d) cone
19. The equation of the cone with vertex at the origin and direction cosines of its generators satisfy the relation $2l^2 + 3m^2 - n^2 = 0$ is
 (a) $3x^2 + 2y^2 + z^2 = 0$
 (b) $2x^2 + 3y^2 = 0$
 (c) $2x^2 + 3y^2 - z^2 = 0$
 (d) $3y^2 - z^2 = 0$
20. The equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ represents a cone if
 (a) $u + v + w = d$
 (b) $\frac{1}{u} + \frac{1}{v} + \frac{1}{w} = d$
 (c) $u^2 + v^2 + w^2 = d$
 (d) $u + v + w = d^2$
21. The equation of the tangent plane at the point $(1, 0, 0)$ of the cone $x^2 + y^2 + z^2 + 2yz + 2zx + 2xy = 0$ is
 (a) $x + y + z = 0$
 (b) $x + 2y + 3z = 0$
 (c) $2x + y + 3z = 0$
 (d) $x - y - z = 0$
22. The tangent plane at any point of a cone always passes through its
 (a) guiding curve (b) generator
 (c) guiding surface (d) vertex

23. If the angle between the lines given by $x + y + z = 0$, $ayz + bzx + cxy = 0$ is $\frac{\pi}{2}$ then
- $a + b + c = 0$
 - $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$
 - $a^2 + b^2 + c^2 = 0$
 - $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 0$
24. The equation of the cone whose vertex is $(1, 2, -2)$ and guiding curve is the parabola $y^2 = 4ax$, $z = 0$ is
- $(y + 2)^2 = a(z - 2x)(z - 2)$
 - $(y + 2)^2 = a(z + 2x)(z + 2)$
 - $(y + 2)^2 = 4a(z - 2x)(z - 2)$
 - $(y + 2)^2 = 4a(z + 2x)(z + 2)$
25. If the plane $2x - y + pz = 0$ cuts the cone $yz + zx + xy = 0$ in perpendicular lines then the value of p is
- 1
 - 2
 - 3
 - 4
26. If the plane $3x - y + az = 0$ cuts the cone $yz + zx + xy = 0$ in perpendicular lines then the value of a is
- 2
 - 3
 - $\frac{3}{2}$
 - $\frac{2}{3}$
27. The vertex of the cone through which all the tangent planes pass is called
- nodal point
 - conjugate point
 - limit point
 - singular point
28. The equation $\sqrt{fx} + \sqrt{gy} + \sqrt{hz} = 0$ represents a
- plane
 - cone
 - cylinder
 - conicoid
29. The locus of the perpendiculars drawn from the origin to the tangent planes to the cone $ax^2 + by^2 + cz^2 = 0$ is
- cone
 - circle
 - sphere
 - cylinder
30. The number of sets of three mutually perpendicular generators of the cone are
- unique
 - finite
 - infinite
 - only 1
31. If the cone $2x^2 - 5y^2 + az^2 + 3yz + 4zx + 5xy = 0$ have three mutually perpendicular generators then a is equal to
- 4
 - 3
 - 5
 - 2
32. The equation of right circular cone where vertex is at the origin, axis the x -axis and semi-vertical angle is 30° is
- $3(y^2 + z^2) = x^2$
 - $y^2 + z^2 = x^2$
 - $3(x^2 + y^2) = z^2$
 - $3(x^2 + z^2) = y^2$
33. The locus of the tangent from a given point to a given surface is called
- reciprocal cone
 - cylinder
 - right circular cone
 - enveloping cone
34. The equation of the cone reciprocal to the cone $x^2 + y^2 + z^2 = 0$ is
- $x^2 + y^2 + z^2 + xy + yz + zx = 0$
 - $x^2 + y^2 + z^2 = 0$
 - $xy + yz + zx = 0$
 - $x^2 + y^2 + z^2 + 2xy + 2yz + 2zx = 0$

35. The equation of the cone formed by rotating the line $2x + 3y = 6, z = 0$ about the y -axis is
- (a) $x^2 + 9y^2 + 4z^2 - 36y - 36 = 0$
 (b) $4x^2 + 9y^2 + 2z^2 + 36y - 36 = 0$
 (c) $x^2 + 9y^2 + 4z^2 - y - 4 = 0$
 (d) $4x^2 - 9y^2 + 4z^2 + 36y - 36 = 0$
36. The condition that the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ have three mutually perpendicular generators is
- (a) $a + b + c = 0$ (b) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$
 (c) $ab + bc + ca = 0$ (d) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 0$
37. If a right circular cone has three mutually perpendicular generators then its semi-vertical angle is
- (a) $\cot^{-1}(\sqrt{2})$ (b) $\sin^{-1}(\sqrt{2})$
 (c) $\tan^{-1}(\sqrt{2})$ (d) $\cos^{-1}(\sqrt{2})$
38. If the generator of the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ is y -axis then the value of b is
- (a) 0 (b) 1
 (c) -1 (d) any number
39. The equation of the tangent plane at the point $(1, 1, 1)$ of the cone $x^2 + y^2 + z^2 + 4yz + 4zx + 4xy = 0$ is
- (a) $4x + 2y + 3z = 0$ (b) $x + y + z = 0$
 (c) $x + 4y + 2z = 0$ (d) $2x + 3y + 4z = 0$
40. The equation $ax^2 + by^2 + cz^2 + 2x + 2y + 2z + 1 = 0$ represents a cone if
- (a) $a + b + c = 1$
 (b) $a^2 + b^2 + c^2 = 1$
 (c) $a + b + c = 0$
 (d) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$
41. If the plane $x + y + z = 1$ meets the co-ordinate axes in A, B, C then the equation of the cone generated by the lines drawn from O to meet the circle ABC is
- (a) $x^2 + y^2 + z^2 + xy + yz + zx = 0$
 (b) $xy + yz + zx = 0$
 (c) $x^2 + y^2 + z^2 = xy$
 (d) none of these
42. If a plane $x + y + z = 0$ cuts the cone $ax^2 + by^2 + cz^2 = 0$ in perpendicular generators then
- (a) $a + b + c = 0$
 (b) $a + b + c = 1$
 (c) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$
 (d) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$
43. If the plane $ax - by + cz = 0$ cut the cone $x^2 + y^2 + z^2 = 0$ in perpendicular generator then
- (a) $a + b + c = 0$
 (b) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$
 (c) $a^2 + b^2 + c^2 = 0$
 (d) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 0$
44. If the cone $2x^2 + y^2 + az^2 + 2yz + 6zx + 4xy = 0$ have three mutually perpendicular tangent planes only when
- (a) $a = 2$ (b) $a = 3$
 (c) $a = 4$ (d) $a = 6$
45. If the angle between the lines in which the plane $ux + vy + wz = 0$ cuts the cone $x^2 + y^2 + z^2 + 2yz + 2zx + 2xy = 0$ is $\frac{\pi}{2}$ then
- (a) $f(u, v, w) = uvw$
 (b) $f(u, v, w) = u + v + w$
 (c) $f(u, v, w) = u^2 + v^2 + w^2$
 (d) $f(u, v, w) = 0$

46. The vertex of the cone $2x^2 + 2y^2 + 7z^2 - 10yz - 10zx + 2x + 2y + 26z - 17 = 0$ is
 (a) (2, 1, 2) (b) (1, 2, 2)
 (c) (2, 2, 1) (d) (2, 2, 2)
47. The vertex of the cone $xy + 2yz + 3zx = 0$ is
 (a) (0, 0, 0) (b) (1, 2, 3)
 (c) (1, 0, 0) (d) (0, 2, 0)
48. If a cone have three mutually perpendicular tangent planes then its reciprocal cone has three mutually
 (a) parallel generators
 (b) perpendicular generators
 (c) perpendicular tangent planes
 (d) parallel tangent planes
49. The straight lines $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is a generator of the cone $3x^2 - 2y^2 + 4z^2 = 0$ if l, m, n satisfy the condition
 (a) $l^2 + m^2 + n^2 = 1$
 (b) $l^2 + m^2 + n^2 = 0$
 (c) $3l^2 - 2m^2 + 4n^2 = 0$
 (d) none of these
50. A straight line lying on the surface of a cone is called :
 (a) tangent (b) generator
 (c) normal (d) axis
51. The tangent plane at any point of a cone always pass through its
 (a) guiding curve (b) tangent plane
 (c) normal plane (d) vertex
52. The number of arbitrary constants in the homogeneous equation of second degree in x, y, z of the cone are
 (a) 4 (b) 5
 (c) 6 (d) 3
53. The number of concurrent lines that must be required to determine a cone are
 (a) 2 (b) 3
 (c) 4 (d) 5
54. If $ax^2 + by^2 + cz^2 + d = 0$ represents a cone then
 (a) $a = b = c$ (b) $a = b = d$
 (c) $b = c = d$ (d) $d = 0$
55. The equation of the cone with vertex at origin semi-vertical angle is $\frac{\pi}{4}$ and axis as z -axis is
 (a) $x^2 + y^2 - z^2 = 0$
 (b) $y^2 + z^2 - x^2 = 0$
 (c) $z^2 + x^2 - y^2 = 0$
 (d) $x^2 + y^2 + z^2 = 0$
56. The section of a right circular by any plane is
 (a) circle (b) ellipse
 (c) hyperbola (d) (a) or (b)
57. The equation $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ represents a
 (a) plane (b) line
 (c) cylinder (d) cone
58. The coordinates of vertex of the cone $4x^2 - y^2 + 2z^2 + 2xy - 3yz + 12x - 11y + 6z + 4 = 0$ is
 [Kanpur 2018]
 (a) $(-1, 2, 3)$ (b) $(-1, -2, 3)$
 (c) $(-1, -2, -3)$ (d) $(1, -2, -3)$
59. Enveloping cone of the sphere $x^2 + y^2 + z^2 + 2x - 2y = 2$ with its vertex at $(1, 1, 1)$ is
 [Kanpur 2018]
 (a) $3x^2 - y^2 + 4zx + 10x + 2y - 4z + 6 = 0$
 (b) $3x^2 - y^2 + 4zx + 10x - 2y - 4z - 6 = 0$
 (c) $3x^2 - y^2 + 4zx + 10x - 2y + 4z - 6 = 0$
 (d) $3x^2 - y^2 + 4zx - 10x + 2y - 4z + 6 = 0$

- # ANSWERS

[illegible]

HINTS AND SOLUTIONS

3. Let $f = 4x^2 - y^2 + 2z^2 + 2xy - 3yz + 12xt - 11yt + 6zt + 4t^2$

$$\frac{\partial f}{\partial x} = 8x + 2y + 12t = 0$$

$$\frac{\partial f}{\partial y} = -2y + 2x - 3z - 11t = 0$$

$$\frac{\partial f}{\partial z} = 4z - 3y + 6t = 0$$

$$\frac{\partial f}{\partial t} = 12x - 11y + 6z + 8t = 0$$

After solving these equations we get $x = -1$,

$$y = -2, z = -3$$

5. Given $\sqrt{fx} \pm \sqrt{gy} \pm \sqrt{hz} = 0$

$$\Rightarrow \sqrt{fx} \pm \sqrt{gy} = \mp \sqrt{hz}$$

Squaring both sides we get $fx + gy - hz = \mp 2\sqrt{fgxy}$ again squaring we get $f^2x^2 + g^2y^2 + h^2z^2 - 2fgxy - 2fhxz - 2ghyz = 0$

it represents a cone with vertex at origin.

Let its reciprocal cone is

$$Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gzx + 2Hxy = 0$$

$$\text{where } A = bc - f^2 = 0, \quad B = C = 0,$$

$$F = gh - af^2 = 2f^2gh, G = 2g^2hf, H = 2h^2fg$$

So reciprocal cone is $fyz + gzx + hxy = 0$.

7. If the cone

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0 \quad \text{has three mutually perpendicular generator then } a + b + c = 0 \text{ so by given cone } 1 + a - 3 = 0 \Rightarrow a = 2$$

9. Let the cone is

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

If x-axis is its generator then d.c. of x-axis 1, 0, 0 satisfy it i.e. $a = 0$. Similarly if generator pass

through y-axis and z-axis then $b = 0$ and $c = 0$. Thus the required cone is $fyz + gzx + hxy = 0$

11. Let the equation of a line of section of given cone by the plane is $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ then $mn + nl + lm = 0$ and

$al + bm + cn = 0$. By last two equations we get

$$a\left(\frac{l}{m}\right)^2 + (a + b - c)\left(\frac{l}{m}\right) + b = 0 \text{ i.e. } \frac{l_1}{m_1} \cdot \frac{l_2}{m_2} = \frac{b}{a}$$

Thus $\frac{l_1 l_2}{\frac{1}{a}} = \frac{m_1 m_2}{\frac{1}{b}} = \frac{n_1 n_2}{\frac{1}{c}}$. So the lines of section will

be perpendicular if $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$ i.e.

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$$

13. Equation of line through (1, 1, 0) is $\frac{x-1}{l} = \frac{y-1}{m} = \frac{z}{n}$

it meets at $y = 0$ at the point $\frac{x-1}{l} = \frac{-1}{m} = \frac{z}{n}$ i.e.

$\left(1 - \frac{l}{m}, 0, \frac{-n}{m}\right)$ This point lie on $x^2 + z^2 = 4$. So

$\left(1 - \frac{l}{m}\right)^2 + \frac{n^2}{m^2} = 4$, by the line $\frac{l}{m} = \frac{x-1}{y-1}$ and

$\frac{n}{m} = \frac{z}{y-1}$. Put in above equation we get

$$x^2 - 3y^2 + z^2 - 2xy + 8y = 4$$

15. The cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx$

$2hxy = 0$ has three mutually perpendicular tangent plane when $ab + bc + ac = f^2 + g^2 + h^2$ so by

given cone $p + 2 + 2p = 1 + 4 + 1 \Rightarrow 3p = 4 \Rightarrow$

$$p = \frac{4}{3}$$

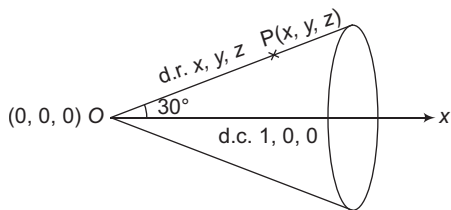
26. Required condition is $(a + b + c)(u^2 + v^2 + w^2)$

$$= f(u, v, w). \text{ So we get } 0 = -3 - a + 3a \Rightarrow a = \frac{3}{2}$$

28. After solving it we get a homogeneous equation of degree two so it represents a cone.

31. $2 - 5 + a = 0 \Rightarrow a = 3$

32. The direction ratios of OP and OX are x, y, z and $1, 0, 0$ respectively



$$\text{So } \cos 30^\circ = \frac{x \cdot 1 + y \cdot 0 + z \cdot 0}{\sqrt{1 + 0 + 0} \sqrt{x^2 + y^2 + z^2}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow 3(y^2 + z^2) = x^2$$

39. Tangent plane at (x_1, y_1, z_1) of the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ is $x(ax_1 + hy_1 + gz_1) + y(hx_1 + by_1 + fz_1) + z(gx_1 + fy_1 + cz_1) = 0$.

So required plane is

$$x(1 + 2 + 2) + y(2 + 1 + 2) + z(2 + 2 + 1) = 0$$

$$\Rightarrow x + y + z = 0$$

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THE CYLINDER

1. A cylinder is a surface generated by a straight line which is parallel to a fixed line, and satisfies one more condition is it may intersect a fixed curve or touch a given surface. Any line on the surface of the cylinder is called its **generator**. The fixed line is called the **axis** of the cylinder. If all the generators of a cylinder intersect a curve, then curve is called the **guiding curve**.
2. The equation of a cylinder whose generators are parallel to the line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ and intersect the conic $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0, z = 0$ is $a(nx - lz)^2 + 2h(nx - lz)(ny - mz) + b(ny - mz)^2 + 2gn(nx - lz) + 2fn(ny - mz) + cn^2 = 0$
3. If z-axis be the axis of cylinder then by point 2 the equation of cylinder becomes $a^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
4. The equation of the cylinder whose axis is z-axis and whose generators intersect the circle $x^2 + y^2 = a^2, z = 0$ is $x^2 + y^2 = a^2$
5. The equation $f(x, y) = 0, f(y, z) = 0$ and $f(z, x) = 0$ respectively represents a cylinder with its generators parallel to z-axis, x-axis and y-axis respectively.

RIGHT CIRCULAR CYLINDER

1. A cylinder whose guiding curve is a fixed circle and whose generator is normal to the plane of the circle is called right circular cylinder. The radius of the circle is called the radius of the right circular cylinder.
2. The equation of the right circular cylinder whose radius is r and axis the line

$$\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n} \text{ is}$$

$$[(y - \beta)n - (z - \gamma)m]^2 + [(z - \gamma)l - (x - \alpha)n]^2 + [(x - \alpha)m - (y - \beta)l]^2 = r^2(l^2 + m^2 + n^2)$$

3. The equation of the right circular cylinder whose axis is the z-axis, and radius r is

$$x^2 + y^2 = r^2$$

ENVELOPING CYLINDER

1. The enveloping cylinder is the locus of the tangents to a surface (sphere or conicoid) which is parallel to a given line.
2. The equation of the enveloping cylinder of the sphere $x^2 + y^2 + z^2 = a^2$, whose generator are parallel to the line $\frac{x}{a} = \frac{y}{m} = \frac{z}{n}$ is $(x^2 + y^2 + z^2 - a^2)(l^2 + m^2 + n^2) = (lx + my + nz)^2$

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The equation of the cylinder in three-dimensions whose generators are parallel to the z -axis is
 - (a) $F(x, y) = 0$
 - (b) $F(y, z) = 0$
 - (c) $F(z, x) = 0$
 - (d) $F(x, y, z) = 0$
2. In three-dimension geometry $x^2 + y^2 = a$ always represents a
 - (a) circle
 - (b) cone
 - (c) sphere
 - (d) cylinder
3. The equation of the cylinder whose axis is z -axis and whose generator intersect the circle $x^2 + y^2 = a^2, z = 0$ is
 - (a) $x^2 + y^2 + z^2 = a^2$
 - (b) $x^2 + z^2 = a^2$
 - (c) $x^2 + y^2 = a^2$
 - (d) $y^2 + z^2 = a^2$
4. The generators of the cylinder $y^2 + z^2 = a^2$ are straight lines parallel to
 - (a) y -axis
 - (b) z -axis
 - (c) yz -axis
 - (d) x -axis
5. The equation of a right circular cylinder whose axis is y -axis is **[Avadh 2018]**
 - (a) $x^2 + y^2 = r^2$
 - (b) $y^2 + z^2 = r^2$
 - (c) $x^2 + z^2 = r^2$
 - (d) $x^2 + y^2 + z^2 = r^2$
6. The equation of the cylinder with generator parallel to z -axis and passing through the curve $x^2 + y^2 = 2z, x + y + z = 1$ is
 - (a) $x^2 + y^2 + z^2 + x + y = 2$
 - (b) $x^2 + y^2 + 2x + 2y = 2$
 - (c) $x^2 + y^2 + z^2 + 2x + 2y = 2$
 - (d) $x^2 + y^2 - 2x - 2y = 2$
7. The equation of the right circular cylinder of radius 2 whose axis is the y -axis is
 - (a) $x^2 + y^2 = 4$
 - (b) $y^2 + z^2 = 4$
 - (c) $x^2 + y^2 + z^2 = 4$
 - (d) $x^2 + z^2 = 4$
8. The surface generated by a variable line which moves parallel to a fixed line and intersect a given cone is called
 - (a) plane
 - (b) sphere
 - (c) cone
 - (d) cylinder
9. The equation of the right circular cylinder described on the circle through the points $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ as the guiding curve
 - (a) $x^2 + y^2 + z^2 + xy + yz + zx + 1 = 0$
 - (b) $x^2 + y^2 + z^2 - xy - yz - zx - 1 = 0$
 - (c) $x^2 + y^2 + z^2 + xy - yz - zx - 1 = 0$
 - (d) $x^2 + y^2 + z^2 - xy - yz + zx + 1 = 0$
10. In three-dimension geometry $F(y, z) = 0$ represents
 - (a) sphere
 - (b) cylinder
 - (c) hyperboloid
 - (d) ellipse
11. The envelope of the sphere $x^2 + y^2 + z^2 = a^2$ whose generators are parallel to the line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ represents
 - (a) cylinder
 - (b) sphere
 - (c) cone
 - (d) circle
12. The equation of the cylinder with generator parallel to z -axis and passing through two curves $x^2 + y^2 = z, x + y + z = 1$ is
 - (a) $x^2 + y^2 + z^2 + x + y = 2$
 - (b) $x^2 + y^2 + x + y = 1$
 - (c) $x^2 + y^2 + z^2 + 2x + 2y = 2$
 - (d) $x^2 + y^2 - 2x - 2y = 2$

13. All the generators of a cylinder are straight line which are
 (a) perpendicular
 (b) parallel
 (c) intersect at a point
 (d) intersect at infinite point
14. The surface generated by the tangent's lines to the sphere $x^2 + y^2 + z^2 = a^2$ which are parallel to the line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is a
 (a) sphere (b) cone
 (c) cylinder (d) conicoid
15. The equation of the cylinder which intersect two curves and whose generators are parallel to z -axis is :
 (a) $f(y, z) = 0$ (b) $f(x, z) = 0$
 (c) $f(x, y) = 0$ (d) $f(x, y, z) = 0$
16. The equation of the cylinder with generators parallel to y -axis and intersect the curve $x^2 + y^2 + 2z^2 = 12$, $x - y + z = 1$ is
 (a) $x^2 + y^2 + 2z^2 = 12$
 (b) $x^2 + (x + z + 1)^2 + z^2 = 12$
 (c) $x^2 + (x + z - 1)^2 + 2z^2 = 12$
 (d) $x^2 + (x + z - 1)^2 - z^2 = 12$
17. In three-dimension geometry $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a :
 (a) cone (b) cylinder
 (c) sphere (d) conic
18. The equation of the cylinder whose generators are parallel to the line $x = -y = z$ and whose guiding curve is the ellipse $x^2 + y^2 = 1, z = 0$ is
 (a) $x^2 + y^2 + 2z^2 + 2yz - 2zx = 1$
 (b) $x^2 + y^2 + z^2 + 2yz + 2zx + 2xy = 1$
 (c) $x^2 + y^2 + 2z^2 + 2yz + 2zx + 2xy = 0$
 (d) $x^2 + y^2 - z^2 + 2yz + 2zx - 2xy = 1$
19. If the equation of enveloping cylinder of conicoid $x^2 + y^2 + z^2 = 1$ whose generators are parallel to the line $x = y = z$ is given by $x^2 + y^2 + z^2 - xy - yz - zx = p$ then p is
 (a) 1 (b) $\frac{3}{2}$
 (c) $\frac{2}{3}$ (d) 2
20. The equation of the right circular cylinder whose axis is $x = z + 2, y = 0$ and which pass through the point $(3, 0, 0)$ is
 (a) $x^2 + y^2 + z^2 - 2zx + 4x + 4z + 3 = 0$
 (b) $x^2 + y^2 + 3zx - 4x + y + 3 = 0$
 (c) $x^2 + 2y^2 + z^2 - 2zx - 4x + 4z + 3 = 0$
 (d) $x^2 + y^2 + z^2 - 4xy + y + 3 = 0$
21. The surface generated by a variable line which moves parallel to a fixed line and intersects a given curve is
 (a) $x^2 + y^2 + z^2 = 0$
 (b) $x^2 + y^2 + 2hxy = 0$
 (c) $x^2 + y^2 - z^2 = a^2$
 (d) $x^2 + y^2 = r^2$
22. The equation of the cylinder whose generators are parallel to the line $\frac{x}{0} = \frac{y}{1} = \frac{z}{-1}$ and whose guiding curve is $x^2 + 2z^2 = 1, y = 0$, is
 (a) $x^2 + 2(y - z)^2 = 1$
 (b) $y^2 + 2(z - y)^2 = 1$
 (c) $x^2 + 2(y + z)^2 = 1$
 (d) $x^2 + 2(y + z)^2 = 2$
23. The equation of the right circular cylinder of radius 1 whose axis is the line $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$ is
 (a) $y^2 + z^2 + 2y + 2z - 2 = 0$
 (b) $y^2 + z^2 + x + 2y - 2 = 0$
 (c) $y^2 + z^2 + 2x + 2y - 2 = 0$
 (d) $y^2 + z^2 + 2x - 2 = 0$

24. The equation of the cylinder with generator parallel to z -axis and passing through the curve $x^2 + y^2 = 3z, x + y + z = 1$ is
- (a) $x^2 + y^2 + 3x + 3y + 2 = 0$
 (b) $x^2 + y^2 + 3y + 3z = 3$
 (c) $x^2 + y^2 + 3x + 3y = 3$
 (d) $x^2 + y^2 + 3x - y = 3$
25. The equation of the cylinder whose axis is x -axis and whose generators intersect the circle $y^2 + z^2 = a^2$ and $x = 0$ is
- (a) $x^2 + y^2 = a^2$
 (b) $y^2 + z^2 = a^2$
 (c) $z^2 + x^2 = a^2$
 (d) $x^2 + y^2 + z^2 = 0$
26. The equation of the cylinder which intersect two curves and whose generators are parallel to x -axis is :
- (a) $f(x, y) = 0$ (b) $f(y, z) = 0$
 (c) $f(z, x) = 0$ (d) $f(x, y, z) = 0$
27. $F(y, z) = 0$ represents a cylinder in three-dimension geometry, whose generators are parallel to the
- (a) x -axis (b) y -axis
 (c) z -axis (d) any axis
28. The equation of a right circular cylinder whose axis is z -axis is
- (a) $x^2 + y^2 + z^2 = a^2$
 (b) $x^2 + y^2 = a^2$
 (c) $y^2 + z^2 = a^2$
 (d) $z^2 + x^2 = a^2$
29. The envelope of any sphere whose generators are parallel to any line represents a
- (a) cone (b) sphere
 (c) cylinder (d) circle
30. The section of a right circular cylinder by any plane perpendicular to its axis is
- (a) parabola (b) hyperbola
 (c) ellipse (d) circle
31. The locus of the lines parallel to a given line r or to touch a given surface is called
- (a) circle
 (b) enveloping cylinder
 (c) cylinder
 (d) sphere
32. The normal to the plane of guiding circle through its centre for a cylinder is called
- (a) generator
 (b) guiding curve
 (c) axis
 (d) envelope
33. The equation of the cylinder whose generators are parallel to z -axis and which pass through the circle whose centre is $(a, 0, 0)$ and radius a lying in the plane $z = 0$ is
- (a) $x^2 + y^2 + z^2 + ax = 0$
 (b) $x^2 + y^2 + ax = 0$
 (c) $x^2 + y^2 = a^2$
 (d) $x^2 + y^2 - ax = 0$
34. The equation of the cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and whose guiding curve is ellipse $x^2 + 2y^2 = 1, z = 3$ is
- [Kanpur 2018]**
- (a) $3x^2 + 6y^2 + 3z^2 + 8yz + 2zx + 6x - 24y - 18z + 24 = 0$
 (b) $3x^2 + 6y^2 + 3z^2 + 8yz - 2zx + 6x - 24y - 18z + 24 = 0$
 (c) $3x^2 + 6y^2 + 3z^2 + 8yz - 2zx - 6x - 24y + 18z - 24 = 0$
 (d) $3x^2 + 6y^2 + 3z^2 + 8yz - 2zx - 6x - 24y + 18z - 24 = 0$
35. Guiding curve of a right circular cylinder is
- [Avadh 2018]**
- (a) Circle
 (b) Ellipse
 (c) Any closed curve
 (d) none of these

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(a)	2.	(d)	3.	(c)	4.	(d)	5.	(c)	6.	(b)	7.	(d)	8.	(d)	9.	(b)	10.	(b)
11.	(a)	12.	(b)	13.	(b)	14.	(c)	15.	(c)	16.	(c)	17.	(b)	18.	(a)	19.	(b)	20.	(b)
21.	(d)	22.	(c)	23.	(d)	24.	(c)	25.	(b)	26.	(b)	27.	(a)	28.	(b)	29.	(c)	30.	(d)
31.	(b)	32.	(c)	33.	(d)	34.	(b)	35.	(a)										

HINTS AND SOLUTIONS

6. Since the generator of the cylinder are parallel to z -axis so it does not contain the terms of z so eliminate z between two curves i.e.

$$\frac{x^2 + y^2}{2} = 1 - x - y$$

$$\Rightarrow x^2 + y^2 + 2x + 2y = 2$$

9. Equation of sphere $OABC$ is

$$x^2 + y^2 + z^2 - x - y - z = 0 \quad \dots(1)$$

$$\text{and plane } ABC \text{ is } x + y + z = 1 \quad \dots(2)$$

Let (x_1, y_1, z_1) lies on the cylinder then its generator is

$$\frac{x - x_1}{1} = \frac{y - y_1}{1} = \frac{z - z_1}{1} = r$$

i.e. a point on generator is $(r + x_1, r + y_1, r + z_1)$ it lies (1) and (2)

$$\text{So } (r + x_1)^2 + (r + y_1)^2 + (r + z_1)^2 - (r + x_1) - (r + y_1) - (r + z_1) = 0$$

$$\text{and } (r + x_1) + (r + y_1) + (r + z_1) = 1$$

$$\Rightarrow 3r^2 + r(2x_1 + 2y_1 + 2z_1 - 3) + (x_1^2 + y_1^2 + z_1^2 - x_1 - y_1 - z_1) = 0 \quad \dots(3)$$

$$\text{and } r = \frac{1}{3}(1 - x_1 - y_1 - z_1) \quad \dots(4)$$

Eliminating r between (3) and (4) we get

$$x_1^2 + y_1^2 + z_1^2 - y_1z_1 - z_1x_1 - x_1y_1 = 1$$

so required locus is

$$x^2 + y^2 + z^2 - xy - yz - zx = 1$$

12. Since generator is parallel to z -axis so eliminate z between two equations we get

$$x^2 + y^2 = 1 - x - y$$

$$\Rightarrow x^2 + y^2 + x + y = 1$$

16. Since the axis of cylinder is y -axis so eliminate y between two equation, the required equation of cylinder is

$$x^2 + (x + z - 1)^2 + 2z^2 = 12$$

18. Let (x_1, y_1, z_1) be the point lies on cylinder then equation of generator is $\frac{x - x_1}{1} = \frac{y - y_1}{-1} = \frac{z - z_1}{1}$ it

meets $z = 0$

$$\text{So, } \frac{x - x_1}{1} = \frac{y - y_1}{-1} = -z_1$$

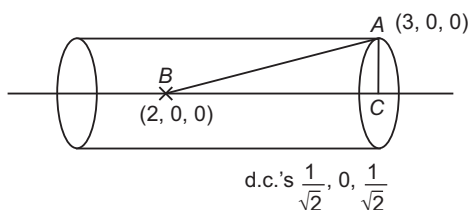
i.e. $(x_1 - z_1, y_1 + z_1, 0)$ lies on $x^2 + y^2 = 1$ i.e.

$$(x_1 - z_1)^2 + (y_1 + z_1)^2 = 1$$

$$\Rightarrow \text{so, locus is } (x - z)^2 + (y + z)^2 = 1$$

$$\Rightarrow x^2 + y^2 + 2z^2 - 2xz + 2yz = 1$$

20. Axis of cylinder is $\frac{x-2}{1} = \frac{y}{0} = \frac{z}{1}$



It pass through $B(2, 0, 0)$ so $AB = 1, BC = \text{projection of } AB \text{ on } BC \text{ i.e. } BC = \frac{1}{\sqrt{2}} + 0 + 0 = \frac{1}{\sqrt{2}}$

$$\therefore \text{Radius } AC = \sqrt{AB^2 - BC^2} = \frac{1}{\sqrt{2}}$$

Consider a point $P(x, y, z)$ on surface of cylinder then by projector rule we get

$$(x-2)^2 + y^2 + z^2 = \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}(x-2) + \frac{1}{\sqrt{2}}(2)\right)^2$$

After solving we get

$$x^2 + y^2 + 3zx - 4x + y + 3 = 0$$

22. If $P(x_1, y_1, z_1)$ lies on cylinder then equation of generator is

$$\frac{x-x_1}{0} = \frac{y-y_1}{1} = \frac{z-z_1}{-1}, \text{ it lies on } y=0 \text{ so}$$

$$\frac{x-x_1}{0} = \frac{-y_1}{1} = \frac{z-z_1}{-1} \text{ i.e. } P \text{ is } (x_1, 0, z_1 + y_1) \text{ it also}$$

lies on $x^2 + 2z^2 = 1$ so $x_1^2 + 2(z_1 + y_1)^2 = 1$, i.e.

$$\text{locus is } x^2 + 2(z+y)^2 = 1]$$

○○○

The Central Conicoids

EQUATION OF CENTRAL CONICOID

1. The surface $ax^2 + by^2 + cz^2 = 1$ is called a conicoid because the sections of this surface by planes parallel to co-ordinate planes are conics. The conicoid is called a central conicoid because every surface has a centre.
2. The origin is the centre of the central conicoid $ax^2 + by^2 + cz^2 = 1$ and the co-ordinate planes bisect all chords perpendicular to them.
3. The central conicoid is symmetric with respect to all the three co-ordinate planes which are called the principal planes. The three principal planes taken in pairs intersect in three lines called the principal axes of the central conicoid.
4. In the central conicoid $ax^2 + by^2 + cz^2 = 1$, there are four cases with respect to the signs of a, b, c :
 - (i) If all are positive then surface is an ellipsoid.
 - (ii) If only one is negative, the surface is hyperboloid of one sheet.
 - (iii) Two are negative, the surface is hyperboloid of two sheets.
 - (iv) All are negative, the surface is virtual quadric.

THE TANGENT AND NORMAL PLANES

1. The equation of the tangent plane to the central conicoid $ax^2 + by^2 + cz^2 = d$ at the point (x_1, y_1, z_1) is $axx_1 + byy_1 + czz_1 = d$
 So the equation of the tangent plane to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}$ at the point (x_1, y_1, z_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} + \frac{zz_1}{c^2} = 1$

2. The condition that the plane $lx + my + nz = p$ may touch the central conicoid $ax^2 + by^2 + cz^2 = 1$ is $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$

Then the tangent plane becomes

$$lx + my + nz = \pm \sqrt{\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c}}$$

3. The condition that the plane $lx + my + nz = p$ may touch the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is $a^2l^2 + b^2m^2 + c^2n^2 = p^2$

4. The line perpendicular to the tangent plane to a conicoid $ax^2 + by^2 + cz^2 = 1$ at any point P on it and passing through P called the normal to the conicoid at P .

The equation of the normal to the central conicoid $ax^2 + by^2 + cz^2 = 1$ at the point (x_1, y_1, z_1) is

$$\frac{x - x_1}{ax_1} = \frac{y - y_1}{by_1} = \frac{z - z_1}{cz_1}$$

In terms of d.c.'s its equation is

$$\frac{x - x_1}{ax_1p} = \frac{y - y_1}{by_1p} = \frac{z - z_1}{cz_1p}$$

5. The equation of the normal to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at (x_1, y_1, z_1) is

$$\frac{x - x_1}{\frac{x_1p}{a^2}} = \frac{y - y_1}{\frac{y_1p}{b^2}} = \frac{z - z_1}{\frac{z_1p}{c^2}}$$

The number of normals that can be drawn to the conicoid which pass through a given point are six.

7. There are S_{1x} feet of normals that can be drawn to a conicoid (respectively ellipsoid) from a given point are the points of intersection of a certain cubic curve with the conicoid (respectively ellipsoid).

THE DIRECTOR SPHERE

1. The director sphere of a central conicoid is the locus of the points of intersection of three mutually perpendicular tangent planes to that central conicoid.
2. The equation of the director sphere of the central conicoid $ax^2 + by^2 + cz^2 = 1$ is $x^2 + y^2 + z^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$

Similarly director sphere of the ellipsoid is

$$x^2 + y^2 + z^2 = a^2 + b^2 + c^2$$

THE POLAR PLANE AND CONJUGATE PLANES

1. Let $ax^2 + by^2 + cz^2 = 1$ be the conicoid and $A(x_1, y_1, z_1)$ be any point. Draw line APQ to meet the conicoid in the points P and Q , then the locus of the point R such that AR is the harmonic mean of AP and AQ is called the polar plane of the point A with respect to the conicoid. The equation of the polar plane is

$$axx_1 + byy_1 + czz_1 = 1$$

The pole of the polar plane is $\frac{l}{ap}, \frac{m}{bp}, \frac{n}{cp}$.

2. Let $u_1 = 0$ and $u_2 = 0$ be the equations of two planes. If the pole of the plane $u_1 = 0$ with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ lies on the plane $u_2 = 0$ then the pole of the plane $u_2 = 0$ will lie on the plane $u_1 = 0$. Then two such planes $u_1 = 0$ and $u_2 = 0$ are called conjugate planes.
3. If the polar plane of the point P with respect to a central conicoid pass through a point Q , then the polar plane of Q pass through P . The points

P and Q satisfying this property are called the conjugate points.

4. Two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are the conjugate points with respect to $ax^2 + by^2 + cz^2 = 1$ if and only if

$$ax_1x_2 + by_1y_2 + cz_1z_2 = 1$$

5. The polar line or polar plane of a given line AB is another line PQ such that the polar planes of all points on AB pass through PQ .
6. Two lines AB and PQ are called the conjugate lines if when PQ intersects the polar line of the line AB then the line AB also intersects the polar line of PQ .

DIAMETRAL PLANE

1. A diametral plane for a conicoid is the locus of the middle points of a system of parallel chords.
2. If the equation of central conicoid is $ax^2 + by^2 + cz^2 = 1$ and equations of a system of parallel chords are $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ then the diametral plane is $alx + bmy + cnz = 0$. In case of ellipsoid, diametral plane is $\frac{lx}{a^2} + \frac{my}{b^2} + \frac{nz}{c^2} = 0$
3. Every plane through the centre is a diametral plane of the central conicoid corresponding to some direction.
4. If the diametral plane of a line OP passes through the point Q then the diametral plane of the line OQ will also pass through the point P .
5. The three semi-diameters OP , OQ and OR of an ellipsoid which are such that the plane containing any two is the diametral plane of the third, are called the conjugate semi-diameters.
6. The three planes POQ , QOR and ROP which are such that each is the diametral plane of the line of intersection of the other two, are called

- the conjugate planes or conjugate diametral planes.
7. The sum of the squares of any three conjugate semi-diameters of an ellipsoid is constant i.e.
- $$OP^2 + OQ^2 + OR^2 = a^2 + b^2 + c^2$$
8. The sum of the squares of the projection of three conjugate semi-diameters of an ellipsoid on a line is constant.
9. The sum of the squares of projections of three semi-diameters of an ellipsoid on any plane is constant.

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The surface represented by the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ is [Avadh 2018]
- (a) ellipsoid
(b) sphere
(c) hyperboloid of one sheet
(d) hyperboloid of two sheets
2. The centre of the central conicoid $ax^2 + by^2 + cz^2 = 1$ is
- (a) (0, 0, 0) (b) (a, 0, 0)
(c) (0, b, 0) (d) (a, b, c)
3. The plane $lx + my + nz = p$ touches the conicoid $ax^2 + by^2 + cz^2 = 1$ if [Avadh 2018]
- (a) $al^2 + bm^2 + cn^2 = 1$
(b) $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$
(c) $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p$
(d) $al^2 + bm^2 + cn^2 = p^2$
4. The centre of the director sphere of the central conicoid $3x^2 + 4y^2 - 5z^2 = 1$ is
- (a) (3, 4, -5) (b) (3, 4, 5)
(c) (-3, 4, 5) (d) (0, 0, 0)
5. The equation of the director sphere of the central conicoid $x^2 + y^2 + z^2 = \frac{1}{3}$ is
- (a) $x^2 + y^2 + z^2 = 3$ (b) $x^2 + y^2 + z^2 = 1$
(c) $x^2 + y^2 + z^2 = \frac{1}{3}$ (d) none of these
6. If the plane $x + y + z = 2$ touches the central conicoid $px^2 + y^2 + z^2 = 1$ the value of p is
- (a) 2 (b) $\frac{1}{2}$
(c) 3 (d) $\frac{1}{3}$
7. If a tangent plane to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ meets the co-ordinate axes in A, B and C then the locus of the centroid of the tetrahedron $OABC$ is
- (a) $\frac{a^2}{x^2} + \frac{b^2}{y^2} + \frac{c^2}{z^2} = 16$ (b) $\frac{a^2}{x^2} + \frac{b^2}{y^2} + \frac{c^2}{z^2} = 4$
(c) $\frac{a^2}{x^2} + \frac{b^2}{y^2} + \frac{c^2}{z^2} = 1$ (d) $\frac{a^2}{x^2} + \frac{b^2}{y^2} + \frac{c^2}{z^2} = 2$
8. The equation of tangent plane to the surface $\frac{x^2}{4} + \frac{y^2}{a} - \frac{z^2}{25} = 1$ at (2, 3, 5) is
- (a) $15x + 10y - 6z = 10$
(b) $15x + 10y - 6z = 20$
(c) $15x + 10y - 6z = 30$
(d) $15x + 10y - 6z = 40$
9. The equation of the plane which cuts $3x^2 + 2y^2 - 15z^2 = 4$ in a conic whose centre is $(-2, -3, -1)$ is given by
- (a) $2x + 2y + z + 2 = 0$
(b) $2x - 2y + 5z + 2 = 0$
(c) $2x + 2y - 5z + 2 = 0$
(d) $2x - 2y - 5z + 2 = 0$

10. If OP , OQ and OR are the conjugate semi-diameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ then $OP^2 + OQ^2 + OR^2$ is
- (a) $a + b + c$ (b) $\frac{1}{a + b + c}$
 (c) $a^2 + b^2 + c^2$ (d) $\frac{1}{a^2 + b^2 + c^2}$
11. The number of normals that can be drawn to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ from a given point is
- [Kanpur 2018]**
- (a) 1 (b) 2
 (c) 3 (d) 6
12. If the normal at $p(\alpha, \beta, \gamma)$ of a central conicoid $ax^2 + by^2 + cz^2 = 1$ meets the three principal planes at A, B, C then $PA : PB : PC$ is
- (a) $a : b : c$ (b) $\frac{1}{a} : \frac{1}{b} : \frac{1}{c}$
 (c) $a^2 : b^2 : c^2$ (d) $\frac{1}{a^2} : \frac{1}{b^2} : \frac{1}{c^2}$
13. The equation of the polar plane of the point (α, β, γ) with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ is
- (a) $ax + by + cz = 1$
 (b) $\alpha x + \beta y + \gamma z = 1$
 (c) $\alpha\alpha x + b\beta y + c\gamma z = 1$
 (d) none of these
14. The pole of the plane $lx + my + nz = p$ with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ is
- (a) $\frac{a}{lp}, \frac{b}{mp}, \frac{c}{np}$ (b) $\frac{p}{al}, \frac{p}{bm}, \frac{p}{cn}$
 (c) $\frac{l}{a}, \frac{m}{b}, \frac{n}{c}$ (d) $\frac{l}{ap}, \frac{m}{bp}, \frac{n}{cp}$
15. If the planes $l_1x + m_1y + n_1z = p_1$ and $l_2x + m_2y + n_2z = p_2$ are the conjugate planes with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ then $\frac{l_1l_2}{a} + \frac{m_1m_2}{b} + \frac{n_1n_2}{c}$ is equal to
- (a) p_1p_2 (b) p_1
 (c) p_2 (d) $p_1 + p_2$
16. The equation of the tangent plane at $(1, 1, -1)$ to the conicoid $2x^2 + 3y^2 - z^2 = 4$ is
- (a) $2x + 3y - z = 4$ (b) $2x - 3y + z = 4$
 (c) $-2x + 3y + z = 4$ (d) $2x + 3y + z = 4$
17. The equation of the polar plane of the point $(1, 2, 3)$ with respect to the conicoid $x^2 + 2y^2 + z^2 = 1$ is
- (a) $x + 4y + 3z = 1$ (b) $x + 2y + 3z = 1$
 (c) $x + 2y + z = 2$ (d) $x + 2y + 3z = 2$
18. The pole of the plane $2x - 3y + 2z = 1$ with respect to the conicoid $x^2 - y^2 + z^2 = 1$ is
- (a) $2, -3, 2$ (b) $2, 3, 2$
 (c) $1, -1, 1$ (d) $2, -3, 1$
19. The radius of the director sphere of the ellipsoid $\frac{x^2}{6} + \frac{y^2}{4} + \frac{z^2}{15} = 1$ is
- (a) 5 (b) 6 (c) 4 (d) 15
20. If the plane $lx + my + nz = p$ touches the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ then the value of $(lx + my + nz)^2$ is
- (a) $al + bm + cn$ (b) $\frac{a^2}{l} + \frac{b^2}{m} + \frac{c^2}{n}$
 (c) $a^2l^2 + b^2m^2 + c^2n^2$ (d) $al^2 + bm^2 + cn^2$
21. The equation of the director sphere of the central conicoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is
- (a) $x^2 + y^2 + z^2 = a + b + c$
 (b) $x^2 + y^2 + z^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$
 (c) $x^2 + y^2 + z^2 = a^2 + b^2 + c^2$
 (d) $x^2 + y^2 + z^2 = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$
22. If $P(x_1, y_1, z_1), Q(x_2, y_2, z_2)$ and $R(x_3, y_3, z_3)$ be the extremities of three conjugate semi-diameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ then $x_1z_1 + x_2z_2 + x_3z_3$ is equal to
- (a) a^2 (b) b^2
 (c) c^2 (d) 0

23. The diametral plane of the central conicoid $ax^2 + by^2 + cz^2 = 1$, while l, m, n are d.c.'s of parallel chords is
 (a) $lx + my + nz = 0$
 (b) $alx + bmy + cnz = 0$
 (c) $alx + bmy + cnz = 1$
 (d) none of these
24. The volume of the parallel piped formed by three conjugate semi-diameters as coterminal edges in an ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is
 (a) abc (b) $a^2b^2c^2$
 (c) $a + b + c$ (d) $a^2 + b^2 + c^2$
25. The pole of the plane $4x + 8y - 3z = 1$ with respect to the conicoid $2x^2 + 4y^2 - z^2 = 1$ is
 (a) $(2, 4, -1)$ (b) $(4, 8, -3)$
 (c) $(2, 2, 3)$ (d) $(2, 3, 4)$
26. The point $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ are the conjugate points with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ if $ax_1x_2 + by_1y_2 + cz_1z_2$ is
 (a) 0 (b) 1 (c) 2 (d) 3
27. The equation of the normal to the central conicoid $x^2 + 2y^2 + 3z^2 = 1$ at the point $(1, 0, 1)$ is
 (a) $\frac{x-1}{1} = \frac{y-0}{0} = \frac{z-1}{3}$
 (b) $\frac{x-1}{1} = \frac{y-0}{2} = \frac{z-1}{3}$
 (c) $\frac{x-1}{1} = \frac{y-0}{0} = \frac{z-1}{1}$
 (d) none of these
28. If $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$, and $R(x_3, y_3, z_3)$ be the extremities of three conjugate semi-diameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ then $z_1^2 + z_2^2 + z_3^2$ is equal to
 (a) a^2 (b) b^2
 (c) c^2 (d) $b^2 + c^2$
29. In the conicoid the number of diameters are
 (a) 1 (b) 3
 (c) finite (d) infinite
30. The equation of the director sphere of the central conicoid $x^2 + 2y^2 + 3z^2 = 1$ is
 (a) $x^2 + y^2 + z^2 = 0$
 (b) $x^2 + y^2 + z^2 = 1$
 (c) $x^2 + y^2 + z^2 = \frac{11}{6}$
 (d) $x^2 + 2y^2 + 3z^2 = \frac{11}{6}$
31. The direction cosines of the normals to the conicoid $ax^2 + by^2 + cz^2 = d$ at the point (α, β, γ) are
 (a) a, b, c (b) $a\alpha, b\beta, \gamma$
 (c) $a\alpha p, b\beta p, c\gamma p$ (d) $\alpha p, \beta p, \gamma p$
32. The equation of the normal to the ellipsoid $3x^2 + 4y^2 + 5z^2 = 64$ at the point $(1, 2, 3)$ on it is
 (a) $\frac{x-1}{3} = \frac{y-2}{4} = \frac{z-3}{5}$
 (b) $\frac{x-1}{3} = \frac{y-2}{8} = \frac{z-3}{15}$
 (c) $\frac{x-1}{3} = \frac{y-2}{16} = \frac{z-3}{5}$
 (d) none of these
33. The length of the perpendicular drawn from the centre of the conicoid $x^2 - y^2 + z^2 = 1$ to the tangent plane to it at the point $(1, 1, 1)$ is
 (a) 1 (b) 3 (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$
34. The tangent planes at the extremities of any diameter of an ellipsoid are always
 (a) perpendicular (b) parallel
 (c) intersect (d) touches
35. The equation of tangent plane to the conicoid $x^2 + 2y^2 + z^2 = 3$ at the point $(1, 2, 3)$ is
 (a) $x + 4y + 3z = 1$
 (b) $x + 4y + 3z = 0$
 (c) $x + 4y + 3z = 3$
 (d) $x + 2y + z = 3$
36. If the plane $x + y + z = p$ touches the ellipsoid $6x^2 + 3y^2 + 2z^2 = 6$ then p is
 (a) ± 2 (b) $\pm \sqrt{6}$
 (c) $\pm \sqrt{13}$ (d) $\pm \sqrt{14}$

37. The equation of the director sphere of the ellipsoid $\frac{x^2}{2} + \frac{y^2}{3} + \frac{z^2}{4} = 1$ is
 (a) $x^2 + y^2 + z^2 = 1$ (b) $x^2 + y^2 + z^2 = 0$
 (c) $\frac{x^2}{2} + \frac{y^2}{3} + \frac{z^2}{4} = 9$ (d) $x^2 + y^2 + z^2 = 9$
38. If the tangent plane to the conicoid $x^2 + y^2 + z^2 = 1$ meets the co-ordinate axes in P, Q and R then the centroid of the triangle PQR lies on
 (a) $x^2 + y^2 + z^2 = 9$ (b) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 1$
 (c) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 9$ (d) $x^2 + y^2 + z^2 = 1$
39. The direction ratios of the normal to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at the point (α, β, γ) are
 (a) $\frac{\alpha}{a^2}, \frac{\beta}{b^2}, \frac{\gamma}{c^2}$ (b) $\frac{\alpha}{a}, \frac{\beta}{b}, \frac{\gamma}{c}$
 (c) $\frac{a}{\alpha}, \frac{b}{\beta}, \frac{c}{\gamma}$ (d) $\frac{a^2}{\alpha}, \frac{b^2}{\beta}, \frac{c^2}{\gamma}$
40. Two points $P(1, 1, 2)$ and $Q(2, 2, 1)$ are the conjugate points with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ if
 (a) $a + b + 2c = 1$ (b) $2a + 2b + c = 1$
 (c) $a + b + c = \frac{1}{2}$ (d) $a + b + c = 1$
41. If $(1, 1, 1)$ is one conjugate point with respect to the conicoid $x^2 + y^2 + z^2 = 1$ then the other one is
 (a) $(0, 1, 1)$ (b) $(1, 1, 0)$
 (c) $(1, 0, 0)$ (d) $(1, 0, 1)$
42. The equation of the director sphere of the central conicoid $36x^2 + 9y^2 + 4z^2 = 36$ is
 (a) $x^2 + y^2 + z^2 = 49$ (b) $x^2 + y^2 + z^2 = 14$
 (c) $x^2 + y^2 + z^2 = 36$ (d) $x^2 + y^2 + z^2 = 1$
43. If the plane $2x + y + 3z = 3$ touches the conicoid $x^2 + y^2 + pz^2 = 1$ then the value of p is
 (a) $\frac{3}{4}$ (b) $\frac{8}{3}$
 (c) $\frac{9}{4}$ (d) $\frac{2}{3}$
44. If $P(x_1, y_1, z_1), Q(x_2, y_2, z_2)$ and $R(x_3, y_3, z_3)$ be the extremities of three conjugate semi-diameters of the ellipsoid $2x^2 + y^2 + 3z^2 = 6$ then $z_1^2 + z_2^2 + z_3^2$ is equal to
 (a) 2 (b) 1 (c) 3 (d) 6
45. The point of contact of the tangent plane $2x + 3y + 10z = 25$ with respect to ellipsoid $2x^2 + 3y^2 + 5z^2 = 25$ is
 (a) $(1, 2, 2)$ (b) $(1, 2, 1)$
 (c) $(1, 1, 2)$ (d) $(2, 2, 1)$
46. A cubic curve meets a plane at
 (a) three points (b) four points
 (c) finite points (d) infinite points
47. The equation of the plane that cuts $2x^2 - 3y^2 + 5z^2 = 1$ in a conic whose centre is $(2, 1, 3)$ is given by
 (a) $4x + 3y + 15z = 50$ (b) $4x + 3y - 15z = 50$
 (c) $-4x + 3y + 15z = 50$ (d) $4x - 3y + 15z = 50$
48. If OP, OQ and OR be the conjugate semi-diameters of the ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$ then $OP^2 + OQ^2 + OR^2$ is
 (a) 4 (b) 9 (c) 16 (d) 29
49. The radius of the director sphere of the conicoid $\frac{x^2}{4} + \frac{y^2}{15} - \frac{z^2}{10} = 1$ is
 (a) 3 (b) 2 (c) 4 (d) 5
50. The radius of the director sphere of $x^2 + 2y^2 + 3z^2 = 1$ is
 (a) 1 (b) $\frac{49}{36}$
 (c) $\sqrt{14}$ (d) $\frac{11}{6}$
51. The diametral plane of the central conicoid $x^2 - 2y^2 + 3z^2 = 1$ whose d.c. of parallel chords are $1, 1, 1$ is
 (a) $x - 2y + 3z = 1$
 (b) $x - 2y + 3z = 0$
 (c) $x - 2y + 3z = 3$
 (d) none of these

52. If (x_1, x_2, x_3) , (y_1, y_2, y_3) and (z_1, z_2, z_3) be the extremities of the three conjugate semi-diameters of the ellipsoid $\frac{x^2}{2} + \frac{y^2}{4} + \frac{z^2}{6} = 1$ then $x_1^2 + x_2^2 + x_3^2$ is :
 (a) 2 (b) 4 (c) 6 (d) 1
53. The equation of tangent plane to the central conicoid $3x^2 - 5y^2 + z^2 + 2 = 0$ at the point $(1, 1, 0)$ is **[Kanpur 2018]**
 (a) $-3x - 3y + 2 = 0$ (b) $3x + 5y + 2 = 0$
 (c) $3x - 5y + 2 = 0$ (d) none of these
54. The pole of $4x + 8y - 3z = 15$ with respect to $3x^2 + 7y^2 + 2z^2 = 12$ is **[Kanpur 2018]**
 (a) $\left(\frac{-16}{15}, \frac{32}{35}, \frac{-6}{5}\right)$ (b) $\left(\frac{16}{15}, \frac{-32}{35}, \frac{6}{5}\right)$
 (c) $\left(\frac{16}{15}, \frac{-32}{35}, \frac{-6}{5}\right)$ (d) $\left(\frac{16}{15}, \frac{32}{35}, \frac{-6}{5}\right)$
55. The equation of the plane which cuts $3x^2 + 2y^2 - 15z^2 = 4$ in a conic whose centre is at the point $(-2, 3, -1)$ is **[Kanpur 2018]**
 (a) $2x + 2y - 5z + 2 = 0$
 (b) $2x - 2y - 5z + 2 = 0$
 (c) $2x + 2y + 2z + 3 = 0$
 (d) $x - y + z - 1 = 0$
56. The surface represented by the equation $5x^2 + 6y^2 - 7z^2 = 1$ is **[Kanpur 2018]**
 (a) Sphere
 (b) Ellipsoid
 (c) Hyperbola of one sheet
 (d) Hyperbolic paraboloid
57. The surface represented by the equation $4(x^2 + y^2) = -9z$ is a **[Kanpur 2018]**
 (a) Parabolic cylinder
 (b) Hyperbolic cylinder
 (c) Elliptic paraboloid
 (d) Paraboloid of revolution
58. The centre of the director sphere of the central conicoid $3x^2 + 4y^2 - 5z^2 = 1$ is **[Avadh 2018]**
 (a) $(0, 0, 0)$ (b) $(1, 1, 1)$
 (c) $(2, 2, 2)$ (d) None of these
59. The equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$ represents **[Avadh 2018]**
 (a) Ellipsoid
 (b) Elliptic paraboloid
 (c) Hyperbolic paraboloid
 (d) Hyperboloid of one sheet
60. The equation of an ellipsoid is **[Agra 2017]**
 (a) $\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
 (c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ (d) None of these

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(c)	2.	(a)	3.	(b)	4.	(d)	5.	(b)	6.	(b)	7.	(a)	8.	(c)	9.	(d)	10.	(c)
11.	(d)	12.	(b)	13.	(c)	14.	(d)	15.	(a)	16.	(d)	17.	(a)	18.	(b)	19.	(a)	20.	(c)
21.	(c)	22.	(d)	23.	(b)	24.	(a)	25.	(c)	26.	(b)	27.	(a)	28.	(c)	29.	(d)	30.	(c)
31.	(c)	32.	(b)	33.	(d)	34.	(a)	35.	(c)	36.	(b)	37.	(d)	38.	(b)	39.	(a)	40.	(c)
41.	(c)	42.	(b)	43.	(c)	44.	(a)	45.	(c)	46.	(a)	47.	(d)	48.	(d)	49.	(a)	50.	(d)
51.	(b)	52.	(a)	53.	(c)	54.	(d)	55.	(b)	56.	(c)	57.	(d)	58.	(a)	59.	(b)	60.	(c)

HINTS AND SOLUTIONS

5. Central conicoid is $3x^2 + 3y^2 + 3z^2 = 1$ so its director sphere is

$$x^2 + y^2 + z^2 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$$
6. $\frac{1}{p} + \frac{1}{1} + \frac{1}{1} = 4 \Rightarrow \frac{1}{p} = 2 \Rightarrow p = \frac{1}{2}$
8. Tangent of $ax^2 + by^2 + cz^2 = 1$ at (x_1, y_1, z_1) is
 $axx_1 + byy_1 + czz_1 = 1$ i.e. $\frac{2x}{4} + \frac{1}{9}3y - \frac{1}{25}5z = 1$
 i.e. $\frac{x}{2} + \frac{y}{3} - \frac{z}{5} = 1$
 $\Rightarrow 15x + 10y - 6z = 30$
17. The equation of the polar plane of (α, β, γ) with respect to $ax^2 + by^2 + cz^2 = 1$ is
 $a\alpha x + b\beta y + c\gamma z = 1$ so required polar plane is
 $1x + 4y + 3z = 1$
25. The pole of the plane $lx + my + nz = p$ with respect to conicoid $ax^2 + by^2 + cz^2 = 1$ is $\left(\frac{l}{ap}, \frac{m}{bp}, \frac{n}{cp}\right)$ so
 required pole is $\left(\frac{4}{2}, \frac{8}{4}, \frac{-3}{-1}\right)$ i.e. $(2, 2, 3)$.
27. The equation of normal of $ax^2 + by^2 + cz^2 = 1$ at (x_1, y_1, z_1) is
 $\frac{x - x_1}{ax_1} = \frac{y - y_1}{by_1} = \frac{z - z_1}{cz_1}$ so required normal is
 $\frac{x - 1}{1} = \frac{y - 0}{2.0} = \frac{z - 1}{3.1}$
 $\Rightarrow \frac{x - 1}{1} = \frac{y - 0}{0} = \frac{z - 1}{3}$
33. The equation of tangent plane of $x^2 - y^2 + z^2 = 1$ at the point $(1, 1, 1)$ is $x - y + z = 1$ and its perpendicular distance from centre $(0, 0, 0)$ is

$$\frac{1}{\sqrt{1 + 1 + 1}} = \frac{1}{\sqrt{3}}$$
36. The condition that the plane $lx + my + nz = p$ touches the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is

$$a^2l^2 + b^2m^2 + c^2n^2 = p^2$$

 so $p^2 = 1.1 + 2.1 + 3.1 = 6$ so $p = \pm \sqrt{6}$
40. Two points (x_1, y_1, z_1) and (x_2, y_2, z_2) are the conjugate points with respect to

$$ax^2 + by^2 + cz^2 = 1$$

 iff $ax_1x_2 + by_1y_2 + cz_1z_2 = 1$
 so we get $a.1.2 + b.1.2 + c.1.2 = 1$
 $\Rightarrow 2a + 2b + 2c = 1$
43. The plane $lx + my + nz = p$ touches the conicoid $ax^2 + by^2 + cz^2 = 1$ if $p^2 = \frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c}$ so we
 get $\frac{4}{1} + \frac{1}{1} + \frac{9}{p} = 9 \Rightarrow \frac{9}{p} = 4 \Rightarrow p = \frac{9}{4}$
44. Since $z_1^2 + z_2^2 + z_3^2 = C^2$ so $z_1^2 + z_2^2 + z_3^2 = 2$
48. Since OP, OQ and OR are conjugate semi-diameters
 so $OP^2 + OQ^2 + OR^2 = a^2 + b^2 + c^2$ i.e.
 $OP^2 + OQ^2 + OR^2 = 4 + 9 + 16 = 29$
50. The director sphere of $x^2 + 2y^2 + 3z^2 = 1$ is
 $x^2 + y^2 + z^2 = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$ so radius of the
 director sphere is $\frac{11}{6}$.

THE ELLIPTIC PARABOLOID

1. The equation of the elliptic paraboloid is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$. The planes $x = 0$ and $y = 0$ are called the principal planes. The sections of the elliptic paraboloid by the planes parallel to the zx -plane ($y = 0$) are parabolas.
2. The equation of the hyperbolic paraboloid is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$. The planes $x = 0$ and $y = 0$ are called the principal planes.
3. The general equation $ax^2 + by^2 = 2cz$. If a and b are both of the same sign then the general equation $ax^2 + by^2 = 2cz$ represents an elliptic parabola, if a and b are of opposite signs then it represents a hyperbolic paraboloid.
4. The equation of the tangent plane at the point (x_1, y_1, z_1) on the paraboloid $ax^2 + by^2 = 2cz$ is $axx_1 + byy_1 = c(z + z_1)$.
5. If the plane $lx + my + nz = p$ touches the paraboloid $ax^2 + by^2 = 2cz$, then the condition for tangency is $\frac{l^2}{a} + \frac{m^2}{b} + \frac{2np}{c} = 0$.
6. The polar plane of the point (x_1, y_1, z_1) with respect to the paraboloid $ax^2 + by^2 = 2cz$ is $axx_1 + byy_1 = c(z + z_1)$.
7. The locus of the middle points of a system of parallel chords is called a diametral plane.
8. Two diametral planes are called conjugate if each bisects the chords parallel to the other.
9. The equation of the normal at any point (x_1, y_1, z_1) of the paraboloid $ax^2 + by^2 = 2cz$ is $\frac{x - x_1}{ax_1} = \frac{y - y_1}{by_1} = \frac{z - z_1}{-c}$.
10. There are five normals can be drawn from a given point to a paraboloid.
11. The five feet of normals from any point to the paraboloid lie on a cubic curve.

Also the point of contact is given by $\left(-\frac{lc}{an}, -\frac{mc}{bn}, -\frac{p}{n}\right)$.

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The equation $ax^2 + by^2 = 2cz$ represents an elliptic paraboloid if a and b are **[Avadh 2018]**
 - (a) +ve sign only
 - (b) -ve sign only
 - (c) same sign
 - (d) none of these
2. The principal planes of the elliptic paraboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$ are
 - (a) $x = 0, y = 0$
 - (b) $z = 0$
 - (c) $x = 0, z = 0$
 - (d) $y = 0, z = 0$

3. The equation of the tangent plane at (x_1, y_1, z_1) on the paraboloid $ax^2 + by^2 = 2cz$ is
- $ax + by = c(z + z_1)$
 - $xx_1 + yy_1 = c(z - z_1)$
 - $axx_1 + byy_1 = c(z + z_1)$
 - $axx_1 + byy_1 = cz_1 + z$
4. The point of contact of the tangent plane $x + y + z = 1$ to the paraboloid $x^2 + y^2 = 2z$ is
- $(1, 1, 1)$
 - $(0, 0, 0)$
 - $(-1, -1, -1)$
 - $(-1, 0, -1)$
5. The polar plane of the point $(1, 1, 1)$ with respect to the paraboloid $2x^2 + y^2 = 2z$ is
- $2x + y + z = 1$
 - $2x + y - z = 1$
 - $x + 2y + 2z = 1$
 - $x + 2y - 2z = 1$
6. The centre of the conic $x^2 + y^2 = 2z, x + y + z = 1$ is
- $(1, 1, 3)$
 - $(1, 1, 1)$
 - $(-1, -1, 3)$
 - $(1, 1, 2)$
7. The vertex of the paraboloid $3x^2 + 2y^2 - 12z - 6x + 8y = 13$ is
- $(0, 0, 0)$
 - $(-2, -2, 1)$
 - $(1, 2, 2)$
 - $(1, -2, -2)$
8. If the plane $lx + my + nz = p$ be the tangent plane of the paraboloid $ax^2 + by^2 = 2z$ at the point (α, β, γ) then the point of contact is
- $\left(\frac{l}{an}, \frac{m}{bn}, \frac{p}{n}\right)$
 - $\left(\frac{al}{n}, \frac{bm}{n}, \frac{p}{n}\right)$
 - $\left(\frac{-al}{n}, \frac{-bm}{n}, \frac{-p}{n}\right)$
 - $\left(\frac{-l}{an}, \frac{-m}{bn}, \frac{-p}{n}\right)$
9. $2x^2 - 3y^2 = 4z$ represents
- ellipse
 - hyperbola
 - elliptic paraboloid
 - hyperbolic paraboloid
10. The equation of the normal at any point (α, β, γ) of the paraboloid is
- $\frac{x - \alpha}{\alpha\alpha} = \frac{y - \beta}{b\beta} = \frac{z - \gamma}{c\gamma}$
 - $\frac{x - \alpha}{\alpha\alpha} = \frac{y - \beta}{b\beta} = \frac{z - \gamma}{-c}$
 - $\frac{x - \alpha}{-a} = \frac{y - \beta}{b\beta} = \frac{z - \gamma}{c\gamma}$
 - none of these
11. The equation $ax^2 + by^2 = 2cz$ represents a hyperbolic paraboloid if a and b are
- same sign
 - opposite sign
 - positive sign only
 - negative sign only
12. If $x + 3y = 3$ be the one conjugate diametral plane of the paraboloid $2x^2 + 3y^2 = 4z$ then the other one is
- $2x + y = 1$
 - $x - 2y = 1$
 - $x + 2y = 1$
 - $2x - y = 1$
13. The condition that the plane $lx + my + nz = p$ touches the paraboloid $ax^2 + by^2 = 2cz$ is
- $\frac{a^2}{l} + \frac{b^2}{m} = \frac{2np}{c}$
 - $\frac{a^2}{l} + \frac{b^2}{m} + \frac{2np}{c} = 0$
 - $\frac{l^2}{a} + \frac{m^2}{b} = \frac{2np}{c}$
 - $\frac{l^2}{a} + \frac{m^2}{b} + \frac{2np}{c} = 0$
14. If the plane $2x + 3y - z = p$ touches the paraboloid $x^2 + y^2 = 2z$ then the value of p is
- $\frac{11}{2}$
 - 2
 - 7
 - $\frac{13}{2}$
15. The locus of the middle points of a system of parallel chords for the paraboloid is called
- conjugate plane
 - semi-conjugate plane
 - diametral plane
 - none of these
16. If the plane $8x - 6y - z = 5$ touches the paraboloid $3x^2 - 2y^2 = 6z$ then the co-ordinates of the point of contact is
- $(9, 8, 5)$
 - $(8, 9, 5)$
 - $(5, 8, 9)$
 - $(3, 2, 6)$
17. The principal planes of the hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$ are
- $x = c, y = c$
 - $x = a, y = b$
 - $x = 0, y = 0$
 - $x = c, y = a$
18. The section of the elliptic paraboloid by the planes parallel to the yz -plane is
- circles
 - ellipses
 - parabolas
 - hyperbolas
19. The equation of the polar plane of the point $(1, 1, 1)$ with respect to the paraboloid $x^2 + y^2 = 2z$ is
- $x + y + z = 0$
 - $x + y + z = 1$
 - $x + y - z = 1$
 - $x + y - z = 0$

20. The section of the surface $\frac{x^2}{a^2} - \frac{y^2}{b^2} = z$ by the planes $x = a$ is
 (a) parabolas (b) hyperbolas
 (c) ellipses (d) cones
21. Two diametral planes $ax + by - cz = 0$ and $a'x + b'y - c'n' = 0$ are conjugate diametric planes only when
 (a) $all' + bmm' = cnn'$ (b) $ll' + mm' = nn'$
 (c) $all' + bmm' = 0$ (d) $ll' + mm' = 0$
22. The equation of the tangent plane to the paraboloid $\frac{x^2}{2} - \frac{y^2}{3} = z$ at the point (8, 9, 5) is
 (a) $8x - 6y + z = 1$ (b) $8x - 6y - z = 5$
 (c) $8x - 6y + z = 5$ (d) $8x - 6y - z = 1$
23. The sections of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$ by the planes $z = k$ is
 (a) parabolas (b) hyperbolas
 (c) circles (d) ellipses
24. The number of normals that can be drawn to the paraboloid $ax^2 + by^2 = 2cz$ from a given point is
 (a) 5 (b) 6 (c) 1 (d) 4
25. The locus of the point of intersection of three mutually perpendicular tangent plane to the paraboloid $x^2 + y^2 + 2z = 1$ is
 (a) $z = -1$ (b) $z = 1$
 (c) $y + z = 1$ (d) $y + z + 1 = 0$
26. The direction ratios of the normal at any point (α, β, γ) of the paraboloid $ax^2 + by^2 = 2cz$ are
 (a) a, b, c (b) $\alpha\alpha, \beta\beta, c$
 (c) α, β, γ (d) $\alpha\alpha, \beta\beta, -c$
27. The sections of the hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$ by the varying plane $z = a$ is
 (a) hyperbolas (b) parabolas
 (c) ellipses (d) circles
28. The point of contact of the plane $x + 2y - 2z = 4$ touches the paraboloid $3x^2 + 4y^2 = 24z$ is
 (a) (2, 3, 2) (b) (2, 2, 3)
- (c) (3, 3, 2) (d) (3, 2, 3)
29. The locus of the mid points of a system of parallel chords of $ax^2 + by^2 = 2cz$ is called
 (a) parallel plane (b) conjugate plane
 (c) diametral plane
 (d) conjugate semi-diameter
30. The locus of the normals from the point (x_1, y_1, z_1) on the paraboloid $x^2 + y^2 = 2az$ is
 (a) cone (b) cylinder
 (c) sphere (d) circle
31. The locus of the point of intersection of three mutually perpendicular tangent planes of the paraboloids
 (a) $2z + \frac{1}{a} + \frac{1}{b} = 0$ (b) $2z + a + b = 0$
 (c) $z + a + b = 0$ (d) $z + \frac{1}{a} - \frac{1}{b} = 0$
32. The surface represented by $x^2 + 2y^2 = 1 - z$ is
 (a) hyperboloid of one sheets
 (b) hyperboloid of two sheets
 (c) elliptic paraboloid (d) ellipsoid
33. The section of the paraboloid $x^2 - 2z^2 = 3y$ by the XOZ plane is
 (a) parabola (b) hyperbola
 (c) circle (d) pair of plane
34. The equation of tangent plane to the paraboloid $\frac{x^2}{2} - \frac{y^2}{3} = 2z$ at the point (8, 9, 5) on it is
 [Kanpur 2018]
 (a) $8x - 6y - z = 5$ (b) $8x + 6y - z = 5$
 (c) $8x + 6y + z = 5$ (d) $8x - 6y + z = 5$
35. The locus of the perpendiculars from the vertex of the paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ to the generators of one system is
 [Kanpur 2018]
 (a) $x^2 + y^2 + 3z^2 \pm \frac{(a^2 + b^2)}{ab} xy = 0$
 (b) $x^2 + y^2 - 3z^2 \pm \frac{(a^2 + b^2)}{ab} xy = 0$

$$(c) x^2 + y^2 + 2z^2 \pm \frac{(a^2 + b^2)}{ab} xy = 0$$

$$(d) x^2 + y^2 - 2z^2 \pm \frac{(a^2 + b^2)}{ab} xy = 0$$

36. The locus of the centre of sections of the paraboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2z$, which are of constant area πx^2 is

[Kanpur 2018]

$$(a) a^2 b^2 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 2 \right)^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + 1 \right) = x^4$$

$$(b) a^2 b^2 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 2 \right)^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} - 1 \right) = x^4$$

$$(c) a^2 b^2 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + 2 \right)^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + 1 \right) = x^4$$

(d) None of these

37. The plane $x + 2y - 2z = 4$ touches the paraboloid $3x^2 + 4y^2 = 24z$ at the following point

[Agra 2017]

(a) (1, 2, 3)

(b) (2, 3, 2)

(c) (3, 2, 2)

(d) None of these

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(c)	2.	(a)	3.	(c)	4.	(c)	5.	(b)	6.	(c)	7.	(d)	8.	(d)	9.	(c)	10.	(b)
11.	(b)	12.	(d)	13.	(d)	14.	(d)	15.	(c)	16.	(b)	17.	(c)	18.	(c)	19.	(c)	20.	(a)
21.	(c)	22.	(b)	23.	(d)	24.	(a)	25.	(a)	26.	(d)	27.	(a)	28.	(a)	29.	(c)	30.	(c)
31.	(a)	32.	(c)	33.	(d)	34.	(a)	35.	(c)	36.	(a)	37.	(b)						

HINTS AND SOLUTIONS

4. The point of contact of the tangent plane $lx + my + nz = p$ with respect to the paraboloid $ax^2 + by^2 = 2cz$ is $\left(\frac{-lc}{an}, \frac{-mc}{bn}, \frac{-p}{n} \right)$ so point of contact is $\left(\frac{-1.1}{1.1}, \frac{-1.1}{1.1}, \frac{-1}{1} \right)$ i.e., $(-1, -1, -1)$.
5. The polar plane of (x_1, y_1, z_1) with respect to $ax^2 + by^2 = cz^2$ is $axx_1 + byy_1 = c(z + z_1)$ so required polar plane is $2x + y = 1(z + 1)$
 $\Rightarrow 2x + y - z = 1$
14. The plane $lx + my + nz = p$ touches the paraboloid $ax^2 + by^2 = 2cz$ then
 $\frac{l^2}{a} + \frac{m^2}{b} + \frac{2np}{c} = 0$ so $\frac{4}{1} + \frac{9}{1} - \frac{2p}{1} = 0$
 $\Rightarrow 2p = 13 \Rightarrow p = \frac{13}{2}$
16. The point of contact of $lx + my + nz = p$ with respect to paraboloid $ax^2 + by^2 = 2cz$ is $\left(\frac{-lc}{an}, \frac{-mc}{bn}, \frac{-p}{n} \right)$ so required point of contact is $\left(\frac{+8.3}{3}, \frac{6.3}{2}, +\frac{5}{1} \right)$ i.e., $(8, 9, 5)$
19. The polar plane of $ax^2 + by^2 = 2cz$ with respect to the point (x_1, y_1, z_1) is $axx_1 + byy_1 = c(z + z_1)$ i.e. $1.1x + 1.1y = 1(z + 1) \Rightarrow x + y - z = 1$
28. Point of contact is $\left(\frac{-lc}{an}, \frac{-mc}{bn}, \frac{-p}{n} \right)$ so $(2, 3, 2)$.

GENERATING LINES OF HYPERBOLOID OF ONE SHEET

1. A ruled surface is defined as a surface which is generated by the motion of one parameter family of straight lines and the straight line itself is called its generating line, generator or ruling.
2. The generating lines of a hyperboloid of one sheet $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ are $\frac{x}{a} - \frac{z}{c} = \lambda \left(1 - \frac{y}{b}\right)$, $\frac{x}{a} + \frac{z}{c} = \frac{1}{\lambda} \left(1 + \frac{y}{b}\right)$ called λ -system family of straight lines and $\frac{x}{a} - \frac{z}{c} = \mu \left(1 + \frac{y}{b}\right)$, $\frac{x}{a} + \frac{z}{c} = \frac{1}{\mu} \left(1 - \frac{y}{b}\right)$ called μ -system family of straight lines.
3. One generator each of λ - and μ - systems passes through every point of the hyperboloid.
4. No two generators of the same systems intersect.
5. Any two generators of the different systems intersect.
6. The tangent plane at a point of a hyperboloid of one sheet meets the hyperboloid in two generators through that point.
7. The locus of the point of intersection of two perpendicular generators of hyperboloid of one sheet is $x^2 + y^2 + z^2 = a^2 + b^2 - c^2$
8. If three points of a straight line lie on a conicoid then the straight line lies wholly on the conicoid.
9. The conditions that the straight line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is a generator of the

conicoid $ax^2 + by^2 + cz^2 = 1$ are $a\alpha^2 + b\beta^2 + c\gamma^2 = 1$, $a\alpha + b\beta + c\gamma = 0$, $al^2 + bm^2 + cn^2 = 0$.

THE GENERATING LINES OF A HYPERBOLIC PARABOLOID

1. The generating lines of a hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ are $\frac{x}{a} - \frac{y}{b} = \lambda^2$, $\frac{x}{a} + \frac{y}{b} = \frac{2}{\lambda}$ called λ -system of generators and $\frac{x}{a} - \frac{y}{b} = \frac{2}{\mu}$, $\frac{x}{a} + \frac{y}{b} = \mu z$ called μ -system of generators.
2. The locus of the point of intersection of two perpendicular generators of a hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ is $a^2 - b^2 + 2z = 0$ together with hyperbolic paraboloid.
3. The generators of the λ - and μ - systems of the hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$ are parallel to the planes $\frac{x}{a} \pm \frac{y}{b} = 0$
4. One generator each of λ - and μ - systems passes through every point of hyperbolic paraboloid.
5. No two generator of the hyperbolic paraboloid of the same system intersect.
6. In the hyperbolic paraboloid any generator of the λ -system intersects any generator of the μ -system.
7. The tangent plane at any point of a hyperbolic paraboloid meets it in two generators through that point.

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. In the hyperboloid of one sheet, one generator each of λ - and μ -systems pass through
 - (a) only one point (b) no point
 - (c) only two points (d) every point
2. The locus of the point of intersection of two perpendicular generators of hyperboloid of one sheet is **[Kanpur 2018]**
 - (a) $x^2 + y^2 + z^2 = a^2 + b^2 - c^2$
 - (b) $x^2 + y^2 + z^2 = a^2 + b^2 + c^2$
 - (c) $x^2 + y^2 + z^2 = a^2 - b^2 - c^2$
 - (d) $x^2 + y^2 + z^2 = -a^2 - b^2 - c^2$
3. If a straight line lies wholly on the conicoid then minimum number of points of straight line that lies on conicoid are
 - (a) 1 (b) 2
 - (c) 3 (d) infinite
4. The locus of the point of intersection of two perpendicular generators of a hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ is
 - (a) $a^2 + b^2 = 2z$
 - (b) $a^2 - b^2 + 2z = 0$
 - (c) $a^2 + b^2 - 2z = 0$
 - (d) $a^2 + b^2 + 2z = 0$
5. The generators of the λ - and μ -systems of the hyperbolic paraboloid $\frac{y^2}{b^2} - \frac{z^2}{c^2} = \frac{2x}{a}$ are parallel to the plane
 - (a) $\frac{x}{a} \pm \frac{y}{b} = 0$ (b) $\frac{y}{b} \pm \frac{z}{c} = a$
 - (c) $\frac{y}{b} \pm \frac{z}{c} = 0$ (d) $\frac{x}{a} \pm \frac{y}{b} = c$
6. One of the generating line of the hyperboloid $\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$ passing through the point (2, 3, 4) is
 - (a) $\frac{x-2}{1} = \frac{y-3}{3} = \frac{z-4}{4}$
 - (b) $\frac{x-2}{1} = \frac{y-3}{0} = \frac{z-4}{4}$
 - (c) $\frac{x-2}{1} = \frac{y-3}{0} = \frac{z-4}{0}$
 - (d) $\frac{x-2}{0} = \frac{y-3}{3} = \frac{z-4}{4}$
7. In the hyperboloid of one sheet two generators of the same system intersect only
 - (a) one point (b) two points
 - (c) infinite points (d) no point
8. If $\frac{x}{a} + \frac{y}{b} = \frac{2}{\lambda}$ be a λ -system generating line of the hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ then other one is
 - (a) $\frac{x}{a} + \frac{y}{b} = \lambda z$
 - (b) $\frac{x}{a} - \frac{y}{b} = \lambda z$
 - (c) $\frac{x}{a} - \frac{y}{b} = \frac{2}{\lambda}$
 - (d) none of these
9. The locus of the point of intersection of two perpendicular generators of $\frac{x^2}{4} + \frac{y^2}{25} - \frac{z^2}{9} = 1$ is
 - (a) $x^2 + y^2 + z^2 = 4$ (b) $x^2 + y^2 + z^2 = 25$
 - (c) $x^2 + y^2 + z^2 = 9$ (d) $x^2 + y^2 + z^2 = 20$
10. The locus of the point of intersection of two perpendicular generators of $\frac{x^2}{16} - \frac{y^2}{9} = 2z$ is
 - (a) $2z + 7 = 0$ (b) $2z + 25 = 0$
 - (c) $2z + 1 = 0$ (d) $2z - 7 = 0$

11. The generators of the λ - and μ -systems of the $y^2 - z^2 = 2x$ are parallel to the planes
 - (a) $y = \pm z$ (b) $y = \pm 2z$
 - (c) $z = \pm 2y$ (d) $x = 0, y = 0$
12. If two generators intersect then systems are
 - (a) same
 - (b) different
 - (c) never intersect
 - (d) may be same or different
13. The generators of the λ - and μ -systems of the hyperbolic paraboloid $x^2 - y^2 = 2z$ are parallel to the planes
 - (a) $x \pm y = 0$
 - (b) $x + y = 0$ only
 - (c) $x - y = 0$ only
 - (d) $x \pm y = 2$
14. The locus of the points of intersection of two perpendicular generators of the hyperboloid $2x^2 + 3y^2 - 4z^2 = 1$ is
 - (a) $x^2 + y^2 + z^2 = 0$
 - (b) $2x^2 + y^2 + z^2 = 1$
 - (c) $2x^2 + 3y^2 - 4z^2 = 1$
 - (d) $x^2 + y^2 + z^2 = \frac{7}{12}$
15. The tangent plane at a point of a hyperboloid of one sheet meets the hyperboloid then the number of generators through that point are
 - (a) 1 (b) 2
 - (c) 3 (d) nil
16. The locus of the point of intersection of two perpendicular generators of a hyperbolic paraboloid $x^2 - y^2 = 2z$ is
 - (a) $2z + 1 = 0$ (b) $2z - 1 = 0$
 - (c) $z = 0$ (d) $z - 1 = 0$
17. One of the generating line of the hyperboloid $yz + 2zx + 3xy + 6 = 0$ which pass through $(-1, 0, 3)$ is
 - (a) $x = -1, z = 3$ (b) $x = 1, z = 3$
 - (c) $x = -1, z = -3$ (d) $x = +1, z = -3$
18. The locus of the point of intersection of two perpendicular generators of hyperboloid of one sheet $x^2 + y^2 - z^2 = 2$ is
 - (a) $x^2 + y^2 + z^2 = 1$
 - (b) $x^2 + y^2 + z^2 = 0$
 - (c) $x^2 + y^2 + z^2 = 2$
 - (d) $x^2 + y^2 + z^2 = 4$
19. The condition that $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ is a generator of $ax^2 + by^2 + cz^2 = 1$ is
 - (a) $a\alpha^2 + b\beta^2 + c\gamma^2 = 1$
 - (b) $a\alpha + b\beta + c\gamma = 0$ only
 - (c) $al^2 + bm^2 + cn^2 = 0$ only
 - (d) All the above
20. The locus of the point of intersection of two perpendicular generators of $x^2 - \frac{y^2}{4} = z$ is
 - (a) $4z - 3 = 0$ (b) $2z - 3 = 0$
 - (c) $4z - 1 = 0$ (d) $4z - 5 = 0$
21. The equation of the generators of hyperboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ which passes through the point $(a \cos \alpha, b \sin \alpha, 0)$ is **[Kanpur 2018]**
 - (a) $\frac{x - \cos \alpha}{a \sin \alpha} = \frac{y - b \sin \alpha}{b \cos \alpha} = \frac{z}{\pm c}$
 - (b) $\frac{x - a \cos \alpha}{a \sin \alpha} = \frac{y - b \sin \alpha}{-b \cos \alpha} = \frac{z}{\pm c}$
 - (c) $\frac{x + a \cos \alpha}{a \sin \alpha} = \frac{y - b \sin \alpha}{b \cos \alpha} = \frac{z}{c}$
 - (d) None of these
22. The generators of the λ - and μ -systems of the hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ are parallel to the planes **[Kanpur 2018]**
 - (a) $\frac{x}{a} \pm \frac{y}{b} = 0$ (b) $\frac{x}{a} \pm \frac{y}{b} = 1$
 - (c) $\frac{x}{a} \pm \frac{y}{b} = 2$ (d) None of these

23. The locus of the point of intersection of two perpendicular generators of a hyperbolic paraboloid is [Avadh 2018]
 (a) $a^2 + b^2 + 2z = 0$
 (b) $a^2 - b^2 + 2z = 0$
 (c) $a^2 - b^2 - 2z = 0$
 (d) $a^2 + b^2 - 2z = 0$
24. Number of system of generating liners in a hyperbolic paraboloid is [Avadh 2018]
 (a) 1 (b) 2
 (c) 3 (d) 4
25. Two generators of the same system of hyperboloid of one sheet [Avadh 2018]
 (a) Intersect at a point
 (b) Never intersect
 (c) Intersect on the hyperboloid
 (d) None of these

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(d)	2.	(a)	3.	(c)	4.	(b)	5.	(c)	6.	(d)	7.	(d)	8.	(b)	9.	(d)	10.	(a)
11.	(a)	12.	(b)	13.	(a)	14.	(d)	15.	(b)	16.	(c)	17.	(a)	18.	(c)	19.	(d)	20.	(a)
21.	(b)	22.	(a)	23.	(b)	24.	(b)	25.	(b)										

HINTS AND SOLUTIONS

9. The locus of the point of intersection of two perpendicular generators of $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ is
 $x^2 + y^2 + z^2 = a^2 + b^2 - c^2$ so required locus is
 $x^2 + y^2 + z^2 = 4 + 25 - 9 = 20$.
10. The locus of the point of intersection of two perpendicular generators of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ is
 $2z = b^2 - a^2$ so required locus is
 $2z = 9 - 16 = -7$ i.e. $2z + 7 = 0$
13. The generators of the λ - and μ - systems of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$ are parallel to the planes $\frac{x}{a} + \frac{y}{b} = 0$.
 So, required plane is $\frac{x}{1} \pm \frac{y}{1} = 0 \Rightarrow x \pm y = 0$
14. The locus of the points of intersection of perpendicular generators of $2x^2 + 3y^2 - 4z^2 = 1$ is
 $x^2 + y^2 + z^2 = a^2 + b^2 - c^2 = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$ ○○○

The Plane Sections of Conicoids

THE PLANE SECTIONS OF CONICOIDS

1. The nature of a plane section (projection) of the central conicoid $ax^2 + by^2 + cz^2 = 1$ by the plane $lx + my + nz = p$ is a hyperbola, a parabola or an ellipse according as $bcl^2 + acm^2 + abn^2 <, =, \text{ or } > 0$.

2. If the central plane section is an ellipse in point 1 then its area is $\pi \left\{ \frac{l^2 + m^2 + n^2}{bcl^2 + acm^2 + abn^2} \right\}^{\frac{1}{2}}$.

If the conicoid be the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ then the area of the central

plane section is $\pi abc \left[\frac{l^2 + m^2 + n^2}{a^2 l^2 + b^2 m^2 + c^2 n^2} \right]^{\frac{1}{2}}$.

3. The section of the conicoid $ax^2 + by^2 + cz^2 = 1$ by any tangent plane to the cone $\frac{x^2}{(b+c)} + \frac{y^2}{(c+a)} + \frac{z^2}{(a+b)} = 0$ is a rectangular hyperbola.

PLANE SECTIONS OF A PARABOLOID

1. The nature of the plane section (projection) of the paraboloid $ax^2 + by^2 = 2cz$ by the plane $lx + my + nz = p$ is a hyperbola, a parabola or an ellipse according as $abn^2 l^2 <, =, \text{ or } > 0$.
2. If the plane section is an ellipse then its area is

$$\frac{\pi c}{\eta^3} \left(\frac{cl^2}{a} + \frac{cm^2}{b} + 2np \right) \sqrt{\frac{l^2 + m^2 + n^2}{ab}}.$$

3. The condition for the section to be a rectangular hyperbola in point 2 is $(a+b)n^2 + am^2 + bl^2 = 0$.
4. The section of the conicoid by the planes are circles since every plane must meet a sphere in a circle.
5. Any two circular section of an ellipsoid which are not parallel always lie on a sphere.
6. The hyperbolic paraboloid has no real circular sections.

UNBILICS

1. Let $P(\alpha, \beta, \gamma)$ be a point on any conicoid. If the planes parallel to the tangent plane at P determine circular sections of the conicoid, then the tangent plane at the point P itself will be a circular section of radius zero. Such a point P is called an unbilic of the conicoid.
2. The number of real unbilics of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ are four.
3. The unbilics of the ellipsoid lie on a sphere.
4. The length p of the perpendicular distance from the centre to the tangent plane at an unbilic of the ellipsoid is $\frac{ac}{b}$.
5. The number of real unbilics of the paraboloid $ax^2 + by^2 = 2cz$ are two.

EXERCISE

MULTIPLE CHOICE QUESTIONS

1. The plane sections of a surface represented by an equation of second degree are called
 - (a) conics
 - (b) conicoids
 - (c) surfaces
 - (d) central conicoids
2. All parallel plane sections of a conicoids are
 - (a) similar
 - (b) different
 - (c) may be similar or may be different
 - (d) none of these
3. The section of the ellipsoid $9x^2 + 6y^2 + 14z^2 = 3$ by the plane $x + y + z = 0$ always represents
 - (a) a circle
 - (b) an ellipse
 - (c) a hyperbola
 - (d) a plane
4. The plane $lx + my + nz = p$, $n \neq 0$, cuts the paraboloid $ax^2 + by^2 = 2z$ in an ellipse if
 - (a) $abn^2 = 0$
 - (b) $abn^2 > 0$
 - (c) $abn^2 < 0$
 - (d) none of these
5. The plane $lx + my + nz = 0$ cuts the conicoid $ax^2 + by^2 + cz^2 = 1$ in a hyperbola if $bcl^2 + cam^2 + abn^2$ is
 - (a) > 0
 - (b) $= 0$
 - (c) < 0
 - (d) none of these
6. If A_1 , A_2 and A_3 are the areas of three mutually perpendicular central sections of an ellipsoid then
 - (a) $A_1 + A_2 + A_3 = \text{constant}$
 - (b) $A_1^2 + A_2^2 + A_3^2 = \text{constant}$
 - (c) $\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} = \text{constant}$
 - (d) $\frac{1}{A_1^2} + \frac{1}{A_2^2} + \frac{1}{A_3^2} = \text{constant}$
7. The plane $x + y + z = 0$, cuts the conicoid $x^2 + 2y^2 + 3z^2 = 1$ in a
 - (a) ellipse
 - (b) parabola
 - (c) hyperbola
 - (d) circle
8. The plane $x + 2y + 3z = 1$, cuts the paraboloid $x^2 + y^2 = 2z$ in a
 - (a) parabola
 - (b) plane
 - (c) ellipse
 - (d) hyperbola
9. The number of real circular sections of the hyperbolic paraboloid are
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) not exist
10. The unilics of the ellipsoid lie on a
 - (a) circle
 - (b) sphere
 - (c) ellipsoid
 - (d) cylinder
11. The plane $lx + my + nz = 0$ cuts the conicoid $ax^2 + by^2 + cz^2 = 1$ in a parabola only when
 - (a) $bcl^2 + cam^2 + abn^2 = 0$
 - (b) $bcl^2 + acm^2 + abn^2 > 0$
 - (c) $bcl^2 + cam^2 + abn^2 < 0$
 - (d) none of these
12. The number of real circular sections of the hyperboloid $4x^2 - 13y^2 = 2z$ are
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) does not exist
13. The real unilics of the ellipsoid are
 - (a) 1
 - (b) 2
 - (c) 4
 - (d) 6
14. The condition that any two lines $x = y = z$ and $2x = 2y = -z$ be the axes of the section of the conicoid $ax^2 + by^2 + cz^2 = 1$ by a plane through them is
 - (a) $a + b - c = 0$
 - (b) $2a + b - c = 0$
 - (c) $a + b - 2c = 0$
 - (d) $a + b + c = 0$
15. The plane $lx + my + nz = 0$ cuts the paraboloid $ax^2 + by^2 = cz$ in a parabola if abn^2
 - (a) > 0
 - (b) < 0
 - (c) equal to 0
 - (d) none of these

16. If the section of the surface $xy + yz + zx = a^2$ by the plane $lx + my + nz = p$ is a parabola then
 - (a) $l + m + n = 0$
 - (b) $l^2 + m^2 + n^2 = 0$
 - (c) $\sqrt{l} + \sqrt{m} + \sqrt{n} = 0$
 - (d) none of these
17. The plane $x - y + z = 2$ cuts the paraboloid $x^2 - 2y^2 = 2z$ in a
 - (a) parabola
 - (b) ellipse
 - (c) hyperbola
 - (d) circle
18. The equation $2x^2 + 3y^2 = 4z$ represents a
 - (a) parabola
 - (b) ellipse
 - (c) paraboloid
 - (d) elliptic paraboloid
19. The plane $lx + my = p$ cuts the paraboloid $ax^2 + by^2 = cz$ in a
 - (a) parabola
 - (b) hyperbola
 - (c) ellipse
 - (d) circle
20. The plane $x + y + z = 2$, cuts the paraboloid $2x^2 + 3y^2 = 2z$ in a
 - (a) parabola
 - (b) ellipse
 - (c) hyperbola
 - (d) hyperboloid
21. The hyperbolic paraboloid has
 - (a) one real circular section
 - (b) many real circular sections
 - (c) no real circular section
 - (d) none of these
22. The section by the plane $3x + 2y - 4z = 1$ on the paraboloid $x^2 + y^2 = 2z$ is
 - (a) hyperbola
 - (b) parabola
 - (c) ellipse
 - (d) circle
23. If r_1 and r_2 be the length of semi axis and plane section be an ellipse then area of plane section is
 - (a) $r_1 r_2$
 - (b) $r_1 + r_2$
 - (c) $\pi r_1 r_2$
 - (d) $\pi(r_1 + r_2)$
24. The section by the plane $x - y + z = 0$ on the conicoid $2x^2 - y^2 + 3z^2 = 1$ represents a
 - (a) ellipse
 - (b) hyperbola
 - (c) parabola
 - (d) conicoid
25. If r_1 and r_2 be the semi axis of a rectangular hyperbola then
 - (a) $r_1 = r_2 = 0$
 - (b) $r_1 + r_2 = 0$
 - (c) $r_1 r_2 = 0$
 - (d) $r_1^2 + r_2^2 = 0$
26. The area of the central plane section of the conicoid $ax^2 + by^2 + cz^2 = 1$ by the plane $lx + my + nz = 1$ is
 - (a) $\frac{\pi}{p\sqrt{abc}}$
 - (b) $\frac{\pi}{\sqrt{abc}}$
 - (c) $\pi\sqrt{abc}$
 - (d) $p\pi\sqrt{abc}$
27. The real central circular sections of the ellipsoid $\frac{x^2}{8} + \frac{y^2}{4} + \frac{3z^2}{4} = 1$ are
 - (a) $z \pm 2x = 0$
 - (b) $x \pm 2z = 0$
 - (c) $y \pm 2z = 0$
 - (d) $z \pm 2y = 0$
28. The circular sections of the paraboloid $\frac{13}{2}y^2 + 2z^2 = x$ through the $(0, 0, 0)$ are
 - (a) $3x \pm 2y = 0$
 - (b) $3x \pm 2z = 0$
 - (c) $2x \pm 3z = 0$
 - (d) $2x \pm 3y = 0$
29. The area of the plane section (central) of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ by the plane $lx + my + nz = p$ is
 - (a) $\pi \frac{abc}{p}$
 - (b) $\frac{pabc}{\pi}$
 - (c) $\frac{\pi\sqrt{abc}}{p}$
 - (d) $\frac{p\sqrt{abc}}{\pi}$
30. The distance from the centre of the tangent plane at an unilic of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is
 - (a) $\frac{bc}{a}$
 - (b) $\frac{ab}{c}$
 - (c) abc
 - (d) $\frac{ac}{b}$
31. The real unilics of the paraboloid $ax^2 + by^2 = 2cz$ are
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
32. The real unilics of hyperboloid of one sheet are
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4

33. The real central circular sections of the ellipsoid $\frac{x^2}{8} + \frac{y^2}{4} + \frac{z^2}{2} = 1$ are

(a) $x \pm z = 0$ (b) $x \pm z\sqrt{2} = 0$
 (c) $x \pm 2z = 0$ (d) $x \pm 2\sqrt{2}z = 0$

34. The circular section of the paraboloid $4x^2 + y^2 - 4z = 0$ are

(a) $z \pm \sqrt{3}x = 0$ (b) $x \pm \sqrt{3}z = 0$
 (c) $x \pm 3z = 0$ (d) $z \pm 3x = 0$

35. The circular section of the hyperbolic paraboloid $x^2 - y^2 = 2z$ are

(a) $x \pm y = 0$ (b) $x \pm 2y = 0$
 (c) $y \pm 2x = 0$ (d) not exist

36. The area of the central plane section of the conicoid $3x^2 + 3y^2 + 4z^2 = 1$ by the plane $x + 2y + z = 2$ is

(a) π (b) $\frac{\pi}{\sqrt{6}}$
 (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{12}$

37. The two circular sections of an ellipsoid which are not parallel lies on a

(a) circle (b) sphere
 (c) ellipse (d) ellipsoid

38. The equation of the real central circular sections of the conicoid $6x^2 + 13y^2 + 6z^2 - 10yz + 4zx - 10xy - 1 = 0$ are **[Kanpur 2018]**

(a) $2(x + 2)^2 - 10(x + 2) - 9y^2 = 0$
 (b) $2(x + 2)^2 + 10(x + 2) + 9y^2 = 0$
 (c) $2(x + 2)^2 + 10(x + 2) - 9y^2 = 0$
 (d) $2(x + 2)^2 - 10(x + 2) + 9y^2 = 0$

39. If the section of conicoid $ax^2 + by^2 + cz^2 = 1$ by the plane $lx + my + nz = 0$ is hyperbola, then its section by the plane $lx + my + nz = p$ is **[Kanpur 2018]**

(a) parabola (b) circle
 (c) ellipse (d) hyperbola

40. The two confocal paraboloids always cut at the angle

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
 (c) π (d) 0

41. The locus of the point of intersection of perpendicular generators of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z$ is

(a) $b^2 - a^2 = 2z$ (b) $b^2 - a^2 = -2z$
 (c) $b^2 - a^2 = 2y$ (d) $b^2 - a^2 = -2y$

42. If A_1, A_2, A_3 are the areas of three mutually perpendicular central reactions of $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ then $\frac{1}{A_1^2} + \frac{1}{A_2^2} + \frac{1}{A_3^2}$ is equal to

(a) constant (b) variable
 (c) abc (d) 0

ANSWERS

MULTIPLE CHOICE QUESTIONS

[illegible]

HINTS AND SOLUTIONS

6. Here $A_1 = \frac{\pi abc(l_1^2 + m_1^2 + n_1^2)^{1/2}}{(a^2l_1^2 + b^2m_1^2 + c^2n_1^2)^{1/2}}$
- So $\frac{1}{A_1^2} = \frac{a^2l_1^2 + b^2m_1^2 + c^2n_1^2}{\pi^2 a^2 b^2 c^2}$, in a similar way A_2
- and A_3 exists so $\frac{1}{A_1^2} + \frac{1}{A_2^2} + \frac{1}{A_3^2} = \frac{1}{\pi^2 a^2 b^2 c^2}$
- $[a^2(l_1^2 + l_2^2 + l_3^2) + b^2(m_1^2 + m_2^2 + m_3^2) + c^2(n_1^2 + n_2^2 + n_3^2)] = \frac{(a^2 + b^2 + c^2)}{\pi^2 a^2 b^2 c^2} = \text{constant}$
7. Here, $bcl^2 + cam^2 + abn^2 = 6 + 3 + 2 = 11 > 0$ so it represents an ellipse.
8. Here, $abn^2l^2 = 1.1.9.1 = 9 > 0$ so it represents an ellipse.
17. Here $abm^2l^2 = (1)(-2)(-1)^2(1)^2 = -2 < 0$ so it represents a hyperbola.
24. $bcl^2 + cam^2 + abn^2 = (-1)(3)(1)^2$
- $+ (3)(2)(-1)^2 + (2)(-1)(1)^2$
 $= -3 + 6 - 2 = +1 > 0$ so it represents an ellipse.
27. The real pair of central circular sections of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ are
- $\frac{x}{a}\sqrt{a^2 - b^2} \pm \frac{z}{c}\sqrt{b^2 - c^2} = 0$
- so we get $\frac{x}{2\sqrt{2}}2 \pm \frac{z\sqrt{3}}{2}\sqrt{4 - \frac{4}{3}} = 0$
- $\Rightarrow \frac{x}{\sqrt{2}} \pm \frac{z\sqrt{3}}{2} \cdot \frac{2\sqrt{2}}{\sqrt{3}} = 0$ i.e. $x \pm 2z = 0$.
28. Real circular section through the origin are given by
- $y\sqrt{a-b} \pm x\sqrt{b} = 0 \Rightarrow y\sqrt{\frac{13}{2}} - 2 \pm x\sqrt{2} = 0$
- $\Rightarrow 2x \pm 3y = 0$.

15

Reduction of General Equation of Second Degree

GENERAL EQUATION OF SECOND DEGREE

- The most general equation of second degree is

$$F(x, y, z) \equiv ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$$

$$+ 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0$$

By the transformation of axes it can be reduced to

$$(i) \quad \lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2 = \mu \quad (\text{First form})$$

$$(ii) \quad \lambda_1 x^2 + \lambda_2 y^2 = 2\mu z \quad (\text{Second form})$$

- First form have the following surfaces

$$(i) \quad Ax^2 + By^2 + Cz^2 = 1 \quad (\text{Ellipsoid})$$

$$(ii) \quad A(x^2 + y^2 + z^2) = 1 \quad (\text{Sphere})$$

$$(iii) \quad Ax^2 + By^2 - Cz^2 = 1$$

(Hyperboloid of one sheet)

$$(iv) \quad Ax^2 - By^2 - Cz^2 = 1$$

(Hyperboloid of two sheet)

$$(v) \quad Ax^2 + By^2 + Cz^2 = 0 \quad (\text{Cone})$$

$$(vi) \quad Ax^2 + By^2 = 1 \quad (\text{Elliptic cylinder})$$

$$(vii) \quad Ax^2 - By^2 = 1 \quad (\text{Hyperbolic cylinder})$$

$$(viii) \quad Ax^2 - By^2 = 0 \quad (\text{Pair of intersecting planes})$$

$$(ix) \quad Ax^2 = 1 \text{ or } By^2 = 1 \text{ or } Cz^2 = 1$$

(Pair of parallel planes)

- Second form have the following surfaces

$$(i) \quad Ax^2 + By^2 = 2Cz \quad (\text{Elliptic paraboloid})$$

$$(ii) \quad Ax^2 - By^2 = 2Cz$$

(Hyperbolic paraboloid)

$$(iii) \quad A(x^2 + y^2) = 2Cz$$

(Paraboloid of revolution)

- By rotation of axes, $f(x, y, z) = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$ transforms to $\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$ where $\lambda_1, \lambda_2, \lambda_3$ are the roots of the equation

$$\begin{vmatrix} a - \lambda & h & g \\ h & b - \lambda & f \\ g & f & c - \lambda \end{vmatrix} = 0$$

It is called the discriminating cubic.

- The roots of the discriminating cubic are all real.
- If the given equation of surface is homogeneous then the centre of the given surface is origin.
- If the given equation of surface $F(x, y, z) = 0$ is not homogeneous then centre of the surface are found by solving the equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0,$

$$\frac{\partial F}{\partial z} = 0.$$

PROCESS OF REDUCING A GENERAL EQUATION TO THE STANDARD FORM

- Form the discriminating cubic and solve it.
- If all the characteristics roots are different from zero, then find the co-ordinates (α, β, γ) of the centre by solving the equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0,$

$$\frac{\partial F}{\partial z} = 0.$$
- Shifting the origin to the centre (α, β, γ) and then rotating the axes, the reduced equation becomes $\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2 = \mu$ where $\mu = -(u\alpha + v\beta + w\gamma + d)$. Thus in this case the surface can be determined.

4. If one root say $\lambda_3 = 0$ then corresponding to this value find the principal direction l_3, m_3, n_3 from any two of the following equations.
- $$al_3 + hm_3 + gn_3 = 0$$
- $$hl_3 + bm_3 + fn_3 = 0$$
- $$gl_3 + fm_3 + cn_3 = 0$$
- and evaluate $k = ul_3 + vm_3 + wn_3$.
5. If $k \neq 0$, the reduced equation is $\lambda_1 x^2 + \lambda_2 y^2 + 2kz = 0$ and represents an elliptic or hyperbolic paraboloid.
6. If $k = 0$, there is a line of centres, given by any two of the equations $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$.
- Let any point (α, β, γ) on the line of centres be chosen or centre; then the reduced equation is
- $$\lambda_1 x^2 + \lambda_2 y^2 = \mu$$
- where $\mu = -(u\alpha + v\beta + w\gamma + d)$.

EXERCISE

MULTIPLE CHOICE QUESTIONS

- In 3-dimension geometry $Ax^2 + By^2 = 1$ represents
 - ellipse
 - cone
 - hyperboloid
 - elliptic cylinder
- The surface of the equation $(3x - 4y + z)^2 + 9x - 12y + 3z - 10 = 0$ represents
 - cone
 - ellipsoid
 - pair of parallel planes
 - hyperboloid
- The discriminating cubic of the conicoid $x^2 + y^2 + z^2 + yz + zx + xy = 1$ is
 - $4\lambda^3 + 2\lambda^2 + 9\lambda - 2 = 0$
 - $4\lambda^3 - 12\lambda^2 + 7\lambda + 1 = 0$
 - $4\lambda^3 + 12\lambda^2 - 9\lambda + 1 = 0$
 - $4\lambda^3 + 12\lambda^2 + 7\lambda + 1 = 0$
- The centre of the conicoid $3x^2 - y^2 + z^2 + 6yz - 6x + 6y - 2z - 2 = 0$ is
 - (1, 1, 1)
 - (1, 0, 1)
 - (1, 0, -1)
 - (-1, 0, 1)
- The surface represented by the equation $ax^2 + by^2 - cz^2 = 1$ is
 - ellipsoid
 - a hyperboloid of one sheet
 - sphere
 - a hyperboloid of two sheet
- The latus rectum of the normal section of the parabolic cylinder $(3x - 4y + z)^2 = 26(x + y + z)$ is
 - 3
 - 2
 - $\sqrt{3}$
 - $\sqrt{2}$
- The centre of the conicoid $3x^2 - y^2 - z^2 + 6yz - 6x + 6y - 2z - 2 = 0$ is
 - (1, 1, 1)
 - (1, 0, 1)
 - (1, 0, -1)
 - (-1, 0, 1)
- The discriminating cubic of the conicoid $2x^2 + 2y^2 + z^2 - 4xy + 2yz - 2zx + x + y = 0$ is
 - $\lambda^3 + \lambda^2 + 2\lambda = 0$
 - $\lambda^3 - 2\lambda^2 + 5\lambda = 0$
 - $\lambda^3 - \lambda^2 + 5\lambda = 0$
 - $\lambda^3 - 5\lambda^2 + 2\lambda = 0$
- The normal form of the equation $3x^2 + 5y^2 + 3z^2 + 2yz + 2zx + 2xy - 4x - 8z + 5 = 0$ is
 - $2x^2 + 3y^2 + 6z^2 = 1$
 - $2x^2 + 3y^2 - 6z^2 = 1$
 - $2x^2 - 3y^2 - 6z^2 = 1$
 - $-2x^2 - 3y^2 + 6z^2 = 1$

10. $Ax^2 = \pm 2cz$ always represents a
 (a) elliptic paraboloid (b) parabolic cylinder
 (c) hyperbolic cylinder (d) elliptic cylinder
11. The centre of the conicoid
 $x^2 + 2yz - 4x + 6y + 2z = 0$ is
 (a) $(2, -1, -3)$ (b) $(2, 1, 3)$
 (c) $(2, -1, 3)$ (d) $(2, 1, -3)$
12. The surface $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ always represents
 (a) paraboloid (b) right circular cylinder
 (c) hyperboloid (d) none of these
13. The surface represented by the equation
 $Ax^2 + By^2 = Cz$ is
 (a) sphere (b) elliptic paraboloid
 (c) ellipse (d) hyperboloid
14. The roots of the discriminating cubic of the conicoid
 $2x^2 + 5y^2 + 10z^2 + 12yz + 6zx + 4xy - 1 = 0$ are
 (a) 1, 1, 5 (b) 1, 5, 5
 (c) 1, 1, 15 (d) 1, 5, 1
15. The line of centre of the hyperbolic cylinder
 $x^2 - y^2 + 4yz + 4zx - 6x - 2y - 8z + 5 = 0$ passes
 through the point
 (a) $(0, 0, 0)$ (b) $(1, 0, 1)$
 (c) $(0, 1, 0)$ (d) $(1, 1, 1)$
16. The centre of the conicoid
 $x^2 + y^2 + z^2 + yz + zx + xy + 3x + y + 4z + 4 = 0$ is
 (a) $(1, 1, 2)$ (b) $(-1, 1, -2)$
 (c) $(1, 1, -2)$ (d) $(-1, 1, 2)$
17. The roots of the discriminating cubic of the surface
 $2y^2 + 4zx - 6x - 8y + 2z + 5 = 0$ are
 (a) 2, 2, 2 (b) 2, 2, -2
 (c) 2, -2, -2 (d) -2, -2, -2
18. The roots of discriminating cubic are
 (a) purely real
 (b) real and imaginary both
 (c) purely imaginary
 (d) none of these
19. The equation $y^2 + z^2 - x^2 = 1$ represents a
 (a) hyperboloid of one sheet
 (b) hyperbolic cylinder
 (c) hyperboloid of revolution
 (d) hyperboloid of two sheets
20. The centre of the surface
 $2x^2 + 5y^2 + 10z^2 + 12yz + 6zx + 4xy - 1 = 0$ is
 (a) $(1, 0, 2)$ (b) $(0, 0, 0)$
 (c) $(0, 1, 2)$ (d) $(1, 2, 0)$
21. The roots of the discriminating cubic of the conicoid
 $x^2 + 2y^2 - z^2 = 1$ are
 (a) 1, 2, 2 (b) 1, 2, 1
 (c) 1, 2, -1 (d) 1, -2, 1
22. The surface $Ax^2 = 1$ or $By^2 = 1$ or $Cz^2 = 1$ always
 represents a
 (a) cone (b) pair of parallel planes
 (c) one plane (d) pair of perpendicular planes
23. The characteristics roots of the equation
 $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ are
 (a) 1, 2, 2 (b) 1, -2, 2
 (c) -1, 2, 2 (d) -1, -2, 2
24. The equation $A(x^2 + y^2 + z^2) = 1$ represents a
 (a) cone (b) cylinder
 (c) ellipsoid (d) sphere
25. The roots of the discriminating cubic of the conicoid
 $2x^2 - 3y^2 + 4z^2 = 5$ are
 (a) 2, 3, 4 (b) -2, 3, -4
 (c) 2, -3, -4 (d) 2, -3, 4
26. The equation $x^2 - y^2 + 2yz - 2zx - x - y + z = 0$
 always represents a
 (a) ellipsoid (b) hyperboloid
 (c) hyperbolic paraboloid
 (d) elliptic paraboloid
27. The surface equation $3x^2 + 4y^2 = 1$ represents a
 (a) circle (b) ellipse
 (c) ellipsoid (d) elliptic cylinder

28. In 3-dimension $Ax^2 - By^2 = 0$ represents
 (a) cone (b) cylinder (c) pair of planes (d) pair of lines
29. The roots of discriminating cubic for $11x^2 + 10y^2 + 6z^2 - 8yz + 4zx - 12xy + 72x - 72y + 362 + 150 = 0$ are
 (a) 3, 4, 6 (b) 3, 6, 12 (c) 3, 6, 18 (d) 3, 4, 12
30. $3x^2 - 4y^2 - 12z^2 = 1$ represents a
 (a) hyperboloid of one sheet (b) hyperbolic paraboloid (c) hyperboloid of two sheets (d) elliptic paraboloid

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(d)	2.	(c)	3.	(a)	4.	(c)	5.	(b)	6.	(c)	7.	(c)	8.	(d)	9.	(a)	10.	(b)
11.	(a)	12.	(b)	13.	(b)	14.	(c)	15.	(d)	16.	(b)	17.	(b)	18.	(a)	19.	(c)	20.	(b)
21.	(c)	22.	(b)	23.	(c)	24.	(d)	25.	(d)	26.	(c)	27.	(d)	28.	(c)	29.	(c)	30.	(c)

HINTS AND SOLUTIONS

2. $(3x - 4y + z)^2 + 3(3x - 4y + z) - 10 = 0$
 so $3x - 4y + z = \frac{-3 \pm \sqrt{9 + 40}}{2} = 2, -5$ i.e. we get
 $3x - 4y + z - 2 = 0$ and $3x - 4y + z + 5 = 0$ a pair of parallel planes.
3.
$$\begin{vmatrix} 1-\lambda & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1-\lambda & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1-\lambda \end{vmatrix} = 0$$
 After solving we get
 $4\lambda^3 - 12\lambda^2 + 9\lambda - 2 = 0$
4. Let $f = 3x^2 - y^2 + z^2 + 6yz - 6x + 6y - 2z - 2$
 $\frac{\partial f}{\partial x} = 6x - 6 = 0, \quad \frac{\partial f}{\partial y} = -2y + 6z + 6 = 0,$
 $\frac{\partial f}{\partial z} = 2z + 6y - 2 = 0$
 Thus $x = 1, y = 0, z = -1$ so centre is $(1, 0, -1)$.
8. The discriminating cubic is

$$\begin{vmatrix} 2-\lambda & -2 & -1 \\ -2 & 2-\lambda & 1 \\ -1 & 1 & 1-\lambda \end{vmatrix} = 0$$
 i.e. $\lambda^3 - 5\lambda^2 + 2\lambda = 0$
11. Let $f = x^2 + 2yz - 4x + 6y + 2z = 0$
 $\frac{\partial f}{\partial x} = 2x - 4 = 0 \Rightarrow x = 2$
 $\frac{\partial f}{\partial y} = 2z + 6 = 0 \Rightarrow z = -3$
 $\frac{\partial f}{\partial z} = 2y + 2 = 0 \Rightarrow y = -1$
 So its centre is $(2, -1, -3)$.
23. $\sqrt{x} + \sqrt{y} = -\sqrt{z}$
 $x + y + 2\sqrt{xy} = z \Rightarrow x + y - z = -2\sqrt{xy}$
 $x^2 + y^2 + z^2 + 2xy - 2yz - 2zx = 4xy$
 i.e. $x^2 + y^2 + z^2 - 2xy - 2yz - 2zx = 0$
 So roots of discriminating cubic are

$$\begin{vmatrix} 1-\lambda & -1 & -1 \\ -1 & 1-\lambda & -1 \\ -1 & -1 & 1-\lambda \end{vmatrix} = 0$$

 i.e. $\lambda^3 - 3\lambda^2 + 4 = 0$ we get $\lambda = -1, 2, 2$
25. The roots of the discriminating cubic are given by

$$\begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & -3-\lambda & 0 \\ 0 & 0 & 4-\lambda \end{vmatrix} = 0$$

 i.e. $(2-\lambda)(-3-\lambda)(4-\lambda) = 0$
 i.e. $\lambda = 2, -3, 4$

16

Vector Calculus

DIFFERENTIATION OF VECTORS

1. If to each point $P(x, y, z)$ of a region R in space there corresponds a unique scalar $f(P)$, then f is called a scalar point function. Similarly if to each point $P(x, y, z)$ of a region R in space there corresponds a unique vector $\vec{f}(P)$, then $\vec{f}$ is called a vector point function.

2. Let $\vec{r} = \vec{f}(t)$ be a vector function of the scalar variable t . We define $\vec{r} + \delta \vec{r} = \vec{f}(t + \delta t)$ so $\delta \vec{r} = f(t + \delta t) - \vec{f}(t)$, consider the vector $\frac{\delta \vec{r}}{\delta t} = \frac{\vec{f}(t + \delta t) - \vec{f}(t)}{\delta t}$.

If $\lim_{\delta t \rightarrow 0} \frac{\delta \vec{r}}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{f(t + \delta t) - f(t)}{\delta t}$ exists, then

the value of this limit, denoted by $\frac{d\vec{r}}{dt}$ is called the derivative of the vector function $\vec{r}$ with respect to the scalar t .

$$\begin{aligned} \text{So } \frac{d\vec{r}}{dt} &= \lim_{\delta t \rightarrow 0} \frac{(\vec{r} + \delta \vec{r}) - \vec{r}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \frac{\vec{f}(t + \delta t) - \vec{f}(t)}{\delta t}. \end{aligned}$$

If $\frac{d\vec{r}}{dt}$ exists, $\vec{r}$ is said to be differentiable. If we

differentiate $\frac{d\vec{r}}{dt}$ again, we get $\frac{d^2\vec{r}}{dt^2}$ which is called the second derivative and so on.

3. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are differentiable vector function of a scalar t and Q is a differentiable scalar function of the same variable t then

$$(i) \quad \frac{d}{dt}(\vec{a} + \vec{b}) = \frac{d\vec{a}}{dt} + \frac{d\vec{b}}{dt}$$

$$(ii) \quad \frac{d}{dt}(\vec{a} \cdot \vec{b}) = \vec{a} \cdot \frac{d\vec{b}}{dt} + \frac{d\vec{a}}{dt} \cdot \vec{b}$$

$$(iii) \quad \frac{d}{dt}(\vec{a} \times \vec{b}) = \vec{a} \times \frac{d\vec{b}}{dt} + \frac{d\vec{a}}{dt} \times \vec{b}$$

$$(iv) \quad \frac{d}{dt}(\phi \vec{a}) = \phi \frac{d\vec{a}}{dt} + \frac{d\phi}{dt} \vec{a}$$

$$(v) \quad \frac{d}{dt}[\vec{a} \vec{b} \vec{c}] = \left\{ \frac{d\vec{a}}{dt} \vec{b} \vec{c} \right\} + \left\{ \vec{a} \frac{d\vec{b}}{dt} \vec{c} \right\} + \left\{ \vec{a} \vec{b} \frac{d\vec{c}}{dt} \right\}$$

$$(vi) \quad \frac{d}{dt} \{ \vec{a} \times (\vec{b} \times \vec{c}) \} = \frac{d\vec{a}}{dt} \times (\vec{b} \times \vec{c}) + \vec{a} \times \left(\frac{d\vec{b}}{dt} \times \vec{c} \right) + \vec{a} \times \left(\vec{b} \times \frac{d\vec{c}}{dt} \right)$$

$$(vii) \quad \frac{d\vec{r}}{dt} = \frac{d\vec{r}}{ds} \frac{ds}{dt} \quad (\text{chain rule})$$

$$(viii) \quad \frac{d\vec{c}}{dt} = 0 \text{ if } \vec{c} \text{ is constant vector}$$

4. Some important results are :

(i) The necessary and sufficient condition for the vector function $\vec{a}(t)$ to be constant is

$$\text{that } \frac{d\vec{a}}{dt} = 0$$

- (ii) If $\vec{a}$ has constant length, then $\vec{a}$ and $\frac{d\vec{a}}{dt}$

are perpendicular provided $\left| \frac{d\vec{a}}{dt} \right| \neq 0$

- (iii) The necessary and sufficient condition for the vector $\vec{a}(t)$ to have constant magnitude

$$\text{is } \vec{a} \cdot \frac{d\vec{a}}{dt} = 0$$

- (iv) The necessary and sufficient condition for the vector $\vec{a}(t)$ to have constant direction is

$$\vec{a} \times \frac{d\vec{a}}{dt} = 0$$

5. If $\vec{r}$ be the position vector of a particle and the scalar variable t be the time, then $\vec{V} = \frac{d\vec{r}}{dt}$ is the velocity vector at instant t and is along the tangent to the path of the particle. The second derivative $\vec{a} = \frac{d^2\vec{r}}{dt^2}$ is the acceleration vector at instant t .

GRADIENT, DIVERGENCE AND CURL

1. Let $\vec{f}$ be a vector function depends on more than one scalar variable and let $\vec{f} = \vec{f}(x, y, z, \dots)$. Suppose the variable x increases from x to $x + \delta x$ while the other remains constant. Let $\delta \vec{f}$ be the corresponding increment in $\vec{f}$ then $\frac{\partial \vec{f}}{\partial x} = \lim_{\delta x \rightarrow 0} \frac{\delta \vec{f}}{\delta x}$
- $$= \lim_{\delta x \rightarrow 0} \frac{\vec{f}(x + \delta x, y, z) - \vec{f}(x, y, z)}{\delta x}$$
- is called the partial derivative of $\vec{f}$ w.r.t. x . The higher partial derivative of $\vec{f}$ can be defined as :

$$\frac{\partial^2 \vec{f}}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial \vec{f}}{\partial x} \right), \frac{\partial^2 \vec{f}}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial \vec{f}}{\partial y} \right),$$

$$\frac{\partial^2 \vec{f}}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial \vec{f}}{\partial x} \right) \text{ etc.}$$

2. The vector differential operator ∇ (del or nabla) is defined as

$$\vec{\nabla} \equiv \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \equiv \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}.$$

3. Let $f(x, y, z)$ be defined and differentiable at each point (x, y, z) in certain region of space. Then the gradient of f (∇f or $\text{grad } f$) is defined as

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

The gradient of a scalar field defines a vector field. Some important results are

$$(i) \quad \nabla(f + g) = \nabla f + \nabla g$$

$$(ii) \quad \nabla(fg) = f\nabla g + g\nabla f$$

$$(iii) \quad \nabla \left(\frac{f}{g} \right) = \frac{g\nabla f - f\nabla g}{g^2}$$

- (iv) The necessary and sufficient condition for a scalar point function to be constant is that $\nabla f = 0$

4. The directional derivative of a scalar field f at a point $P(x, y, z)$ in the direction of a unit vector $\hat{a}$ is given by $\frac{df}{dS} = \nabla f \cdot \hat{a}$

5. The gradient ∇f is a vector in the direction of which the maximum value of directional derivative of f is $\frac{df}{dS}$ occurs.

6. Let $\vec{v}$ be any given differentiable vector point function then the divergence of $\vec{v}$ is

$$\text{div } \vec{v} = \vec{\nabla} \cdot \vec{v} = \hat{i} \cdot \frac{\partial \vec{v}}{\partial x} + \hat{j} \cdot \frac{\partial \vec{v}}{\partial y} + \hat{k} \cdot \frac{\partial \vec{v}}{\partial z}$$

It should be noted that $\text{div } \vec{v}$ is a scalar quantity

A vector $\vec{v}$ is said to be solenoidal if $\vec{\nabla} \cdot \vec{v} = 0$

7. Let $\vec{f}$ be any given differentiable vector point function. Then the curl of $\vec{f}$ is defined as curl

$$\vec{f} = \vec{\nabla} \times \vec{f} = \hat{i} \times \frac{\partial \vec{f}}{\partial x} + \hat{j} \times \frac{\partial \vec{f}}{\partial y} + \hat{k} \times \frac{\partial \vec{f}}{\partial z} \quad \text{curl}$$

$\vec{f}$ is a vector quantity.

A vector $\vec{f}$ is said to be irrotational if $\vec{\nabla} \times \vec{f} = \vec{0}$

8. Some properties of divergence and curl are defined as

$$(i) \quad \vec{\nabla} \cdot (\vec{A} + \vec{B}) = \vec{\nabla} \cdot \vec{A} + \vec{\nabla} \cdot \vec{B}$$

$$(ii) \quad \vec{\nabla} \times (\vec{A} + \vec{B}) = \vec{\nabla} \times \vec{A} + \vec{\nabla} \times \vec{B}$$

$$(iii) \quad \vec{\nabla} \cdot (\phi \vec{A}) = (\vec{\nabla} \cdot \phi) \cdot \vec{A} + \phi (\vec{\nabla} \cdot \vec{A})$$

$$(iv) \quad \vec{\nabla} \times (\phi \vec{A}) = (\vec{\nabla} \cdot \phi) \times \vec{A} + \phi (\vec{\nabla} \times \vec{A})$$

$$(v) \quad \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})$$

$$(vi) \quad \vec{\nabla} \times (\vec{A} \times \vec{B}) = (\vec{B} \cdot \vec{\nabla}) \vec{A} - \vec{B} (\vec{\nabla} \cdot \vec{A}) \\ - (\vec{A} \cdot \vec{\nabla}) \vec{B} + \vec{A} (\vec{\nabla} \cdot \vec{B})$$

$$(vii) \quad \vec{\nabla} \cdot (\vec{\nabla} \phi) = \nabla^2 \phi$$

$$(viii) \quad \vec{\nabla} \times (\vec{\nabla} \phi) = \vec{0}$$

$$(ix) \quad \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

$$(x) \quad \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$$

INTEGRATION OF VECTORS

1. Integration is the reverse process of differentiation. Let $\vec{f}(t)$ and $\vec{F}(t)$ be two vector function of the scalar t such that $\frac{d}{dt} \vec{F}(t) = \vec{f}(t)$.

Then $\vec{F}(t)$ is called the indefinite integral of $\vec{f}(t)$

$\vec{f}(t)$ w.r.t. t and symbolically $\int \vec{f}(t) dt = \vec{F}(t) + C$.

The function $\vec{f}(t)$ to be integrated is called the integrand.

Some important results are given by

$$(i) \quad \int \left(\frac{d\vec{r}}{dt} \cdot \vec{s} + \vec{r} \cdot \frac{d\vec{s}}{dt} \right) dt = \vec{r} \cdot \vec{s} + C$$

$$(ii) \quad \int \left(2\vec{r} \cdot \frac{d\vec{r}}{dt} \right) dt = \vec{r}^2 + C$$

$$(iii) \quad \int \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2} \right) dt = \vec{r} \times \frac{d\vec{r}}{dt} + C$$

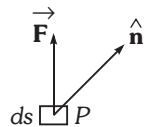
$$(iv) \quad \int \left(\vec{a} \times \frac{d\vec{r}}{dt} \right) dt = \vec{a} \times \vec{r} + C$$

3. Any integral which is to be evaluated along a curve is called a line integral and denoted by $\int_C \vec{F} \cdot d\vec{r}$.

4. The total work done W by the force $\vec{F}$ to displace the particle $d\vec{r}$ along the curve C is $W = \int_C \vec{F} \cdot d\vec{r}$.

5. Any integral which is to be evaluated over a surface is called the surface integral and is denoted by $\iint_S \vec{F} \cdot \hat{n} ds$

$$\text{where } ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|} = \frac{dy dz}{|\hat{n} \cdot \hat{i}|} \\ = \frac{dx dz}{|\hat{n} \cdot \hat{j}|}$$



- If $f(x, y, z)$ is a scalar point function, then its volume integral over a region v is defined as $\iiint_v f dv = \iiint_v f(x, y, z) dx dy dz$.

GREEN'S, GAUSS'S AND STOKE'S THEOREMS

1. Let C be a closed regular curve in the xy -plane bounding a region R and $M(x, y)$ and $N(x, y)$ be continuously differentiable functions inside and on C then

$$\int_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$$

The area bounded by a simple closed curve C is given by $A = \frac{1}{2} \int_C (xdy - ydx)$

The Green's theorem in plane is

$$\iint_R (\vec{\nabla} \times \vec{F}) \cdot \hat{k} dR = \oint_C \vec{F} \cdot d\vec{r}$$

2. Let v is the volume bounded by a closed piecewise smooth surface S . Suppose $\vec{F}(x, y, z)$ is a vector function of position which is

continuous and has continuous first partial derivative in v then

$$\iiint_v \vec{\nabla} \cdot \vec{F} dv = \iint_s \vec{F} \cdot \hat{n} ds$$

where $\hat{n}$ is the outwards drawn unit normal vector to s . This is called Gauss's theorem.

3. Let $\vec{F}$ be a continuous differentiable vector point function and S be an open, two sided surface bounded by a simple closed curve C then

$$\int_C \vec{F} \cdot d\vec{r} = \iint_s \text{curl } \vec{F} \cdot \hat{n} ds$$

Where C is traversed in the positive direction and $\hat{n}$ is the unit normal vector at any point of S .

This is called Stoke's theorem.

EXERCISE

MULTIPLE CHOICE QUESTIONS

Differentiation of Vectors

1. If the vector function $\vec{a}(t)$ is constant then

(a) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$ (b) $\vec{a} \times \frac{d\vec{a}}{dt} = 0$

(c) $\frac{d\vec{a}}{dt} = 0$ (d) $\vec{a} = 0$

2. If $\vec{a}$ has constant length with $\left| \frac{d\vec{a}}{dt} \right| \neq 0$ then

(a) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$ (b) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$

(c) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$ (d) $\vec{a} = 0$

3. The locus of a point whose co-ordinates are function of a single parameter is called

- (a) surface (b) curve
(c) point (d) none of these

4. If $\vec{r} = \sin t \hat{i} + \cos t \hat{j} + t\hat{k}$ then $\left| \frac{d^2\vec{r}}{dt^2} \right|$ is

(a) $-\sin t \hat{i} - \cos t \hat{j}$ (b) $\sqrt{2}$

(c) -1 (d) 1

5. If $\vec{r} = (\cos nt) \hat{i} + (\sin nt) \hat{j}$, where n is a constant

and t varies then $\vec{r} \times \frac{d\vec{r}}{dt}$ is

(a) $\hat{k}$ (b) $\sin nt \hat{k}$

(c) $\cos nt \hat{k}$ (d) $n\hat{k}$

6. If $\vec{r} = e^{nt} \vec{a} + e^{-nt} \vec{b}$, where $\vec{a}$ and $\vec{b}$ are constant vector then $\frac{d^2\vec{r}}{dt^2}$ is equal to

(a) $\vec{0}$ (b) $\vec{r}$

(c) $n\vec{r}$ (d) $n^2 \vec{x}$

7. If a vector $\vec{a}(t)$ have constant magnitude then
- (a) $\vec{a} \times \frac{d\vec{a}}{dt} = 0$ (b) $\frac{d\vec{a}}{dt} = 0$
- (c) $\vec{a} = 0$ (d) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$
8. The value of $\frac{d}{dt} \left[\vec{r}, \frac{d\vec{r}}{dt}, \frac{d^2\vec{r}}{dt^2} \right]$ is
- (a) $\left[\frac{d\vec{r}}{dt}, \frac{d^2\vec{r}}{dt^2}, \frac{d^2\vec{r}}{dt^2} \right]$ (b) $\left[\vec{r}, \frac{d^2\vec{r}}{dt^2}, \frac{d^3\vec{r}}{dt^2} \right]$
- (c) $\left[\vec{r}, \frac{d\vec{r}}{dt}, \frac{d^2\vec{r}}{dt^3} \right]$ (d) $\left[\vec{r}, \frac{d\vec{r}}{dt}, \frac{d^2\vec{r}}{dt^2} \right]$
9. If $\vec{a} = t^2\hat{i} - t\hat{j} + (2t + 1)\hat{k}$ and $\vec{b} = (2t - 3)\hat{i} + \hat{j} - t\hat{k}$ then $\frac{d}{dt}(\vec{a} \cdot \vec{b})$ at $t = 1$ is
- (a) 6 (b) -6
- (c) 5 (d) -5
10. If $\vec{a} = a \cos t \hat{i} + a \sin t \hat{j} + a \tan \alpha \hat{k}$ then $|\dot{\vec{r}} \times \ddot{\vec{r}}|$ is **[Kanpur 2018; Avadh 2018]**
- (a) $a \sec \alpha$ (b) $a \tan \alpha$
- (c) $a^2 \sec \alpha$ (d) $a^2 \tan \alpha$
11. A particle moves along the curve $r = e^{-t}\hat{i} + e^t \sin t \hat{j} + \hat{k}$ then the magnitude of its velocity at $t = 0$ is
- (a) 2 (b) $\sqrt{2}$
- (c) $2\sqrt{2}$ (d) 0
12. The necessary and sufficient condition for the vector $\vec{a}$ to have constant direction is
- (a) $\vec{a} = 0$ (b) $\frac{d\vec{a}}{dt} = 0$
- (c) $\vec{a} \times \frac{d\vec{a}}{dt} = 0$ (d) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$
13. If $\vec{r} = \sin t \hat{i} + \cos t \hat{j} + t\hat{k}$, then $\left| \frac{d\vec{r}}{dt} \right|$ is equal to
- (a) 1 (b) 0 (c) 2 (d) $\sqrt{2}$
14. If $\vec{r} = \cos \omega t \vec{a} + \sin \omega t \vec{b}$, where $\vec{a}, \vec{b}$ are constant vectors and ω is constant then $\frac{d^2\vec{r}}{dt^2} + \omega^2\vec{r}$ is equal to
- (a) $\vec{a} + \vec{b}$ (b) $\omega \vec{a}$
- (c) $\omega \vec{b}$ (d) 0
15. If $\vec{r} = e^{nt} \vec{a} + e^{-nt} \vec{b}$ where $\vec{a}$ and $\vec{b}$ are constant vectors then $\frac{d^2\vec{r}}{dt^2}$ is equal to
- (a) $\vec{r}$ (b) $n^2 \vec{r}$
- (c) $n \vec{r}$ (d) $-n^2 \vec{r}$
16. If $\vec{A} = a\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + b\hat{k}$ and $\vec{A} \times \vec{B} = 0$ then the value of a and b are
- (a) 2 and -1 (b) 1 and -2
- (c) 2 and -2 (d) 2 and 3
17. The necessary and sufficient condition for $\vec{r}$ to have constant magnitude is
- (a) $\frac{d\vec{r}}{dt} = 0$ (b) $\vec{r} \times \frac{d\vec{r}}{dt} = 0$
- (c) $\vec{r} \cdot \frac{d\vec{r}}{dt} = 0$ (d) $\vec{r} = 0$
18. If a particle moves along the curve $x = t^3 + 1, y = t^2, z = 2t + 5$ where t is time, then the components of its velocity at $t = 1$ in the direction $\hat{i} + \hat{j} + 3\hat{k}$ is
- (a) $\sqrt{7}$ (b) $\sqrt{6}$
- (c) $\sqrt{11}$ (d) $\sqrt{3}$
19. If $\vec{r} = a \cos t \hat{i} + a \sin t \hat{j} + a t \tan \alpha \hat{k}$ then $[\dot{\vec{r}} \ddot{\vec{r}} \ddot{\vec{r}}]$ is **[Kanpur 2018]**
- (a) $a \tan \alpha$ (b) $a^3 \tan \alpha$
- (c) $a \sec \alpha$ (d) $a^3 \sec \alpha$
20. If $\vec{r} = (\cos \omega t) \hat{i} + (\sin \omega t) \hat{j}$ then $\vec{r} \times \frac{d^2\vec{r}}{dt^2}$ is **[Avadh 2018]**
- (a) 1 (b) $\sin \omega t$
- (c) $\cos \omega t$ (d) 0

GRADIENT, DIVERGENCE AND CURL

21. If F is a scalar point function, then gradient of f (∇f) is a
 (a) scalar point function
 (b) vector point function
 (c) either scalar or vector point function
 (d) none of these
22. The necessary and sufficient condition for a scalar point function to be constant is that
 (a) $\nabla \cdot F = 0$ (b) $\nabla \times f = \vec{0}$
 (c) $\nabla f = \vec{0}$ (d) none of these
23. If $f(x, y, z) = 3x^2y - y^3z^2$ then $\text{grad } f$ at the point $(1, 1, 1)$ is
 (a) $6\hat{i} + 2\hat{j} - 2\hat{k}$ (b) $6\hat{i} - \hat{j} + 4\hat{k}$
 (c) $6\hat{i} - 2\hat{k}$ (d) none of these
24. If $r = |\vec{r}|$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then the value of ∇r is
 (a) 0 (b) $\frac{r}{\vec{r}}$
 (c) $-\frac{\vec{r}}{r}$ (d) $+\frac{\vec{r}}{r}$
25. The value of $\nabla(\vec{r} \cdot \vec{a})$ is
 (a) $\vec{a}$ (b) $\vec{r}$
 (c) 0 (d) $\nabla \times \vec{a}$
26. For the surface $f(x, y, z) = c$ where c is a constant the vector ∇f is
 (a) parallel vector (b) normal vector
 (c) intersect the surface (d) none of these
27. The vector in the direction of which the maximum value of the directional derivative of f occurs is
 (a) $\text{curl } F$ (b) $\text{div } F$
 (c) $\text{grad } f$ (d) none of these
28. The normal to the surface $x^4 - 3xyz + z^2 + 1 = 0$ at the point $(1, 1, 1)$ is
 (a) $\hat{i} + 3\hat{j} + \hat{k}$ (b) $\hat{i} + 3\hat{j} - \hat{k}$
 (c) $\hat{i} - 3\hat{j} + \hat{k}$ (d) $\hat{i} - 3\hat{j} - 3\hat{k}$
29. $\text{grad} \left(\frac{1}{r} \right)$ is
 (a) $-\frac{\vec{r}}{r}$ (b) $-\frac{\vec{r}}{r^2}$
 (c) $-\frac{\vec{r}}{r^3}$ (d) $-\frac{1}{r^2}$
30. $\text{grad } f(r) \times \vec{r}$ is
 (a) $\vec{r}$ (b) $f(r)$
 (c) $\vec{r} f(r)$ (d) 0
31. The unit vector normal to the level surface $x^2 + y - z = 1$ at the point $(1, 0, 0)$ is
 (a) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}$ (b) $\frac{2\hat{i} + \hat{j} - \hat{k}}{\sqrt{6}}$
 (c) $\frac{\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{6}}$ (d) $\frac{\hat{i} + \hat{j}}{6}$
32. The directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$ is
[Kanpur 2018]
 (a) 3 (b) 37
 (c) $\frac{3}{37}$ (d) $\frac{37}{3}$
33. The greatest value of the direction derivative of the function $2x^2 - y - z^4$ at the point $(2, -1, 1)$ is
 (a) 7 (b) 6
 (c) 8 (d) 9
34. The equation of the tangent plane to the surface $x^2 + 2y^2 + 3z^2 - 12 = 0$ at the point $(1, 2, -1)$ is
 (a) $x + 4y - 3z = 12$ (b) $x + 3y - 4z = 12$
 (c) $3x + y - 4z = 12$ (d) $3x + 4y - z = 12$
35. The directional derivative of $f = xyz$ at the point $(1, 1, 1)$ in the direction $\hat{i}$ is
 (a) 3 (b) $\sqrt{3}$
 (c) 1 (d) $\sqrt{2}$
36. The $\text{div } f$ is a
 (a) scalar
 (b) vector
 (c) may be scalar or vector both
 (d) none of these

37. The vector $\vec{f}$ is called solenoidal if [Avadh 2018]
- (a) $\nabla \vec{f} = 0$ (b) $\nabla \cdot \vec{f} = 0$
 (c) $\nabla \times \vec{f} = 0$ (d) $\vec{f} = 0$
38. The vector $\vec{f}$ is called irrotational if
- (a) $\nabla \vec{f} = 0$ (b) $\nabla \cdot \vec{f} = 0$
 (c) $\nabla \times \vec{f} = 0$ (d) $\vec{f} = 0$
39. $\text{div } \vec{r}$ is [Avadh 2018]
- (a) 1 (b) 2
 (c) 3 (d) 0
40. $\text{Curl } \vec{r}$ is
- (a) 0 (b) 1
 (c) 2 (d) 3
41. If $\vec{f} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$ then $\text{div } \vec{f}$ at the point $(1, -1, 1)$ is
- (a) 0 (b) 3
 (c) 6 (d) 9
42. If $\vec{f}$ is a constant vector then
- (a) $\text{div } \vec{f} = 0$ only
 (b) $\text{curl } \vec{f} = 0$ only
 (c) $\text{div } \vec{f} = 0$ but $\text{curl } \vec{f} \neq 0$
 (d) $\text{div } \vec{f} = 0$ and $\text{curl } \vec{f} = 0$
43. If $\vec{a}$ is a constant vector then $\text{div } (\vec{r} \times \vec{a})$ is
- (a) $\vec{r}$ (b) $\vec{a}$
 (c) $\vec{r} \cdot \vec{a}$ (d) 0
44. $\nabla \times (\nabla f)$ is
- (a) f (b) $\vec{0}$
 (c) 0 (d) 1
45. If $\vec{A}$ and $\vec{B}$ are irrotational then $\vec{A} \times \vec{B}$ is
- (a) solenoidal only
 (b) irrotational only
 (c) either solenoidal or irrotational
 (d) none of these
46. If the vector $\vec{A} = (ax + 2y)\hat{i} + (2y - 3)\hat{j} - 3z\hat{k}$ is solenoidal; then the constant a is
- (a) 0 (b) 1
 (c) 2 (d) 3
47. If the vector $\vec{A} = ay\hat{i} - 3z\hat{j} + 2b\hat{k}$ is irrotational then the constant a is
- (a) 1 (b) 3
 (c) 2 (d) 0
48. The greatest rate of increase of xyz^2 at the point $(1, 0, 3)$ is
- (a) 0 (b) 3
 (c) 1 (d) 9
49. The maximum value of the directional derivative of x^2yz at the point $(1, 4, 1)$ is
- (a) 9 (b) 4
 (c) 1 (d) none of these
50. If $f = x^2 - y^2 + 4z$ then the value of $\nabla^2 f$ is
- (a) 4 (b) 1
 (c) -1 (d) 0
51. The vector $r^n \vec{r}$ is solenoidal only when
- (a) $n = 0$ (b) $n + 1 = 0$
 (c) $n + 2 = 0$ (d) $n + 3 = 0$
52. If $\vec{F} = x^2y + 2xy + z^2$ then $\text{curl grad } \vec{F}$ is
- (a) $x\hat{i} + y\hat{j} + 2z\hat{k}$ (b) $2x\hat{i} + 2y\hat{j} + 2z\hat{k}$
 (c) $2x\hat{i} + y\hat{j}$ (d) 0

Integration of Vectors

53. If $\vec{f}(t) = (t^2 - 1)\hat{i} + t^2\hat{j} - \hat{k}$ then $\int_0^1 f(t) dt$ is
- (a) $\frac{1}{3}(2\hat{i} - \hat{j} + 3\hat{k})$ (b) $\frac{1}{3}(\hat{i} + \hat{j} - 3\hat{k})$
 (c) $-\frac{1}{3}(2\hat{i} - \hat{j} + 3\hat{k})$ (d) $\frac{1}{3}(-\hat{i} - \hat{j} + 3\hat{k})$
54. If $\vec{a} = t\hat{i} - t^2\hat{j} + (t - 1)\hat{k}$ and $\vec{b} = 2t^2\hat{i} + 6t\hat{k}$ then $\int_0^2 \vec{a} \cdot \vec{b} dt$ is
- (a) 10 (b) 12
 (c) -12 (d) -10

55. The solution of the equation $\frac{d^2 \vec{r}}{dt^2} = \vec{a}$, where $\vec{a}$ is a constant vector and it is given that $\vec{r} = \vec{0}$ and $\frac{d\vec{r}}{dt} = \vec{0}$ when $t = 0$ is
- (a) $\frac{1}{2} \vec{a}$ (b) $\frac{1}{2} t \vec{a}$
 (c) $\frac{1}{2} t^2 \vec{a}$ (d) $\frac{1}{2} t \vec{a} + 1$
56. If $\vec{r}(t) = \begin{cases} 2\hat{i} - \hat{j} + 2\hat{k} & \text{when } t = 2 \\ 4\hat{i} - 2\hat{j} + 3\hat{k} & \text{when } t = 3 \end{cases}$ then $\int_2^3 \left(\vec{r} \cdot \frac{d\vec{r}}{dt} \right) dt$ is equal to
- (a) 10 (b) 6
 (c) -7 (d) 0
57. Any integral which is evaluated along curve is called
- (a) volume integral (b) surface integral
 (c) line integral (d) none of these
58. If $\vec{F} = x\hat{i} + \hat{j}$ and curve C is the line $y = x$ from $(0, 1)$ to $(1, 2)$ then $\int_C \vec{F} \cdot d\vec{r}$ is
- (a) $\frac{2}{3}$ (b) 1
 (c) 0 (d) $\frac{3}{2}$
59. The value of $\int_C F \cdot dr$, where $\vec{F} = x^2\hat{i} - xy\hat{j}$, from the point $(0, 0)$ to $(1, 1)$ along the curve $y^2 = x$ is
- (a) 12 (b) -12 (c) $\frac{1}{12}$ (d) $-\frac{1}{2}$
60. The value of the integral $\int (x dy - y dx)$ around the circle $x^2 + y^2 = 1$ is
- (a) π (b) 2π (c) 0 (d) 1
61. If $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ and curve C is the line $y = 0$ from $(0, 0)$ to $(a, 0)$ then the value of $\int_C \vec{F} \cdot d\vec{r}$ is
- (a) $\frac{a}{3}$ (b) $\frac{a^2}{2}$ (c) $\frac{a^3}{3}$ (d) $-\frac{a}{3}$
62. The circulation of $\vec{F}$ round the curve C , where $\vec{F} = (x - y)\hat{i} + (x + y)\hat{j}$ and C is the circle $x^2 + y^2 = 16, z = 0$ is
- (a) 2π (b) 8π
 (c) 16π (d) 32π
63. If $\vec{F} = \cos y\hat{i} - x \sin y\hat{j}$ and curve C is the circle $x^2 + y^2 = 1$ in the xy -plane from $(1, 0)$ to $(0, 1)$ then $\int_C \vec{F} \cdot d\vec{r}$ is
- (a) 0 (b) 1
 (c) -1 (d) π
64. The circulation of $\vec{F}$ round the curve C where $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and C is the circle $x^2 + y^2 = 1, z = 0$ is
- (a) $-\pi$ (b) π (c) 2π (d) 0
65. The value of the integral $\iiint_v 45x^2y$, where v is the closed region bounded by the planes $4x + 2y + z = 8, x = 0, y = 0, z = 0$ is
- (a) 26 (b) 64 (c) 128 (d) 0
66. If $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and surfaces is the cylinder $x^2 + y^2 = 4$ and the circular discs $z = 0$ and $z = 3$ then $\iiint_s \vec{F} \cdot \hat{n} ds$ over the plane face $z = 0$ of the cylinder is
- (a) 36π (b) $x^2 + y^2$
 (c) 2π (d) 0
67. For a closed surface, $\iiint_s \vec{r} \cdot \hat{n} ds$ is [Avadh 2018]
- (a) $3v$ (b) $2v$ (c) v (d) 0
68. If S is the surface of unique sphere and $\vec{F} = ax\hat{i} + by\hat{j} + cz\hat{k}$ where a, b, c are constants then $\iiint_s \vec{F} \cdot \hat{n} ds$ is
- (a) $\frac{\pi}{3}(a + b + c)$ (b) $\frac{4}{3}\pi(a + b + c)$
 (c) $\frac{2\pi}{3}(a + b + c)$ (d) 0

69. If $\vec{a} = t\hat{i} + t\hat{k}$, $\vec{b} = \hat{i} - \hat{j}$, $\vec{c} = -\hat{i} + t\hat{j} + 2\hat{k}$ then the value of $\int_0^1 (\vec{a} \cdot \vec{b} \times \vec{c}) dt$ is

- (a) $\frac{6}{7}$ (b) $\frac{-7}{6}$
(c) $\frac{-6}{7}$ (d) 0

70. If $\vec{F} = 2x\hat{i} - y\hat{j}$ and C is the curve in the xy -plane, $y = x^2$, from $(0, 0)$ to $(2, 4)$ then the value of $\int_C \vec{F} \cdot d\vec{r}$ is

- (a) -2 (b) -6
(c) -4 (d) 0

71. If $\vec{F} = (2x + y)\hat{i} + (3y - x)\hat{j}$ and C is the straight line from $(0, 0)$ to $(2, 0)$ then $\int_C \vec{F} \cdot d\vec{r}$ is

- (a) 1 (b) 2
(c) 4 (d) 8

72. If $\vec{F} = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$ and C is the straight line from $(0, 0, 0)$ to $(0, 0, 1)$ then the value of $\int_C \vec{F} \cdot d\vec{r}$ is

- (a) 0 (b) 4
(c) -5 (d) 10

73. If $\vec{F} = x\hat{i} + y\hat{j}$ and curve C is the arc of the curve $y = x^2$ from $(0, 0)$ to $(1, 1)$ then $\int_C \vec{F} \cdot d\vec{r}$ is

- (a) 0 (b) 1
(c) 2 (d) none of these

74. The work done in moving a particle in a force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + 3\hat{k}$ along the line joining $(0, 0, 0)$ to $(2, 1, 3)$ is

- (a) 0 (b) 24
(c) 10 (d) 16

75. If $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\iint_S \vec{F} \cdot \hat{n} ds$, where S is the surface of unit sphere is

- (a) π (b) 2π
(c) 4π (d) 0

Green's, Gauss's and Stoke's Theorems

76. According to Gauss divergence theorem $\int_S \vec{F} \cdot \hat{n} ds$ is equivalent to

- (a) $\int_V \text{grad } \vec{F} dv$ (b) $\int_V \text{div } \vec{F} dv$
(c) $\int_V \text{curl } \vec{F} dv$ (d) $\int_V \text{curl } \vec{F} \cdot \hat{n} dv$

77. If v is the volume of the surface S then $\int_S \vec{r} \cdot \hat{n} ds$ is equal to

- (a) v (b) $2v$
(c) $3v$ (d) 0

78. The value of $\int_S \vec{r} \times \hat{n} ds$ is

- (a) 1 (b) 2
(c) -1 (d) 0

79. For any closed surface S , $\iint_S \text{curl } \vec{F} \cdot \hat{n} ds$ is

- (a) 1 (b) $\hat{i}$
(c) $2\vec{F}$ (d) 0

80. If v is the region bounded by S and vector $\vec{N}$ is always normal to S then $\int_V \text{curl } \vec{N} dv$ is

- (a) 0 (b) $2v$
(c) $3v$ (d) v

81. If $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ then the value of $\int_V \text{div } \vec{F} dv$ over the rectangular parallelepiped $0 \leq x \leq 9$, $0 \leq y \leq b$, $0 \leq z \leq c$ is

- (a) $a + b + c$ (b) $3abc$
(c) $ab + bc + ac$ (d) 0

82. The value of $\int_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$ is

- (a) $\frac{2}{3}$ (b) $\frac{4}{3}$
(c) $\frac{3}{4}$ (d) $\frac{3}{2}$

83. If $\vec{F} = yz\hat{i} + z\hat{j} + 2xy\hat{k}$ over the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$ then the value of $\int_S \vec{F} \cdot \hat{n} ds$ is
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) 2 (d) 0
84. If $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ then the value of $\int_S \vec{F} \cdot \hat{n} ds$ over the rectangular parallelepiped $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$ is
 (a) abc (b) $2abc$
 (c) 0 (d) $3abc$
85. If $\vec{F} = x\hat{i} + xz\hat{j} + y^2\hat{k}$ over the region bounded by a cube of unit length then $\int_S \vec{F} \cdot \hat{n} ds$ is
 (a) 6 (b) 2
 (c) 1 (d) 0
86. If $\vec{F} = (2x - z)\hat{i} + (3y - z)\hat{j} - z\hat{k}$ taken over the region bounded by $x = 0$, $x = a$, $y = 0$, $y = a$, $z = 0$, $z = a$ then $\int_V \text{div } \vec{F} dv$ is
 (a) $4a^3$ (b) a^3
 (c) $4a$ (d) 0
87. If S is the surface of the unit cube then $\int_S x^2 dydz + y^2 dzdx + 2z(xy - x - y) dxdy$, is
 (a) 1 (b) 4
 (c) 2 (d) $\frac{1}{2}$
88. If $\vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$ and S is the surface of the sphere of radius 3 then $\int_S \vec{F} \cdot \hat{n} ds$ is
 (a) 18π (b) 108π (c) π (d) 0
89. If $\vec{F} = x\hat{i} - y\hat{j} + 2z\hat{k}$ over the sphere $x^2 + (y - 1)^2 + (z - 1)^2 = 1$ then $\int_S \vec{F} \cdot \hat{n} ds$ is
 (a) $\frac{2\pi}{3}$ (b) $\frac{4\pi}{3}$
 (c) $\frac{8\pi}{3}$ (d) π
90. According to Stoke's theorem $\int_C \vec{F} \cdot d\vec{r}$ is
 (a) $\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{S}$ (b) $\int_S (\vec{\nabla} \cdot \vec{F}) \cdot d\vec{S}$
 (c) $\int_S (\vec{\nabla} \times \vec{F}) \cdot dS$ (d) none of these
91. $\int_C \vec{r} \cdot d\vec{r}$ is
 (a) 1 (b) 2
 (c) 3 (d) 0
92. If $\vec{F} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary then $\int_C \vec{F} \cdot d\vec{r}$ is
 (a) -3 (b) π
 (c) 0 (d) 1
93. If $\vec{F} = 2\hat{i} + 3\hat{j} + 9\hat{k}$ and S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 9$ and C be its boundary then $\int_C \vec{F} \cdot d\vec{r}$ is
 (a) 9π (b) 6π
 (c) -12 (d) 0
94. If $\vec{F} = x^2\hat{i} + xy\hat{j}$ and S is the rectangle in xy -plane whose sides are $x = 0$, $y = 0$, $x = 1$, $y = 1$ then $\int_C \vec{F} \cdot d\vec{r}$ is
 (a) 2 (b) $\frac{1}{2}$
 (c) 3 (d) 0
95. The value of $\oint_C (e^x dx + y dy - 3dz)$ over the curve $x^2 + y^2 = 1$, $z = 0$ is
 (a) π (b) 8
 (c) -1 (d) 0
96. If $\vec{F} = 4xy\hat{i} + (2x^2 + y)\hat{j}$ and C is the boundary of the triangle with vertices at $(0, 0)$, $(2, 0)$, $(2, 1)$ then the value of $\oint_C \vec{F} \cdot d\vec{r}$ is
 (a) 0 (b) 2π
 (c) -1 (d) none of these

97. If C be the boundary of the rectangle $0 \leq x \leq 1$, $0 \leq y < 1$, $z = 3$ then the value of $\oint_C (\sin z \, dx - \cos x \, dy + \sin y \, dz)$ is
 (a) 1 (b) 2
 (c) π (d) $-\pi$
98. If $\vec{F} = x^2\hat{i} + xy\hat{j}$ and S is the rectangle in the plane $z = 0$, whose sides are $x = 0$, $y = 0$, $x = 1$, $z = 1$ then $\int_S \text{curl } \vec{F} \cdot \hat{n} \, dS$ is
 (a) $\frac{1}{2}$ (b) 2
 (c) $\frac{1}{3}$ (d) 0
99. If C is the curve $x^2 + y^2 = a$, $z = y^2$ then the value of $\oint_C (yz \, dx + zx \, dy + xy \, dz)$ is
 (a) a (b) $2a$
 (c) $8a$ (d) 0
100. If $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C lying in a simple connected region R then
 (a) $\text{grad } \vec{F} = 0$ (b) $\text{div } \vec{F} = 0$
 (c) $\text{curl } \vec{F} = 0$ (d) $\text{grad div } \vec{F} = 0$
101. If C is the boundary of the region enclosed by $x = y^2$ and $y = x^2$ describe in the positive sense then the value of $\oint ((2xy - x^2) \, dx + (x^2 + y^2) \, dy)$ is
 (a) 2π (b) $-\pi$
 (c) 8 (d) 0
102. The value of $\oint_C ((x^2 - y \cos x) \, dx + (y + \sin x) \, dy)$, where C is the rectangle with vertices $(0, 0)$, $(\frac{\pi}{2}, 0)$, $(\frac{\pi}{2}, 1)$, $(0, 1)$ is
 (a) 0 (b) π
 (c) $\frac{\pi}{2}$ (d) 2π
103. The value of $\oint_C ((x^2 - y^2) \, dx + 2xy \, dy)$ around the rectangular boundary $x = 0$, $x = a$, $y = 0$, $y = b$ is
 (a) $11ab$ (b) $2ab$
 (c) $2ab^2$ (d) $2a^2b$
104. If C is the circle $x^2 + y^2 = 1$ then the value of $\oint_C [(\cos x \sin y - xy) \, dx + \sin x \cos y \, dy]$ is
 (a) 0 (b) $\frac{\pi}{2}$
 (c) π (d) 1
105. If A be the area bounded by a simple closed curve C then $\oint_C (x \, dy - y \, dx)$ is
 (a) A (b) $2A$
 (c) $3A$ (d) $4A$
106. If a vector field $\vec{F}$ is not irrotational then
 (a) $\vec{F} = \nabla\phi$ (b) $\nabla \times \vec{F} = 0$
 (c) $\oint_C \vec{F} \cdot d\vec{r} = 0$ (d) $\nabla \cdot \vec{F} = 0$
107. If $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ such that $\vec{F} = \nabla\phi$ then the value of ϕ is
 (a) $(xy + yz + zx) + c$ (b) $(x^2 + y^2 + z^2) + c$
 (c) $(x + y + z) + c$ (d) $\frac{1}{2}(x^2 + y^2 + z^2) + c$
108. If the vector $\vec{F} = (3x - 3y + az)\hat{i} + (bx + 2y - 4z)\hat{j} + (2x + cy + z)\hat{k}$ is irrotational then (a, b, c) is
 (a) $(2, -3, -4)$ (b) $(-3, 2, -4)$
 (c) $(2, 3, 4)$ (d) $(2, -3, 5)$

MISCELLANEOUS

109. If $\vec{a} = \hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + p\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$ are coplanar then p is given by
 (a) 6 (b) 4
 (c) -4 (d) -6
110. If the vector $\vec{A} = (ax + ay)\hat{i} + 4y\hat{j} + 3\hat{k}$ is solenoidal then a is
 (a) -4 (b) -2
 (c) 0 (d) 2
111. If $\vec{F} = x\hat{i} + 2y\hat{j} + 3z\hat{k}$ then evaluate $\iint_S \vec{F} \cdot \hat{n} \, d\vec{S}$ is
 (a) 3 S (b) 6 V
 (c) 6 S (d) 3 V

112. If C is the curve given by $x^2 + y^2 = 1, y^2 = z$ then $\int_C yz \, dx + zx \, dy + xy \, dz$ is
 (a) 0 (b) 1
 (c) -1 (d) 2
113. If $\vec{r} = xyz\hat{i} + xz^2\hat{j} - y^3\hat{k}$ then $\frac{\partial^2 r}{\partial x \partial y}$ at $(0, 0, 1)$ is
 (a) $\hat{i} + \hat{j}$ (b) $\hat{i} - \hat{j}$
 (c) $\hat{i}$ (d) $\hat{j}$
114. The vector $\vec{A} = e^x \sin y \, \lambda + e^x \cos y \, \hat{j}$ is
 (a) solenoidal
 (b) rotational
 (c) irrotational
 (d) (a) and (c) both
115. If C is the curve $x^2 + y^2 = 4, z = 0$ then $\int_C (e^x dx + 2y dy - dz)$ is
 (a) 1 (b) 2
 (c) 4 (d) 0
116. $\nabla \cdot (\nabla r^n)$ is [Avadh 2018]
 (a) $n(n+1)r^{n-2}$ (b) $n(n+1)r^{n-1}$
 (c) $n(n+3)r^{n-2}$ (d) $n(n+1)r^n$
117. The projection of $\hat{i} - 2\hat{j} + \hat{k}$ on $4\hat{i} - 4\hat{j} + 7\hat{k}$ is
 (a) $\frac{17}{9}$ (b) $\frac{19}{9}$
 (c) $\frac{19}{3}$ (d) $\frac{17}{3}$
118. ∇e^{r^2} is given by
 (a) $2\vec{r} e^{r^2}$ (b) $2e^{r^2}$
 (c) $\vec{r} e^{r^2}$ (d) $\vec{r} e^r$
119. $\nabla f(\vec{r}) \times \vec{r}$ is given by
 (a) 0 (b) $\vec{r} f'(r)$
 (c) $\frac{f'(\vec{r})}{\vec{r}}$ (d) $\vec{r} f(\vec{r})$
120. Green's theorem is a special case of [Avadh 2018]
 (a) Gauss theorem
 (b) Stoke's theorem
- (c) divergence theorem
 (d) none of these
121. The maximum value of directional derivative of $\vec{A} = x^2 - 2y^2 + 4z^2$ at $(1, 1, -1)$ is
 (a) $2\sqrt{\frac{7}{3}}$ (b) $\sqrt{\frac{7}{3}}$
 (c) $2\sqrt{\frac{3}{7}}$ (d) $6\sqrt{\frac{7}{3}}$
122. If $\frac{d\vec{a}}{dt} = \vec{r} \times \vec{a}, \frac{d\vec{b}}{dt} = \vec{r} \times \vec{b}$ then $\frac{d}{dt}(\vec{a} \times \vec{b})$ is
 (a) $\vec{r}(\vec{a} \times \vec{b})$
 (b) $\vec{r} \cdot (\vec{a} \times \vec{b})$
 (c) $\vec{r} \times (\vec{a} \times \vec{b})$
 (d) $\vec{r} \times (\vec{a} \cdot \vec{b})$
123. If the surface $ax^2 - yz = (a+z)x$ is orthogonal to the surface $4x^2y + z^3 = 4$ at $(1, -1, 2)$ then a is
 (a) $\frac{2}{5}$ (b) $\frac{5}{2}$
 (c) $\frac{3}{2}$ (d) 5
124. If $(x+3y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal then a is equal to
 (a) 2 (b) 3
 (c) -2 (d) 0
125. The value of $\int_{(0,2,1)}^{(2,0,1)} (ze^x dx + 3yz dy + e^x + y^2) dz$ is
 (a) $e - 1$ (b) $e^2 + 1$
 (c) $e^2 - 5$ (d) $e + 1$
126. If $\hat{n}$ is the unit outward drawn normal to any closed surface s then $\int_v \text{div} \hat{n} \, dv$ is
 (a) S (b) v
 (c) $S + v$ (d) $S - v$
127. The value of $\int_{(0,0)}^{(1,1)} [(x^2 + y^2)\hat{i} + (x^2 - y^2)\hat{j}] \cdot d\vec{r}$ over $y^2 = x$ is
 (a) 7 (b) 10
 (c) $\frac{7}{10}$ (d) $\frac{10}{7}$

128. A particle moves along the curve $x = 4 \cos t$, $y = 4 \sin t$, $z = 6t$, then the acceleration at time t is **[Kanpur 2018]**
- (a) $4 \cos t \hat{i} + 4 \sin t \hat{j} + \hat{k}$
 (b) $4 \sin t \hat{i} - 4 \cos t \hat{j} + \hat{k}$
 (c) $4 \cos t \hat{i} - 4 \sin t \hat{j} - \hat{k}$
 (d) $-4 \cos t \hat{i} - 4 \sin t \hat{j}$
129. The position vector of a moving particle at time t is given by $\vec{r} = (3t - 4)\hat{i} + (t^2 - 2)\hat{j} + 4t^3\hat{k}$, then the acceleration at time $t = 2$ is **[Kanpur 2018]**
- (a) $2\hat{i} + 2\hat{j} + 48\hat{k}$
 (b) $2\hat{j} + 48\hat{k}$
 (c) $2\hat{i} + \hat{j} + 24\hat{k}$
 (d) $3\hat{j} + 4\hat{k}$
130. For $\vec{r} = 2t^2\hat{i} + t\hat{j} - 3t^3\hat{k}$, the value of $\int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2} \right) dt$ is **[Kanpur 2018]**
- (a) $-42\hat{i} + 90\hat{j} - 6\hat{k}$
 (b) $42\hat{i} - 90\hat{j} - 6\hat{k}$
 (c) $42\hat{i} + 90\hat{j} + 6\hat{k}$
 (d) $-42\hat{i} - 90\hat{j} + 6\hat{k}$
131. A particle moves along the curve $\vec{r} = e^{-t} \cos t \hat{i} + e^{-t} \sin t \hat{j} + e^{-t} \hat{k}$. The magnitude of its velocity at time $t = 0$ is **[Kanpur 2018]**
- (a) $2\sqrt{3}$ (b) $\sqrt{3}/2$
 (c) $\sqrt{3}$ (d) $2/\sqrt{3}$
132. If $\vec{F}(t) = t\hat{i} + (t^2 - 2t)\hat{j} + (3t^2 + 3t^3)\hat{k}$ then the value of $\int_0^1 \vec{F}(t) dt$ is **[Kanpur 2018]**
- (a) $\hat{i}/2 + 2/3\hat{j} + 7/4\hat{k}$
 (b) $\hat{i}/2 - 2/3\hat{j} + 7/4\hat{k}$
 (c) $-1/2\hat{i} + 2/3\hat{j} + 7/4\hat{k}$
 (d) $-1/2\hat{i} - 2/3\hat{j} + 7/4\hat{k}$
133. If $\vec{r} = x\hat{i} + y\hat{j} + 2\hat{k}$, then the value of $\nabla \log |\vec{r}|$ is **[Kanpur 2018]**
- (a) $\frac{\vec{r}}{r}$ (b) $\frac{\vec{r}}{r^2}$
 (c) $\frac{\vec{r}}{r^3}$ (d) None of these
134. The value of grad ϕ , where ϕ is given by $\phi = x^3 - y^3 + xz^2$, at the point $(1, -1, 2)$ is **[Kanpur 2018]**
- (a) $3\hat{i} - 7\hat{j} + 4\hat{k}$
 (b) $-7\hat{i} + 3\hat{j} + 4\hat{k}$
 (c) $-7\hat{i} - 3\hat{j} + 4\hat{k}$
 (d) $7\hat{i} - 3\hat{j} + 4\hat{k}$
135. A unit normal vector to the level surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$ is **[Kanpur 2018]**
- (a) $1/3\hat{i} + 2/3\hat{j} + 2/3\hat{k}$
 (b) $1/3\hat{i} - 2/3\hat{j} + 2/3\hat{k}$
 (c) $-1/3\hat{i} + 2/3\hat{j} + 2/3\hat{k}$
 (d) $1/3\hat{i} + 2/3\hat{j} - 2/3\hat{k}$
136. If $\vec{F} = x^2y\hat{i} - 2x^2z\hat{j} + 2yz\hat{k}$, then $\text{curl } \vec{F}$ is **[Kanpur 2018]**
- (a) $(2x + 2)\hat{i} - (x^2 + 2z)\hat{k}$
 (b) $(2x + 2z)\hat{i} - (x^2 + 2z)\hat{k}$
 (c) $(2x + z)\hat{i} + (x^2 + 2z)\hat{k}$
 (d) $(x + 2z)\hat{i} - (x^2 + 2z)\hat{k}$
137. If $\vec{F} = 2x^2\hat{i} - xy^2z\hat{j} + 3y^2x\hat{k}$ then $\text{div. } \vec{F}$ is **[Kanpur 2018]**
- (a) $2xz(2 - y)$
 (b) $xz(1 - y)$
 (c) $xy(1 - z)$
 (d) $xy(2 - z)$

138. The value of $\text{div.} (r^4 \vec{r})$ is **[Kanpur 2018]**
 (a) $(n+2)r^{4+1}$
 (b) $(n+3)r^{4+2}$
 (c) $(4+3)r^4$
 (d) $(n+2)r^4$
139. The value of $\text{div.} [r \text{ grad } r^{-3}]$ is **[Kanpur 2018]**
 (a) $2r^{-3}$ (b) $3r^4$
 (c) $2r^3$ (d) $3r^{-4}$
140. The value of the line integral $\int (xdy - ydx)$ around the circle $x^2 + y^2 = 1$ is **[Kanpur 2018, Avadh 2018]**
 (a) π (b) 2π (c) 3π (d) 4π
141. The value of $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ and the curve c is the rectangle in the xy -plane bounded by $y = 0, x = a, y = b, x = 0$ is **[Kanpur 2018]**
 (a) $2ab^2$ (b) $2a^2b$
 (c) $-2a^2b$ (d) $-2ab^2$
142. The workdone in moving a particle in a force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + 3\hat{k}$ along the line joining $(0, 0, 0)$ to $(2, 1, 3)$ is **[Kanpur 2018]**
 (a) 12 (b) 16
 (c) 0 (d) 20
143. The value of $\iint_S \vec{F} \cdot \vec{n} ds$, where $\vec{F} = yz\hat{i} + 2x\hat{j} + xy\hat{k}$ and s is the part of surface of the sphere $x^2 + y^2 + z^2 = 1$ which lies in the first octant is **[Kanpur 2018]**
 (a) $3/8$ (b) $3/10$
 (c) $3/11$ (d) $3/13$
144. If $\vec{F} = (2x^2 - 32)\hat{i} - 2xy\hat{j} - 4x\hat{k}$, then the value of $\iiint_V \vec{\nabla} \cdot \vec{F} dv$, where V is the closed region bounded by the planes $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$ is **[Kanpur 2018]**
 (a) $2/3$ (b) $4/3$
 (c) $7/3$ (d) $8/3$
145. By Green's theorem the value of $\oint_C (x^2 - \cos h y)dx + (y + \sin x)dy$, where C is the rectangle with vertices $(0, 0), (\pi, 0), (\pi, 1), (0, 1)$
 (a) $\cos 1 - 1$ (b) $\sin 1 - 1$
 (c) $\cosh 1 - 1$ (d) $\sinh 1 - 1$
146. By Green's theorem the value of $\oint [(\cos x \sin y - xy)dx + \sin x \cos y dy]$, where ρ is the circle $x^2 + y^2 = 1$
 (a) 0 (b) 1
 (c) 2 (d) 3
147. If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, then $\text{curl } \vec{a}$ is equal to **[Avadh 2018]**
 (a) 0 (b) $\vec{0}$
 (c) $\sqrt{14}$ (d) $\vec{a}$
148. The value of $\frac{d}{dt}(\vec{a} \times \vec{b})$ is **(Avadh 2018)**
 (a) $\vec{a} \times \frac{d\vec{b}}{dt} + b \times \frac{d\vec{a}}{dt}$
 (b) $\frac{d\vec{a}}{dt} \times \frac{d\vec{b}}{dt}$
 (c) $\vec{a} \times \frac{d\vec{b}}{dt}$
 (d) $\frac{d\vec{a}}{dt} \times \vec{b} + \vec{a} \times \frac{d\vec{b}}{dt}$
149. If ϕ is a constant function, then the value of $\text{grad } \phi$ is
 (a) $|\text{grad } \phi|$
 (b) zero vector
 (c) zero scalar
 (d) None of these
150. The vector $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k})$ is equal to
 (a) $\vec{0}$ (b) $\vec{a}$
 (c) $2\vec{a}$ (d) None of these
151. The Green's theorem establishes a relation between **[Avadh 2018]**
 (a) Line integral and surface integral in a plane
 (b) Line integral and volume integral

- (c) Surface integral and volume integral
(d) None of these
152. The value of $[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}]$ is **[Avadh 2018]**
 (a) 0 (b) 1
 (c) 2 (d) 3
153. The value of $\nabla^2\left(\frac{1}{r}\right)$ is **[Avadh 2018]**
 (a) $2/r^3$ (b) $-2/r^3$
 (c) 0 (d) None of these
154. The directional derivative of $\phi = xy + yz + zx$ in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$ at $(1, 2, 0)$ **[Avadh 2018]**
 (a) $5/3$ (b) $8/3$
 (c) $10/3$ (d) $7/3$
155. If $\vec{r} \times d\vec{r} = 0$ then unit vector $\hat{r}$ is **[Avadh 2018]**
 (a) constant (b) zero
 (c) one (d) None of these
156. The unit normal vector to the surface $z = x^2 + y^2$ at the point $(1, 1, 2)$ is **[Avadh 2018]**
 (a) $2\hat{i} + 2\hat{j} - \hat{k}$ (b) $2\hat{i} + 2\hat{j} + \hat{k}$
 (c) $\frac{\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{6}}$ (d) $\frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$
157. The formula $\iiint_V \vec{\nabla} \cdot \vec{F} dv = \iint_S \vec{F} \cdot \hat{n} ds$ is governed by **[Avadh 2018]**
 (a) Stoke's theorem
 (b) Gauss's theorem
- (c) Green's theorem
(d) None of these
158. A necessary and sufficient condition that the line integral $\int_C \vec{A} \cdot d\vec{V} = 0$ for any closed curve C is that **[Avadh 2018]**
 (a) $\text{div. } \vec{A} = 0$ (b) $\text{curl } \vec{A} = 0$
 (c) $\text{div. } \vec{A} \neq 0$ (d) $\text{curl } \vec{A} \neq 0$
159. If $\vec{F} = y\hat{i} + (x - 2xz)\hat{j} - xy\hat{k}$, then the value of $\iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} ds$ is where S is the surface of sphere $x^2 + y^2 + z^2 = a^2$ above the xy -plane **[Kanpur 2018]**
 (a) 0 (b) 1
 (c) 2 (d) 3
160. The value of $\iint_S (x^2 + y^2) ds$ is where S is the surface of the cone $z^2 = 3(x^2 + y^2)$ bonded by $z = 0$ and $z = 3$. **[Kanpur 2018]**
 (a) 3π
 (b) 6π
 (c) 9π
 (d) 11π
161. If S denotes the surface of the cube bonded by the planer $x = 0, x = a, y = 0, y = a, z = 0, z = a$, then by the application of Gauss divergence theorem the value of $\iint_S (x\hat{i} + y\hat{j} + z\hat{k}) \cdot \hat{n} ds$ is **[Kanpur 2018]**
 (a) a^3 (b) $2a^3$
 (c) $3a^3$ (d) 0

ANSWERS

MULTIPLE CHOICE QUESTIONS

[illegible]

HINTS AND SOLUTIONS

4. $\vec{r} = \sin t \hat{i} + \cos t \hat{j} + t \hat{k}$

$$\frac{d\vec{r}}{dt} = \cos t \hat{i} - \sin t \hat{j} + \hat{k}$$

and $\frac{d^2\vec{r}}{dt^2} = -\sin t \hat{i} - \cos t \hat{j}$

so $\left| \frac{d^2\vec{r}}{dt^2} \right| = \sqrt{\sin^2 t + \cos^2 t} = 1.$

5. If $\vec{r} = \cos nt \hat{i} + \sin nt \hat{j}$ then

$$\vec{r} \times \frac{d\vec{r}}{dt} = \begin{vmatrix} i & j & k \\ \cos nt & \sin nt & 0 \\ -n \sin nt & n \cos nt & 0 \end{vmatrix}$$

$$= \hat{k} |n \cos^2 nt + n \sin^2 nt| = n \hat{k}$$

11. Here $\vec{r} = e^{-t} \hat{i} + e^t \sin t \hat{j} + \hat{k}$ then velocity

$$\vec{v} = \frac{d\vec{r}}{dt} = -e^{-t} \hat{i} + (e^t \cos t + e^t \sin t) \hat{j}$$

at $t = 0, \vec{v} = -\hat{i} + \hat{j}$ so $v = |\vec{v}| = \sqrt{1+1} = \sqrt{2}$

16. Here $\vec{A} = a\hat{i} + \hat{j} - 2\hat{k}$ & $\vec{B} = 2\hat{i} - \hat{j} + b\hat{k}$

Given $\vec{A} \times \vec{B} = 0$

$$\text{i.e. } \begin{vmatrix} i & j & k \\ a & 1 & -2 \\ 2 & -1 & b \end{vmatrix} = 0$$

$$\Rightarrow i(b-2) + j(-4-ab) + k(-a-2) = 0$$

i.e. $b = 2, ab = -4$

and $a = -2$ so $(a, b) = (-2, 2)$

19. $[\dot{r} \ddot{r} \ddot{r}] = \begin{vmatrix} -a \sin t & a \cos t & a \tan \alpha \\ -a \cos t & -a \sin t & 0 \\ a \sin t & -a \cos t & 0 \end{vmatrix}$

$$= a \tan \alpha (a^2) = a^3 \tan \alpha$$

23. Here $f = 3x^2y - y^3z^2$

$$\therefore \text{grad } f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z}$$

$$\therefore \nabla f = i(6xy) + j(3x^2 - 3y^2z^2) + k(-2y^3z)$$

At $(1, 1, 1), \nabla f = 6\hat{i} - 2\hat{k}$

31. Let $f = x^2 + y - z - 1$, normal vector is ∇f

$$\text{so } \nabla f = i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} = 2x\hat{i} + \hat{j} - \hat{k}$$

At $(1, 0, 0), \nabla f = 2\hat{i} + \hat{j} - \hat{k}$, unit vector is $\frac{2\hat{i} + \hat{j} - \hat{k}}{\sqrt{6}}$

32. The directional derivative is $\nabla f \cdot \hat{a}$

Here $\text{grad } f = \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$

$$\nabla f = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}$$

At $(1, -2, -1) \nabla f = 8\hat{i} - \hat{j} - 10\hat{k}$

and $\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3}$

So directional derivative

$$= (8\hat{i} - \hat{j} - 10\hat{k}) \cdot \left(\frac{2\hat{i} - \hat{j} - 2\hat{k}}{3} \right) = \frac{37}{3}$$

34. $f = x^2 + 2y^2 + 3z^2 - 12$

$$\text{grad } f = \nabla f = \hat{i}(2x) + \hat{j}(4y) + \hat{k}(6z)$$

At $(1, 2, -1), \nabla f = 2\hat{i} + 8\hat{j} - 6\hat{k}$

equation of tangent is $(R\vec{r}) \cdot \text{grad } f = 0$

where $R = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{r} = \hat{i} + 2\hat{j} - \hat{k}$

So we get

$$[(x-1)\hat{i} + (y-2)\hat{j} + (z+1)\hat{k}] \cdot (2\hat{i} + 8\hat{j} - 6\hat{k}) = 0$$

$$\text{i.e. } 2(x-1) + 8(y-2) - 6(z+1) = 0$$

$$\text{i.e. } 2x + 8y - 6z = 24 \Rightarrow x + 4y - 3z = 12$$

$$44. \quad \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$\begin{aligned} \text{Curl } (\nabla f) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} = \hat{i} \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \\ &+ \hat{j} \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) + \hat{k} \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) = 0 \end{aligned}$$

$$46. \quad \vec{A} \text{ is solenoidal then } \text{div. } \vec{A} = 0$$

$$\text{Thus } \frac{\partial}{\partial x}(ax + 2y) + \frac{\partial}{\partial y}(2y - 3) + \frac{\partial}{\partial z}(-3z) = 0$$

$$\text{i.e. } a + 2 - 3 = 0 \Rightarrow a = 1$$

$$47. \quad \vec{A} \text{ is irrotational then } \text{curl } \vec{A} = 0$$

$$\text{Thus } \text{curl } A = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ay & -3z & 2y \end{vmatrix} = 0$$

$$\text{i.e. } \hat{i}(2+3) + \hat{j}(0-0) + \hat{k}(0-a) = 0$$

$$54. \quad \vec{a} \cdot \vec{b} = 2t^3 + (6t^2 - 6t)$$

$$\therefore \int_0^2 \vec{a} \cdot \vec{b} = \int_0^2 (2t^3 + 6t^2 - 6t) dt$$

$$= \left(\frac{t^4}{2} + 2t^3 - 3t^2 \right)_0^2$$

$$= (8 + 16 - 12) = 12$$

$$58. \quad \int_c \vec{F} \cdot d\vec{r} = \int (x dx + y dy)$$

$$\therefore y = x \text{ so } dy = dx$$

$$= \int_0^1 x dx + \int_0^1 1 dx = \int_0^1 (1+x) dx$$

$$= \left[x + \frac{x^2}{2} \right]_0^1 = 1 + \frac{1}{2} = \frac{3}{2}$$

$$60. \quad \text{For circle } x^2 + y^2 = 1, x = \cos t, y = \sin t$$

$$dx = -\sin t dt, dy = \cos t dt \text{ so}$$

$$\begin{aligned} \int x dy - y dx &= \int_0^{2\pi} (\cos t \cdot \cos t dt + \sin t \cdot \sin t dt) \\ &= \int_0^{2\pi} dt = 2\pi \end{aligned}$$

$$64. \quad \text{The circulation is } \int \vec{F} \cdot d\vec{r} = \int y dx + z dy + x dz$$

$$\text{where curve is } x^2 + y^2 = 1, z = 0$$

$$\text{so } x = \cos t, y = \sin t, z = 0$$

$$\text{i.e. } dx = -\sin t dt, dy = \cos t dt, dz = 0$$

$$\begin{aligned} \text{So } \int \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} (-\sin^2 t dt + 0 + 0) \\ &= -\int_0^{2\pi} \sin^2 t dt = -\pi \end{aligned}$$

$$66. \quad \text{Over a plane face } z = 0, \text{ normal vector is } -\hat{k} \text{ so}$$

$$\iint \vec{F} \cdot \hat{n} ds = \iint (4x\hat{i} - 2y^2\hat{j}) \cdot (-\hat{k}) ds = \iint 0 ds = 0$$

$$70. \quad \therefore y = x^2 \text{ so } dy = 2x dx$$

$$\begin{aligned} \int_c (\vec{F} \cdot d\vec{r}) &= \int_c (2x\hat{i} - y\hat{j}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int (2x dx - y dy) \end{aligned}$$

$$\begin{aligned}
 &= \int 2x \, dx - x^2 2x \, dx \\
 &= \int_0^2 (2x - 2x^3) \, dx = -4
 \end{aligned}$$

74. The work done $w = \int \vec{F} \cdot d\vec{r}$

$$= \int [3x^2 dx + (2xz - y)dy + 3dz]$$

(2, 1, 3)

Line is $\frac{x}{2} = \frac{y}{1} = \frac{z}{3}$ (0, 0, 0)

$$\therefore y = \frac{x}{2}, z = \frac{3}{2}x$$

i.e. $dy = \frac{1}{2}dx, dz = \frac{3}{2}dx$

$$\therefore \int \vec{F} \cdot d\vec{r} = \int_0^2 \left[3x^2 dx + \left(3x^2 - \frac{x}{2} \right) \frac{dx}{2} + \frac{9}{2} dx \right] = 16$$

81. $\iiint \text{div} \cdot \vec{F} \, dv = \int_0^a \int_0^b \int_0^c 3 \, dx \, dy \, dz = 3abc$

83. $\iiint_s \vec{F} \cdot \hat{n} \, ds = \iiint \nabla \cdot \vec{F} \, dv$
(by Gauss divergence theorem)

$$= \iiint \left(\frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(2xy) \right) = 0$$

85. $\iint \vec{F} \cdot \hat{n} \, ds = \iiint \text{div} \cdot \vec{F} \, dv$
(By Gauss divergence theorem)

$$\begin{aligned}
 \text{so } \iint_s \vec{F} \cdot \hat{n} \, ds &= \iiint (1 + 0 + 0) \, dx \, dy \, dz \\
 &= \int_0^1 \int_0^1 \int_0^1 dx \, dy \, dz = 1
 \end{aligned}$$

92. By Stoke's theorem $\int_c \vec{F} \cdot d\vec{r} = \iint_s \text{curl} \vec{F} \cdot \hat{n} \, ds$

$$\text{where } \text{curl} \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix} = \hat{k}$$

$$\text{So } \int_c \vec{F} \cdot d\vec{r} = \iint_s \hat{k} \cdot \hat{k} \, ds = \iint ds = \pi$$

94. By Stoke's theorem $\int_c \vec{F} \cdot d\vec{r} = \iint_s \text{curl} \vec{F} \cdot \hat{n} \, ds$

$$\text{curl} \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & xy & 0 \end{vmatrix} = y\hat{k}$$

$$\text{So } \int_c \vec{F} \cdot d\vec{r} = \iint_s y\hat{k} \cdot \hat{k} \, ds = \int_0^1 \int_0^1 y \, dx \, dy = \frac{1}{2}$$

99. Here $\text{curl} \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & zx & xy \end{vmatrix} = 0$

$$\text{So by Stoke's theorem } \oint_c \vec{F} \cdot d\vec{r} = \iint_s \text{curl} \vec{F} \cdot \hat{n} \, ds = 0$$

104. $\therefore \oint (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy$

$$M = (\cos x \sin y - y), N = \sin x \cos y \hat{j}$$

$$\frac{\partial M}{\partial y} = \cos x \cos y - x, \frac{\partial N}{\partial x} = \cos x \cos y$$

$$\text{Hence } \iint_R dx \, dy = \int_0^{2\pi} \int r \cos \theta r d\theta dr = 0$$

108. Since $\vec{F}$ is irrotational $\text{curl} \vec{F} = 0$

$$\text{curl} \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x - 3y + az & bx + 2y - 4z & 2x + cy + z \end{vmatrix}$$

$$= 0$$

$$\hat{i}(c + 4) + \hat{j}(a - 2) + \hat{k}(b + 3) = 0$$

$$\Rightarrow a = 2, b = -3, c = -4$$

111. By Gauss divergence theorem

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint \text{div} \cdot \vec{F} dv$$

$$\text{div} \cdot \vec{F} = 1 + 2 + 3 = 6$$

$$\text{So } \iint_S \vec{F} \cdot \hat{n} ds = \iiint 6 dv = 6v$$

114. $\therefore \text{div } \vec{A} = e^x \sin y - e^x \sin y = 0$

$$\text{and } \text{curl } \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x \sin y & e^x \cos y & 0 \end{vmatrix} = 0$$

So $\vec{A}$ is both solenoidal and irrotational.

115. By Stoke's theorem $\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} ds$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x & 2y & 0 \end{vmatrix}$$

$$= \hat{i}(0) + \hat{j}(0 - 0) + \hat{k}(0 - 0) = 0$$

$$\text{So } \int_C \vec{F} \cdot d\vec{r} = \iint_S 0 ds = 0.$$

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BRA University, Agra, Examination-2019

Paper-III Geometry and Vector Calculus

[Time : 2 Hours]

[Maximum Marks : 70]

Note : Attempt all questions. Each question carries equal marks.

1. The greater rate of increase of $\phi = xyz^2$ at $(1, 0, 3)$ is :
(a) 0 (b) 1 (c) 3 (d) 9
2. The angle between the surface $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 + z - 3 = 0$ at the point $(2, -1, 2)$ is :
(a) $\cos^{-1}\left(\frac{4}{3\sqrt{21}}\right)$ (b) $\cos^{-1}\left(\frac{8}{5\sqrt{21}}\right)$
(c) $\cos^{-1}\left(\frac{2}{3\sqrt{21}}\right)$ (d) $\cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$
3. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{div } \vec{r}$ is equal to :
(a) 0 (b) 3
(c) Can't find (d) None of these
4. If $\vec{R} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$ then the value of $\text{divgrad}\left(\frac{1}{r}\right)$ is :
(a) 0 (b) 1
(c) $\frac{2}{r}$ (d) None of these
5. For what value of 'a' so that the vector $\vec{f} = (ax + 2y + 4z)\hat{i} + (2x - 3y + 3z)\hat{j} + (3x + 2y - 2z)\hat{k}$ is solenoidal :
(a) 3 (b) 4 (c) 5 (d) 6
6. If $\vec{f} = (x + y + 1)\hat{i} + \hat{j} + (-x - y)\hat{k}$, then $\text{Curl } \vec{f}$ is :
(a) $\hat{i} + \hat{j} + \hat{k}$ (b) $-\hat{i} - \hat{j} + \hat{k}$
(c) $-\hat{i} + \hat{j} - \hat{k}$ (d) $-\hat{i} - \hat{j} - \hat{k}$
7. If $\phi = x^2y + y^2z + z^2x$, then $\text{Curl grad } \phi$ is
(a) $2(z\hat{i} + x\hat{j} + y\hat{k})$
(b) $y\hat{i} + (4xz - 3z^3)\hat{j} + 2z\hat{k}$
(c) $2x\hat{i} + 3y\hat{j} + 2z\hat{k}$
(d) 0
8. For what value of a, b, c such that $\vec{f} = (6xy + az^3)\hat{i} + (bx^2 - z)\hat{j} + (3xz^2 + cy)\hat{k}$ is irrotational?
(a) $a = 1, b = -1, c = 3$
(b) $a = 1, b = +3, c = -1$
(c) $a = -1, b = 3, c = -1$
(d) $a = -1, b = -3, c = -1$
9. If $\vec{F} = x^2y\hat{i} + y^2z\hat{j} + z^2x\hat{k}$, then $\text{Curl } \vec{F}$ is
(a) $2(z\hat{i} + x\hat{j} + y\hat{k})$ (b) $z\hat{i} + x\hat{j} + y\hat{k}$
(c) $3(z\hat{i} + x\hat{j} + y\hat{k})$ (d) None of these
10. If $\vec{F} = x^2\hat{i} + xy\hat{j}$, then the value of $\int_c \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, 1)$ along the straight line joining these points :
(a) 1 (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{4}{3}$
11. The value of the integral $\int \int_S \vec{F} \cdot \hat{n} ds$ on the yz -plane is :
(a) $\int \int_S \vec{F} \cdot \hat{n} \frac{dydz}{[\hat{n} \cdot \hat{j}]}$ (b) $\int \int_S \vec{F} \cdot \hat{n} \frac{dydz}{[\hat{n} \cdot \hat{i}]}$
(c) $\int \int_S \vec{F} \cdot \hat{n} \frac{dydz}{[\hat{n} \cdot \hat{k}]}$ (d) None of these

12. By Stoke's theorem, $\int_C \vec{F} \cdot d\vec{r}$ is equal to :
- (a) $\int_C (\nabla \times \vec{F}) \cdot \hat{n} d\vec{r}$ (b) $\int \int_S (\nabla \times \vec{F}) \cdot \hat{n} ds$
 (c) $\int \int_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} ds$ (d) None of these
13. By Gauss divergence theorem, the value of $\int_S (ax \hat{i} + by \hat{j} + cz \hat{k}) \cdot \hat{n} ds$ is :
- (a) $\frac{2\pi}{3}(a+b+c)$ (b) $\frac{4\pi}{3}(a+b)$
 (c) $\frac{4\pi}{3}(b+c)$ (d) $\frac{4\pi}{3}(a+b+c)$
14. By Green's theorem, the value of $\oint_C (2x-y) dx + (x+3y) dy$ where C is $\frac{x^2}{4} + \frac{y^2}{1} = 1$:
- (a) π (b) 2π (c) 4π (d) 6π
15. The work done in moving a particle in a force field $\vec{F} = 3x^2 \hat{i} + (2xz-y) \hat{j} + 3 \hat{k}$ along the line joining $(0, 0, 0)$ to $(2, 1, 3)$ is
- (a) 0 (b) 8 (c) 16 (d) 20
16. The equation $\frac{l}{r} = 1 + e \cos \theta$ and $\frac{l}{r} = -1 + e \cos \theta$ represent :
- (a) Different conics
 (b) Two straight lines
 (c) Tangents of each other
 (d) The same conic
17. The line $\frac{l}{r} = A \cos \theta + B \sin \theta$ touches the conic $\frac{l}{r} = 1 + e \cos \theta$, if :
- (a) $(A-e)^2 + B^2 = 1$ (b) $(A+e)^2 + B^2 = 1$
 (c) $(A+e)^2 - B^2 = 1$ (d) None of these
18. The polar equation $\frac{4}{r} = 3 + 4 \cos \theta + 3 \sin \theta$ represents :
- (a) An ellipse (b) An parabola
 (c) A hyperbola (d) A circle
19. PSP' is the focal chord of the conic $\frac{\ell}{r} = 1 + e \cos \theta$, then the value of $\frac{1}{SP} + \frac{1}{SP'}$ is :
- (a) 1 (b) 0
 (c) $\frac{1}{\ell}$ (d) $\frac{2}{\ell}$
20. The equation of tangent at a point ' α ' on the conic $\frac{\ell}{r} = 1 + e \cos \theta$ is :
- (a) $\frac{\ell}{r} = e \sin \theta + \sin (\theta - \alpha)$
 (b) $\frac{\ell}{r} = e \cos \theta + \cos (\theta - \alpha)$
 (c) $\frac{\ell}{r} = \cos \theta + e \cos (\theta - \alpha)$
 (d) None of the above
21. If S is a pole, SS' is the initial line, $SS' = 2ae$ and a point P moves in such a way that $SP + S'P = 2a$, where $0 < e < 1$, then the locus of P is :
- (a) $\frac{a(1-e^2)}{r^2} = 1 - e \cos \theta$
 (b) $\frac{a(1-e^2)}{r^2} = 1 + e \cos \theta$
 (c) $\frac{a(1-e^2)}{r} = (1 + e \cos \theta)^2$
 (d) $\frac{a(1-e^2)}{r} = (1 - e \cos \theta)$
22. If the normals at the points whose vectorial angles are α, β and γ on the parabola $\frac{\ell}{r} = 1 + \cos \theta$ meet in a point (ρ, ϕ) then :
- (a) $\phi = \alpha + \beta + \gamma$ (b) $2\phi = \alpha + \beta + \gamma$
 (c) $3\phi = \alpha + \beta + \gamma$ (d) $4\phi = \alpha + \beta + \gamma$
23. The curve represented by the equation $r = 2a \cos (\theta - \alpha)$ is :
- (a) A circle (b) A parabola
 (c) An ellipse (d) A hyperbola
24. The latus rectum of the conic $(\operatorname{cosec} \alpha + \cos \theta) = 1$ is :
- (a) $2 \cos \alpha$ (b) $2 \operatorname{cosec} \alpha$
 (c) $2 \sin \alpha$ (d) $\sin \alpha$

25. The equation $\sqrt{ax} + \sqrt{by} = 1$ represents :
- (a) Ellipse
(b) Pair of straight line
(c) Parabola
(d) Pair of perpendicular straight line
26. The centre of the conic $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$ is :
- (a) (2, 2) (b) (3, 2)
(c) (3, 3) (d) (2, 3)
27. The following equation represents : $3x^2 + 12xy + 12y^2 + 435x - 9y + 72 = 0$ is:
- (a) Ellipse (b) Parabola
(c) Hyperbola (d) Circle
28. The confocal conics of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is :
- (a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (b) $\frac{x^2}{a\lambda} + \frac{y^2}{b\lambda} = 1$
(c) $\frac{x^2}{a^2\lambda^2} + \frac{y^2}{b^2\lambda^2} = 1$ (d) None of these
29. Confocals cut at :
- (a) Right angle (b) Acute angle
(c) Obtuse angle (d) None of these
30. The length of the major axis of the conic $22x^2 - 12xy + 17y^2 - 112x + 92y + 178 = 0$ is:
- (a) 2 (b) $2\sqrt{2}$
(c) $\sqrt{2}$ (d) $3\sqrt{2}$
31. If the coordinates of the points A, B, C and D are (3, 4, 5), (4, 6, 3), (-1, 0, 4) and (1, 2, 5) respectively the projection of CD on AB is :
- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$
32. Direction cosines of the lines equally inclined to the coordinates axes, are :
- (a) $\left(\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}\right)$ (b) $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
(c) $\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$ (d) (1, 1, 1)
33. The angle between two lines whose direction ratios are : (1, 1, 2) and $(\sqrt{3} - 1, -\sqrt{3} - 1, 4)$ is :
- (a) 30° (b) 45°
(c) 90° (d) None of these
34. If a line makes angles α, β, γ with the coordinates axes, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ is equal to :
- (a) 0 (b) 1
(c) 2 (d) 3
35. The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $2lm + 2nl - mn = 0$ is :
- (a) 60°
(b) 90°
(c) 120°
(d) 150°
36. The shortest distance between the lines $\frac{x - 1/2}{1} = \frac{y - 1/3}{2} = \frac{z - 1/4}{3}$ and $\frac{3y - 1}{3} = \frac{4z - 1}{4} = \frac{2x - 1}{2}$ is :
- (a) 0 (b) $\frac{4}{5}$
(c) $\frac{4}{\sqrt{3}}$ (d) None of these
37. The shortest distance between the z-axis and the line $x + y + 2z - 3 = 0, 2x + 3y + 4z - 4 = 0$ is :
- (a) 1 (b) 2
(c) 3 (d) 4
38. The lines $x = py + q, z = ry + s$ and $x = p'y + q', z = r'y + s'$ are perpendicular if the value of $pp' + rr'$ is equal to :
- (a) 0 (b) 1
(c) -1 (d) -2

39. A plane cuts the axes in A, B, C and the centroid of the triangle ABC is (a, b, c) . The equation of plane is :
- (a) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (b) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 2$
 (c) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$ (d) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0$
40. The shortest distance between the lines $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$ and $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$ is :
- (a) 81 (b) 9
 (c) 27 (d) 1
41. The straight line $x = 2y = \lambda z$ is parallel to the plane $3x - 4y + 5z - 10 = 0$ if λ is equal to :
- (a) 0 (b) -3
 (c) -4 (d) -5
42. The equation of the plane passing through the point $(1, -2, 3)$ and perpendicular line $\frac{x-2}{3} = \frac{y-4}{5} = \frac{z-6}{7}$ is :
- (a) $3x - 5y + 7z = 0$ (b) $3x - 5y + 7z = 8$
 (c) $3x - 5y + 7z = 14$ (d) None of these
43. The distance of the origin from the plane $2x + 6y - 3z + 7 = 0$ is :
- (a) 0 (b) 1
 (c) 2 (d) 3
44. If the lines $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$ lies on the plane $2x - 4y + z = 7$, then the value of K is :
- (a) 7 (b) -7
 (c) 0 (d) None of these
45. The acute angle between two diagonals of a cube is :
- (a) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$
 (c) $\cos^{-1}\left(\frac{-1}{3}\right)$ (d) $\cos^{-1}\left(\frac{1}{3}\right)$
46. If the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are perpendicular to each other then :
- (a) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (b) $a_1a_2 + b_1b_2 + c_1c_2 = 0$
 (c) $a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2 = 0$ (d) None of the above
47. The angle between the planes $3x + 4y - 5z = 9$ and $2x + 6y + 6z = 7$ is :
- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{2}$ (d) None of these
48. If the cone $x^2 + y^2 - z^2 \tan^2 \alpha = 0$ has three mutually perpendicular generators, then α equals :
- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{4}$ (d) $\tan^{-1} \sqrt{2}$
49. If the equation $\alpha x^2 + \beta y^2 + 2z^2 - 2\alpha x - 4\beta y - 40 = 0$ represents a sphere, its radius is :
- (a) $2\sqrt{30}$ (b) $2\sqrt{10}$
 (c) $3\sqrt{5}$ (d) 5
50. If the plane $lx + my + nz = p$ touches the sphere $x^2 + y^2 + z^2 = a^2$ then $l^2 + m^2 + n^2$ is equal to :
- (a) $\frac{p^2}{a^2}$ (b) p^2
 (c) a^2 (d) $\frac{a^2}{p^2}$
51. The angle between the radical plane of two spheres and the line joining their centres is :
- (a) 0° (b) 30°
 (c) 90° (d) None of these
52. The equation $ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + d = 0$ represents a cone if :
- (a) $d = au^2 + bv^2 + cw^2$ (b) $d = \frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c}$
 (c) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} + d = 0$ (d) $au^2 + bv^2 + cw^2 + d = 0$

53. The equation $\frac{a}{y-z} + \frac{b}{z-x} + \frac{c}{x-y} = 0$ represents
a :
(a) Cylinder (b) Cone
(c) Pair of planes (d) Sphere
54. If a tangent plane to the sphere $x^2 + y^2 + z^2 = r^2$ makes intercepts a, b, c on the coordinate axes, then
(a) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{r}$ (b) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{r^2}$
(c) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = r$ (d) None of these
55. The vertex of the cone $(x-1)^2 + (y-3z)(2y+z) - (x-1)(4y+6z) = 0$ is :
(a) $(-1, 0, 0)$ (b) $(1, 0, 0)$
(c) $(-1, 2, 3)$ (d) $(1, 2, -3)$
56. The plane $\alpha x + \beta y + \gamma z = 0$ cuts the cone $yx + zx + xy = 0$ in perpendicular generators if :
(a) $\frac{\alpha}{\beta} + \frac{\beta}{\gamma} + \frac{\gamma}{\alpha} = 0$ (b) $\alpha + \beta + \gamma = 0$
(c) $\beta\gamma + \gamma\alpha + \alpha\beta = 0$ (d) None of these
57. The radius of the smallest spheres passing through the points $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ is :
(a) $\sqrt{\frac{2}{3}}$ (b) $\frac{2}{3}$
(c) $\frac{\sqrt{3}}{2}$ (d) 1
58. Two spheres of radii 3 and 4 cut orthogonally. The radius of common circle is :
(a) 12 (b) $\frac{\sqrt{12}}{5}$
(c) $\sqrt{12}$ (d) $\frac{12}{5}$
59. The pole of the plane $3x + 4y + 5z = 6$ with respect to the sphere $x^2 + y^2 + z^2 = 12$ is :
(a) $(1, 1, 0)$ (b) $(0, 0, 0)$
(c) $(6, 8, 10)$ (d) $(3, 4, 5)$
60. The number of radical planes and radical centres of n spheres are equal if n is equal to :
(a) 6 (b) 12
(c) 18 (d) 24
61. The vertex of the cone $xy + 2yz + 3zx = 0$ is the point :
(a) $(-1, 0, 0)$ (b) $(0, 0, 0)$
(c) $(-1, -2, 0)$ (d) $(-1, 2, 2)$
62. The equation of the right circular cylinder whose axis is z -axis and radius ' a ' is:
(a) $y^2 + z^2 = a^2$ (b) $z^2 + x^2 = a^2$
(c) $x^2 + y^2 = a^2$ (d) $x^2 + y^2 + z^2 = a^2$
63. The equation of the cylinder with generator parallel to x -axis and passing through the circle $x^2 + y^2 + z^2 = 9, 2x = y + z$ is :
(a) $5y^2 + 2yz + 4z^2 = 36$
(b) $5y^2 + 2yz + 5z^2 = 36$
(c) $x^2 + z^2 + 2yz = 81$
(d) $3y^2 + z^2 + 2yz = 81$
64. Which one of the following equations represents a cone not having three mutually perpendicular generators :
(a) $xy + 2yz + 3zx = 0$ (b) $xy + 2yz - 3zx = 0$
(c) $x^2 + 2y^2 - 3z^2 = 0$ (d) $x^2 - 2y^2 + 3z^2 = 0$
65. Two spheres $x^2 + y^2 + z^2 = 25$ and $x^2 + y^2 + z^2 - 24x - 40y - 18z + 225 = 0$:
(a) Touch internally
(b) Touch externally
(c) Intersect in two points
(d) Do not intersect
66. The number of normals that can be drawn to the Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ from any given point (α, β, γ) is :
(a) 2 (b) 4
(c) 6 (d) 8

67. The equation of the normal to the conicoid $ax^2 + by^2 + cz^2 = 1$ at (α, β, γ) is :
- (a) $\frac{x - \alpha}{a\alpha} = \frac{y - \beta}{b\beta} = \frac{z - \gamma}{c\gamma}$
 (b) $a\alpha x + b\beta y + c\gamma z = 1$
 (c) $\frac{x - \alpha}{\alpha/a^2} = \frac{y - \beta}{\beta/b^2} = \frac{z - \gamma}{\gamma/c^2}$
 (d) None of the above
68. The equation of tangent plane to the central conicoid $5x^2 - 3y^2 + z^2 + 2 = 0$ at the point $(1, 1, 0)$ is :
- (a) $5x - 3y + 2z + 2 = 0$
 (b) $5x - 3y + 2 = 0$
 (c) $5x + 2z + 2 = 0$
 (d) $-3y + 2z + 2 = 0$
69. The centre of the central conicoid $ax^2 + by^2 + cz^2 = 1$ is :
- (a) $(0, 0, 0)$ (b) $(0, 1, 0)$
 (c) $(0, 0, 1)$ (d) $(1, 0, 0)$
70. The equation of director sphere of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is :
- (a) $x^2 + y^2 + z^2 = a^2 + b^2 + c^2$
 (b) $x^2 + y^2 + z^2 = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$
 (c) $x^2 + y^2 + z^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$
 (d) None of these
71. The condition that the plane $x \cos \alpha + y \cos \beta + z \cos \gamma = p$ touches the Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is :
- (a) $a \cos \alpha + b \cos \beta + c \cos \gamma = p^2$
 (b) $a^2 \cos^2 \alpha + b^2 \cos^2 \beta + c^2 \cos^2 \gamma = p^2$
 (c) $a \cos \alpha + b \cos \beta + c \cos \gamma = p$
 (d) None of these
72. The surface represented by the equation $5(x^2 + y^2) = -6z$ is a :
- (a) Parabolic cylinder
 (b) Elliptic paraboloid
 (c) Right circular cylinder of revolution
 (d) Paraboloid of revolution
73. If the section of conicoid $ax^2 + by^2 + cz^2 = 1$ by the plane $lx + my + nz = 0$ is hyperbola, then its section by the plane $lx + my + nz = p$ is :
- (a) A hyperbola (b) A circle
 (c) A ellipse (d) Pair of straight line
74. The surface represented by the equation $2x^2 + 5y^2 - 9z^2 = 1$ is :
- (a) An ellipsoid
 (b) A hyperboloid of one sheet
 (c) Elliptic paraboloid
 (d) Elliptic cylinder
75. If $\vec{r} = 3t^2 \hat{i} + (t^2 - 2t) \hat{j} + t^3 \hat{k}$ then the value of $\frac{d^2 \vec{r}}{dt^2}$ at $t = 1$ is :
- (a) $6\hat{i} + 2\hat{j} - 6\hat{k}$ (b) $6\hat{i} - 2\hat{j} + 6\hat{k}$
 (c) $6\hat{i} + 2\hat{j} + 6\hat{k}$ (d) None of these
76. If $\vec{r} = a \cos \lambda t + b \sin \lambda t$, where a and b are constant vectors, then $\frac{d^2 \vec{r}}{dt^2} + \lambda^2 \vec{r}$ is equal to :
- (a) $-\lambda^2 \vec{r}$ (b) $\lambda^3 \vec{r}$
 (c) $\lambda^4 \vec{r}$ (d) 0
77. If $\vec{r} = t \hat{i} - t^2 \hat{j} + (t - 1) \hat{k}$ and $\vec{S} = 2t^2 \hat{i} + 6t \hat{j}$, then the value of $\int_0^2 (\vec{r} \cdot \vec{S}) dt$ is :
- (a) 0 (b) 4
 (c) 8 (d) 12

78. If $\vec{r} = 5t^2\hat{i} + t\hat{j} - t^3\hat{k}$, then the value of

$$\int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2} \right) dt \text{ is :}$$

(a) $14\hat{i} + 75\hat{j} + 15\hat{k}$ (b) $14\hat{i} - 75\hat{j} + 15\hat{k}$

(c) $-14\hat{i} + 75\hat{j} - 15\hat{k}$ (d) $-14\hat{i} - 75\hat{j} - 15\hat{k}$

79. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and

$$r = |\vec{r}| = (x^2 + y^2 + z^2)^{1/2}$$

then the value of $\text{grad } \frac{1}{r}$ is :

(a) $\frac{\vec{r}}{r}$

(b) $-\frac{\vec{r}}{r^3}$

(c) $\frac{\vec{r}}{r^2}$

(d) None of these

80. Find the directional derivative of $\frac{1}{r}$ in the direction of

$\vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is :

(a) $-\frac{1}{r^2}$

(b) $-\frac{1}{r^3}$

(c) $-\frac{1}{r^4}$

(d) None of these

ANSWERS

1.	(c)	2.	(b)	3.	(d)	4.	(c)	5.	(c)	6.	(c)	7.	(d)	8.	(a)	9.	(a)	10.	(d)
11.	(b)	12.	(c)	13.	(d)	14.	(c)	15.	(c)	16.	(d)	17.	(a)	18.	(a)	19.	(d)	20.	(b)
21.	(d)	22.	(b)	23.	(a)	24.	(a)	25.	(d)	26.	(b)	27.	(b)	28.	(c)	29.	(a)	30.	(b)
31.	(b)	32.	(b)	33.	(d)	34.	(c)	35.	(b)	36.	(c)	37.	(d)	38.	(c)	39.	(c)	40.	(b)
41.	(a)	42.	(d)	43.	(b)	44.	(b)	45.	(d)	46.	(c)	47.	(b)	48.	(d)	49.	(b)	50.	(a)
51.	(c)	52.	(b)	53.	(c)	54.	(b)	55.	(d)	56.	(c)	57.	(c)	58.	(d)	59.	(d)	60.	(a)
61.	(d)	62.	(c)	63.	(a)	64.	(c)	65.	(a)	66.	(c)	67.	(a)	68.	(c)	69.	(a)	70.	(b)
71.	(b)	72.	(c)	73.	(d)	74.	(b)	75.	(c)	76.	(d)	77.	(c)	78.	(c)	79.	(b)	80.	(a)

MULTIPLE CHOICE QUESTIONS

HINTS AND SOLUTIONS

$$\begin{aligned}
 3. \quad \cos \alpha &= \frac{(\sqrt{3}-1) + 1(\sqrt{3}-1) + 2 \times 4}{\sqrt{1+1+4} \sqrt{(\sqrt{3}-1)^2 + (-\sqrt{3}-1)^2 + 16}} \\
 &= -\frac{-1 + \sqrt{3} - \sqrt{3} - 1 + 8}{\sqrt{6} \sqrt{3+1-2\sqrt{3}+3+1+2\sqrt{3}+16}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{6}{\sqrt{6}\sqrt{24}} = \frac{1}{2} = 0 \\
 \theta &= \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}
 \end{aligned}$$

5. Vector $\vec{F}$ will be solenoidal then $\nabla \cdot \vec{F} = 0$ or

$$\left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right) \cdot \{ (dx + 2y + 4z) i + (2x - 3y + 3z) j + (3x + 2y - 2z) k \} = 0$$

$$\Rightarrow a - 3 - 2 = 0 \Rightarrow a = 5$$

9. $\text{Curl } \vec{F} = \nabla \times F = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x^2 y & y^2 z & z^2 x \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix}$

$$= -y^2 \hat{i} - z^2 \hat{j} - x^2 \hat{k}$$

$$\text{Curl } \text{Curl } F = \nabla \times (\nabla \times F) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & -z^2 & -x^2 \end{vmatrix}$$

$$= \hat{i} \{0 + (2z)\} + \hat{j} \{0 + 2x\} + \hat{k} \{0 + 2y\}$$

$$= 2(3\hat{i} + x\hat{j} + y\hat{k})$$

66. $\frac{l}{r} = 1 + e \cos \theta \quad \dots(1)$

$$\frac{l}{r} = -1 + e \cos \theta \quad \dots(2)$$

Let $p(r_1, \theta_1)$ for any point on the curve (1) so that

$$\frac{l}{r_1} = 1 + e \cos \theta_1$$

$$\frac{l}{-r_1} = -1 + e \cos (\theta_1 + \pi)$$

$$\text{i.e., } \frac{l}{r_1} = 1 + e \cos \theta_1$$

Thus every point p on the curve (1) also lies on the curve (2). Hence the equation (1) and (2) represent the same conic

67. Suppose the line

$$\frac{l}{r} = A \cos \theta + B \sin \theta \quad \dots(1)$$

is a tangent to the conic

$$\frac{l}{r} = 1 + e \cos \theta \quad \dots(2)$$

at the point whose vectorial angle is α . The equation of the tangent to (2) at the point ' α ' is

$$\frac{l}{r} = \cos (\theta - \alpha) + e \cos \theta$$

$$\frac{l}{r} = (e + \cos \alpha) \cos \theta + \sin \theta \cdot \sin \alpha$$

the equation (1) and (3) should represent the same line so comparing the coefficients of $\frac{1}{r}$, $\cos \theta$ and

$$\sin \theta \text{ we have } 1 = \frac{e + \cos \alpha}{A} = \frac{\sin \alpha}{B}$$

$$\cos \alpha = A - e \text{ and } \sin \alpha = B$$

$$\text{Squaring and adding, we have } (A - e)^2 + B^2 = 1$$

86. When you will draw an ellipse and a hyperbola in which the focus and incident point has to be same, the angle between the tangents at the incident point will be 90° which is right angle.

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Allahabad State University, Examination-2019

Paper-III Analytical Geometry

[Time : 2 Hours]

[Maximum Marks : 50]

Note : All questions are **compulsory**. This question paper consists of 80 objective questions.

1. The distance between the points $(2, 90^\circ)$ and $(3, 30^\circ)$ is :
(a) 7 (b) $\sqrt{7}$
(c) 13 (d) $\sqrt{13}$
2. The perpendicular distance from the pole to the line $\frac{10}{r} = 3 \cos \theta + 4 \sin \theta$ is :
(a) 1 (b) 2 (c) 3 (d) 5
3. The cartesian equation of polar equation $r^2 = a^2 \cos 2\theta$ is :
(a) $x^2 - y^2 = a^2(x^2 + y^2)$
(b) $x^2 + y^2 = a^2(x^2 - y^2)$
(c) $(x^2 - y^2)^2 = a^2(x^2 + y^2)$
(d) $(x^2 + y^2)^2 = a^2(x^2 - y^2)$
4. If (x, y) and (r, θ) be the cartesian and polar coordinates of a point, then :
(a) $x = r \sin \theta, y = r \cos \theta$
(b) $x = r \cos \theta, y = r \sin \theta$
(c) $r = x \sin \theta, r = y \cos \theta$
(d) $r = x \cos \theta, r = y \sin \theta$
5. The polar co-ordinates of the point whose cartesian co-ordinates are $(1, \sqrt{3})$ are :
(a) $(2, \pi/3)$ (b) $(2, \pi/6)$
(c) $(1, \pi/3)$ (d) $(1, \pi/6)$
6. The polar equation $\frac{5}{r} = 3 \cos \theta + 5 \sin \theta$ represents :
(a) A straight line (b) A circle
(c) A parabola (d) A hyperbola
7. The centre of the circle $r = 2a \cos (\theta - \alpha)$ is:
(a) (a, a) (b) (a, α)
(c) $(a, 0)$ (d) $(0, \alpha)$
8. The radius of the circle $r = 6 \cos \alpha$ is :
(a) 2 (b) 4 (c) 6 (d) 3
9. The equation of a directrix of the conic $\frac{l}{r} = 1 - e \cos \theta$ is :
(a) $\frac{l}{r} = e \cos \theta$ (b) $\frac{l}{r} = -e \cos \theta$
(c) $\frac{l}{r} = e \cos^2 \theta$ (d) $\frac{l}{r} = -e \cos^2 \theta$
10. The conic $\frac{7}{r} = 5 + 4 \cos \theta$ represents :
(a) A circle (b) A parabola
(c) A hyperbola (d) An ellipse
11. The latus rectum of the conic $\frac{6}{r} = 3 + 4 \cos \theta$ is :
(a) 2 (b) 3
(c) 4 (d) 6
12. The line $\frac{1}{r} = A \cos \theta + B \sin \theta$ will touch the conic $\frac{1}{r} = 1 + e \cos \theta$ if :
(a) $(A - e)^2 + B^2 = 1$ (b) $(A - e)^2 + B^2 = 0$
(c) $(A + e)^2 + B^2 = 1$ (d) $A^2 + (B - e)^2 = 0.3$
13. The equation of tangent to the conic $\frac{1}{r} = 1 - e \cos \theta$ at point 'a' is :
(a) $\frac{1}{r} = e \cos \theta + \cos (\theta - \alpha)$
(b) $\frac{1}{r} = -e \cos \theta + \cos (\theta - \alpha)$
(c) $\frac{1}{r} = e \cos (\theta - \alpha)$
(d) $\frac{1}{r} = e \cos \theta + \cos^2 (\theta - \alpha)$

14. Which one of the following conics have asymptote :
 (a) Circle (b) Parabola
 (c) Ellipse (d) Hyperbola
15. The equation of normal to the conic $\frac{1}{r} = 1 + e \cos \theta$ at point ' α ' is :
 (a) $\frac{le \sin \alpha}{r(1 + e \cos \alpha)} = e \sin \theta + e \cos (\theta - \alpha)$
 (b) $\frac{le \sin \alpha}{r(1 + e \cos \alpha)} = e \sin \theta + \sin (\theta - \alpha)$
 (c) $\frac{le \sin \alpha}{r(1 + e \cos \alpha)} = \sin \theta + e \cos (\theta - \alpha)$
 (d) None of the above
16. The equation of normal to the parabola $\frac{1}{r} = 1 + \cos \theta$ at point $\pi/2$ is :
 (a) $\frac{1}{r} = \cos \theta + \sin \theta$ (b) $\frac{1}{r} = \sin \theta - \cos \theta$
 (c) $\frac{1}{r} = \cos \theta$ (d) $\frac{1}{r} = 2 \cos \theta$
17. The equation of the director circle of the conic $\frac{1}{r} = 1 + e \cos \theta$ is :
 (a) $r^2(1 - e^2) + 2ler \cos \theta + 2l^2 = 0$
 (b) $r^2(1 + e^2) - 2ler \cos \theta - 2l^2 = 0$
 (c) $r^2(1 - e^2) + 2ler \cos \theta - 2l^2 = 0$
 (d) None of the above
18. For the conic $\frac{1}{r} = 1 + e \cos \theta$, the sum of the reciprocals of the segment of any focal chord is equal to :
 (a) l (b) $2l$
 (c) $1/l$ (d) $2/l$
19. If PSP' is a focal chord of a conic, then the tangents P at and P' intersect on :
 (a) Initial line (b) Vertex
 (c) Directrix (d) None of these
20. The point on the conic $\frac{4}{r} = 1 + 2 \cos \theta$ is:
 (a) $(2, \pi/3)$ (b) $(4, \pi/6)$
 (c) $(1, \pi/2)$ (d) None of these
21. The direction ratios of the line joining of the points $(2, 3, 5)$ and $(6, -5, 8)$ are :
 (a) $(4, -8, 3)$ (b) $(8, -2, 13)$
 (c) $(4, 6, 7)$ (d) None of these
22. The direction cosines of the line equally inclined to the positive direction of coordinate axes are :
 (a) $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ (b) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$
 (c) $1, 1, 1$ (d) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$
23. The sum of intercepts made by the plane $2x + 3y + 4z = 12$ on axes is :
 (a) 9 (b) 10
 (c) 12 (d) 13
24. The equation of plane passing through the points $(2, 3, -4)$ and $(1, -1, 3)$ and parallel to z -axis is :
 (a) $4x + y = 5$ (b) $4x - y = 5$
 (c) $2x + 3z = 7$ (d) None of these
25. A plane meets the co-ordinate axes at points A, B, C such that the centroid of $\triangle ABC$ is the point (a, b, c) . Then the equation of plane is :
 (a) $x/a + y/b + z/c = 2$ (b) $x/a + y/b + z/c = 3$
 (c) $ax + by + cz = 2$ (d) $ax + by + cz = 3$
26. The length of perpendicular drawn from origin to the plane $3x + 4y - 5z = 50$ is:
 (a) 12 (b) 25
 (c) $5\sqrt{2}$ (d) None of these
27. Planes $a_1x - b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ will be perpendicular if :
 (a) $a_1a_2 + b_1b_2 + c_1c_2 = 0$
 (b) $\frac{a_1}{a_2} + \frac{b_1}{b_2} + \frac{c_1}{c_2} = 0$
 (c) $a_1a_2 + b_1b_2 + c_1c_2 = 1$
 (d) $\frac{a_1}{a_2} + \frac{b_1}{b_2} + \frac{c_1}{c_2}$
28. Planes $2x - 3y - 5z = 6$ and $-8x + 12y + 20z = 13$ are :
 (a) Parallel (b) Perpendicular
 (c) Equal (d) None of these

29. The angle between the planes $2x - y + z = 5$ and $x + y + 2z = 7$ is :
- (a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$
30. The line $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-2}{3}$ meets the plane $2x - 3y + z + 5 = 0$ at point :
- (a) (2, 3, 3) (b) (1, 2, 3)
(c) (2, -3, 1) (d) (7, 10, 11)
31. The equation of straight line passing through the points (2, 3, 4) and (3, 4, 5) is :
- (a) $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{1}$
(b) $\frac{x-3}{1} = \frac{y-4}{-1} = \frac{z-5}{1}$
(c) $\frac{x+2}{1} = \frac{y+3}{-1} = \frac{z+4}{-1}$
(d) None of the above
32. The lines $x = a_1y + b_1, z = c_1y + d_1$ and $x = a_2y + b_2, z = c_2y + d_2$ will be perpendicular if :
- (a) $a_1a_2 + b_1b_2 = 1$ (b) $b_1b_2 + c_1c_2 = 1$
(c) $a_1a_2 + c_1c_2 + 1 = 0$ (d) None of these
33. The image of the point (3, 5, 7) in the plane $2x + y + z = 6$ is :
- (a) (-1, 3, 5) (b) (1, 3, 5)
(c) (3, 5, 7) (d) None of these
34. The angle between the line $\frac{x-1}{2} = \frac{y-3}{2} = \frac{z-4}{-1}$ and $\frac{x+1}{1} = \frac{y+5}{2} = \frac{z-7}{-2}$ is :
- (a) $\pi/3$ (b) $\cos^{-1}(1/3)$
(c) $\pi/2$ (d) None of these
35. The co-ordinates of foot of perpendicular drawn from the point (1, 3, 7) on the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ is :
- (a) (1, 2, 3) (b) (2, 4, 6)
(c) (3, 6, 9) (d) (1, 0, 4)
36. The equation of plane containing the line $\frac{x-1}{0} = \frac{y-2}{3} = \frac{z-3}{4}$ and point (0, 0, 5) is :
- (a) $14x - 4y + 3z = 15$ (b) $x + 2y + 5z = 6$
(c) $3x + 4y + 5z = 9$ (d) None of these
37. The equation of line passing through the point (1, -1, 2) and parallel to line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+3}{4}$ is :
- (a) $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-2}{4}$
(b) $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$
(c) $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z+3}{2}$
(d) None of the above
38. The equation of plane passing through the line of intersection of planes $2x - 2y + z = 1$ and $x - 2y + 3z = 2$ and point (1, -1, 1) is:
- (a) $x + 2z = 1$ (b) $y + 2z = 1$
(c) $y - 2z = 1$ (d) $x - 2z + 1 = 0$
39. The shortest distance between the lines $\frac{x-2}{1} = \frac{y}{2} = \frac{z-1}{2}$ and $\frac{x-4}{2} = \frac{y-3}{2} = \frac{z}{3}$ is :
- (a) 1 (b) 2
(c) 3 (d) 4
40. The shortest distance between the lines $\frac{x-1}{0} = \frac{y}{0} = \frac{z}{3}$ and $\frac{x}{2} = \frac{y-2}{0} = \frac{z}{0}$ is :
- (a) 1 (b) 2
(c) 3 (d) 5
41. The radius of the sphere $3x^2 + 3y^2 + 3z^2 + 12x + 18y - 24z - 12 = 0$ is :
- (a) 2 (b) 3
(c) 4 (d) 5
42. The equation of sphere passing through origin and points (a, 0, 0), (0, b, 0) and (0, 0, c) is:
- (a) $x^2 + y^2 + z^2 + ax + by + cz = 0$
(b) $x^2 + y^2 + z^2 - ax + by - cz = 0$
(c) $x^2 + y^2 + z^2 - ax - by - cz = 0$
(d) $x^2 + y^2 + z^2 = a + b + c$

43. If two spheres of radii 3 and 4 cut orthogonally. Then the radius of common circle is :
 (a) 12/5 (b) 5/12
 (c) 5 (d) 12
44. If the co-ordinates of ends of diameters of a sphere are (1, 2, 3) and (2, -3, 4), then the equation of sphere is :
 (a) $x^2 + y^2 + z^2 - 3x + y - 7z + 8 = 0$
 (b) $x^2 + y^2 + z^2 + 3x - y + 7z + 8 = 0$
 (c) $x^2 + y^2 + z^2 - 3x + y - 7z = 0$
 (d) None of the above
45. The pole of the plane $lx + my + nz = p$ w.r.t the sphere $x^2 + y^2 + z^2 = r^2$ is :
 (a) $\left(\frac{lr}{p^2}, \frac{mr}{p^2}, \frac{nr}{p^2}\right)$ (b) $\left(\frac{lr^2}{p}, \frac{mr^2}{p}, \frac{nr^2}{p}\right)$
 (c) $\left(\frac{l^2r}{p}, \frac{m^2r}{p}, \frac{n^2r}{p}\right)$ (d) None of these
46. The point of contact of plane $2x + y - 2z = 9$ and sphere $x^2 + y^2 + z^2 = 9$ is :
 (a) (2, 1, 2) (b) (-2, 1, 2)
 (c) (1, 1, 2) (d) (2, 1, -2)
47. The radius of the circle $x^2 + y^2 + z^2 - 2y - 4z - 11 = 0$, $x + 2y + 2z = 15$ is :
 (a) 7 (b) 4
 (c) $\sqrt{7}$ (d) None of these
48. The equation of cone which has vertex at origin and passing through the curve $ax^2 + by^2 = z$, $lx + my + nz = p$ is :
 (a) $p(ax^2 + by^2) = z(lx + my + nz)$
 (b) $p^2(ax^2 + by^2) = z(lx + my + nz)$
 (c) $p(ax^2 + by^2) = z(lx + my + nz)^2$
 (d) None of the above
49. The cone has three mutually perpendicular generators is :
 (a) $x^2 - y^2 + 2z^2 = 0$ (b) $x^2 + y^2 + 2z^2 = 0$
 (c) $x^2 + 2y^2 - 3z^2 = 0$ (d) $x^2 + y^2 + z^2 = 0$
50. The equation $ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + d = 0$ represents a cone if :
 (a) $au^2 + bv^2 + cw^2 = d$
 (b) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = d$
 (c) $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = d$
 (d) None of these
51. The equation of right circular cone whose vertex is at origin, x-axis and semivertical angle ' α ' is :
 (a) $y^2 + z^2 = x^2 \tan^2 \alpha$
 (b) $x^2 + y^2 = z^2 \tan^2 \alpha$
 (c) $x^2 + z^2 = y^2 \tan^2 \alpha$
 (d) $x^2 + y^2 + z^2 = \tan^2 \alpha$
52. The vertex of the cone $x^2 + 2y^2 + 3z^2 - 2xy + 3yz + 2xz = 0$ is :
 (a) (0, 0, 0) (b) (1, 1, 1)
 (c) (1, 2, 3) (d) (1, -2, -3)
53. The cone, reciprocal to the cone $ax^2 + by^2 + cz^2 = 0$ is :
 (a) $ax^2 + by^2 + cz^2 = 0$
 (b) $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 0$
 (c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$
 (d) None of these
54. The equation of right circular cylinder whose axis is x-axis and radius ' r ' is :
 (a) $x^2 + y^2 = r^2$ (b) $y^2 + z^2 = r^2$
 (c) $x^2 + z^2 = r^2$ (d) None of these
55. The equation of cylinder with generator parallel to x-axis and passing through the circle $x^2 + y^2 + z^2 = 4$, $x - y - z = 0$ is :
 (a) $y^2 + z^2 + yz = 2$ (b) $y^2 + z^2 - 2yz = 4$
 (c) $y^2 + z^2 + 4yz = 0$ (d) None of these

56. The equation of axis of right circular cylinder $x^2 + z^2 = 4$ is :
- (a) $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$ (b) $\frac{x}{0} = \frac{y}{1} = \frac{z}{0}$
- (c) $\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$ (d) $x = y = z$
57. The equation of cylinder whose generators are parallel to line $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$ and passes through the curve $x^2 + y^2 = 3, z = 0$ is :
- (a) $x^2 + y^2 + 5z^2 - 2zx - 4yz = 3$
- (b) $x^2 - y^2 - 5z^2 + 2zx + 4yz = 3$
- (c) $x^2 + y^2 - 5z^2 + 2zx + 4yz = 3$
- (d) None of these
58. The surface generated by a straight line which is always parallel to a fixed line and intersect a given curve or touches a given surface is called a :
- (a) Sphere (b) Cone
- (c) Cylinder (d) Central conicoid
59. The equation of cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and passing through the curve $x^2 + 2y^2 = 1, z = 3$ is :
- (a) $3x^2 + 6y^2 + 3z^2 - 2xz + 8yz - 3 = 0$
- (b) $3x^2 - 6y^2 - 3z^2 + 2xz + 8yz = 3$
- (c) $3x^2 + 6y^2 - 3z^2 - 2xz + 8yz - 3 = 0$
- (d) None of the above
60. The equation of cone reciprocal to the cone $x^2 - y^2 + z^2 = 0$:
- (a) $x^2 + y^2 + z^2 = 0$ (b) $x^2 - y^2 - z^2 = 0$
- (c) $x^2 - y^2 + z^2 = 0$ (d) None of these
61. The locus of point of intersection of three mutually perpendicular tangent planes to a conicoid is a :
- (a) Circle (b) Sphere
- (c) Conicoid (d) Cone
62. The equation of tangent plane to the conicoid $2x^2 + 6y^2 - 3z^2 = 5$ at point $(-1, 0, 1)$ is :
- (a) $2x + 3z + 5 = 0$ (b) $2x - 3y = 4$
- (c) $2x + 3y - 5z = 7$ (d) $2x + 5y = 3z$
63. The condition that the plane $lx + my + nz = p$ touches the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is :
- (a) $\frac{a^2}{l^2} + \frac{b^2}{m^2} + \frac{c^2}{n^2} = p^2$
- (b) $\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2} = p^2$
- (c) $a^2l + b^2m + c^2n = p^2$
- (d) $a^2l^2 + b^2m^2 + c^2n^2 = p^2$
64. The equation of normal to the conicoid $ax^2 + by^2 + cz^2 = 1$ at point (α, β, γ) is :
- (a) $\frac{x - \alpha}{a} = \frac{y - \beta}{b} = \frac{z - \gamma}{c}$
- (b) $\frac{x - a}{\alpha} = \frac{y - b}{\beta} = \frac{z - c}{\gamma}$
- (c) $\frac{x - \alpha}{a\alpha} = \frac{y - \beta}{b\beta} = \frac{z - \gamma}{c\gamma}$
- (d) None of these
65. The feet of six normals that can be drawn from a point on a conicoid lie on :
- (a) Sphere (b) Cone
- (c) Cylinder (d) Conicoid
66. The sum of squares of any three conjugate semidiameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is equal to :
- (a) $a + b + c$ (b) $2abc$
- (c) $ab + bc + ca$ (d) $a^2 + b^2 + c^2$
67. The equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ represents :
- (a) Ellipsoid
- (b) Hyperboloid of one sheet
- (c) Hyperboloid of two sheet
- (d) Elliptic paraboloid

68. The equation of director sphere of the conicoid $2x^2 + 3y^2 + 4z^2 = 1$ is :
- (a) $x^2 + y^2 + z^2 = 9$ (b) $x^2 + y^2 + z^2 = 16$
 (c) $x^2 + y^2 + z^2 = 29$ (d) $x^2 + y^2 + z^2 = 1$
69. The pole of the plane $lx + my + nz = p$ w.r.t the conicoid $ax^2 + by^2 + cz^2 = 1$ is :
- (a) $\left(\frac{al}{p}, \frac{bm}{p}, \frac{cn}{p}\right)$ (b) $\left(\frac{a}{lp}, \frac{b}{mp}, \frac{c}{np}\right)$
 (c) $\left(\frac{l}{ap}, \frac{m}{bp}, \frac{n}{cp}\right)$ (d) None of these
70. The volume of the parallelopiped formed with three conjugate semi-diameter as cotermious edges of $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is equal to :
- (a) $ab + bc + ca$ (b) abc
 (c) $a^2b^2c^2$ (d) $(a + b + c)^2$
71. The condition that the plane $lx + my + nz = p$, may touch the paraboloid $ax^2 + by^2 = 2cz$ is :
- (a) $c \left(\frac{a^2}{l^2} + \frac{b^2}{m^2} \right) + 2pn = 0$
 (b) $c \left(\frac{a^2}{l} + \frac{b^2}{m} \right) + 2pn = 0$
 (c) $c \left(\frac{a^2}{l} + \frac{m^2}{b} \right) + 2pn = 0$
 (d) None of these
72. The locus of mid point of a system of parallel chords of a paraboloid is called :
- (a) Conjugate planes
 (b) Diametral planes
 (c) Parallel planes
 (d) None of these
73. The feet of the normals from any point on the paraboloid $x^2 + y^2 = 2az$ lie on :
- (a) A sphere (b) A cone
 (c) A cylinder (d) A cube
74. If the plane $x + 2y - 2z = 4$, touches the paraboloid $3x^2 + 4y^2 = 24z$, then the point of contact is :
- (a) (2, 3, 2) (b) (3, 3, 2)
 (c) (-2, 3, -2) (d) (2, -3, -3)
75. The equation of tangent plane to the paraboloid $2x^2 - 5y^2 = 10z$ at (5, -2, 3) is :
- (a) $x + y - z = 3$ (b) $x + 2y - 3z = 4$
 (c) $x + y - z = 4$ (d) $2x + 2y - z = 3$
76. How many normals can be drawn from any point to a paraboloid?
- (a) 2 (b) 4
 (c) 5 (d) 6
77. The equation of normal to the paraboloid $x^2 - y^2 = 2z$ at point (2, 3, 4) is :
- (a) $\frac{x-2}{2} = \frac{y-3}{-3} = \frac{z-4}{-1}$
 (b) $\frac{x-2}{2} = \frac{y-3}{-3} = \frac{z-4}{4}$
 (c) $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z-4}{1}$
 (d) None of these
78. The locus of chord of the paraboloid $x^2 - y^2 = 2z$ bisected at point (2, -3, 4) is :
- (a) $2x + 3y - z + 9 = 0$ (b) $2x - 3y + z = 9$
 (c) $2x - 3y + 2z = 10$ (d) None of these
79. The polar of point (1, -2, 3) w.r.t the paraboloid $3x^2 + 2z^2 = 4y$ is :
- (a) $3x + 2y + 6z = 4$ (b) $3x - 2y + 6z + 4 = 0$
 (c) $2x + 3y + 6z = 4$ (d) None of these
80. The equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$ represents a :
- (a) Elliptic paraboloid
 (b) Parabolic paraboloid
 (c) Hyperbolic paraboloid
 (d) None of these

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(b)	2.	(d)	3.	(d)	4.	(b)	5.	(a)	6.	(a)	7.	(c)	8.	(d)	9.	(b)	10.	(d)
11.	(c)	12.	(a)	13.	(b)	14.	(d)	15.	(b)	16.	(b)	17.	(c)	18.	(d)	19.	(c)	20.	(a)
21.	(a)	22.	(b)	23.	(d)	24.	(b)	25.	(b)	26.	(c)	27.	(a)	28.	(a)	29.	(c)	30.	(d)
31.	(a)	32.	(c)	33.	(a)	34.	(d)	35.	(b)	36.	(a)	37.	(b)	38.	(d)	39.	(c)	40.	(b)
41.	(d)	42.	(c)	43.	(a)	44.	(a)	45.	(b)	46.	(d)	47.	(c)	48.	(d)	49.	(c)	50.	(d)
51.	(a)	52.	(d)	53.	(c)	54.	(b)	55.	(a)	56.	(b)	57.	(b)	58.	(c)	59.	(a)	60.	(c)
61.	(b)	62.	(a)	63.	(d)	64.	(c)	65.	(b)	66.	(d)	67.	(b)	68.	(c)	69.	(c)	70.	(b)
71.	(d)	72.	(b)	73.	(a)	74.	(a)	75.	(d)	76.	(c)	77.	(b)	78.	(c)	79.	(c)	80.	(c)

HINTS AND SOLUTIONS

1. The distance between the two points in polar form

$$= \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}$$

$$\text{Here } r_1 = 2, \theta_1 = 90, r_2 = 3, \theta_2 = 30$$

$$= \sqrt{4 + 9 - 2 \times 2 \times 3 \cos 60^\circ} = \sqrt{7}$$

3. We have $x = r \cos \theta$, $y = r \sin \theta$

$$x^2 + y^2 = r^2$$

$$\therefore r^2 = a^2 \cos 2\theta$$

$$r^2 = a^2(\cos^2 \theta - \sin^2 \theta)$$

$$\Rightarrow r^4 = a^2(r^2 \cos^2 \theta - r^2 \sin^2 \theta)$$

$$\Rightarrow (x^2 + y^2)^2 = a^2(x^2 - y^2)$$

6. $l = r \cos \theta$, $\sqrt{3} = r \sin \theta$

$$r^2 = 1 + 3 = 4, r = 2, \tan \theta = \sqrt{3}$$

$$\theta = \tan^{-1}(\sqrt{3})$$

$$= \frac{\pi}{3}$$

$$10. \frac{2}{r} = 1 + \frac{4}{3} \cos \theta$$

$$\text{latus rectum} = 2l = 2 \times 2 = 4$$

24. The equation of plane passing $(2, 3, -4)$ is $a(x - 2) + b(y - 3) + c(z + 4) = 0$ passed $(1, -1, 3)$

$$-a - 4b + 7c = 0, \text{parallel to } z\text{-axis with dc. } (0, 0, 1)$$

$$\Rightarrow c = 0 \quad a = -4b$$

$$-4b(x - 2) + b(y - 3) = 0$$

$$-4x + 8 + y - 3 = 0$$

$$4x - y = 5$$

$$27. \frac{50}{\sqrt{16 + 9 + 25}} = \frac{50}{\sqrt{50}} = \frac{50}{\sqrt{50}} \times \frac{\sqrt{50}}{\sqrt{50}} = \sqrt{50} = 5\sqrt{2}$$

$$29. \cos \theta = \frac{2 - 1 + 2}{\sqrt{4 + 1 + 1} \sqrt{1 + 1 + 4}} = \frac{3}{6} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

$$30. \frac{x - a_1}{a_2 - a_1} = \frac{y - b_1}{b_2 - b_1} = \frac{z - c_1}{c_2 - c_1}$$

$$34. \quad \cos\theta = \frac{2+4+2}{\sqrt{9}\sqrt{9}} = \frac{8}{9}$$

$$38. \quad (2x - 2y + z - 1) + \lambda(x - 2y + 3z - 2) = 0$$

$$\Rightarrow (2 + 2 + 1 - 1) + \lambda(1 + 2 + 3 - 2) = 0$$

$$4 + 4\lambda = 0, \lambda = -1$$

$$\Rightarrow x - 2z + 1 = 0$$

39. Shortest distance between two lines is

$$\frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - a_1c_2)^2 + (a_1b_2 - a_2b_1)^2}}$$

$$\Rightarrow \frac{\begin{vmatrix} 4-2 & 3-0 & -1 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{vmatrix}}{\sqrt{(6-4)^2 + (4-3)^2 + (2-4)^2}} = \frac{\begin{vmatrix} 2 & 3 & -1 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{vmatrix}}{\sqrt{4+9+4}} = \frac{2(6-4) + 3(4-3) - 1(2-4)}{\sqrt{17}} = \frac{4+3+2}{\sqrt{17}} = \frac{9}{\sqrt{17}}$$

$$41. \quad x^2 + y^2 + z^2 + 4x + 6y - 8z - 4 = 0$$

$$\text{radius} = \sqrt{u^2 + v^2 + w^2 - d} = \sqrt{4 + 9 + 16 + 4}$$

43. Two spheres of radius r_1 and r_2 then radius of common circle is

$$\frac{r_1 r_2}{\sqrt{r_1^2 + r_2^2}} = \frac{12}{5}$$

Here $r_1 = 3, r_2 = 4$

$$44. \quad (x-1)(x-2) + (y-2)(y+3) + (z-3)(z-4) = 0$$

$$\Rightarrow x^2 + y^2 + z^2 - 3x + y - 7z + 8 = 0$$

$$55. \quad x^2 + y^2 + z^2 = 4 \quad \dots(1)$$

$$x - y - z = 0 \quad \dots(2)$$

make homogenous equation of (1) with the help of (2) from (2) $x = y + z$

$$\Rightarrow (y+z)^2 + y^2 + z^2 = 4$$

$$2y^2 + 2z^2 + 2yz = 4$$

$$y^2 + z^2 + yz = 2$$

62. The equation of tangent plane to the conic $2x^2 + 6y^2 - 3z^2 = 5$ at $(-1, 0, 1)$ is

$$2x(-1) + 6(y)(0) - 3(z)(1) = 5$$

$$-2x - 3z = 5$$

$$\Rightarrow 2x + 3y + 5 = 0$$

68. The equation of director circle of conicoid $ax^2 + by^2 + cz^2 = 1$ is $x^2 + y^2 + z^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{6+4+3}{12} = \frac{13}{12}$$

75. The equation of tangent plane to $2x^2 - 5y^2 = 10z$ at $(5, -2, 3)$ is

$$2(5x) - 5(-2y) = 5(z + 3)$$

$$\Rightarrow 2x + 2y - z = 3$$

77. The equation of normal at (α, β, γ) of paraboloid $ax^2 + by^2 = 2cz$ is $\frac{x-\alpha}{a\alpha} = \frac{y-\beta}{b\beta} = \frac{z-\gamma}{c\gamma}$

Here $a = 1, b = -1, c = 1$

$$\alpha = 2, \beta = 3, \gamma = 4$$

$$\frac{x-2}{2} = \frac{y-3}{-3} = \frac{z-4}{4}$$

CSJM University, Kanpur, Examination-2019

Paper-III Geometry and Vector Calculus

[Time : 2 Hours]

[Maximum Marks : 70]

Note : All questions are **compulsory**. This question paper consists of 80 objective type questions.

1. If $\vec{a}$ is any vector, then the value of $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{a})$ is equal to :
(a) $-\nabla^2 \vec{a}$ (b) 0
(c) $2\vec{a}$ (d) $-2\vec{a}$
2. If $\text{curl } \vec{a} = 0$ and $\text{curl } \vec{b} = 0$, then $\text{div } (\vec{a} \times \vec{b})$ is equal to :
(a) $\text{grad } (\vec{a} \cdot \vec{b})$ (b) $\vec{a} \times \vec{b}$
(c) 0 (d) $|\vec{a} \times \vec{b}| \text{div } \vec{a}$
3. Find $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and C is the circle $x^2 + y^2 = 1, z = 0$:
(a) 0 (b) $\frac{\pi}{2}$
(c) π (d) $-\pi$
4. For any closed surface S the value $\iint_S \text{curl } \vec{F} \cdot \hat{n} dS$ is :
(a) 0 (b) $\text{div } \vec{F}$
(c) $\text{curl } \vec{F}$ (d) $-\vec{F}$
5. Evaluate $\iint_S (ax^2 + by^2 + dz^2) dS$ over the sphere $x^2 + y^2 + z^2 = 1$:
(a) 0 (b) abc
(c) $\frac{4}{3}\pi(a^2 + b^2 + c^2)$ (d) $\frac{4}{3}\pi(a + b + c)$
6. The conic $9x^2 + 25y^2 - 90x - 150y + 225 = 0$ represent a :
(a) Pair of straight lines (b) Parabola
(c) Ellipse (d) Hyperbola
7. The equation of the asymptotes of the conic $x^2 - 3xy + y^2 + 10x - 10y + 21 = 0$ is
(a) $x^2 - 3xy + y^2 + 10x - 10y + 20 = 0$
(b) $x^2 + 3xy + y^2 + 10x + 10y + 20 = 0$
(c) $x^2 - 3xy + y^2 + 10x - 10y - 20 = 0$
(d) $x^2 - 3xy + y^2 + 10x + 10y - 20 = 0$
8. The equation $\frac{1}{x+y-a} + \frac{1}{x-y+a} + \frac{1}{y-x+a} = 0$ represents a :
(a) Circle (b) Pair of straight lines
(c) Parabola (d) Ellipse
9. The principal axes of the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ are parallel to the lines ;
(a) $(x^2 - y^2)h = (a + b)xy$
(b) $(x^2 + y^2) = h(a - b)xy$
(c) $(x^2 - y^2) = h(a + b)xy$
(d) None of these
10. There pass two confocals through any point in the plane of an ellipse of which :
(a) One is ellipse and other is parabola
(b) One is ellipse and other is also ellipse
(c) One is ellipse and other is a hyperbola
(d) One is parabola and other is a hyperbola
11. Passing through a given point two confocals cut at angle :
(a) 45° (b) 30° (c) 60° (d) 90°

12. The point of intersection of the two perpendicular tangents one to each of two given confocals lie on a :
 (a) Circle (b) Parabola
 (c) Ellipse (d) Hyperbola
13. The equation to the directrix of the conic $\frac{l}{r} = 1 + e \cdot \cos \theta$ is :
 (a) $\frac{l}{r} = e$ (b) $\frac{l}{r} = e \cos \theta$
 (c) $\frac{l}{r} = 1 - e \cos \theta$ (d) $\frac{l}{r} = e \tan \theta$
14. The equation of the asymptotes of the conic $\frac{l}{r} = 1 + e \cos \theta$ is :
 (a) $\frac{le}{r} = (e^2 - 1) \cos \theta \pm \sqrt{e^2 - 1} \sin \theta$
 (b) $\frac{le}{r} = (1 - e^2) \cos \theta \pm \sqrt{1 + e^2} \sin \theta$
 (c) $\frac{l}{r} = (e^2 - 1) \cos \theta \pm \sqrt{e^2 - 1} \sin \theta$
 (d) $\frac{l}{r} = (1 - e^2) \cos \theta \pm \sqrt{1 - e^2} \sin \theta$
15. If the line $\frac{l}{r} = A \cos \theta + B \sin \theta$ is tangent to the conic $\frac{l}{r} = 1 + e \cos \theta$ then :
 (a) $(A - e)^2 + B^2 = 1$ (b) $A^2 + (B - e)^2 = 1$
 (c) $A^2 + B^2 = e^2$ (d) None of these
16. The perpendicular distance of the point (x, y, z) from the z -axis is :
 (a) Z (b) $\sqrt{x^2 + y^2}$
 (c) $\sqrt{x^2 + z^2}$ (d) $\sqrt{y^2 + z^2}$
17. The distance between the points $(1, -1, 3)$ and $(2, 3, -5)$ is :
 (a) 2 (b) 5
 (c) 12 (d) 9
18. If a straight line is equally inclined to the three co-ordinate axes, then the direction cosines of the line are :
 (a) 1, 1, 1 (b) $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$
 (c) $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ (d) $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$
19. The length of a segment of a line whose projections on the axes are 2, 3 and 6 is :
 (a) 2 (b) 3
 (c) 6 (d) 7
20. The angle between any two diagonals of a cube is :
 (a) 90° (b) 30°
 (c) 60° (d) $\cos^{-1}\left(\frac{1}{3}\right)$
21. The co-ordinates of a point A are $(2, 3, -5)$. If O is the origin, then the equation of the plane through A and perpendicular to OA is ;
 (a) $2x + 3y + 5z = 38$ (b) $2x + 3y - 5z = 38$
 (c) $x + y + z = 18$ (d) $2x + 3y - 5z = 20$
22. The distance between the parallel planes $2x - y + 3z - 4 = 0$ and $6x - 3y + 9z + 13 = 0$ is :
 (a) $\frac{1}{\sqrt{126}}$ (b) $\frac{10}{\sqrt{126}}$
 (c) $\frac{25}{\sqrt{126}}$ (d) None of these
23. The equation of the plane through the point $(0, 7, -7)$ and containing the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$ is :
 (a) $x + y + z = 1$ (b) $x - y + z = 0$
 (c) $x + y + z = 0$ (d) $x + y + z = 7$
24. The lines $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$ and $\frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$ intersect at the point :
 (a) $(0, 0, 0)$ (b) $(1, -1, 1)$
 (c) $(-1, 3, -2)$ (d) $(2, 1, -3)$
25. If the distance of a point from the x -axis is twice of its distance from the yz plane, then the locus of the point is :
 (a) $4x^2 - y^2 - z^2 = 0$ (b) $x^2 - 4y^2 - z^2 = 0$
 (c) $x^2 - y^2 - 4z^2 = 0$ (d) $x^2 + y^2 + z^2 = 4$

26. The length of the shortest distance between the z -axis and the line $x + y + 2z - 3 = 0 = 2x + 3y + 4z - 4$ is :
 (a) 0 (b) 1
 (c) 2 (d) $\frac{1}{\sqrt{2}}$
27. The co-ordinates of the foot of the perpendicular from the point $(2, 3, 7)$ to the plane $3x - y - z = 7$ are :
 (a) $(-1, 1, 1)$ (b) $(0, 1, 1)$
 (c) $(5, 2, 6)$ (d) $(1, 1, 0)$
28. The equation of the straight lines which bisect the angles between the lines $\frac{x}{l_1} = \frac{y}{m_1} = \frac{z}{n_1}$, $\frac{x}{l_2} = \frac{y}{m_2} = \frac{z}{n_2}$ is :
 (a) $\frac{l_1 x}{l_1 \pm l_2} = \frac{m_1 y}{m_1 \pm m_2} = \frac{n_1 z}{n_1 \pm n_2}$
 (b) $\frac{x}{l_1 \pm l_2} = \frac{y}{m_1 \pm m_2} = \frac{z}{n_1 \pm n_2}$
 (c) $\frac{x}{l_1^2 \pm l_2^2} = \frac{y}{m_1^2 \pm m_2^2} = \frac{z}{n_1^2 \pm n_2^2}$
 (d) None of the above
29. The co-ordinates of the point of intersection of the line $\frac{x+1}{1} = \frac{y+3}{3} = \frac{z-2}{2}$ with the plane $3x + 4y + 5z = 20$ is :
 (a) $(1, 0, 0)$ (b) $(0, 1, 0)$
 (c) $(0, 0, 1)$ (d) $(0, 0, 4)$
30. The angle between the planes $3x - 4y + 5z = 0$ and $2x - y - 2z = 5$ is :
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$
31. The normal form of the plane $x + 2y - 2z - 9 = 0$ is :
 (a) $\frac{1}{3}x + \frac{2}{3}y - \frac{2}{3}z = 3$
 (b) $\frac{1}{3}x + \frac{2}{3}y - \frac{2}{3}z = 1$
 (c) $\frac{1}{3}x + \frac{2}{3}y + \frac{2}{3}z = 3$
 (d) $x + y + z = \frac{1}{\sqrt{3}}$
32. The equation of the plane which is horizontal and passes through the point $(1, -2, -5)$ is :
 (a) $x - 2y - 5z = 0$ (b) $2y + 5z = 0$
 (c) $z + 5 = 0$ (d) $x - 2y = 0$
33. A plane meets the co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r) , then the equation of the plane is :
 (a) $px + qy + rz = 1$ (b) $px + qy + rz = 3$
 (c) $\frac{p}{x} + \frac{q}{y} + \frac{r}{z} = 3$ (d) $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$
34. If the lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d'$ are perpendicular then the condition is :
 (a) $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$
 (b) $aa' + bb' + cc' = d + d'$
 (c) $aa' + cc' + 1 = 0$
 (d) $aa' + bb' + cc' + dd' = 0$
35. The equation of the plane through the point $(2, -1, 1)$ and the line $4x - 3y + 5 = 0 = y - 2z - 5$ is :
 (a) $4x - y - 4z = 5$ (b) $x - 4y + 4z = 1$
 (c) $4x - 4y + z = 5$ (d) $4x + y + 4z = 5$
36. The number of spheres that can be made to pass through the three given points $(1, 0, 0), (0, 1, 0)$ and $(0, 0, 1)$ is :
 (a) 1 (b) 2
 (c) 3 (d) Infinite
37. A sphere is described on the line joining $A(2, -1, 4)$ and $B(-2, 2, -2)$ as diameter. Then the co-ordinates of the centre of the sphere are :
 (a) $(0, 0, 0)$ (b) $\left(0, \frac{1}{2}, 1\right)$
 (c) $(0, 1, 2)$ (d) $\left(0, -\frac{1}{2}, -1\right)$
38. The condition that the plane $lx + my + nz = p$ touches the sphere $x^2 + y^2 + z^2 = a^2$ is :
 (a) $p^2 = a^2(l^2 + m^2 + n^2)$
 (b) $p = a lmn$
 (c) $p^2 = a^2 - (l^2 + m^2 + n^2)$
 (d) $p^2 = a^2 + (l^2 + m^2 + n^2)$

39. If the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z - 3 = 0$ is touched by the plane $2x - 2y + z + 12 = 0$ then the point of contact is :
 (a) $(-1, 4, -2)$ (b) $(1, -4, 2)$
 (c) $(1, 2, 1)$ (d) $(-1, 2, -1)$
40. The equation of the sphere through the circle $x^2 + y^2 + z^2 = 9$, $x + y - 2z + 4 = 0$ and the origin is :
 (a) $x^2 + y^2 + z^2 + x + y - 2z = 0$
 (b) $x^2 + y^2 + z^2 - x - y + 2z = 0$
 (c) $4(x^2 + y^2 + z^2) - 9x - 9y + 18z = 0$
 (d) $4(x^2 + y^2 + z^2) + 9x + 9y - 18z = 0$
41. The angle of intersection of the sphere $x^2 + y^2 + z^2 + 6y + 2z + 8 = 0$ and $x^2 + y^2 + z^2 + 6x + 8y + 4z + 20 = 0$ is given by :
 (a) $\cos^{-1}\left(\frac{2}{3}\right)$ (b) $\frac{\pi}{2}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
42. The pole of the plane $lx + my + nz = p$ with respect to the sphere $x^2 + y^2 + z^2 = a^2$ is :
 (a) $\left(-\frac{a^2}{p}, -\frac{a^2}{p}, -\frac{a^2}{p}\right)$
 (b) $\left(-\frac{la^2}{p}, -\frac{ma^2}{p}, -\frac{na^2}{p}\right)$
 (c) $\left(\frac{la^2}{p}, \frac{ma^2}{p}, \frac{na^2}{p}\right)$
 (d) $\left(\frac{l^2}{ap}, \frac{m^2}{ap}, \frac{n^2}{ap}\right)$
43. The centres of the spheres of a coaxial system which have zero radius are called the :
 (a) Point of contact
 (b) Limiting points
 (c) Boundary points
 (d) End points
44. The equation $2x^2 + 2y^2 + 2z^2 - x - y - z = 0$ represents a :
 (a) Sphere (b) Pair of planes
 (c) Cone (d) End points
45. The equation of the cone whose vertex is origin and which passes through the curve given by $x^2 + y^2 = 2z$, $x + y + z = 1$:
 (a) $x^2 + y^2 - 2z + (x + y + z)^2 = 0$
 (b) $x^2 + y^2 = 2z$
 (c) $x^2 + y^2 + z^2 = 1$
 (d) $x^2 + y^2 = 2z(x + y + z)$
46. If the line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is a generator of the cone $2x^2 + 3y^2 - 5z^2 = 0$ then the condition is :
 (a) $2l^2 + 3m^2 - 5n^2 = 0$
 (b) $\frac{l^2}{2} + \frac{m^2}{3} - \frac{n^2}{5} = 0$
 (c) $2l^2 + 3m^2 - 5n^2 = 1$
 (d) $\frac{l^2}{2} + \frac{m^2}{3} - \frac{n^2}{5} = 1$
47. The reciprocal cone of $ax^2 + by^2 + cz^2 = 0$ is given by :
 (a) $ax^2 + by^2 + cz^2 = 1$
 (b) $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1$
 (c) $a^2x^2 + b^2y^2 + c^2z^2 = 0$
 (d) $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 0$
48. If the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gxz + 2hxy = 0$ have three mutually perpendicular generators, then condition is :
 (a) $a + b + c = 0$ (b) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$
 (c) $fa + bg + ch = 0$ (d) $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$

49. If the vertex of the right circular cone is origin, axis is z -axis and semivertical angle is θ , then the equation of this cone is :
- (a) $x^2 + z^2 = y^2 \tan^2 \theta$
 (b) $x^2 + y^2 = z^2 \tan^2 \theta$
 (c) $y^2 + z^2 = x^2 \tan^2 \theta$
 (d) $x^2 + y^2 + z^2 = \cot^2 \theta$
50. The equation of the enveloping cone of the sphere $x^2 + y^2 + z^2 = 1$ with the vertex at the point (α, β, γ) is :
- (a) $\alpha^2 x^2 + \beta^2 y^2 + \gamma^2 z^2 = 1$
 (b) $(\alpha x + \beta y + \gamma z)^2 = \alpha^2 + \beta^2 + \gamma^2$
 (c) $(x^2 + y^2 + z^2 - 1)(\alpha^2 + \beta^2 + \gamma^2 - 1) = (\alpha x + \beta y + \gamma z - 1)^2$
 (d) None of the above
51. The equation of the right circular cylinder whose axis is the z -axis and the radius r is :
- (a) $x^2 + y^2 = r^2$ (b) $x^2 + y^2 + z^2 = r^2$
 (c) $y^2 + z^2 = r^2$ (d) $x^2 + z^2 = r^2$
52. Any line on the surface of the cylinder is called :
- (a) Axis (b) Generator
 (c) Guiding curve (d) None of these
53. The equation of the director sphere of the central conicoid $ax^2 + by^2 + cz^2 = 1$ is :
- (a) $x^2 + y^2 + z^2 = 1$
 (b) $x^2 + y^2 + z^2 = a + b + c$
 (c) $x^2 + y^2 + z^2 = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$
 (d) $ax^2 + by^2 + cz^2 = (ax + by + cz)^2 - 1$
54. The equation of the tangent plane to the central conicoid $3x^2 - 5y^2 + z^2 + 2 = 0$ at the point $(1, 1, 0)$ is :
- (a) $3x - 5y + 2 = 0$ (b) $3x - 5y + z + 2 = 0$
 (c) $3x + 5y + z + 2 = 0$ (d) $3x - 5y - z + 2 = 0$
55. In general from a given point the number of normals which can be drawn to the conicoid $ax^2 + by^2 + cz^2 = 1$ are :
- (a) 3 (b) 4
 (c) 5 (d) 6
56. The sum of the squares of any three conjugate semidiameters of an ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is :
- (a) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$ (b) $a^2 + b^2 + c^2$
 (c) abc (d) $a^2 b^2 c^2$
57. The surface represented by the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ is :
- (a) Sphere
 (b) Ellipsoid
 (c) Paraboloid
 (d) Hyperboloid of one sheet
58. The equations of the generators of the hyperboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ which pass through the point $(a \cos \alpha, b \sin \alpha, 0)$ are :
- (a) $\frac{x + a \cos \alpha}{a \cos \alpha} = \frac{y + b \sin \alpha}{b \sin \alpha} = \frac{z}{\pm c}$
 (b) $\frac{x - a \cos \alpha}{a \cos \alpha} = \frac{y - b \sin \alpha}{-b \sin \alpha} = \frac{z}{\pm c}$
 (c) $\frac{x - a \cos \alpha}{a \sin \alpha} = \frac{y - b \sin \alpha}{-b \cos \alpha} = \frac{z}{\pm c}$
 (d) None of the above
59. Any two generators of the different systems :
- (a) Intersect (b) Do not intersect
 (c) Are parallel (d) None of these
60. The plane section of the central conicoid $ax^2 + by^2 + cz^2 = 1$ by the plane $lx + my + nz = p$ is an ellipse, if :
- (a) $bcl^2 + cam^2 + abn^2 < 0$
 (b) $bcl^2 + cam^2 + abn^2 = 0$
 (c) $bcl^2 + cam^2 + abn^2 > 0$
 (d) $al^2 + bm^2 + cn^2 > 0$

61. The area of plane section of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ by the central plane $lx + my + nz = 0$ is given by :
- (a) $\pi abc \left\{ \frac{a^2 l^2 + b^2 m^2 + c^2 n^2}{l^2 + m^2 + n^2} \right\}^{\frac{1}{2}}$
 (b) $\pi abc \left\{ \frac{l^2 + m^2 + n^2}{a^2 l^2 + b^2 m^2 + c^2 n^2} \right\}^{\frac{1}{2}}$
 (c) $\frac{\pi}{abc} \left\{ \frac{a^2 l^2 + b^2 m^2 + c^2 n^2}{l^2 + m^2 + n^2} \right\}^{\frac{1}{2}}$
 (d) $\frac{\pi}{abc} \left\{ \frac{l^2 + m^2 + n^2}{a^2 l^2 + b^2 m^2 + c^2 n^2} \right\}^{\frac{1}{2}}$
62. If the section of the conicoid $ax^2 + by^2 + cz^2 = 1$ by the plane $lx + my + nz = 0$ is a hyperbola then the section by the plane $lx + my + nz = p$ is :
- (a) A parabola (b) A circle
 (c) An ellipse (d) A hyperbola
63. The equation $(3x - 4y + 5z - 1)^2 + 4(3x - 4y + 5z - 1) + 1 = 0$ represents :
- (a) A pair of parallel planes
 (b) Hyperboloid
 (c) Paraboloid
 (d) Ellipsoid
64. The surface $\sqrt{x} + \sqrt{y} + \sqrt{z} = 0$ represents :
- (a) A paraboloid
 (b) An ellipsoid
 (c) A hyperboloid
 (d) A right circular cone with vertex $(0, 0, 0)$
65. The general equation to the cone of second degree which passes through the three co-ordinate axes is :
- (a) $fy^2 + gx^2 + hx^2 = 0$
 (b) $fyx + gzx + hxy = 0$
 (c) $kxyz = \text{constant}$
 (d) None of the above
66. The necessary and sufficient condition for a vector $\vec{a}(t)$ to have a constant direction is :
- (a) $\vec{a} \times \frac{d\vec{a}}{dt} = 0$ (b) $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$
 (c) $\vec{a} = \text{constant}$ (d) $\left(\frac{d\vec{a}}{dt} \right)^2 = 0$
67. If $\vec{r} = \sin t \hat{i} + \cos t \hat{j} + t \hat{k}$, then the value of $\left| \frac{d^2 \vec{r}}{dt^2} \right|$ is :
- (a) $\sqrt{2}$ (b) 1
 (c) 0 (d) 2
68. If $\frac{d\vec{u}}{dt} = \vec{w} \times \vec{u}$, $\frac{d\vec{v}}{dt} = \vec{w} \times \vec{v}$ then $\frac{d}{dt}(\vec{u} \times \vec{v})$ is :
- (a) 0 (b) $[\vec{u} \vec{v} \vec{w}]$
 (c) $\vec{w} \times (\vec{u} \times \vec{v})$ (d) 1
69. A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 5$ where t is time. Find the component of acceleration in the direction $\hat{i} + \hat{j} + 3\hat{k}$ at $t = 1$:
- (a) $\sqrt{11}$ (b) $\frac{8}{\sqrt{11}}$
 (c) 8 (d) 6
70. Given that $\vec{r}(t) = \begin{cases} 2\hat{i} - \hat{j} + 2\hat{k}, & \text{when } t = 2 \\ 4\hat{i} - 2\hat{j} + 3\hat{k} & \text{when } t = 3 \end{cases}$ then the value of $\int_2^3 \left(\vec{r} \cdot \frac{d\vec{r}}{dt} \right) dt$ is :
- (a) 0 (b) 2
 (c) 5 (d) 10
71. If $r = |\vec{r}|$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $\nabla f(r) \times \vec{r}$ is equal to :
- (a) $\vec{0}$ (b) $-\frac{1}{r^3} \vec{r}$
 (c) $\frac{1}{r^3} \vec{r}$ (d) $\vec{r}$

72. A unit normal vector to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$ is :
- (a) $-\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$ (b) $\frac{1}{3}\hat{i} - \frac{2}{3}\hat{j} - \frac{2}{3}\hat{k}$
- (c) Both (a) and (b) (d) $-2\hat{i} + 4\hat{j} + 4\hat{k}$
73. If $\vec{f} = x^2y\hat{i} - 2xz\hat{j} + 2yz\hat{k}$, then divergence $\vec{f}$ is equal to :
- (a) 0 (b) $2y$
- (c) $2xy$ (d) $2y(x + 1)$
74. If $\vec{a}$ is constant vector, then the value of curl $(\vec{r} \times \vec{a})$ is equal to :
- (a) $\vec{0}$ (b) $2\vec{a}$
- (c) $-2\vec{a}$ (d) $\vec{a}$
75. If $u = x^2 - y^2 + 4z$, then $\nabla^2 u$ is equal to :
- (a) 0 (b) 1
- (c) $2x$ (d) $-2y$
76. The equation $\sqrt{x} + \sqrt{y} = \sqrt{a}$ represent a :
- (a) Straight line (b) Circle
- (c) Parabola (d) Ellipse
77. The centre of the curve $x^2 - xy - 2y^2 - x - 4y - 2 = 0$ is :
- (a) $(0, -1)$ (b) $(-1, 0)$
- (c) $(-1, 1)$ (d) $(1, 0)$
78. The general equation of the second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent a conic with centre at $(0, 0)$ if :
- (a) $f = 0, c = 0$ (b) $f = 0, g = 0$
- (c) $g = 0, c = 0$ (d) Only $c = 0$
79. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola, if :
- (a) $\Delta = 0, h^2 > ab$
- (b) $\Delta \neq 0, h^2 < ab$
- (c) $\Delta \neq 0, h^2 > ab$
- (d) $\Delta \neq 0, h^2 > ab$ and $a + b = 0$
80. The eccentricity of the hyperbola $9x^2 - 16y^2 = 144$ is :
- (a) $\frac{4}{5}$ (b) $\frac{5}{4}$
- (c) $\frac{9}{2}$ (d) $\frac{2}{9}$

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(b)	2.	(c)	3.	(d)	4.	(a)	5.	(d)	6.	(c)	7.	(a)	8.	(c)	9.	(d)	10.	(c)
11.	(d)	12.	(a)	13.	(b)	14.	(b)	15.	(a)	16.	(b)	17.	(d)	18.	(d)	19.	(d)	20.	(d)
21.	(b)	22.	(c)	23.	(b)	24.	(d)	25.	(a)	26.	(c)	27.	(c)	28.	(b)	29.	(d)	30.	(b)
31.	(a)	32.	(c)	33.	(c)	34.	(c)	35.	(a)	36.	(d)	37.	(b)	38.	(a)	39.	(a)	40.	(d)
41.	(b)	42.	(a)	43.	(b)	44.	(a)	45.	(d)	46.	(a)	47.	(d)	48.	(a)	49.	(c)	50.	(c)
51.	(a)	52.	(b)	53.	(c)	54.	(a)	55.	(d)	56.	(b)	57.	(d)	58.	(b)	59.	(a)	60.	(c)
61.	(b)	62.	(c)	63.	(a)	64.	(c)	65.	(b)	66.	(a)	67.	(b)	68.	(b)	69.	(b)	70.	(d)
71.	(a)	72.	(b)	73.	(d)	74.	(a)	75.	(a)	76.	(c)	77.	(a)	78.	(b)	79.	(d)	80.	(b)

HINTS AND SOLUTIONS

1. Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$. Then

$$\begin{aligned}\text{curl } \vec{a} &= \nabla \times \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix} \\ &= \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) \vec{i} + \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} \right) \vec{j} \\ &\quad + \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \right) \vec{k}\end{aligned}$$

Now $\text{div curl } \vec{a} = \nabla \cdot (\nabla \times \vec{a})$

$$\begin{aligned}&= \frac{\partial}{\partial x} \left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial a_1}{\partial z} - \frac{\partial a_3}{\partial x} \right) \\ &\quad + \frac{\partial}{\partial z} \left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y} \right)\end{aligned}$$

2. $\text{div} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot \text{curl } \vec{a} - \vec{a} \cdot \text{curl } \vec{b}$
 $= 0 - 0 = 0$

3. $\oint_C F \cdot dr = - \oint_C y dx = \int_0^{2\pi} \sin\theta (-\sin\theta) d\theta = -\pi$

5. $\iint_S (ax^2 + by^2 + cz^2) ds = \iiint_V F \cdot nds$
 $= \iiint_Q \text{div} \cdot (F) dr$
 $= \iiint_r (a + b + c) dv$
 $= (a + b + c) v = (a + b + c) \frac{4\pi}{3}$

6. Compare $9x^2 + 25y^2 - 90x - 150y + 225 = 0$ with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ we get $a = 9$, $h = 0$, $b = 25$, $g = -45$, $f = -75$, $c = 225$,

$$\begin{aligned}\Delta &= abc + 2fgh - af^2 - bg^2 - ch^2 \\ &= 9 \times 25 \times -750 + 0 - 9 \times (750)^2 \\ &\quad - 25(-45)^2 \neq 0\end{aligned}$$

$$h^2 = 0, ab = 25 \times 9 \neq h^2 2ab$$

$\Rightarrow$ Ellipse

15. The equation of tangent at α to conic is $\frac{l}{r} = \cos(\theta - \alpha) + e \cos \theta$

compare with $\frac{l}{r} = A \cos \theta + B \sin \theta$

then eliminate we get

$$(A - e)^2 + B^2 = 1$$

17. $PQ = \sqrt{(1-2)^2 + (-1-3)^2 + (3+5)^2}$
 $= \sqrt{1 + 16 + 64} = 9$

18. $\alpha = \beta = \gamma$,
 $\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$
or $l = m = n$, $\Rightarrow l^2 + m^2 + n^2 = 1$
 $\Rightarrow 3l^2 = 1$ or $l = \pm \frac{1}{\sqrt{3}}$

$$\therefore \left(\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}} \right)$$

26. The S.D. between two plane

$$ax + by + cz - d = 0, a'x + b'y + c'z - d' = 0$$

from z-axis is $\frac{dc' - cd'}{\sqrt{(ac' - a'c)^2 + (bc' - b'c)^2}}$

Here, $a = 1, b = 1, c = 2, d = -3$

$$a' = 2, b' = 3, c' = 4, d' = -4$$

$$= \frac{-12 + 8}{\sqrt{(4-4)^2 + (4-6)^2}} = \frac{-4}{2} = 2 \text{ (Numerically)}$$

27. The equation of plane is $3x - y - z - 7 = 0 \dots (1)$

The d.r. of normal to plane (1) are $(3, -1, -1)$

Hence, the equations of the line passing through $(2, 3, 7)$ and perpendicular to plane (1) is

$$\frac{x-2}{3} = \frac{y-3}{-1} = \frac{z-7}{-1} = r$$

$$\Rightarrow x = 3r + 2, y = -r + 3, z = -r + 7$$

lies on plane (1)

$$3(3r + 2) - (-r + 3) - (-r + 7) - 7 = 0$$

$$9r + 6 + r - 3 + r - 7 - 7 = 0$$

$$11r - 11 = 0 \Rightarrow r = 1$$

Then co-ordinate are (5, 2, 6)

$$29. \quad \frac{x+1}{1} = \frac{y+3}{3} = \frac{z-2}{2} = r$$

$$\Rightarrow P(-1+r, -3+3r, 2+2r)$$

Put in plane $3x + 4y + 5z = 20$

$$\Rightarrow r = 1, \text{ then } P(0, 0, 4)$$

$$37. \quad \text{The equation of sphere through diameter is} \\ (x-2)(x+2) + (y+1)(y-2) + (z-4)(z+2) = 0$$

$$x^2 - 4 + y^2 - y - 2 + z^2 - 2z - 8 = 0$$

$$x^2 + y^2 + z^2 - y - 2z - 14 = 0$$

$$\text{centre} \left(0, \frac{1}{2}, 1 \right)$$

$$40. \quad (x^2 + y^2 + z^2 - 9) + \lambda(x + y - 2z + 4) = 0$$

$$\text{At origin } x = 0, y = 0, z = 0 \Rightarrow 4\lambda = 9, \lambda = \frac{9}{4}$$

$$4(x^2 + y^2 + z^2 - 9) + 9(x + y - 2z + 4) = 0$$

$$4(x^2 + y^2 + z^2) + 9x + 9y - 18z = 0$$

$$42. \quad \text{The equation of the polar plane at } (x_1, y_1, z_1)$$

$$\text{w.r.t to sphere is } xx_1 + yy_1 + zz_1 = a^2$$

$$\text{by comparing } \frac{x_1}{e} = \frac{y_1}{m} = \frac{z_1}{n} = \frac{a^2}{p}$$

$$\Rightarrow \left(\frac{1}{p} a^2, \frac{m}{p} b^2, \frac{n}{p} a^2 \right)$$

$$54. \quad 3x \cdot (1) - 5y(1) + z \cdot 0 + 2 = 0 \text{ or } 3x - 5y + 2 = 0$$

$$56. \quad OP^2 + OQ^2 + OP^2 = (x_1^2 + y_1^2 + z_1^2) \\ + (x_2^2 + y_2^2 + z_2^2) + (x_3^2 + y_3^2 + z_3^2) \\ = a^2 + b^2 + c^2$$

$$72. \quad \text{Unit normal} = \frac{\nabla f}{|\nabla f|} = \frac{-2i + 4j + 4k}{\sqrt{4 + 16 + 16}} \\ = \frac{-1}{3}i + \frac{2}{3}j + \frac{2}{3}k$$

$$73. \quad \text{div. } f = \nabla \cdot f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right)$$

$$\cdot (x^2 yi - 2xzj + 2yzk)$$

$$= 2y(x + 1)$$

$$75. \quad \nabla^2 x = \nabla \cdot \nabla x = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (2xi - 2yj + 4z) \\ = 2 - 2 = 0$$

$$76. \quad \sqrt{x} + \sqrt{y} = \sqrt{a}$$

$$x + y + 2\sqrt{xy} = 9$$

$$2\sqrt{xy} = (a - x - y) \text{ square}$$

$$4xy = a^2 + x^2 + y^2 - 2ax - 2ya + 2xy$$

$$x^2 + y^2 - 2xy + 2ax + 2ay = a^2$$

$$(x - y)^2 + 2ax + 2ay = a^2$$

$\Rightarrow$ Parabola

$$77. \quad \frac{\partial F}{\partial n} = 0, \frac{\partial F}{\partial y} = 0$$

$$\Rightarrow 2x - y - 1 = 0$$

$$-4y - 4 = 0 \Rightarrow y = -1 \text{ then } x = 0$$

$$(0, -1)$$

$$80. \quad \frac{x^2}{\frac{144}{9}} - \frac{y^2}{\frac{144}{16}} = 1$$

$$a^2 = \frac{144}{9}, b^2 = \frac{144}{16}$$

$$\Rightarrow a = \frac{12}{3}, b = \frac{12}{4}$$

$$a = 4, b = 3$$

$$a^2 = b^2(1 - e^2)$$

$$16 = 9(1 - e^2)$$

$$1 - e^2 = \frac{16}{9} \quad e^2 = 1 + \frac{16}{9} = \frac{25}{9}, e = 5/3$$

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RML Avadh University, Examination-2019

Paper-III Geometry and Vector Calculus

[Time : 2 Hours]

[Maximum Marks : 70]

Note : Attempt all questions. Each question carries equal marks.

1. The conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents an ellipse when :
 - (a) $h^2 = ab, \Delta \neq 0$
 - (b) $h^2 - ab < 0, \Delta = 0$
 - (c) $h^2 - ab < 0, \Delta \neq 0$
 - (d) $h^2 - ab > 0, \Delta \neq 0$
2. The centre of the conic $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$ is :
 - (a) (2, 3)
 - (b) (3, 2)
 - (c) (4, 5)
 - (d) (-5, 4)
3. Polar equation of the circle when pole lying on the circumference of the circle is :
 - (a) $r = 2a \cos(\theta - \alpha)$
 - (b) $r = 2a \sin(\theta - \alpha)$
 - (c) $r = 2a \cos \theta$
 - (d) $r = 2a \sin \theta$
4. In any conic the sum of reciprocals of the segment of focal chord is :
 - (a) $1/\ell$
 - (b) $2/\ell$
 - (c) $4/\ell$
 - (d) ℓ
5. The equation of tangent to the conic $\frac{\ell}{r} = 1 + e \cos \theta$ at any point ' α ' is :
 - (a) $\frac{\ell}{r} = e \cos \theta + \cos(\theta + \alpha)$
 - (b) $\frac{\ell}{r} = e \cos \theta + \sin(\theta + \alpha)$
 - (c) $\frac{\ell}{r} = e \cos \theta + \cos(\theta - \alpha)$
 - (d) $\frac{\ell}{r} = e \sin \theta + \sin(\theta - \alpha)$
6. The equation of normal to the curve $\frac{\ell}{r} = 1 + e \cos \theta$ at any point ' α ' is :
 - (a) $\frac{\ell}{r} \frac{e \sin \alpha}{1 + e \cos \alpha} = e \sin \theta + \sin(\theta - \alpha)$
 - (b) $\frac{\ell}{r} \frac{e \cos \alpha}{1 + e \cos \alpha} = e \sin \theta + \sin(\theta - \alpha)$
 - (c) $\frac{\ell}{r} \frac{e \sin \alpha}{1 + e \sin \alpha} = e \cos \theta + \cos(\theta - \alpha)$
 - (d) $\frac{\ell}{r} \frac{e \sin \alpha}{1 + e \cos \alpha} = e \sin \theta - \sin(\theta - \alpha)$
7. The nature of conic $17x^2 - 12xy + 8y^2 + 46x - 28y + 17 = 0$ is :
 - (a) Parabola
 - (b) Pair of straight lines
 - (c) Hyperbola
 - (d) Ellipse
8. The equation of y-axis is :
 - (a) $z = 0$
 - (b) $y = 0, z = 0$
 - (c) $z = 0, x = 0$
 - (d) $x = 0$
9. The angle between two diagonals of cube is :
 - (a) $\cos^{-1} 1/3$
 - (b) $\sin^{-1} 1/3$
 - (c) $\cos 1/3$
 - (d) $\tan^{-1} 1/3$
10. The general equation of plane passing through a point (x_1, y_1, z_1) is :
 - (a) $A(x - x_1) + B(y - y_1) + C(z - z_1) + D = 0$
 - (b) $Ax_1 + By_1 + Cz_1 + D = 0$
 - (c) $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$
 - (d) $Ax \cdot x_1 + By \cdot y_1 + Cz \cdot z_1 + D = 0$

11. The equation of plane through $(-1, 1, 1)$ and $(1, -1, 1)$ and perpendicular to the plane $x + 2y + 2z = 7$ is :
- (a) $x + y + z + 3 = 0$
 (b) $2x + 2y - 3z + 3 = 0$
 (c) $2x + 2y - 3z + 8 = 0$
 (d) $x - y - z + 3 = 0$
12. A variable plane is at a constant distance $3p$ from the origin and meets the axes in A, B and C , then the locus of the centroid of the triangle ABC is :
- (a) $x^2 + y^2 + z^2 = p^2$
 (b) $x^2 + y^2 + z^2 = \frac{1}{p^2}$
 (c) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 9$
 (d) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$
13. The co-ordinates of the points where the line joining the points $(2, -3, 1), (3, -4, -5)$ cuts the plane $2x + y + z = 7$ is :
- (a) $(1, 3, 6)$ (b) $(1, -2, 5)$
 (c) $(2, -3, 6)$ (d) $(1, -2, 7)$
14. The equation of sphere which passes through origin and intercepts length a, b and c on the axis respectively is :
- (a) $x^2 + y^2 + z^2 - ax - by - cz + d = 0$
 (b) $x^2 + y^2 + z^2 - ax - by + d = 0$
 (c) $x^2 + y^2 + z^2 - ax - cz + d = 0$
 (d) $x^2 + y^2 + z^2 - ax - by - cz = 0$
15. A plane passes through a fixed point (p, q, r) and cuts the axis at A, B, C then the locus of centre of the sphere $OABC$ is :
- (a) $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 2$ (b) $\frac{p}{x} + \frac{q}{y} + \frac{r}{z} = 1$
 (c) $\frac{p}{x} + \frac{q}{y} + \frac{r}{z} = 2$ (d) $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 1$
16. The distance between the planes $2x + 2y - z + 2 = 0$ and $4x + 4y - 2z + 5 = 0$ is :
- (a) $1/2$ (b) $1/6$
 (c) $1/4$ (d) $2/7$
17. The image of point $(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$ is :
- (a) $(1, 3, 3)$ (b) $(3, 5, 2)$
 (c) $(3, -5, -2)$ (d) $(-3, 5, 2)$
18. The equation of straight line passing through $(a, 0, b)$ and parallel to x -axis is :
- (a) $\frac{x-a}{0} = \frac{y}{1} = \frac{z-b}{0}$ (b) $\frac{x-a}{1} = \frac{y}{0} = \frac{z-b}{0}$
 (c) $\frac{x-a}{0} = \frac{y}{0} = \frac{z-b}{1}$ (d) $\frac{x-b}{1} = \frac{y}{0} = \frac{z-a}{0}$
19. If two spheres $S_1 \equiv 0$ and $S_2 \equiv 0$ cuts orthogonally then :
- (a) $u_1u_2 + v_1v_2 + 2w_1w_2 = d_1 + d_2$
 (b) $2u_1u_2 + v_1v_2 + w_1w_2 = d_1 + d_2$
 (c) $u_1u_2 + v_1v_2 + w_1w_2 = d_1 + d_2$
 (d) $2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$
20. The equation of radical plane of two spheres $S_1 \equiv x^2 + y^2 + z^2 + 2u_1x + 2v_1y + 2w_1z + d_1 = 0$ and $S_2 \equiv x^2 + y^2 + z^2 + 2u_2x + 2v_2y + 2w_2z + d_2 = 0$
- (a) $x(u_1 - u_2) + y(v_1 - v_2) + z(w_1 + w_2) + (d_1 - d_2) = 0$
 (b) $x(u_2 - u_1) + y(v_2 - v_1) + z(w_2 - w_1) + (d_1 - d_2) = 0$
 (c) $2x(u_1 - u_2) + 2y(v_1 - v_2) + 2z(w_1 - w_2) + 2(d_1 - d_2) = 0$
 (d) $2x(u_1 - u_2) + 2y(v_1 - v_2) + 2z(w_1 - w_2) + (d_1 - d_2) = 0$
21. If $S_1 \equiv 0$ and $S_2 \equiv 0$ be the equation of two spheres, then the equation $S_1 + \lambda S_2 = 0$ represents a sphere except only for :
- (a) $\lambda = 1$ (b) $\lambda = -1$
 (c) $\lambda \neq 1$ (d) $\lambda \neq -1$
22. Every homogenous equation of second degree, $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ where $\Delta \neq 0$, represents :
- (a) A pair of planes (b) A sphere
 (c) A cylinder (d) A cone

23. The equation of cone whose vertex is origin and passes through the curve of intersection of the plane $lx + my + nz = p$ and the surface $ax^2 + by^2 + cz^2 = 1$ is :
- (a) $(ax^2 + by^2 + cz^2) = (lx + my + nz)^2$
 (b) $p^2(ax^2 + by^2 + cz^2) = (lx + my + nz)^2$
 (c) $p^2(ax^2 + by^2 + cz^2)^2 = (lx + my + nz)$
 (d) $p^2(ax^2 + by^2 + cz^2)^2 = (lx + my + nz)^2$
24. The equation $ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + d = 0$ represents a cone if :
- (a) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = d$
 (b) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = \frac{1}{d}$
 (c) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = \frac{1}{d^2}$
 (d) $\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = d^2$
25. The equation of right circular cone whose vertex is origin 0 and axis is z axis and semi vertical angle α :
- (a) $x^2 + z^2 = y^2 \tan^2 \alpha$
 (b) $x^2 + y^2 = z^2 \tan^2 \alpha$
 (c) $x^2 + z^2 = x^2 \tan^2 \alpha$
 (d) None of the above
26. The condition that the plane $lx + my + nz = p$ be the tangent plane to the conicoid $ax^2 + by^2 + cz^2 = 1$ is :
- (a) $p^2 = \frac{\ell^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}$
 (b) $p = \frac{\ell^2}{a} + \frac{m^2}{b} + \frac{n^2}{c}$
 (c) $p^2 = a^2 \ell^2 + b^2 m^2 + c^2 n^2$
 (d) $p^2 = \frac{\ell^2}{a} + \frac{m^2}{b} + \frac{n^2}{c}$
27. Polar plane of (α, β, γ) with respect to ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is :
- (a) $\frac{x}{\alpha a} + \frac{y}{\beta b} + \frac{z}{\gamma c} - 1 = 0$
 (b) $\alpha ax + \beta by + \gamma cz - 1 = 0$
 (c) $\frac{\alpha x}{a^2} + \frac{\beta y}{b^2} + \frac{\gamma z}{c^2} - 1 = 0$
 (d) $\frac{\alpha x^2}{a^2} + \frac{\beta y^2}{b^2} + \frac{\gamma z^2}{c^2} - 1 = 0$
28. The equation of normal to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at (α, β, γ) is :
- (a) $\frac{x - \alpha}{a^2} = \frac{y - \beta}{b^2} = \frac{z - \gamma}{c^2}$
 (b) $\frac{x - \alpha}{\alpha/a^2} = \frac{y - \beta}{\beta/b^2} = \frac{z - \gamma}{\gamma/c^2}$
 (c) $\frac{x - \alpha}{\alpha a^2} = \frac{y - \beta}{\beta b^2} = \frac{z - \gamma}{\gamma c^2}$
 (d) $\frac{x - \alpha}{\alpha^2 a^2} = \frac{y - \beta}{\beta^2 b^2} = \frac{z - \gamma}{\gamma^2 c^2}$
29. The equation of plane parallel to x-axis is
- (a) $ax + by + cz + d = 0$ (b) $ax + by + d = 0$
 (c) $by + cz + d = 0$ (d) $ax + cz + d = 0$
30. The equation of right circular cylinder whose axis is z-axis is :
- (a) $x^2 + y^2 + z^2 = a^2$ (b) $x^2 + y^2 = a^2$
 (c) $y^2 + z^2 = a^2$ (d) $z^2 + x^2 = a^2$
31. The radius of director sphere of the conicoid $\frac{x^2}{4} + \frac{y^2}{15} - \frac{z^2}{10} = 1$ is :
- (a) 2 (b) 3 (c) 4 (d) 5
32. If lines $\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}$ and $\frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$ are coplanar then their point of intersection is :
- (a) (2, 4, -3) (b) (1, -5, 3)
 (c) (1, 3, -2) (d) None of these

33. If the lines $\frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}$, $\frac{x}{a\alpha} = \frac{y}{b\beta} = \frac{z}{c\gamma}$, $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ are coplanar if :
- (a) $\alpha(b - c) + \beta(c - a) + \gamma(a - b) = 0$
 (b) $\frac{l}{\alpha}(b - c) + \frac{m}{\beta}(c - a) + \frac{n}{\gamma}(a - b) = 0$
 (c) $\frac{\alpha}{l}(b - c) + \frac{\beta}{m}(c - a) + \frac{\gamma}{n}(a - b) = 0$
 (d) $a + b + c = 0$
34. The surface $x^2 - y^2 + 2yz - 2zx - x - y + z = 0$ represents :
- (a) A hyperbolic paraboloid
 (b) An elliptic paraboloid
 (c) A parabolic cylinder
 (d) None of the above
35. The angle between the planes $2x - y + z = 7$ and $x + y + 2z = 9$ is :
- (a) $\pi/2$ (b) $\pi/4$
 (c) $\pi/3$ (d) π
36. If the vectors $\hat{i} - 2x\hat{j} + 3y\hat{k}$ and $\hat{i} + 2x\hat{j} - 3y\hat{k}$ are perpendicular then the locus of (x, y) is :
- (a) Straight line
 (b) Pair of straight line
 (c) Rectangular hyperbola
 (d) None of the above
37. $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j}(\vec{a} \times \hat{j}) + \hat{k}(\vec{a} \times \hat{k}) :$
- (a) 0 (b) $\vec{a}$
 (c) $2\vec{a}$ (d) $4\vec{a}$
38. If $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}| = 3, |\vec{b}| = 5, |\vec{c}| = 7$ then the angle between $\vec{a}$ and $\vec{b}$ is :
- (a) $\pi/6$ (b) $2\pi/3$
 (c) $\pi/3$ (d) $4\pi/3$
39. The unit vector perpendicular to the vector $\vec{a} = 2\hat{i} - 6\hat{j} - 3\hat{k}$ and $\vec{b} = 4\hat{i} + 3\hat{j} - \hat{k}$ is :
- (a) $\frac{4\hat{i} + 3\hat{j} - 5\hat{k}}{\sqrt{18}}$ (b) $\frac{3\hat{i} - 6\hat{j} + 5\hat{k}}{7}$
 (c) $\frac{3\hat{i} + 2\hat{j} + 6\hat{k}}{7}$ (d) $\frac{3\hat{i} - 2\hat{j} + 6\hat{k}}{7}$
40. If $\vec{r} = \sin t \hat{i} + \cos t \hat{j} + t \hat{k}$ then $\left| \frac{d^2 \vec{r}}{dt^2} \right|$ is :
- (a) 1 (b) 2
 (c) -1 (d) 0
41. A particle moves along the curve $x = 4 \cos t, y = 4 \sin t, z = 6t$. The magnitude of acceleration at any time t is :
- (a) 2 (b) 1
 (c) 4 (d) 5
42. If $r = |\vec{r}|$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\nabla f(r)$ will be :
- (a) $f(r)\nabla\vec{r}$ (b) $f'(r)\nabla\vec{r}$
 (c) $\frac{1}{2}\nabla f(r)$ (d) $\frac{1}{f'(r)}\nabla\vec{r}$
43. The maximum value of directional derivative of $\phi = x^2yz$ at point $(1, 4, 1)$ is :
- (a) 8 (b) 9
 (c) 6 (d) 7
44. The equation of normal to the surface $xyz = 4$ at $(1, 2, 2)$ is :
- (a) $\frac{x-1}{4} = \frac{y-2}{2} = \frac{z-2}{2}$
 (b) $\frac{x-1}{4} = \frac{y-2}{4} = \frac{z-2}{3}$
 (c) $\frac{x-1}{4} = \frac{y-2}{2} = \frac{z}{3}$
 (d) $\frac{x-1}{4} = \frac{y-2}{2} = \frac{z-1}{2}$
45. If $\vec{F} = yz^2\hat{i} - zx^2\hat{k}$ then $\text{div. } \vec{F} :$
- (a) $-zx^2$ (b) $-x^2$
 (c) yz^2 (d) z^2

46. Find $\text{curl } \vec{F}$ if $F = e^x \sin y \hat{i} + e^x \cos y \hat{j}$:
 (a) 1 (b) 2
 (c) -1 (d) 0
47. For any closed surface $S \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$:
 (a) 0 (b) F
 (c) $2F$ (d) 1
48. Using Stroke' theorem $\oint_C \vec{r} \cdot d\vec{r}$:
 (a) $2\vec{r}$ (b) 0
 (c) $\vec{r}$ (d) 1
49. The shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$, $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is :
 (a) $3/\sqrt{6}$ (b) $\sqrt{6}$
 (c) $1/\sqrt{6}$ (d) 0
50. The enveloping cone of the sphere $x^2 + y^2 + z^2 + 2x - 2y = 2$ with its vertex at (1, 1, 1) is :
 (a) $4x^2 + 3y^2 + 5z^2 + 6yz + 8x + 16z = 0$
 (b) $4x^2 + 3y^2 + 6yz + 8x = 0$
 (c) $4x^2 + 3y^2 - 5z^2 = 0$
 (d) None of the above
51. The pole of the plane $lx + my + nz = p$ with respect to sphere $x^2 + y^2 + z^2 = a^2$ is :
 (a) $\left(\frac{a^2}{pl}, \frac{a^2}{pm}, \frac{a^2}{pn}\right)$ (b) $\left(\frac{a^2l}{p}, \frac{a^2m}{p}, \frac{a^2n}{p}\right)$
 (c) $\left(\frac{l}{a^2p}, \frac{m}{a^2p}, \frac{n}{a^2p}\right)$ (d) None of these
52. The equation of hyperboloid of two sheet is :
 (a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
 (b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
 (c) $\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
 (d) $a^2x^2 + b^2y^2 + c^2z^2 = 1$
53. Guiding curve of a right circular cylinder is :
 (a) Circle (b) Ellipse
 (c) Parabola (d) Pair of straight lines
54. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$ then ∇r^n :
 (a) $nr^{n-1}\vec{r}$ (b) nr^{n-2}
 (c) $nr^{n-2}\vec{r}$ (d) $n(n-2)r^{n-2}\vec{r}$
55. If $\vec{r} = ae^{wt} + be^{-wt}$ where a, b are constant vectors then $\frac{d^2\vec{r}}{dt^2} - w^2\vec{r}$ will be :
 (a) $a b$ (b) $a^2 + b^2$
 (c) a^2b^2 (d) 0
56. Confocal cut at an angles :
 (a) $\pi/4$ (b) $\pi/3$
 (c) $\pi/6$ (d) None of these
57. The foci of the confocal $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$ are :
 (a) $(\pm \sqrt{a^2 - b^2}, 0)$ (b) $(0, \pm \sqrt{a^2 - b^2})$
 (c) $(\pm \sqrt{a^2 - b^2 - 2\lambda}, 0)$ (d) None of these
58. The number of real umbilics of an ellipsoid is :
 (a) Two (b) Three (c) Four (d) Zero
59. The slope of tangent at ' α ' to the conic $\frac{1}{r} = 1 + e \cos \theta$ is :
 (a) $-\frac{\cos \theta + e}{\sin \alpha}$ (b) $\frac{\cos \alpha + e}{\sin \alpha}$
 (c) $\frac{\sin \alpha}{\cos \alpha + e}$ (d) $-\frac{\sin \alpha}{\cos \alpha + e}$
60. The semi-latus rectum of the conic $\frac{15}{r} = 5 - 4 \cos \theta$ is :
 (a) 3 (b) 15 (c) 6 (d) 5
61. The direction-cosines of a line whose dr's are 2, 3, -6 are :
 (a) $2, 3, -6$ (b) $\frac{2}{7}, \frac{3}{7}, \frac{-5}{7}$
 (c) $\frac{2}{7}, \frac{3}{7}, \frac{-6}{7}$ (d) $\frac{1}{7}, \frac{2}{7}, \frac{3}{7}$

62. If $l_1 = l_2$, $m_1 = m_2$ and $n_1 = n_2$, where $l_1, m_1, n_1, l_2, m_2, n_2$ are dc's of two lines, then the lines are :
 (a) Perpendicular (b) Parallel
 (c) Make an angle 30° (d) None of these
63. The angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$ is :
 (a) $\pi/6$ (b) $\pi/3$
 (c) $\pi/4$ (d) $\pi/2$
64. The intercept on x-axis of the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ is :
 (a) b (b) a
 (c) 1 (d) c
65. The angle between the lines $\pi/4$, then the lines are :
 (a) Coplanar (b) Non-Coplanar
 (c) Collinear (d) None fo these
66. The centres of all the spheres of coaxial system lie on :
 (a) Straight line (b) Sphere
 (c) Cone (d) None of these
67. The number of spheres that are touching the Co-ordinate axis are :
 (a) Infinite (b) 8
 (c) 1 (d) None of these
68. The radical plane of two spheres cuts the line joining their centres at the angle :
 (a) $\pi/4$ (b) $\pi/3$
 (c) $\pi/2$ (d) $\pi/6$
69. Two spheres with centres C_1 and C_2 of radii r_1 and r_2 respectively touch externally if C_1C_2 equals :
 (a) r_1 (b) r_2
 (c) $r_1 + r_2$ (d) $r_1 - r_2$
70. The equation of a cone reciprocal to the cone $\sqrt{fx} + \sqrt{gy} + \sqrt{hz} = 0$ is :
 (a) $fyz + gzx + hxy = 0$ (b) $fyz - gzx - hxy = 0$
 (c) $fx + gy + hz = 0$ (d) None of these
71. The vertex of the cone $fyz + gzx + hxy = 0$ is :
 (a) (0, 0, 0) (b) (1, 1, 1)
 (c) (0, 1, 0) (d) (0, 0, 1)
72. The generators of the cylinder $f(y, z) = 0$ are parallel to the axis :
 (a) x-axis (b) y-axis
 (c) z-axis (d) None of these
73. The number of normals that can drawn to an ellipsoid from a given point are :
 (a) 4 (b) 2
 (c) 6 (d) 3
74. The equation $ax^2 + by^2 + cz^2 = 1$ represents an ellipsoid if :
 (a) a, b are positive (b) $a \geq 0, b > 0, c > 0$
 (c) $a < 0, b \geq 0, c \geq 0$ (d) None of these
75. The centre of the surface $x^2 + y^2 + z^2 = 1$ is :
 (a) (1, 1, 3) (b) (1, 1, 1)
 (c) (0, 0, 0) (d) (3, 1, 1)
76. Ruled surfaces are generated by a :
 (a) Moving line (b) Moving point
 (c) Fixed line (d) Fixed point
77. Locus of the point of intersecion of perpendicular generators of the hyperboloid of one sheet is a :
 (a) Hyperboloid of two sheets
 (b) Director sphere
 (c) Paraboloid
 (d) Plane
78. If a and b are irrotational, then $\text{div. } (a \times b)$ is equal to :
 (a) a (b) b
 (c) 0 (d) $a + b$
79. The vector $r^n \underline{r}$ is Solenoidal if n equals :
 (a) 3 (b) -3 (c) 2 (d) 0
80. The $\text{div. curl } \underline{a}$ is equal to :
 (a) 0 (b) 1
 (c) -1 (d) a
81. Gradient of a constant scalar field is equal to :
 (a) 0 (b) 1
 (c) -1 (d) None of these

82. If a vector V is Solenoidal the div. V is equal to :
 (a) 1 (b) -1 (c) 2 (d) 0
83. $\int_s F \times n ds$ is equal to :
 (a) $\int_v \nabla \times F dV$ (b) $-\int_v \nabla \times F dV$
 (c) $\int_v \nabla F dV$ (d) $-\int_v \nabla F dV$
84. Value of $\int_C r \cdot dr$ is :
 (a) 0 (b) $2\int_s ndr$
 (c) Unit vector (d) Indeterminate
85. A relation between surface and volume integrals is :
 (a) Stoke's Th^m (b) Gauss Th^m
 (c) Green's Th^m (d) Bernoulli's Th^m
86. Curl grad ϕ is equal to :
 (a) 2 (b) 3 (c) 0 (d) 4
87. div. $\underline{r}$ is equal to :
 (a) 2 (b) 3 (c) 3.5 (d) 4
88. To the surface $f(x, y, z) = c$, ∇f represents :
 (a) Vector Normal (b) Vector Tangent
 (c) Unit Normal (d) None of these
89. A central conic has four foci of which :
 (a) Two are real and two imaginary
 (b) All the four are real
 (c) All the four are imaginary
 (d) None of the above
90. The locus of the point from which two perpendicular tangents can be drawn one to each of two given confocals :
 (a) Pair of straight line (b) Circle
 (c) Confocal conic (d) An Ellipse
91. The line $\frac{l}{r} = \sin \theta + e \cos \theta$ is a tangent line to the conic $\frac{l}{r} = \underline{r} + e \cos \theta$ at the point :
 (a) $\pi/6$ (b) $\pi/3$
 (c) $\pi/4$ (d) None of these
92. The plane $ax + by + cz = 1$ meets the co-ordinates axes in A, B, C. The centroid of the triangle is :
 (a) $(3a, 3b, 3c)$ (b) $(a/3, b/3, c/3)$
 (c) $(3/a, 3/b, 3/c)$ (d) $(1/3a, 1/3b, 1/3c)$
93. The equation of the plane through the Intersection of two planes $P = 0$ and $Q = 0$ will be :
 (a) $PQ = 0$ (b) $P = \lambda Q = 0$
 (c) $P/\lambda Q = 0$ (d) $P = Q = 0$
94. The angle between the planes $2x - y + z = 16$, $x + y + 2z = 7$ is :
 (a) $\cos^{-1}(4/\sqrt{21})$ (b) $\pi/3$
 (c) $\pi/4$ (d) $\pi/2$
95. Two lines which do not lie in the same plane are called :
 (a) Parallel (b) Intersecting
 (c) Co-incident (d) Skew
96. Find the equation of the sphere described on the join $(2, -3, 4)$ and $(-5, 6, -7)$ as diameter :
 (a) $x^2 + y^2 + z^2 = 1$
 (b) $x^2 + y^2 + z^2 + 2x - 3y + 3z = 0$
 (c) $x^2 + y^2 + z^2 + 3x - 3y + 3z - 56 = 0$
 (d) $x^2 + y^2 + z^2 + 5x + 7z - 56 = 0$
97. The radius of the sphere $2x^2 + 2y^2 + 2z^2 = 50$ is :
 (a) 25 (b) 5 (c) $5\sqrt{5}$ (d) 2
98. The centre of the sphere $x^2 + y^2 + z^2 - ax - by - cz = 0$ is :
 (a) $(0, 0, 0)$ (b) $(a/2, b/2, c/2)$
 (c) $(\frac{-a}{2}, \frac{-b}{2}, \frac{-c}{2})$ (d) (a, b, c)
99. The plane section of a sphere cut by the plane through the centre of the sphere is :
 (a) Circle (b) Sphere
 (c) Parabola (d) Great Circle
100. The semi vertical angle of a right circular cone having a set of three mutually perpendicular generators is :
 (a) $\tan^{-1} 2$ (b) $\tan^{-1} \sqrt{2}$
 (c) $\tan^{-1} \frac{1}{\sqrt{2}}$ (d) $\pi/4$

ANSWERS

MULTIPLE CHOICE QUESTIONS

1.	(c)	2.	(a)	3.	(c)	4.	(b)	5.	(c)	6.	(a)	7.	(d)	8.	(c)	9.	(a)	10.	(c)
11.	(b)	12.	(d)	13.	(d)	14.	(d)	15.	(b)	16.	(b)	17.	(d)	18.	(b)	19.	(d)	20.	(d)
21.	(d)	22.	(d)	23.	(b)	24.	(a)	25.	(b)	26.	(a)	27.	(c)	28.	(b)	29.	(c)	30.	(b)
31.	(b)	32.	(d)	33.	(c)	34.	(a)	35.	(c)	36.	(d)	37.	(c)	38.	(c)	39.	(c)	40.	(a)
41.	(c)	42.	(b)	43.	(b)	44.	(a)	45.	(b)	46.	(d)	47.	(a)	48.	(b)	49.	(d)	50.	(d)
51.	(b)	52.	(c)	53.	(d)	54.	(c)	55.	(d)	56.	(d)	57.	(a)	58.	(c)	59.	(a)	60.	(b)
61.	(c)	62.	(b)	63.	(b)	64.	(b)	65.	(d)	66.	(a)	67.	(b)	68.	(c)	69.	(c)	70.	(a)
71.	(a)	72.	(a)	73.	(c)	74.	(d)	75.	(c)	76.	(a)	77.	(b)	78.	(c)	79.	(b)	80.	(a)
81.	(a)	82.	(d)	83.	(a)	84.	(a)	85.	(a)	86.	(c)	87.	(b)	88.	(a)	89.	(a)	90.	(c)
91.	(d)	92.	(b)	93.	(b)	94.	(b)	95.	(d)	96.	(d)	97.	(b)	98.	(b)	99.	(a)	100.	(b)

HINTS AND SOLUTIONS

2. Let $f(x, y) = 14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$

$$\frac{\partial F}{\partial x} = 28x - 4y - 44 = 0$$

$$\frac{\partial F}{\partial y} = -4x + 22y - 58 = 0$$

solve we get $x = 2, y = 3$

3. The equation of circle with pole on its circumference is $r = 2a \cos \theta$.

4. We have $\frac{1}{2} \left[\frac{1}{SP} + \frac{1}{SP'} \right] = \frac{1}{l}$

or $\frac{1}{SP} + \frac{1}{SP'} = \frac{2}{l}$

6. The equation of normal at the point P in term of α is

$$\frac{l e \sin \alpha}{r(1 + e \cos \alpha)} = \sin(\theta - \alpha) + e \sin \theta$$

7. Compare $17x^2 - 12xy + 8y^2 + 46x - 28y + 17 = 0$

where $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$a = 17, h = -6, b = 8, g = 23, f = -14, c = 17$

and $h^2 < ab, \Delta \neq 0$ so it is ellipse.

9. Let θ be angle between two diagonal OP and AC with d.c are $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$ $\left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$ then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{1}{3} \right)$$

11. The equation of plane passing through $(-1, 1, 1)$ is

$$a(x + 1) + b(y - 1) + c(z - 1) = 0 \quad \dots(1)$$

It passed $(1, -1, 1)$

$$\Rightarrow 2a - 2b + oc = 0 \quad \dots(2)$$

and perpendicular to $x + 2y + 2z = 0$

then $a + 2b + 2c = 0$

$\therefore 2a - 2b + 2c = 0$

$a + 2b + 2c = 0$

$$\frac{a}{-4} = \frac{b}{-4} = \frac{c}{4+2}$$

$$\frac{a}{-4} = \frac{b}{-4} = \frac{c}{6} \text{ or } \frac{a}{2} = \frac{b}{2} = \frac{c}{-3}$$

$\therefore 2(x+1) + 2(y-1) - 3(z-1) = 0$

$\Rightarrow 2x + 2 + 2y - 2 - 3z + 3 = 0$

$\Rightarrow 2x + 2y - 3z + 3 = 0$

16. The distance between the given parallel plane = length of perpendicular from P to the plane

$$= \left| \frac{2}{3} - \frac{5}{6} \right| = \left| \frac{4-5}{6} \right| = \left| \frac{-1}{6} \right| = \frac{1}{6}$$

17. The given plane is $2x - y + z + 3 = 0$... (1)

d.c. of line perpendicular to the plane (1) are $(2, -1, 1)$ let Q be image of given point. Then the equation of line PQ passing $P(1, 3, 4)$ perpendicular to plane (1) are

$$\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = \lambda \quad \dots (2)$$

$\Rightarrow Q(2\lambda + 1, -\lambda + 3, \lambda + 4) \quad \dots (3)$

$\therefore$ co-ordinate of middle point PQ is

$$\left(\lambda + 1, \frac{-1}{2}\lambda + 3, \frac{\lambda}{2} + 4 \right)$$

put in (3) $\lambda = -2$

$\therefore Q(-3, 5, 2)$

23. $ax^2 + by^2 + cz^2 = 1 \quad \dots (1)$

make homogenous equation of two degree with the help $lx + my + nz = p$

$\Rightarrow \frac{lx + my + nz}{p} = 1$

$$\therefore ax^2 + by^2 + cz^2 = \left(\frac{lx + my + nz}{p} \right)^2$$

$\Rightarrow (ax^2 + by^2 + cz^2)p^2 = (lx + my + nz)^2$

24. Let $F(x, y, z) = ax^2 + by^2 + cz^2 + 24x + 2y + 2wz + d = 0 \quad \dots (1)$

$$\Rightarrow F(x, y, z, t) = ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + dt^2$$

Put $t = 0$ in

$$\therefore \frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0, \frac{\partial F}{\partial t} = 0$$

we get $x = -u/a, y = -v/b, z = -w/c$

put these values in $\frac{\partial F}{\partial t} = 0$ we get

$$\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = d$$

25. Here $l = 0 = m, l = 1$

$\therefore$ Then from Lagrange's identity

$$x^2(x^2 + y^2 + z^2) \sin^2 \theta = x^2 y^2 + x^2 z^2$$

or $x^2 + y^2 = z^2 \tan^2 \theta$

26. The equation of tangent plane at (x_1, y_1, z_1) to conic $ax^2 + by^2 + cz^2 = 1$ is

$$axx_1 + byy_1 + czz_1 = 1 \quad \dots (1)$$

compare with $lx + my + nz = p$

we get $x_1 = \frac{l}{ap}, y_1 = \frac{m}{bp}, z_1 = \frac{n}{cp}$

put in $ax_1^2 + by_1^2 + cz_1^2 = 1$

we get $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$

28. The equation of normal to the given ellipsoid at (α, β, γ) are

$$\frac{x-\alpha}{\alpha/a^2} = \frac{y-\beta}{\beta/b^2} = \frac{z-\gamma}{\gamma/c^2}$$

29. If the plane $ax + by + cz + d = 0$ parallel to x -axis then normal to the plane will be right angle to x -axis, hence

$$a \times 1 + b \times 0 + c \times 0 \text{ or } a = 0$$

then $by + cz + d = 0$

33.
$$\begin{vmatrix} \alpha & \beta & \gamma \\ l & m & n \\ a\alpha & b\beta & c\gamma \end{vmatrix} = 0$$

$$\Rightarrow \frac{l}{\alpha}(b-c) + \frac{m}{\beta}(c-a) + \frac{n}{\gamma}(a-b) = 0$$

$$34. \quad \cos \theta = \frac{(2)(1) + (-1)(1) + (1)(2)}{\sqrt{(2)^2 + (-1)^2 + (1)^2} \sqrt{1^2 + 1^2 + 2^2}} = \frac{1}{2}$$

$$\theta = \pi/3$$

$$37. \quad \vec{i} \times (\vec{a} \times \vec{i}) + \vec{j} \times (\vec{a} \times \vec{j}) + \vec{k} \times (\vec{a} \times \vec{k})$$

$$= \begin{vmatrix} a & i \\ i & a & 1 \end{vmatrix} + \begin{vmatrix} a & j \\ j & a & 1 \end{vmatrix} + \begin{vmatrix} \vec{a} & k \\ k & a & 1 \end{vmatrix}$$

$$= a = a_1i + a_2j + a_3k$$

$$= a - i(i \cdot a) + a - j(j \cdot a) + a - k(k \cdot a)$$

$$= 3a - a_1i - a_2j - a_3k$$

$$= 3a - a = 2a$$

$$38. \quad a + b = -c$$

$$(a + b) \cdot (a + b) = c \cdot c$$

$$|\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}|\cos\theta = |c|^2$$

$$9 + 25 + 2 \times 3 \times 5 \cos\theta = 49$$

$$\cos\theta = 1/2, \theta = 60 = \pi/3$$

$$39. \quad \text{Let } c = c_1i + c_2j + c_3k$$

$$\Rightarrow |c| = \sqrt{c_1^2 + c_2^2 + c_3^2}$$

$$\Rightarrow c_1^2 + c_2^2 + c_3^2 = 1 \quad \dots(1)$$

$$c \text{ is perpendicular to } A \text{ and } B, A = 2i - 6j - 3k$$

$$B = 4i + 3j - k$$

$$\Rightarrow C \cdot A = 0, C \cdot B = 0$$

$$\Rightarrow 2c_1 - 6c_2 - 3c_3 = 0$$

$$4c_1 + 3c_2 - c_3 = 0$$

$$\frac{c_1}{6+9} = \frac{c_2}{-12+2} = \frac{c_3}{6+24}$$

$$\frac{c_1}{15} = \frac{c_2}{10} = \frac{c_3}{30} = k$$

$$\text{or } \frac{c_1}{3} = \frac{c_2}{2} = \frac{c_3}{6} = k$$

$$\Rightarrow c_1 = 3k, c_2 = 2k, c_3 = 6k$$

$$9x^2 + 4x^2 + 36x^2 = 1$$

$$49x^2 = 1, x = \frac{1}{7}$$

$$\therefore c = \frac{3i + 2j + 6k}{7}$$

$$40. \quad r = \sin ti + \cos tj + kt$$

$$\frac{dr}{dt} = \cos t i - \sin t j + k$$

$$\frac{d^2r}{dt^2} = -\sin t i - \cos t j$$

$$\left| \frac{d^2r}{dt^2} \right| = \sqrt{\sin^2 t + \cos^2 t} = 1$$

$$41. \quad r = 4\cos ti + 4\sin tj + 6tk$$

$$v = \frac{dr}{dt} = -u \sin ti + 4\cos tj + 6k$$

$$\vec{a} = \frac{d^2r}{dt^2} = -4\cos ti - 4\sin tj + 0$$

$$|\vec{a}| = \sqrt{16\cos^2 t + 16\sin^2 t} = 4$$

$$43. \quad \phi = x^2yz, \frac{\partial \phi}{\partial x} = 2xyz, \frac{\partial \phi}{\partial y} = x^2z, \frac{\partial \phi}{\partial z} = x^2y$$

$$\text{at } (1, 4, 1), \frac{\partial \phi}{\partial x} = 8, \frac{\partial \phi}{\partial y} = 1, \frac{\partial \phi}{\partial z} = 4$$

maximum value of directional derivative

$$|\nabla \phi| = |8i + j + 4k| = \sqrt{64 + 1 + 16} = 9$$

$$45. \quad F = yz^2i - zx^2k$$

$$\nabla \cdot F = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right) \cdot (yz^2i - zx^2k) = 0 - x^2 = -x^2$$

$$46. \quad F = e^x \sin yi + e^x \cos yj$$

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x \sin y & e^x \cos y & 0 \end{vmatrix}$$

$$= i(0 - 0) + j(0 - 0) + k(e^x \cos y - e^x \sin y)$$

$$= 0$$

$$52. \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \text{ (Ellipsoid)}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \text{ (Hyperboloid in one sheet)}$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \text{ (Hyperboloid in two sheets)}$$

60. $\frac{l}{r} = 1 + e \cos \theta, l = \text{semi latus rectum.}$

63. $\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} = \frac{1}{2}$

$$\theta = \pi / 6$$

71. Here $a = f^2, b = g^2, c = h^2, f = -gh$

$$g = -fh, h = -fg$$

$$A = bc - f^2 = g^2 h^2 - g^2 h^2 = 0, B = 0, C = 0$$

$$F = gh - af = 2f^2 gh, G = 2g^2 hf, H = 2h^2 fg$$

Put in

$$Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gzx + 2Hxy = 0$$

$$\Rightarrow fyz + gzx + hxy = 0$$

73. 6 normals can be drawn to conicoids

98. The centre of sphere

$$x^2 + y^2 + z^2 + 24x + 2vy + 2wz + d = 0 \text{ is}$$

$$\left(\frac{-u}{2}, \frac{-v}{2}, \frac{-w}{2} \right)$$

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